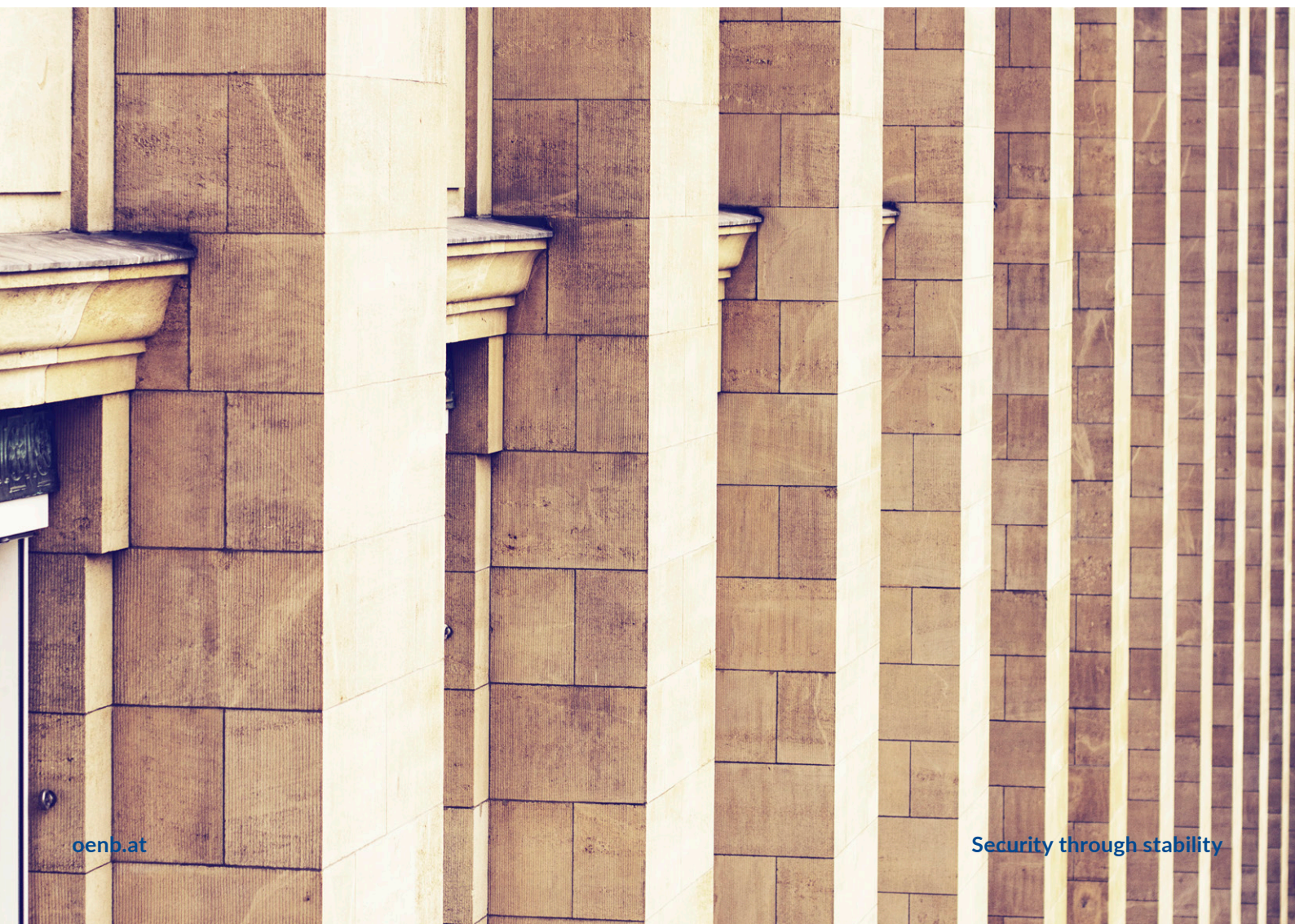


OeNB Financial Literacy Evaluation Series

Rigorous and Comprehensive Evaluation of Financial Education and Literacy –
Collected Articles from 2024–2025



Acknowledgements

The OeNB Financial Literacy Evaluation Series provides researchers, policymakers, and educators with insights into current developments in financial literacy and education. It offers guidance for the design and implementation of comprehensive evaluation studies. This volume brings together contributions written in 2024 and 2025. For more information and additional articles in this series, visit <https://finanzbildung.oenb.at/evaluationstoolkit>.

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Foreword

Robert Holzmann, Governor of the Oesterreichische Nationalbank

Financial literacy is an essential skill for the 21st century. We live in an era of growing globalization and digitalization, with increasingly complex financial products and heightened economic and financial uncertainty – navigating the financial landscape has become more challenging than ever. Policymakers around the world are placing great emphasis on initiatives and programs that equip individuals with the knowledge and skills needed to manage their finances effectively, so they can build financial resilience and ultimately achieve financial well-being.

However, financial literacy is not just a personal asset – it is a collective necessity. When individuals understand fundamental financial and economic concepts such as inflation, interest rates and investments, they make more informed decisions that enhance the effectiveness of monetary policy. Central banks can and should play a critical role in building financial literacy. After all, interest rates are among the key tools of monetary policy. Their effectiveness in part relies on businesses' and households' understanding of interest rates and their influence on borrowing, saving spending and inflation.

Moreover, by promoting financial literacy, central banks can help mitigate risks associated with panic-driven market reactions, speculative bubbles and unsustainable debt accumulation. As independent institutions, central banks are uniquely positioned to develop and implement trustworthy financial education initiatives without commercial bias.

Between 2008 and 2013, during my tenure at the World Bank, I led its most comprehensive financial literacy initiative to date. The project's primary objective was to generate knowledge and evidence to support low- and middle-income countries in the implementation of national financial literacy strategies. As part of this effort, we developed a financial capability framework, identified four key content domains of financial capability – managing money, planning ahead, making choices and getting help – and promoted rigorous measurement, monitoring and evaluation of financial education programs.

Since then, the financial literacy landscape has evolved significantly, gaining even more attention from policymakers and researchers. The number of countries with national strategies for financial literacy has grown substantially and research in this field has expanded rapidly, as has the number of impact evaluations. Moreover, financial literacy is now widely understood as a component of financial well-being in a broader sense, which has become the primary objective for many financial education providers.

In response to these developments, I initiated the OeNB Financial Literacy Evaluation Series – a comprehensive guide that aims to inform researchers, policymakers and educators about the current state of evaluation research on financial literacy and education. It provides an overview on the concept of financial literacy, reviews literature addressing the effectiveness of financial education programs and gives guidelines for designing and implementing comprehensive evaluation studies.

The relationship between evaluation research and policy progress may sometimes appear paradoxical. The more rigorous the evaluation, the more inconclusive or even disappointing results sometimes may seem. This paradox is partly due to the fact that evaluations often employ sophisticated econometric models but frequently lack critical insights into the processes and experiences behind the measured effects. Fundamental questions – such as whether the intervention was implemented as intended and whether it addressed relevant needs – are often insufficiently explored. The same holds for underlying causal mechanisms that link program inputs to outcomes.

I therefore advocate more comprehensive evaluation approaches, where monitoring and process evaluation are integrated alongside impact assessments, also incorporating qualitative and mixed-methods approaches. I am certain that such a broader approach to monitoring and evaluation will provide the critical feedback needed to adapt strategies and refine interventions, ultimately leading to more effective, evidence-based financial literacy policies. In this spirit, the OeNB Financial Literacy Evaluation Series consolidates research insights, best practices and guidelines for rigorous and comprehensive evaluations and aims to serve as an essential tool to support evidence-based decision-making and advance the field of financial literacy.

Vienna, May 2025

A handwritten signature in black ink, appearing to read 'R. Holzmann', with a long horizontal flourish extending to the right.

Robert Holzmann, Governor of the Oesterreichische Nationalbank

Part I:

Introduction to financial literacy and related concepts



Financial literacy in research and policymaking: the concept at the core

Maximilian Zieser, Valentin Voith, Theresa Lorenz and Sofia Anyfantaki¹

Financial literacy has become an increasingly important research and policy topic in the past decades and has been adopted by a multitude of stakeholders. However, there is little consensus on how to define financial literacy, how to measure it or how empirically relevant it is. For related financial education interventions, evaluations and research to be successful, however, it is essential to have a clear idea of the underlying concept and its potential implications. This paper aims to provide both an overview of and guide to the concept by exploring six perspectives of financial literacy: its historical origins, its key stakeholders, its various definitions, the approaches to its measurement, its determinants and its potential impact. Indeed, we find that concepts of financial literacy, and thus its implications for research and policymaking, vary considerably. We therefore recommend using coherent definitions and a rigorous approach to measurement and reporting to ensure that future evaluation and research projects produce meaningful, interpretable and comparable results.

Many public and international institutions such as central banks, the Organisation for Economic Co-operation and Development (OECD) or the World Bank have embraced financial education and financial literacy as policy goals, stressing their importance for people's personal financial well-being and their potential wider implications for economic and financial stability. There has also been a notable surge in academic interest in financial literacy, with a tenfold increase in related publications between 2014 and 2022 (Zaimovic et al., 2023). Moreover, attention to the topic extends far beyond policymaking and research, encompassing the media (e.g. Barrett, 2023; Pfluger, 2023), private banks (e.g. Erste Bank und Sparkasse, 2023; Santander, 2023) and even social media platforms ("finfluencers" in particular, see e.g. Haase et al., 2023).

At the same time, related research is characterized by considerable heterogeneity (Goyal and Kumar, 2021; Ingale and Paluri, 2022), with a large number of different definitions having been proposed by researchers and policymakers alike (see Cude, 2022). In addition, a great variety of measurement approaches are in use (Ouachani et al., 2021). While these developments have greatly extended the scope of the concept of financial literacy, they may prevent the direct comparison of empirical results and the drawing of clear conclusions on the effectiveness of financial education and the relevance of financial literacy. Researchers, policymakers and educators thus face fundamental questions: What exactly is financial literacy? How can it be measured? How can it be improved? How does it affect people's lives?

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While this paper cannot provide conclusive answers to these questions, it aims to distinguish different definitions of financial literacy and approaches to measure it, help assess its empirical relevance and enable targeted decisions in defining and quantifying it. To this end, we explore six different perspectives of financial literacy, namely its historical origins, its key stakeholders, its various definitions, the approaches to its measurement, its determinants and its potential impact.

In section 1, we establish the historical and institutional background to help readers understand the circumstances under which the concept of financial literacy has been shaped and is being promoted. We find that questions of economically efficient and sustainable decision-making were already of interest to ancient Greek philosophers. The concept of financial literacy in its modern sense, however, has only recently emerged as a focal point of international organizations, national institutions and empirical research.

Building on section 1, section 2 introduces the major players in the field of financial literacy and financial education. Besides discussing related work by influential international organizations such as the OECD and the World Bank, we provide brief insights into public efforts at the national level and critically examine the role of private financial education providers.

In section 3, we consider fundamental theoretical questions concerning financial literacy, namely its different definitions and their potential implications. We find that the concept of financial literacy has evolved to include a wide range of traits and behaviors, with many definitions now asserting financial well-being as the central outcome of financial literacy. We provide an overview of popular definitions of financial literacy and closely related concepts and conclude with a summary of key differences in the central theoretical assumptions underlying these definitions.

Section 4 shows how different notions of financial literacy are also reflected in the range of measurement approaches used. For example, well-established knowledge tests have been adapted to also cover respondents' financial behavior and attitude, as in the International Survey of Adult Financial Literacy (ISAFL) of the OECD's International Network on Financial Education (INFE). On the other hand, competence-oriented instruments such as the OECD's Programme for International Student Assessment (PISA) financial literacy test focus on capturing financial literacy as a latent trait. We also show that seemingly minor differences in methodology can, in the aggregate, produce strikingly different results. We conclude section 4 with critical methodological considerations regarding the measurement of financial literacy.

Sections 5 summarizes the existent research on the most common determinants of financial literacy. We find that various sociodemographic and socioeconomic factors, along with macro-level factors, appear to significantly affect financial literacy. In addition, empirical studies paint a mixed picture of the role financial education plays: While the effects of financial education on people's financial behavior have been described as small and declining quickly over time (Fernandes et al., 2014), more current evidence points to significant and meaningful effects (Kaiser et al., 2022). Any such conclusions, however, depend heavily on the studies and financial education interventions considered.

Section 6 briefly explores effects of financial literacy on financial behavior, financial resilience and financial vulnerability as well as its impact on financial well-being. While significant links have been found between financial literacy and financial well-being, other context factors appear to be the predominant determinants of financial well-being (e.g. ANZ Bank New Zealand Limited, 2021). For reliable causal conclusions concerning the effects of financial education on financial well-being, more research and rigorous evaluations are needed.

Although we find remarkable progress in the field, it comes with considerable ambiguity in the definitions of financial literacy itself, in measurement approaches and in empirical results. With this paper, we

therefore aim to provide a starting point and guide for researchers, policymakers and educators working in this field. We do not attempt to replicate the scope or depth to be found in handbooks on financial literacy (in particular Aprea et al., 2016; Nicolini and Cude, 2022) or topic-specific literature reviews (e.g. Compen et al., 2019; Lyons and Kass-Hanna, 2021). Instead, we aim to give a structured, accessible overview of the concept at the core of a vast field of research and policymaking.

1 Origins and evolution of the concept of “financial literacy”

Questions on the efficient use of personal resources to achieve, maintain and increase individual prosperity are as old as market economy itself. One of the earliest systematic discussions of the subject is the philosophical dialogue *Oikonomikos* (*The Economist*) by Greek philosopher Xenophon, which dates back to the 4th century BCE, presumably. In line with the original meaning of the word “economics,” this work deals with the proper management of a house or estate (Xenophon, ca. 360 BCE [2008]). While it discusses fundamental concepts in the context of an ancient society, it provides timeless insights that still appear relevant today.

Throughout the modern era, practice-oriented scholars occasionally gave advice on financial affairs as well: In one of his famous essays, Renaissance philosopher Francis Bacon highlighted the importance of controlling one’s expenses and regularly saving a certain share of one’s income for a rainy day (Bacon, 1596 [2018]). In a similar vein, Benjamin Franklin published “Necessary hints to those that would be rich,” pointing, inter alia, to the additional costs that arise when buying things on credit (Franklin, 1736 [1794]).

There is also a long tradition of advice literature on the proper handling of personal finances for different fields of application, including guidebooks for entrepreneurs, e.g. *The Art of Money Getting* (Barnum, 1880 [2013]), for administrators, e.g. *The Young Clerk’s Manual* (Unknown, 1848), or for housekeepers, e.g. *The American Frugal Housewife* (Child, 1828 [2020]). Laying the cornerstone of a whole new genre which enjoys great popularity to this day, George S. Clason’s book *The Richest Man in Babylon* (1926) can be regarded as the first modern classic of personal financial advice literature. There, by means of illustrative parables, readers learn about the “Seven Cures for a Lean Purse” and the “Five Laws of Gold,” which address, for example, the temptation of lifestyle inflation, the use of compound interest in investing or the dangers of scams. Like in its contemporary counterparts, the rules and principles taught are mainly based on anecdotal evidence and practical experience rather than on a rigorous scientific approach.

Throughout the 20th century, the introduction of consumer protection regulations was accompanied by consumer education programs, e.g. by the Federal Trade Commission in the USA or the Citizens Advice Bureau in the UK, which also aimed to promote prudent financial behaviors among the wider population. Before the term “financial literacy” was coined, pioneering academic work was carried out in university programs comprising research and courses on *consumer economics* or *household finances* (Hira, 2009). It was not until the late 1990s and early 2000s that financial literacy was gradually established as an autonomous research field (Faulkner, 2015).

The Anglo-Saxon countries led the way in developing financial literacy, both as a concept and as a topic in national policies. In New Zealand, coordinated financial literacy policies were first sparked in 1992 through a Taskforce on Private Saving for Retirement (Cameron and Wood, 2016). One of the first financial literacy initiatives by name, dating back to 1995, was the Jump\$tart Coalition for Personal Financial Literacy, which brought together private, nonprofit and public stakeholders to foster the integration of financial education in school curricula and provide expertise for policymakers in the USA (Jump\$tart, 2024). In 2003, the UK Financial Services Authority began to develop the first national strategy for financial

capability. While these early initiatives shared the common goal of empowering people to make informed financial decisions, their emergence is also closely linked to the growing trend of household savings being invested in the stock market during and after the economic boom of the 1990s (Wolf, 2018).

The idea of educating consumers in financial literacy quickly gained momentum among established international organizations, especially the World Bank and the OECD, with the latter establishing its International Network on Financial Education (INFE) in 2008 (see Kovács and Terták, 2019) and each organization developing and further refining the concept of financial literacy as such (see sections 2 and 3).

During this period, the significance of financial literacy increased further as financial products became increasingly complex and digitalized (Faulkner, 2015). While not isolated from other developments, the global financial crisis of 2007 and 2008 can be viewed as an important catalyst of financial literacy policies as private households were seen not only as victims but, given their poor understanding of financial products, also as an important factor contributing to the crisis (see Kovács and Terták, 2019).

Besides the fact that attention in financial literacy increased, the concept itself also underwent significant progress in terms of definition and scope. Until the early 2000s, financial literacy was largely understood as *financial knowledge*. However, it continuously evolved to comprise a much broader concept (see, e.g., Holzmann, 2010). Moreover, public bodies and international organizations like the OECD have considerably expanded the concept to include components beyond knowledge. Others, like the World Bank with its concept of *financial capability*, developed their own approach.

Recently, a broader concept of financial literacy has been integrated into a comprehensive framework of *financial well-being*, which now represents the overarching goal of financial literacy for many organizations and their initiatives, such as PISA (OECD, 2012d), OECD/INFE (OECD, 2020c) and the US Consumer Financial Protection Bureau (CFPB) (CFPB, 2015). In parallel, national strategies are shifting their focus toward financial well-being, replacing the previous emphasis on financial literacy (see OECD, 2022a).

Naturally, the overall evolution of and growing attention to financial literacy are reflected in its complexity and nuances as well as a drastically increasing volume of related publications and citations (see Goyal and Kumar, 2021). Indeed, the annual volume of published works on financial literacy increased tenfold between 2014 and 2022 (Zaimovic et al., 2023), reflecting a stronger focus on the various subdomains, assessments, effects and determinants of financial literacy as well as the growing number of stakeholders involved in both financial literacy and financial education.

2 Institutions and providers of financial education

2.1 OECD

The OECD promotes and assesses financial literacy predominantly within its International Network on Financial Education (INFE) and through its Programme for International Student Assessment (PISA). While INFE and PISA are both coordinated by the OECD and follow a similar understanding of financial literacy, they are based on rather different concepts and measurement approaches. While INFE measures financial literacy as a combination of knowledge, attitudes and behaviors, for PISA it is a latent competence that is visible in effective financial decision-making. Further insights into different definitions of financial literacy are discussed in section 3. Large-scale measurement efforts, including the OECD/INFE International Survey of Adult Financial Literacy (ISAFL) and the PISA financial literacy tests, are presented in section 4.

The OECD established INFE in 2008. With over 130 participating economies, this is the most comprehensive international effort to foster financial literacy in the world. Its main activities include the systematic

collection and analysis of cross-country financial literacy data on a regular basis (see subsection 4.2), the formulation of related policy instruments and evaluation methodologies and the exchange on good practices in financial education (OECD, 2023b). Full membership is open to public authorities and is typically held by central banks or ministries closely associated with the topic, which can actively participate in specialized working groups.

Currently, the INFE working groups address the implementation, evaluation, digitalization and sustainable financing of financial literacy and education (OECD, 2023b). The comprehensive reports prepared by the working groups are intended to serve as guidance for policymakers. Influential examples of OECD/INFE outputs that shaped the understanding of financial literacy and education in different contexts include the High-level Principles on National Strategies for Financial Education (OECD, 2012c), the Core Competency Frameworks for youths and adults (OECD, 2015b, 2016a), the OECD/INFE Guidelines for Financial Education in Schools (OECD, 2012b) as well as the OECD/INFE Survey of Adult Financial Literacy (OECD, 2023c) including a measurement toolkit (OECD, 2022c).

2.2 World Bank

As part of its development mandate, the World Bank's commitment to financial education is primarily intended to contribute to effective financial intermediation as an important prerequisite for reducing poverty and increasing prosperity, particularly in low- and middle-income countries (Holzmann, 2010; Holzmann et al., 2013). Strengthening financial capability should counteract undesirable phenomena, such as overindebtedness and fraud, and thereby increase individual well-being as well as the stability of financial markets (World Bank, 2014). Besides financial education, financial inclusion and financial consumer protection are assigned critical roles in achieving these goals (Perotti et al., 2013, p. 7).

From 2008 to 2013, the World Bank conducted its most comprehensive financial literacy project under the Russia Financial Literacy and Education Trust Fund. Its main objective was to provide knowledge and evidence in order to guide low- and middle-income countries in their implementation of national financial education strategies. To this end, the World Bank developed its own positivist financial capability framework and measurement approach based on empirical research that was carried out in different regions of the world (see subsections 3.2 and 4.3). Using this framework and measurement approach, it was possible to detect weaknesses in the financial capability for various population segments across different regions and cultures (Kempson et al., 2013).

The World Bank also addressed evaluation and impact assessment as a crucial part of financial education interventions to determine if and how financial education can enhance financial capability (Holzmann et al., 2013). As a result of these efforts, an evaluation toolkit was created that is specifically aimed at financial education programs in low- and middle-income countries (Yoong et al., 2013). Its main purpose is to provide guidance to those wanting to perform high-quality evaluation that can causally attribute possible changes in financial literacy or capability to the corresponding interventions. The World Bank also accompanied and advised a number of evaluation exercises for financial education measures (Holzmann et al., 2013).

The World Bank's outcome-oriented approach implies that the reasons behind advantageous financial decision-making are secondary. The World Bank is thus open to interventions beyond traditional financial education, including edutainment, social marketing and behavioral economics-based approaches (Holzmann, 2010). Despite being strong supporters of financial education in principle, the World Bank

has expressed strong scientific skepticism about the actual effectiveness of financial education interventions: “One general observation seems to be that the more rigorous the evaluation, the less likely the program is to demonstrate a positive impact” (Holzmann et al., 2013, p. xxiii).

2.3 Other international efforts

As the UN agency dedicated to promoting decent work for all, the International Labour Organization (ILO) started a financial education program back in 2006, with the twofold aim of expanding access to financial services and assisting people in making informed financial decisions. Bringing together governments, employers, employees and workers, the ILO adopts a holistic approach, utilizing various channels to enhance financial literacy on a global scale. On the one hand, through a network of trainers the ILO conducts workshops that directly target the ultimate beneficiaries of financial education (e.g. youths, migrants or entrepreneurs). On the other, it provides materials, tools and expertise to various stakeholders that want to promote financial education within their sphere of influence as providers or multipliers (ILO, 2024a, 2024b).

While the movement is no longer active itself, the influence of Child & Youth Finance International (CYFI), established in 2011, is still felt through its initiatives that are now continued by other major players. CYFI can be regarded as the largest private international effort to make financial education for young people a policy priority. It achieved this by creating a social movement advocating financial inclusion and financial education, with the aim of bringing about a “system change” in both the finance and education sector (CYFI, 2019). Its most prominent initiative is the *Global Money Week*, an awareness-raising campaign that mobilizes thousands of organizations in over 100 participating countries and is currently coordinated by the OECD/INFE (OECD, 2022b).

Other important stakeholders in financial education are private banks, which, especially in Europe, form international networks to represent common interests in face of supranational legislature and regulation. Notably, the European Banking Federation (EBF) advocates policies that promote financial literacy and education and runs its own initiatives such as the *European Money Quiz* and the *European Money Week*, targeting young people as potential beneficiaries. Furthermore, the EBF pleads for better cooperation between public and private stakeholders, referring to positive examples like the International Federation of Finance Museums, where private and public museums engage in productive exchanges. The EBF itself does not see any inherent contradiction in private stakeholders assuming a dual role as promoters of financial education on the one hand and profit-seeking providers of financial products on the other (Frenken and Folcher, 2020).

2.4 National strategies

Driven by the OECD’s financial education initiative in the early 2000s and further fueled by the 2008 financial crisis, which was interpreted by some as a consequence of consumers’ financial incompetence (e.g. Dinwoodie, 2010; Shiller, 2008), public institutions in many countries began to become increasingly active in financial education. Their efforts frequently resulted in the drafting of national strategies for financial education, i.e. coordinated approaches aimed at achieving common goals related to financial literacy and financial well-being at the national level (OECD, 2012c). Such strategies are often led by public authorities responsible for financial and/or educational matters, such as ministries of finance or education, financial regulatory authorities or central banks.

The OECD defines certain principles that should be adhered to when creating such national strategies (OECD, 2012c, 2015a): In a preparatory phase, the financial education initiatives that already exist in a country should be mapped and representative surveys should be carried out to establish the level of financial literacy and people's financial situation. In a next step, in a coordinated process of exchange, different stakeholders from the public, private and nonprofit sectors should agree on overarching goals, develop a common code of conduct and establish quality standards. Resources should also be allocated to monitoring and evaluating the national strategy to determine whether progress is actually being made with respect to predefined performance indicators.

The national financial education strategies of developed countries are substantially similar in content as they often address similar challenges. Population aging, for example, which causes the number of workers to decrease against that of retirees, creates pressures to reform public pension systems, which already record significant budgetary deficits (Arrondel et al., 2022; Letkiewicz, 2022; Van der Schors and Simonse, 2016). It is anticipated that future generations will increasingly need to rely on themselves for retirement planning and, therefore, require greater financial competence to independently build assets that ensure a comfortable retirement.

Another issue is that a substantial share of the population in Western industrialized nations has limited or no financial buffers (Arrondel et al., 2022; Letkiewicz, 2022; Stillwell, 2016; Van der Schors and Simonse, 2016), which makes them vulnerable to financial hardship in case of unexpected expenditure shocks or loss of income. Here, national financial education strategies seek to raise awareness for the importance of making emergency savings to be able to cope with unexpected financial shocks. Many developed countries also record large numbers of cases of overindebtedness, particularly among young people (Arrondel et al., 2022; Heath, 2016; Letkiewicz, 2022; OECD, 2021a; Stillwell, 2016; Van der Schors and Simonse, 2016). In view of easy access to credit, many national strategies aim to help people understand the business interests behind financial products and make people aware that financial decisions that provide short-term gratification (such as impulse purchases) threaten their long-term financial well-being.

Despite their substantial overlap in terms of topics and content, national financial education strategies are flexible enough to address country-specific challenges, as the following examples show:

- The *US National Strategy for Financial Literacy 2020* emphasizes the role of wide-spread high-quality housing counseling in assisting home buyers, homeowners and renters in making sound financial decisions (US Financial Literacy and Education Commission, 2020).
- Spain's *Financial Education Plan 2022–2025* focuses on university students studying subjects that are not directly or indirectly related to economics or finance (Comisión Nacional del Mercado de Valores, Banco de España and Ministerio de Asuntos Económicos y Transformación Digital, 2022).
- The *UK Strategy for Financial Wellbeing 2020–2030* includes mental health as a cross-cutting perspective, as people with mental health problems are particularly vulnerable to financial hardship (Money & Pensions Service, 2020).
- Canada's *National Financial Literacy Strategy 2021–2026* declared “reducing barriers [...] that hinder people from accessing, understanding, and using appropriate financial products, services, and information to their benefit” (Financial Consumer Agency of Canada, 2021, p. 17) a strategic priority to ensure that the financial world is more inclusive toward people with disabilities, the elderly, immigrants and linguistic minorities.

- Austria's current *National Financial Literacy Strategy* gives great attention to gender equality issues, seeing financial education as a tool to counteract the persistent gender gaps in pension benefits, income and wealth (OECD, 2021a).
- Greece's *National financial literacy strategy for Greece* pursues seven main objectives: reducing overindebtedness, encouraging long-term financial planning, promoting the safe use of digital financial services, informed participation in capital markets and tax compliance, preventing gambling and supporting financial literacy stakeholders in implementing education programs (OECD, 2024a, p. 14).

Advocacy groups such as trade unions, employers' organizations and other professional associations are sometimes directly involved in the design and implementation of national financial education strategies. However, they also frequently address financial education in their own domains, targeting their stakeholders or the general public. The potentially political nature of financial education may be reflected in the way the prioritization of these topics differs across institutions. In Austria, for instance, employers' organizations tend to focus more on the management of personal finances and the functioning of financial products (e.g. Wirtschaftskammer Wien, 2023), while trade unions or related organizations may put a stronger emphasis on employee rights and social inequality (e.g. Arbeiterkammer Wien, 2023).

2.5 Financial education in schools

Numerous national strategies, alongside initiatives led by nonprofit organizations (NPOs) and private entities, share the common objective of integrating financial education more fully into school curricula. In line with a future-oriented focus of financial education, the common reasoning is that enhancing financial literacy among young people is likely to have a bigger influence on their long-term financial security and well-being, given that major life decisions are still ahead of them and they are not yet responsible for earning their living. Making financial education a mandatory part of the standard school curriculum ensures that all students have access to this knowledge. This counteracts disparities in financial literacy resulting from different social backgrounds.

As to the extent to which financial education is currently integrated into the school systems of developed countries, one pattern becomes apparent. If a country's political system is characterized by strong federalism, the amount and content of financial education taught at school usually varies substantially within that country. Primary examples are Germany and the USA. Without a single body at the national level that has the authority to determine if and how financial education is taught at school, pursuing a uniform and coordinated approach in financial education is a lot more difficult in federalist countries than in centralized ones (Frühauf and Retzmann, 2016; Heath, 2016).

In many countries at least some form of basic financial literacy is taught at elementary school. Teaching content usually includes everyday economic relationships, the concept of money and the benefits of saving (Ackermann and Eberle, 2016; Greimel-Fuhrmann et al., 2016; Lacatus, 2016; Van der Schors and Simonse, 2016). In lower secondary school, elements of financial education are often integrated into other subjects. Typically, these subjects are mathematics (e.g. percentage calculation, interest calculation and business calculations), (social) sciences (usually comprising not only financial education but also economic education) or "general knowledge" subjects imparting useful practical knowledge (Ackermann and Eberle, 2016; Faulkner, 2022; Greimel-Fuhrmann et al., 2016; Van der Schors and Simonse, 2016).

Regarding upper secondary education, students in some developed countries can choose between continuing their general education at a higher level or enrolling in a vocational school to be trained in a specific vocation. Vocational schools, particularly those dedicated to business administration and commerce,

assign significantly more time and resources to financial and economic content than general-education upper secondary schools (Ackermann and Eberle, 2016; Frühauf and Retzmann, 2016; Greimel-Fuhrmann et al., 2016). Despite being likely to foster consumer financial literacy, the curriculum at vocational schools is dominated by the supply-side perspective. For those students who continue general education at the upper secondary level, financial education is rarely a compulsory subject. Some schools draw on external financial education providers, which may be public, nonprofit or private organizations.

In view of the relatively low emphasis placed on financial education, many supporters of financial education advocate a redesign of school curricula. Redesign proposals range from giving personal finance-related topics more weight within existing subjects to establishing a subject exclusively dedicated to financial or economic education (Faulkner, 2022; Henager and Kabaci, 2022). Even though the comparatively recent popularity of financial literacy has resulted in the setting-up of many projects aimed at potentially adapting school curricula, there are still substantial challenges to implementing financial education at schools. First, there is little consensus regarding the scope and content of financial literacy education (Frühauf and Retzmann, 2016). Second, standardized teacher training in financial or economic education is scarce, with some countries also lacking officially approved high-quality teaching materials (Faulkner, 2022; Frühauf and Retzmann, 2016; Henager and Kabaci, 2022; Stillwell, 2016). Last but not least, people's personal financial affairs may still be considered taboo, which makes the implementation of related content a delicate affair (Stillwell, 2016).

2.6 Private financial education providers

There is a wide range of private stakeholders active in financial education. A significant group are for-profit financial sector companies, typically banks and insurance companies, which provide financial education in addition to their core business activities (Frenken and Folcher, 2020). In addition to these, there are for-profit companies that specialize in financial education itself, making profit by marketing related seminars, counseling sessions or digital content. While some of these specialized providers focus on classical personal finance issues, many more focus on wealth creation, entrepreneurship and investing. Finally, an increasing number of “finfluencers” reach large audiences by providing finance-related content on social media platforms (Guan, 2023; Haase et al., 2023).

Another type of private stakeholders providing financial education are NPOs. NPOs are funded by public subsidies, private donations or corporate sponsorships. While they may be regarded as impartial and objective (Collins, 2011), their sponsors may still have an influence on their way of providing financial education. Again, NPOs active in financial education differ in terms of focus, format and target groups, with their services ranging from offering one-on-one counseling for working adults to conducting workshops for financially vulnerable young persons in “problem schools.”

Despite being pursued by public authorities, national financial education strategies generally encourage involving private financial education providers to help reach the overarching goal of strengthening financial literacy and, ultimately, the financial well-being of the population (OECD, 2012c). To ensure certain quality standards and avoid open conflicts of interests, such private actors are frequently required to commit to a code of conduct that prohibits advertising, ensures the accuracy of information and demands a certain degree of impartiality (OECD, 2015a). Although public-private partnerships can effectively promote financial literacy, many public bodies are reluctant to collaborate with private institutions, given concerns about aligning objectives and maintaining public trust.

In particular, educational efforts of financial service providers may give rise to conflicts of interest. While there is general consensus that the label “financial education” should not be misused for advertisement, financial service providers frequently consider providing financial education a win-win scenario (Frenken and Folcher, 2020). While consumers may benefit from financial literacy in terms of their financial well-being, financial service providers may benefit from consumers who demand more financial products and are more reliable in their loan repayments (Kuchciak and Wiktorowicz, 2021). However, some scholars doubt whether financial sector firms are genuinely committed to financial education beyond using it for their own gains (e.g. Willis, 2008).

Fraudulent activities have been reported with regard to private finance or investment coaches. In some cases, entrepreneurship or investment education served as a guise for what amounted to de facto Ponzi schemes (Europäisches Verbraucherzentrum Österreich, 2023b). Moreover, some finance coaches were found to use questionable sales tactics and exploit consumers’ unrealistic desires and expectations about wealth and prosperity (Europäisches Verbraucherzentrum Österreich, 2023a). Consumers should thus be cautious about the credibility of financial education providers and the promises they make. Indeed, being able to identify and access trustworthy sources of information has been identified as a key component of financial literacy (European Union and OECD, 2022; Holzmann et al., 2013).

3 Definitions of financial literacy, financial capability and financial competence

Financial literacy can be considered the core concept of a field of research concerned with the knowledge, abilities and behaviors necessary for positive financial outcomes and well-being. Indeed, Goyal and Kumar (2021, p. 89) find in a keyword analysis that “financial literacy” has become a termed concept and by far the most prevalent term in the relevant literature. However, despite or perhaps due to the deceptively self-explanatory nature of financial literacy, scholars have not yet agreed on a consistent definition (Cude, 2022; Huston, 2010; Remund, 2010).

In fact, there seem to be as many definitions of financial literacy as there are organizations promoting it. Moreover, related terms, such as *financial capability*, are sometimes used as synonyms for financial literacy, yet sometimes also to clearly distinguish other concepts from the well-established term of financial literacy or to highlight distinct aspects. The scope of definitions may also have to be adapted regularly to reflect the changing financial landscape, e.g. to take into account aspects of digitalization (Koskelainen et al., 2023) or sustainable finance products (OECD, 2023a). To illustrate how complex the concept of financial literacy is and to provide insights into popular definitions and related concepts employed in research and policymaking, this section explores the term “financial literacy” and other terms used in the literature.

Goyal and Kumar (2021) find that besides financial literacy, the terms *financial education*, *financial capability*, *financial knowledge* and *financial behavior* are most commonly used in the literature. “Financial capability” as an alternative term and concept is discussed in detail in a separate subsection below. “Financial knowledge” and “financial behavior” are explored as parts of the superordinate definitions. “Financial education,” on the other hand, is presented briefly in subsection 5.4 as a determinant of financial literacy. We also include *financial competence*, a term that is rarely used in the field of financial literacy but that may be significant for conceptualizing or operationalizing financial literacy. Moreover, we briefly touch on *digital financial literacy* as it is becoming more and more congruent with financial literacy, and on *economic literacy*, which has become increasingly important in some countries’ school curricula.

3.1 Financial literacy

For quite some time, financial literacy was commonly understood as financial knowledge. Its measurement, especially, used to consist of a few short knowledge questions on topics such as (compound) interest, inflation and risk diversification. Subsequently, such questions, in particular the “Big Three” and “Big Five” (see subsection 4.1), were used as the basis of numerous studies and have, thus, considerably shaped our understanding of financial literacy (see Haupt, 2022).

However, semantically, the term “literacy” suggests a distinction from “knowledge,” in our case indicating a deeper understanding of personal finance and the capacity to make prudent financial decisions – just like the original meaning of literacy suggests the ability to read *and* write. This notion is already evident in one of the earliest definitions of financial literacy by Noctor et al. (1992), who define it as the “ability to make informed judgments and effective decisions regarding the use and management of money” (p. 4).

Indeed, the common understanding of financial literacy has evolved toward a broader concept extending far beyond mere knowledge (Holzmann, 2010). This conceptual expansion may, in part, stem from the observation that (1) knowledge about financial concepts and principles alone does not automatically imply applying that knowledge and translating it into informed financial behavior (Kaiser and Menkhoff, 2017) and (2) psychological traits such as self-efficacy or attitudes appear to be key in translating knowledge into financial behavior (Fernandes et al., 2014).

Definitions of financial literacy from the literature exemplify this observation: Remund (2010) used an extensive literature review to develop a detailed definition, characterizing financial literacy as “a measure of the degree to which one understands key financial concepts and possesses the ability and confidence to manage personal finances through appropriate, short-term decision-making and sound, long-term financial planning, while mindful of life events and changing economic conditions” (p. 284). Lusardi and Mitchell (2014) describe financial literacy as the “ability to process economic information and make informed decisions about financial planning, wealth accumulation, debt, and pensions” (p. 6), thus emphasizing both the economic nature of financial knowledge and its primary domains of application. Both definitions explicitly refer to having the ability and confidence to make informed or appropriate decisions and are, thus, not limited to the notion of knowledge per se.

The most recent definition of financial literacy by the OECD/INFE regards financial knowledge, attitudes and behaviors as components of financial literacy, referring to financial literacy as “a combination of financial awareness, knowledge, skills, attitudes and behaviors required to make sound financial decisions and ultimately achieve individual financial well-being” (OECD, 2020c, p. 6). This definition includes *behaviors* as components of financial literacy and explicitly connects the concept to the overarching goal of financial well-being (see subsection 6.3). This current definition is also reflected in recent international efforts by the OECD/INFE to measure financial literacy (see subsection 4.2).

In a school education context, PISA’s Financial Literacy Analytical and Assessment Framework (OECD, 2023d) does not include behaviors or context factors but refers to financial literacy as a prerequisite of effective financial decision-making: “Financial literacy is knowledge and understanding of financial concepts and risks, as well as the skills and attitudes to apply such knowledge and understanding in order to make effective decisions across a range of financial contexts, to improve the financial well-being of individuals and society, and to enable participation in economic life” (p. 112). Although the PISA framework only occasionally refers to *financial competencies*, it may be regarded as a competence framework in a psychometric tradition, as described in subsection 3.3. In contrast to the OECD definition presented above, PISA’s definition not only names individual but also societal financial well-being as a goal.

3.2 Financial capability

Use of the term *financial capability* remains inconsistent in the literature. For instance, in the national financial literacy strategy of New Zealand, the term is used synonymously with financial literacy (New Zealand Commission for Financial Capability, 2021). Other authors use it to highlight the crucial role of behavior as opposed to that of mere knowledge (Serido et al., 2013; Xiao and O'Neill, 2016). As regards international organizations, the term is strongly associated with the World Bank's approach, which adopts a strong stance concerning the distinction between financial literacy and financial capability with regard to their practical application and outcomes.

Concerning fundamental assumptions such as goals and content, the World Bank has employed a positivist approach in defining financial capability by not assuming any normative criteria of "good" behavior. Instead, it defines people's optimal behavior only through the prevalent subjective and social judgment of their peers. Moreover, it has operated under an "agnostic" research paradigm by not assuming any mechanisms of how financial capability may lead to desirable outcomes. By collecting qualitative and quantitative data among the general population, the World Bank thus identifies ten components of financial capability (Kempson et al., 2013; see subsection 4.3).

The World Bank regards financial literacy mainly as financial knowledge, i.e. as a necessary but not sufficient requirement for reaching financial capability. Financial capability itself is defined as "the internal capacity to act in one's best financial interest, given socioeconomic and environmental conditions. Financial capability encompasses the knowledge (literacy), attitudes, skills, and behaviors of consumers regarding understanding, selecting, and using financial services and the ability to access financial services that fit their needs" (World Bank, 2014, p. 1). This definition emphasizes the perspective of the individual, who eventually should be able to act in their best financial interest and choose financial services according to their needs (World Bank, 2014).

The World Bank's definition of financial capability also sets itself apart by acknowledging the role of external factors such as socioeconomic and environmental conditions that are beyond a person's immediate control. Similarly, Johnson and Sherraden (2007) explicitly include the element of "opportunity to act" (p. 122) in their definition. This perspective puts further emphasis on the potential systemic barriers and the support and resources necessary to benefit from financial capability. The argument that it is important to create an enabling environment by enhancing access to financial services can also be found in the 2015 Financial Capability Strategy for UK (Money Advice Service, 2015).

3.3 Financial competence

Historically, the concept of competence and its assessment emerged, in part, as a response to the common practice of intelligence assessments carried out to predict academic and professional performance (e.g. McClelland, 1973). In contrast to intelligence, competence is commonly regarded as acquired (e.g. through education and training) and specific to real-world situations and tasks. Contrary to definitions of knowledge, definitions of competence typically include a wider range of performance prerequisites, such as cognitive processes, motivation or self-efficacy (e.g. Hartig, 2008; Klieme et al., 2008; Le Deist and Winterton, 2005; Westera, 2001). Notably, unlike observable behavior or environmental context, these characteristics lie within individuals.

However, the scopes of the terms "competence" and "competency" depend on the context and field of research. In general, *competence* on its own can be regarded as a more general ability, and has even been

proposed “as a symbol for an alternative approach to traditional intelligence testing” (McClelland, 1973, p. 7). *Competencies* often refer to more task-specific characteristics such as knowledge, skills and values, while they may even solely describe behaviors in a human resources context (see Drisko, 2014). Overall, there is little consensus on what is covered by the terms, and definitions and terminology can vary widely depending on disciplines, cultures and translations.

Reflecting a structured approach to defining characteristics of individuals that lead to positive outcomes, e.g. a certain performance in tasks or informed decisions, “competence models” are often used for educational assessments. In German-speaking countries, for example, competence models regularly serve as the basis for defining competence domains used in school curricula, including “economic competence”, and for developing corresponding tests (e.g. Eberle et al., 2016; Kaiser et al., 2020; Retzmann and Seeber, 2016; see also subsections 3.5 and 4.4).

On an international level, PISA’s Financial Literacy Analytical and Assessment Framework (OECD, 2023d) only occasionally refers to “competency.” Nevertheless, the PISA framework may count as a competence model as described above as it quite clearly defines intrapersonal prerequisites for effective decision-making without including behaviors or context factors. PISA also further defines financial literacy along the dimensions of content, contexts, cognitive processes and noncognitive factors in order to translate the concept into test items, an approach typical of competence models and assessments.

In contrast, the European Union’s and OECD’s Financial Competence Framework for Adults (European Union and OECD, 2022) follows a different approach. It lists a staggering 564 granular “competences” under four content areas: (1) money and transactions, (2) planning and managing finances, (3) risk and reward and (4) financial landscape. Behavior is explicitly included in three dimensions: (1) awareness, knowledge and understanding, (2) skills and behaviors and (3) confidence, motivation and attitudes. Additional emphasis is placed on competences related to sustainable and digital finance. A similar framework is also available for children and youths (European Union and OECD, 2023).

3.4 Digital financial literacy

Digitalization has greatly expanded the range of financial services available and made the financial landscape even more complex. With reliance on digital platforms increasing, people must not only navigate novel financial services but also be aware of new phenomena, such as data privacy issues or online fraud (Lyons and Kass-Hanna, 2021). Understanding digital financial products and instruments thus appears crucial to making informed financial decisions and avoiding potential risks (Koskelainen et al., 2023; Morgan et al., 2019; Yakoboski et al., 2018). On the other hand, digital technology may also provide tools to improve financial behaviors. For example, budgeting apps and fintech products that use behavioral nudging may foster beneficial savings habits and help prevent overspending (Koskelainen et al., 2023; OECD, 2018).

Overall, the importance of digitalization has been recognized in financial literacy research in two ways: First, financial literacy definitions and competence frameworks have been expanded to explicitly include digital competences. Particularly worth mentioning are the *Financial competence framework for adults in the European Union* by the European Union and OECD (2022) and the OECD/INFE International Survey of Adult Financial Literacy, which consider digital competencies and online behavior related to financial services and fraud prevention (OECD, 2022c). Second, the stand-alone concept of *digital financial literacy* has emerged in the scientific literature, describing a skill set that complements traditional financial literacy

and particularly considers the peculiarities of the digital realm (Lyons and Kass-Hanna, 2022; Morgan et al., 2019). Similar to some definitions of financial literacy, it has been defined as a multidimensional concept that may include knowledge, skills, awareness, know-how, attitudes, behavior and self-protection (Lyons and Kass-Hanna, 2021).

3.5 Economic literacy

Economic literacy or *economic competence* are terms related to financial literacy. While economic literacy is sometimes used as a synonym for financial literacy (Jappelli, 2010; Prete, 2013), it tends to have a broader scope than personal finance, comprising a deeper understanding of economic concepts, theories and principles such as economic growth, unemployment, inflation, fiscal and monetary policies and international trade (McCowage et al., 2022; Walstad et al., 2013). It may also include the ability to develop a critical awareness of economic systems and policies as well as their socioeconomic implications both at an individual and societal level (Soroko, 2022).

Like financial education, the term “economic education” (or “economics education”) has sparked some debate on its definition and goals, particularly regarding the corresponding content in school curricula (e.g. Fridrich, 2019). As mentioned in subsection 3.3, there are various efforts in German-speaking countries to define, foster and measure economic competence (e.g. Eberle et al., 2016; Kaiser et al., 2020; Retzmann and Seeber, 2016). While many considerations concerning financial literacy may also apply to economic literacy and vice versa, a detailed description of the concept of economic literacy and its implications is beyond the scope of this paper.

3.6 Differences in central theoretical assumptions

Overall, as shown above, contemporary definitions of financial literacy and similar concepts always imply elements such as knowledge, understanding or comprehension. Moreover, virtually all stakeholders and their definitions of financial literacy appear to consider financial literacy not an innate trait but, for the most part, the result of education, training, practice or experience. While not all organizations explicitly define financial literacy as acquirable, acquirability can still be regarded as a central assumption of financial literacy definitions that is universally accepted.

Nevertheless, definitions may differ with regard to other central theoretical assumptions, which may not be obvious when looking at the terms alone. In short, definitions of financial literacy and related concepts may vary with regard to the following five key aspects: (1) goals, (2) traits considered, (3) observable behaviors, (4) thematic focus and (5) context factors. These five aspects are discussed below in further detail.

First, definitions may focus on different goals or outcomes of financial literacy. Some emphasize the quality of people’s behavior, referring e.g. to informed, prudent or sound decision-making and implying a rational or utility-maximizing perspective. What qualifies as *good* decisions, however, may continue to be a judgment call. In this regard, the World Bank employs a positivist rather than a normative approach, explicitly defining the quality of decisions and behavior through peer judgment. Many definitions include specific individual or societal outcomes as the ultimate goal of financial literacy. This, in turn, requires clear definitions of the outcomes themselves and ultimately builds on empirical evidence. Possible definitions of the most prominent goal, financial well-being, and its determinants within and beyond the scope of financial literacy are discussed briefly in subsection 6.3.

Second, many definitions are not limited to financial knowledge but include factors such as attitudes, motivation or self-efficacy. This feature can be considered the deciding factor that distinguishes *financial literacy* or *financial competence* from mere *financial knowledge*. The World Bank, however, sees *financial literacy* as a form of knowledge and developed the concept of *financial capability* to capture components beyond mere knowledge. The inclusion of behaviors as components of financial literacy in some definitions further complicates matters, as will be outlined below.

Third, financial literacy could be considered a latent (unobservable), i.e. an *intrapersonal*, ability. As such a latent psychological *construct*, it would be limited to cognitive and attitudinal components only, where behaviors represent potential indicators and likely outcomes but not components. Such an understanding appears to be in line with the way PISA defines and measures financial literacy (see also subsection 5.4). Other definitions, particularly the one given by the OECD/INFE, explicitly include behaviors as components of the concept of financial literacy, which may complicate the distinction of components, outcomes and indicators in theory, measurements and research. It may also diminish a (hypothetical) element of choice by assuming that all people who *are* financially literate must also necessarily *behave* in a financially literate way.

Fourth, concerning their thematic focus, definitions generally appear to rely on a common understanding of the terms “financial” or “finances.” More detailed descriptions of topics covered by the required knowledge and skills or of the situations in which financial literacy applies shape definitions considerably and thus determine assessment instruments and education and policy measures. In their definitions of financial literacy, PISA and the World Bank, for example, both specify relatively concrete knowledge areas, attitudes or behaviors. As extreme examples, the financial competence frameworks presented by the European Union and the OECD contain comprehensive lists of granular competences required in specific situations. In other cases, both topics and usage situations may only become apparent through education materials or questions in tests or surveys.

Fifth, context and environmental factors play a central role in shaping people’s financial literacy and financial well-being. Most definitions presented above do not explicitly name individuals’ (financial) context, situation or opportunities (such as socioeconomic, structural or cultural factors) as parts of the applicable concept of financial literacy. However, the World Bank’s definition considers these factors integral parts of financial capability. Just like including behaviors as components of financial literacy, including context factors distinguishes financial capability according to the World Bank from a purely intrapersonal ability.

The five key aspects outlined above offer a structured starting point for developing and interpreting financial literacy definitions. When designing or employing definitions of financial literacy or related concepts, these aspects should be carefully considered as they can affect not only the design of financial education interventions but also their evaluation through measurement and impact assessment. To reduce ambiguity and enhance the comparability of interventions and evaluations across studies and contexts, we generally recommend that educators and evaluators transparently report the underlying assumptions of the definitions they adopt.

4 Measuring financial literacy

Measuring financial literacy is a crucial part of effective financial education interventions and educational programs and of their evaluations. There are countless methodological approaches to measuring financial literacy levels, ranging from qualitative, interview-based methods to quantitative competence tests. However, as outlined in section 3, definitions of financial literacy vary across disciplines, stakeholders and

researchers and there is no common understanding of the concept. Moreover, the advancing digitalization of financial services and the increasing complexity of financial products and processes are constantly adding to the potential set of elements that make up financial literacy. This lack of conceptual uniformity has implications for the development of measurement tools and benchmarks.

Given the lack of consensus regarding definitions and measurement methods, the question of who is financially literate can therefore not be addressed without simultaneously asking how their financial literacy was measured. Using concrete examples, this section focuses on *quantitative* measurement approaches that involve tests and questionnaires employed at the international level, namely the “Big Three” and “Big Five” (Lusardi, 2011; Lusardi and Mitchell, 2011), the World Bank’s measurement approach (Kempson et al., 2013), the PISA financial literacy test (OECD, 2024b) and the OECD/INFE International Survey of Adult Financial Literacy (ISAFL) (OECD, 2023c). We then compare results from two seemingly similar international assessments, the ISAFL and the Standard & Poor’s Ratings Services Global Financial Literacy Survey (Klapper et al., 2015; hereafter S&P survey). This section concludes with an overview of methodological considerations on measuring financial literacy.

4.1 The “Big Three” and “Big Five”

Focusing solely on the knowledge dimension of financial literacy (Haupt, 2022), the Big Three items were developed by Lusardi and Mitchell as a short and simple instrument to be applied in telephone surveys, face-to-face interviews and online surveys (Lusardi and Mitchell, 2014). First employed in the US Health and Retirement Study in 2004 (see Lusardi and Mitchell, 2006), the Big Three and their variants and adaptations have since become the most widely used items in measuring financial literacy (Haupt, 2022) and have served as the basis for most empirical studies that have been conducted on financial literacy so far.

The Big Three comprise three single-choice questions on compound interest, inflation and risk diversification (Lusardi and Mitchell, 2011, p. 499). A popular variant of the instrument, the Big Five, add the concepts of bond prices and mortgages to the Big Three (see Lusardi, 2011). Questions are either phrased as statements which respondents need to judge as correct or incorrect, or as close-ended questions with up to four answer options, one of which is correct. The score is typically calculated as the sum of correct responses.

Concerning the psychometric properties of the Big Five when used in the USA and in the Netherlands, Lusardi and Mitchell (2011) report that different wording in questions can lead to pronounced differences in results and that the questions may suffer from significant measurement errors. Knoll and Houts (2012) also find some limitations of the test instrument: While the Big Five were deemed suitable as part of the authors’ 20-item test instrument that was based on item response theory (IRT, see subsection 4.6), the 20-item instrument more accurately predicted self-reported retirement planning than the Big Five alone. Nicolini and Haupt (2019) also find that the Big Five are not always the best instrument to predict financial behaviors when compared to longer or more specific tests and that their predictive qualities can vary across countries.

Applying IRT to data from the UK, Ranyard et al. (2020) find that the Big Five item on bonds is too difficult and reduces the reliability of the measurement. Nevertheless, the Big Five overall exhibited good sensitivity, particularly through the easier items on interest, inflation and mortgages. However, like the results reported by Knoll and Houts (2012), the authors find that their newly constructed extended instrument predicts financial well-being better than the Big Five. Moreover, they construct a five-item alternative to the Big Five that shows slight improvements over the originals regarding reliability and difficulty.

While their small number of questions cannot assess financial literacy with the same accuracy as test instruments using much longer questionnaires, Lusardi's and Mitchell's Big Three and Big Five proved viable in many contexts despite, or perhaps owing to, their brevity and limited scope. These questions have shaped the understanding of financial literacy on an international scale and may even, in part, be responsible for the dominance of the concepts of "inflation," "interest" and "risk diversification" in the common understanding of financial literacy. Indeed, other ways of operationalizing financial literacy, such as the OECD/INFE ISAFL detailed below, also include these concepts as the main focus of their financial knowledge questions.

4.2 OECD/INFE International Survey of Adult Financial Literacy

The OECD/INFE International Survey of Financial Literacy (ISAFL) aims to systematically assess financial literacy across participating countries in an effort to pinpoint areas for improvement and to inform policymaking accordingly (OECD, 2023c). The survey has been carried out at semiregular intervals in a growing number of countries since 2010, with 39 countries having participated in the most recent wave and Austria having made an additional contribution. A comparison of results from the 40 countries (Austria included) can be found in Voith and Zieser (2024).

Following the OECD/INFE definition of financial literacy, the ISAFL questionnaire covers financial knowledge, attitudes and behaviors and indeed allows for calculating distinct knowledge, attitude and behavior scores. The total financial literacy score it produces is the sum of these three individual scores, giving the most weight to behavior followed by knowledge and attitude (see OECD, 2022c, for the full list of questions). During the design of financial literacy items and corresponding scores, the survey underwent several reliability and validity tests. The OECD reports pilot testing in several countries, cognitive interviews to refine question comprehensibility and statistical analyses to ensure that questions reliably measure the intended concept (Atkinson and Messy, 2012). Besides financial literacy items, the latest questionnaire of the ISAFL test includes digital financial literacy and financial well-being items and proposes methods for constructing indices that reflect these concepts (OECD, 2022c).

In total, seven ISAFL questions contribute to the financial knowledge score, covering the topics of inflation, interest (calculation), compound interest and risk and return in investments. De Clercq (2019) evaluates these financial knowledge questions with 2015 ISAFL data, using IRT. While the results indicate that the knowledge questions indeed appear to capture a unidimensional construct, they also show lacking reliability and discriminatory power, which means that the knowledge questions used may not be able to always reliably discern high levels of knowledge from low levels. De Clercq also finds considerable differential item functioning, which means that difficulty levels of some questions vary disproportionately across countries. This challenges the assumption that the questions are able to capture financial knowledge equally across countries and highlights the risk of misinterpreting country rankings and misdiagnosing problem areas. The author concludes that the ISAFL may not meet the necessary requirements to be considered a proper "international large-scale assessment" (ILSA) in terms of the validity, reliability and comparability of results.

In a study on data from the ISAFL version conducted in Italy in 2013, Bongini et al. (2018) employ IRT and confirm that, overall, the three scores (knowledge, attitudes and behavior) as well as the total score appear to be appropriate measures of (the three dimensions of) financial literacy. However, the authors find that using IRT as opposed to conventional analyses leads to different results with regard to which groups are in need of better financial literacy, concluding that IRT is better suited as an analysis framework

for the survey data than other methods. Moreover, they find that typical demographic groups, e.g. women, cannot be treated as homogeneous with regard to survey results and financial literacy needs and should be viewed in a more nuanced way, e.g. in conjunction with their education.

The ISAFL can be considered a cornerstone of financial literacy policymaking in participating countries and constitutes one of the few international efforts to measure financial literacy systematically across a large number of countries. This, however, may also raise requirements concerning psychometric properties such as validity, reliability and cultural fairness, which are already difficult to fulfill in assessments on a much smaller scale. Naturally, however, some compromises concerning psychometric properties appear necessary in light of the administrative challenges of coordinating an assessment in dozens of countries at the same time, producing results that must not only be comparable across countries but also across time.

To illustrate the extent to which methodological differences can affect the outcomes of international surveys, in subsection 4.5 we compare the results of the 2015 ISAFL with another international financial literacy assessment, the S&P survey from 2014. Although both surveys use similar methodological approaches and questions to gauge financial knowledge, we find results to be surprisingly heterogeneous.

4.3 The World Bank

Building on research conducted in the UK (Personal Finance Research Centre, 2005), the World Bank regards *financial capability* as an extension of financial literacy that also includes behaviors and knowledge, skills and attitudes and their interactions (see also subsections 2.2 and 3.2). In a research project focusing on low- and middle-income countries (Kempson et al., 2013) the World Bank developed a measurement approach that differs significantly from the other ways of operationalizing financial literacy presented here.

Concerning the development of the World Bank's financial capability survey instrument, the authors explicitly state the fundamental assumptions underlying their measurement approach: (1) Financial capability is to be considered a latent, underlying capability only measurable through indicators (manifestations) found in the behavior or responses of people, which corresponds to reflective measurement (see also subsection 4.6). (2) Financial capability may be sufficiently described by one domain (or dimension) or could comprise several components, each to be assessed separately. (3) Optimal financial behavior is to be defined only through peer judgement in a positivist as opposed to a normative approach (Kempson et al., 2013). Moreover, no assumptions were made on determinants or processes causing improved patterns or behavior. The World Bank thus operated under an "agnostic" (as opposed to, for example, a cognitive) research paradigm (Holzmann et al., 2013).

In line with these assumptions, the World Bank followed a rigorous bottom-up process by developing its concept of financial capability based on a phase of qualitative exploration. In a first step, the World Bank collected data from among the general population via focus groups in eight countries. It then identified and developed questions suited to capture the manifestations of financial capability as expressed in the focus groups. Questions were subsequently tested through in-depth interviews and pilot surveys, and necessary adaptations were made. Unlike most financial literacy assessments, the resulting instrument does not contain any knowledge questions. Variables for the corresponding analysis were constructed using a considerable number of questions on respondents' behaviors, attitudes and financial situation. Ultimately, the questionnaire was deployed in seven countries with approximately 20,000 participants (Kempson et al., 2013).

Reporting detailed results of factor analyses on the survey questions, Kempson et al. (2013) conclude that financial capability as operationalized in the World Bank questionnaire is not captured by one single construct, but rather by ten components, namely: budgeting, living within one's means, monitoring expenses, using information, not overspending, covering unexpected expenses, saving, attitude toward the future, not being impulsive and achievement orientation. Two additional components emerged for certain subgroups, namely covering old-age expenses for people aged under 60 and choosing financial products for people who personally chose a financial product in the past five years. Importantly, the authors identified no higher-level domains based on intercorrelations between components.

In five countries, participants completed five knowledge questions on numeracy, inflation and (compound) interest in addition to the main questionnaire. As a result, Kempson et al. (2013) were able to explore links between financial capability and financial literacy (in terms of knowledge). While correlations between the two concepts were generally positive, the authors also found instances of strong negative correlations, which may corroborate the assumption that financial literacy and financial capability are indeed different concepts.

4.4 PISA

Particularly in the context of school education, program evaluations and international large-scale assessments (ILSAs) increasingly rely on accurate assessments of the state and development of competencies through standardized tests (see Baartman et al., 2007). The standardized assessment of students' financial literacy has been part of the OECD's Programme for International Student Assessment (PISA) since 2012. PISA financial literacy assessments were carried out in 2012, 2015, 2018 and 2022 (OECD, 2023d). Like the assessments of the regular domains covered by PISA (reading, mathematics, and science), the financial literacy assessment is used to gauge the related performance of 15-year-olds in many countries around the world, allowing for comprehensive comparisons between countries and assessment periods. At the time of writing, the most current PISA financial literacy results available are from 2022 (OECD, 2024b).

In principle, PISA relies on a competence-based notion of financial literacy with a focus on knowledge, understanding, skills, motivation and confidence (see subsection 3.3). With the goal of assessing these factors, selections of 43 items overall were presented to students in the 2018 PISA assessments (OECD, 2020b). While PISA items still in use are not published to avoid students' prior knowledge of questions, PISA regularly publishes items excluded from the test to illustrate item content and format. Published items address, for example, issues such as distinguishing between fixed and variable costs, understanding loan conditions, reading bank statements or selling something online (see OECD, n.d.).

By using IRT, PISA ensures sufficient psychometric properties, particularly the reliability and fairness of test questions. This IRT-based analysis framework allows PISA to systematically quantify both students' abilities and the difficulty level of test questions and to assess differential item functioning, i.e. whether items behave similarly in different student groups or countries. Moreover, tests designed under the IRT framework can be highly flexible regarding question selection: Only a subset of questions needs to overlap between students or assessment waves, which enables PISA to introduce new questions and still be able to compare financial literacy over time (see OECD, 2020b).

The OECD reports going to great lengths to ensure precise, flexible measurements in PISA that capture abilities rather than focus solely on subject knowledge. However, as for other large-scale student assessments, the test instruments need to find the correct balance between narrowly assessing subject-specific knowledge and overemphasizing general academic abilities, such as reading, or general cognitive ability.

Indeed, the validity of PISA results has been met with doubts as scores in the different domains show high intercorrelations, suggesting the dominance of a common underlying factor, e.g. general cognitive ability (see Luo et al. 2003; Rindermann, 2007). Financial literacy scores have also exhibited strong correlations of over $r = .80$ with the domains of mathematics and reading (OECD, 2014; OECD, 2024b). In situations where the same concept or domain is tested multiple times, such high correlations between test scores would indicate sufficient retest reliability in many measurement contexts.

Some studies on the three general domains in PISA (mathematics, reading and science) conclude that students' abilities in these domains indeed explain much less of their performance in the related PISA test items than their general cognitive or academic ability would (Pokropek et al., 2022a; Pokropek et al., 2022b). Pokropek et al. (2022a) attribute these results to the type of questions used in PISA, which attempt to test relatively broad competencies in various specific contexts. Other authors, however, emphasize the crucial contribution each PISA domain makes to testing students' performance and that the concept of general intelligence (or cognitive ability) should be reconsidered in an educational context (Baumert et al., 2009).

4.5 Discrepancies in financial literacy assessments

All of the financial literacy tests or questionnaires described above have been used in national or international studies to benchmark financial literacy or to serve as a basis for policy recommendations or development. However, as is evident, financial literacy assessments strongly depend on the underlying definitions and methodological approaches used. To illustrate potential discrepancies in financial literacy assessments related to these dependencies and other methodological differences, we briefly compare two international financial literacy surveys among adults that were conducted in the same 28 countries at around the same time between 2014 and 2016 – the OECD/INFE ISAFL (OECD, 2016b) and the S&P survey (Klapper et al., 2015). After comparing their results, we explore possible causes of the differences between them.

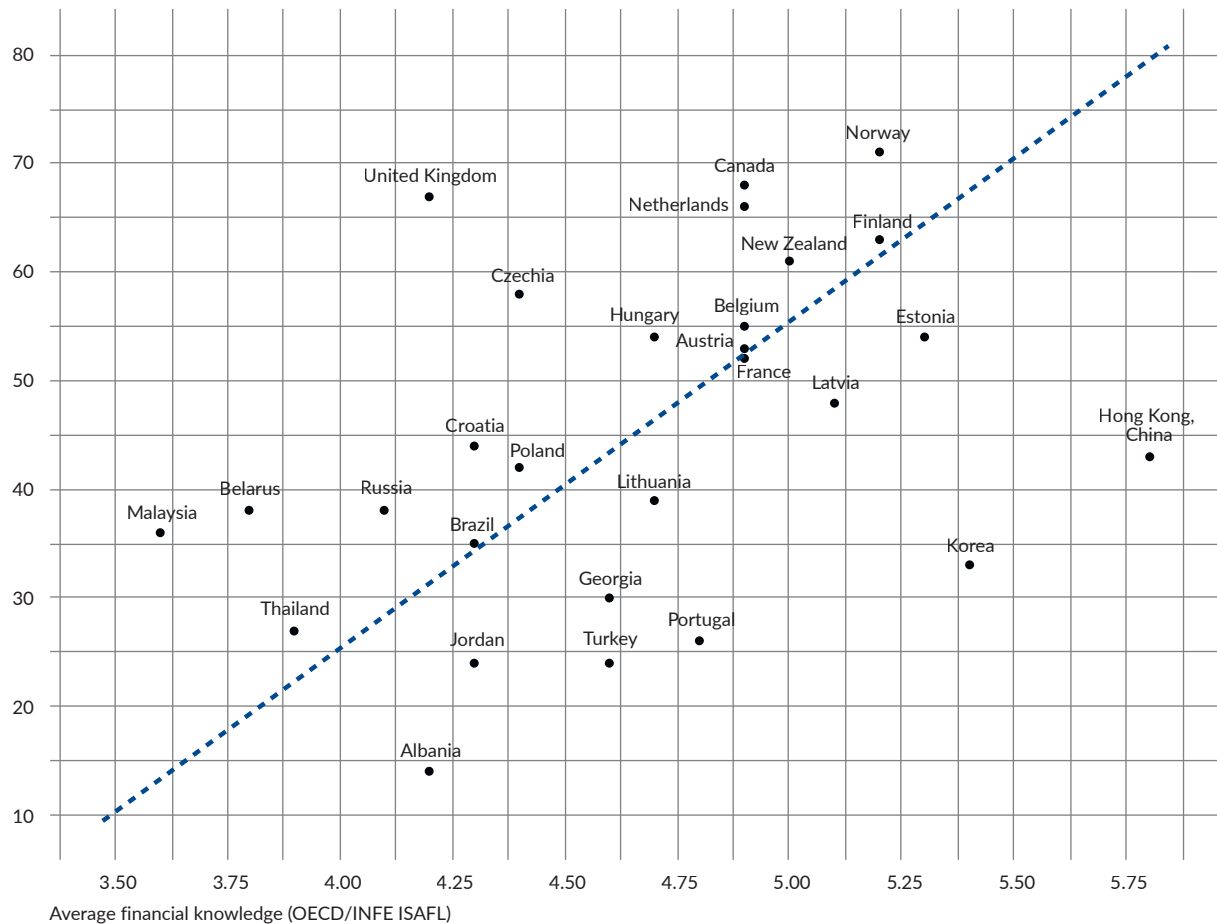
Thirty countries participated in the 2015 ISAFL, with most of the data being collected in 2015. Only some countries collected ISAFL data in 2014 or 2016. All in all, responses from 51,650 adults were collected in this ISAFL wave. Details on the survey's background and questionnaire are presented in subsection 4.2. The S&P survey took place in 2014 as part of the Gallup World Poll survey. It measured people's financial knowledge with regard to four basic financial concepts: risk diversification, inflation, numeracy (interest calculation) and compound interest. Data were collected in 143 countries from over 150,000 adults (Demirgüç-Kunt et al., 2015; Klapper et al., 2015). The questions used in the S&P survey are similar in format and content to the items introduced by Lusardi and Mitchell (see subsection 4.1) and to the financial knowledge questions used in the ISAFL, but they differ in focus and phrasing.

Chart 1 shows the results of the S&P survey and the ISAFL for those 28 countries that participated in both. We present the average financial knowledge score from the ISAFL as reported in OECD (2016b) and the percentage of participants that answered more than three of four questions correctly in the S&P survey as reported in Klapper et al. (2015). For the ISAFL, we show the financial knowledge score (as opposed to the total financial literacy score) as it corresponds more closely to the questions in the S&P survey.

Chart 1

OECD/INFE ISAFL knowledge score and S&P survey results

Financially literate population, % (S&P survey)



Source: OECD (2016b); Klapper et al. (2015).

Note: This chart shows countries' mean financial knowledge scores according to the OECD/INFE ISAFL and the share of financially literate persons in countries' total population according to the S&P survey. The dashed diagonal line represents a hypothetical perfect correlation. It connects the points of ± 2.3 standard deviations from the mean of both variables.

As shown in chart 1, the two surveys did not find the same relative financial literacy levels in many countries. The correlation between countries' S&P percentages and countries' average ISAFL knowledge scores is $r = .41$ (rank correlation $r = .46$), despite the seemingly similar set of questions used in both surveys. Surprisingly, the correlation between the overall S&P survey percentage and the total ISAFL financial literacy score (also reported in OECD, 2016b), which includes attitudes and behaviors not covered in the S&P survey, is marginally higher with $r = .48$ (rank correlation $r = .47$). Neither correlation would indicate sufficient retest reliability in most measurement and assessment contexts.

The reasons for these differences are likely to be manifold. First, the target population of the S&P survey was the civilian population aged 15 and above (Demirgüç-Kunt et al., 2015), whereas that of the ISAFL was the adult population aged 18 to 79 (OECD, 2016b). Moreover, we cannot rule out that financial literacy levels actually changed in the time between the two surveys.

Second, differences in survey methodology are likely to have contributed to differences in results. As the ISAFL is conducted and adapted by separate commissioning bodies in the participating countries (e.g. central banks or ministries of finance), there may be significant differences in sampling, data collection and data weighting methods as well as in translations and country-specific adaptations of the questionnaire (OECD, 2016b). Data collection varied across countries also in the S&P survey, depending, for example, on survey participants having telephone access (Demirgüç-Kunt et al., 2015).

Third, the questionnaires themselves may have caused differences. While the knowledge-specific questions in the ISAFL cover similar topics as the S&P survey (inflation, interest, compound interest, risk diversification), the wording of questions differs considerably. For example, the ISAFL asks, “Suppose you put \$ 100 into a savings account with a guaranteed interest rate of 2 % per year. You don’t make any further payments into this account and you don’t withdraw any money. How much would be in the account at the end of the first year, once the interest payment is made?” (OECD, 2016b, p. 20). The S&P survey asks, “Suppose you need to borrow 100 US dollars. Which is the lower amount to pay back: 105 US dollars or 100 US dollars plus three percent?” (Klapper et al., 2015, p. 6).

Finally, the way the final financial literacy levels are calculated may also affect survey results. While the OECD (2016b) presents average (mean) scores as the main result, Klapper et al. (2015) report the percentage of participants per country who correctly answer questions on “at least three out of the four financial concepts” (p. 7). Calculating population percentages above a certain score threshold as opposed to score means may significantly influence final rankings and may thus be another reason for differing results.

4.6 Methodological considerations on measuring financial literacy

Test instruments not only need to be based on a clear understanding of the measured concept (i.e. they must have construct validity; see Schuhen and Schürkmann, 2014), they also have to be tailored to the intended target group and use case. As shown above, financial literacy assessments thus vary widely. In some instances, financial literacy may also be operationalized as part of more general economic literacy (Kaiser et al., 2020, see also subsection 3.5) or may be further specified as digital financial literacy (Lyons and Kass-Hanna, 2021; see also subsection 3.4).

In a recent overview of financial literacy measurements, Haupt (2022) concludes that capturing the concept of financial literacy presents a major challenge, citing reviews that find common issues regarding the consistency of definitions and the coverage of financial literacy components (e.g. Huston, 2010; Remund, 2010; Rieger, 2020). The small set of knowledge questions that dominates the research field (see subsection 4.1) may indeed not be sufficient to cover the range of knowledge or other individual traits relevant to fully capturing the intended concept of financial literacy. As with all test instruments, challenges thus include ensuring adequate psychometric properties, i.e. sufficient validity and reliability (see e.g. Kimberlin and Winterstein, 2008), adequate difficulty (Knoll and Houts, 2012) and test fairness across subpopulations (De Clercq, 2019).

Concerning the statistical analysis of test instruments, the analysis framework based on item response theory (IRT; see De Ayala, 2013) has become a popular alternative to using classical test theory. In the IRT framework, the probability of responses is explicitly modeled on the basis of question (item) and participant characteristics. One of the simpler IRT models, the Rasch or one parameter logistic (1-PL) model, for example, is similar to a logistic regression. Computed on answers to a set of knowledge questions with only right or wrong options, a 1-PL model can be used to obtain difficulty estimates for each question and an ability estimate for each participant (Mair, 2018). The IRT framework also offers a host of tests and

indicators to assess various quality criteria and assumptions of test instruments. For example, the fairness of questions across groups can be assessed in differential item functioning tests, i.e. tests of whether items behave similarly across different groups (e.g. De Clercq, 2019).

Despite the significant progress made both in the design and validation of test instruments, quantifying financial literacy relies on basic statistical assumptions that are not always made explicit. First and foremost, the dimensionality of the construct of financial literacy must be considered. Financial literacy could, for example, be regarded as a single, coherent construct or as a multidimensional construct that covers several components, each of which is to be measured separately and which are only combined at a later stage under certain conditions.

Moreover, one central assumption of most measurement instruments is the existence of a *latent* (unobservable) *construct* that manifests itself in observable and measurable *indicators*. For example, the correct responses to financial knowledge questions (the indicators) are interpreted as manifestations of financial literacy (the latent construct). This assumption is the basis of *reflective measurement*, which underlies the vast majority of tests and survey scales. Other approaches, in particular *formative measurement*, do not assume that indicators are manifestations of a latent construct but rather reverse the assumed causal direction by forming the construct through measurement itself (see Coltman et al., 2008). In some instances, such a formative approach might be better suited to assess a complex construct such as financial literacy (e.g. Warmath and Zimmerman, 2019).

No matter what approach is used, it is not possible to confirm the validity of a measurement instrument directly. Validity can only be inferred through the interrelationship of indicators or their correlations with other observable variables. All measurement instruments and analytical methods thus come with significant limitations that must be considered when interpreting assessment results.

Independently of the measurement instrument used, sampling, interview modes and weighting may also influence measurement results and should thus be deliberated and reported transparently. Indeed, from the initial understanding of financial literacy and the design of assessment instruments to methodological factors such as question phrasing, target population, sampling strategies, data collection methods, interviewer effects and statistical analysis, every stage of the assessment process can cause considerable disparity between measurement efforts (see subsection 4.5).

Researchers and evaluators should thus be cautious in their methodological choices and, if possible, use state-of-the-art measurement instruments, sampling strategies, interview methods and analysis frameworks. IRT, in particular, appears to be well-suited to analyze test instruments. As differences in methodological approaches will remain unavoidable, the transparent reporting of validation efforts and their results, e.g. reliability metrics, factor loadings and similar parameters, will facilitate comparisons between methods, measurements and, ultimately, evaluation results.

5 Determinants of financial literacy

Identifying determinants of financial literacy is the basis for designing evidence-based, targeted and effective financial education measures. In recent years, several comprehensive studies on this topic have synthesized and empirically tested findings from dozens of research articles exploring the relationship between financial literacy and specific variables (Klapper and Lusardi, 2020; Potrich et al., 2015; Zaimovic et al., 2023).

Based on this literature, the determinants of financial literacy can be categorized into four main groups: (1) sociodemographic factors, (2) socioeconomic factors, (3) macro-level factors and (4) financial education

interventions. These determinants have various implications for policies aimed at enhancing financial literacy levels beyond financial education as they prompt considerations of structural inequality within society (O'Connor et al., 2019; Salignac et al., 2019). However, any interpretation of these findings should consider that determinants of financial literacy may vary considerably according to the specific definitions adopted in the respective studies (see also section 3).

5.1 Sociodemographic determinants

A prominent finding in the literature shows that, on average, men tend to exhibit higher levels of financial literacy than women, even after controlling for factors like income or employment status (Klapper and Lusardi, 2020; Potrich et al., 2018; Preston and Wright, 2019). However, gender roles influencing participation in economic decision-making may account for a considerable proportion of this gender gap (Cupák et al., 2018). Several researchers have suggested that the gender gap in financial literacy may not solely be due to an actual lack of financial knowledge among women but rather to a disparity in *confidence* when dealing with financial matters (Klapper and Lusardi, 2020; Lusardi and Mitchell, 2011). It has been noted that even in cases where women would have guessed the correct answer, they often refrain from responding, indicating a tendency to underestimate their level of financial literacy (Bucher-Koenen et al., 2021). Interestingly, the gender gap in financial literacy appears to exist already during adolescence (Driva et al., 2016), suggesting that socialization practices and cultural norms possibly contribute to instilling an overall more passive approach to money matters in girls from an early age (Agnew and Cameron-Agnew, 2015; Calamato, 2010; Rink et al., 2021).

In developed countries, financial literacy shows an inverted U-shape relationship with age, suggesting that both young and old people tend to have lower average financial literacy levels than middle-aged people (Boisclair et al., 2017; Kadoya and Khan, 2020; Klapper and Lusardi, 2020). Although cohort effects, particularly in single-country studies, might have a certain influence on such results, studies featuring cross-country comparisons and multivariate analyses point to a consistent age effect. Following the notion of “learning-by-doing,” young people might simply have less experience in dealing with more sophisticated financial matters than older age groups (Frijns et al., 2014). Also, people’s willingness to acquire financial knowledge is likely to increase with the amount of available resources they can use for making investments. This amount is typically highest at the end of their professional life, around the age of 60 to 65 (Lusardi et al., 2017). However, as people enter their 60s and beyond, they again tend to exhibit lower levels of financial knowledge (Agarwal et al., 2007; Finke et al., 2017).

A well-documented positive relationship exists between formal education levels and financial literacy, suggesting that people with higher educational attainment tend to understand fundamental financial concepts better. This holds true even when factors like income are considered (Boisclair et al., 2017; Garg and Singh, 2018; Kadoya and Khan, 2020; Klapper and Lusardi, 2020; Van Rooij et al., 2011). This might be due to the fact that higher exposure to formal education implies greater chances of attending courses where relevant financial knowledge is actively taught. Notably, students who have participated in economics, finance or business classes tend to score better in financial literacy tests (Akben-Selcuk and Altıok-Yılmaz, 2014; Paraboni et al., 2020). Moreover, more years of formal education, including mathematics and science classes, can improve people’s numeracy, i.e. “the ability to access, use, interpret, and communicate mathematical information” (OECD, 2012a, p. 33). Numeracy, in turn, plays a crucial role in financial literacy (Cole et al., 2016; Darriet et al., 2022), suggesting that it might act as a mediator between formal education and financial literacy (Gan et al., 2019). Although substantive evidence is still lacking,

Gan et al. (2019) also discuss the potential influence of increased social capital associated with high levels of education, which facilitates financial knowledge acquisition and exchange through larger social networks.

5.2 Socioeconomic determinants

Income and wealth have been found to be major determinants of financial literacy, as *ceteris paribus* people with higher incomes and/or greater financial assets tend to exhibit higher levels of financial knowledge (Almenberg and S  ve-S  derbergh, 2011; Guiso and Jappelli, 2009; Kadoya and Khan, 2020; Klapper and Lusardi, 2020; Potrich et al., 2015).

Particularly as regards the relationship between financial literacy and wealth, there is empirical evidence of bidirectional causality. On the one hand, financial literacy is likely to facilitate wealth accumulation as it fosters prudent saving behavior and investment decisions (Lusardi and Mitchell, 2007; Van Rooij et al., 2012). On the other hand, people with higher incomes and greater wealth have a stronger incentive to acquire financial literacy as they can effectively use their financial knowledge to maximize their available capital and gain substantial returns by making better investment decisions (Lusardi et al., 2010; Monticone, 2010).

Conversely, people with limited assets and income might rationally choose to be ignorant toward certain types of financial knowledge and financial education measures since their primary focus is on meeting basic needs and they lack the financial resources to reap the benefits of thoroughly understanding concepts like inflation or compound interest (Lusardi et al., 2017; Son and Park, 2019).

5.3 Macro-level determinants

One main argument discussed in the literature concerns the relationship between financial literacy and the generosity of a country's welfare regime or public pension system. In countries where social security systems are highly developed – a situation which is usually associated with higher tax rates – individual responsibility for retirement planning or making emergency provisions is considerably reduced. As a result, people may have fewer incentives or even opportunities to acquire (a certain degree of) financial knowledge or behavior (see Almenberg and S  ve-S  derbergh, 2011; Crossan et al., 2011; Fessler et al., 2020; Jappelli and Padula, 2015).

In countries like the USA, by contrast, where social security is less extensive, people have greater personal responsibility as they must *privately* provide for their retirement and hedge against various risks. Consequently, they face more financial decisions that may have severe consequences which, in turn, means that they can literally not afford to be ignorant of at least basic financial literacy if they want to avoid financial difficulties (Finlayson, 2009; Lusardi and Mitchell, 2011; Oehler and Werner, 2008).

Concerns about financial illiteracy and an increased emphasis on financial literacy are often linked to future challenges of sustaining public pension and social security systems, implying that people may need to assume more responsibility for their own financial well-being instead of relying on public systems (Almenberg and S  ve-S  derbergh, 2011; Arrondel et al., 2014; Lusardi and Mitchell, 2011; Oehler and Werner, 2008). However, it is essential to note that financial literacy is not an end in itself but should ultimately contribute to financial well-being. While giving more responsibility to individuals by cutting social security benefits may possibly increase their financial literacy, it might have negative consequences for the financial well-being of large parts of the population, especially those with low incomes and low educational attainment who are prone to also lack financial literacy.

Experience from people's own history may also play a role in shaping their financial knowledge (Klapper and Lusardi, 2020). Some scholars report evidence that people who witnessed their countries undergo hyperinflation tend to understand the concept of inflation better than those who live in countries with stable prices (Hanke and Krus, 2013). Others report no such effect (Beckmann, 2013; Moure, 2016). This suggests that even if "inflation literacy" is enhanced in the short term by corresponding events, it declines again when the issue of inflation becomes less urgent.

Finally, culture and social context may also influence people's financial literacy. Klapper and Lusardi (2020) highlight the possible impact of cultural norms and attitudes related to managing money or incurring debt. Especially regarding people's risk propensity, long-term orientation and ease of discussing personal finance issues, cultural differences are prone to explain a certain degree of intercountry (De Beckker et al., 2020) and intracountry variation (Brown et al., 2018; Davoli and Rodriguez-Planas, 2020) in financial literacy levels.

5.4 The role of financial education

Financial education must be considered the main intervenable factor of financial literacy, i.e. the primary way to improve financial literacy through interventions. Indeed, Huston (2010) describes financial education as an "input intended to increase a person's human capital, specifically financial knowledge and/or application (i.e., financial literacy)" (p. 308, emphasis added), emphasizing those providing the input, i.e. the educators.

In contrast, the OECD defined financial education as "the *process* by which financial consumers/investors improve their understanding of financial products, concepts and risks and, through information, instruction and/or objective advice, develop the skills and confidence to become more aware of financial risks and opportunities, to make informed choices, to know where to go for help, and to take other effective actions to improve their financial well-being" (OECD, 2005, p. 13, emphasis added). This definition highlights that financial education can also be considered an active process by which people acquire financial literacy.

One subject of discussion is the *content* of financial education measures, which, as some authors claim, should go beyond personal finance. Retzmann and Seeber (2016) e.g. argue that "being financially educated means more than being financially literate and should be seen as a proper subset of economic education" (p. 9). This understanding also refers to financial education as an "end," rather than an input or a process. Others criticize a narrow understanding of financial education as a form of "consumer training" and want financial education to contribute to the development of a "critical reflective-reflexive capacity to make informed choices in life" (Baumann and Hall, 2012, p. 513).

While it is evident that financial education influences financial knowledge, there has been mixed evidence regarding its impact on actual financial behaviors (Fernandes et al., 2014; Kaiser and Menkhoff, 2017). This "knowledge-behavior gap" can be deemed highly problematic as it is only through decision-making, actions and behaviors that financial literacy has the potential to contribute to financial well-being. Moreover, any impact may be considerably time sensitive. Fernandes et al. (2014) find in their analysis that the effect of financial education measures wears off rather quickly, and they thus conclude that financial education could be effective when it takes place "just in time" and is tied to specific decisions or behaviors.

Relatedly, there is little agreement on whether financial education should be primarily about teaching knowledge that is later applied in a financial decision-making context or whether it should try to directly foster prudent financial behaviors (Bartholomae and Fox, 2022). The World Bank, for example, has been open toward direct, behavioral approaches (see subsection 2.2). Indeed, combinations of conventional

financial education and direct behavioral interventions (“nudging”) may prove particularly effective (e.g. García and Vila, 2020).

Given the limited comparability and validity of studies (see Willis, 2022), drawing a definite conclusion on the effectiveness of financial education will remain challenging. Nevertheless, more recent meta studies provided evidence that financial education does have an impact on people’s financial behavior (Kaiser et al., 2022). Overall, there is indeed growing causal evidence that financial education and related methods positively affect behavior in certain settings, particularly large-scale school programs and “innovative” methods for adults (Kaiser and Lusardi, 2024).

As to the causal mechanisms that translate financial literacy into financial behavior, psychological traits, such as self-efficacy in financial matters (see Goyal and Kumar, 2021; Lučić et al., 2023), are likely to play an important role. However, the causal mechanisms underlying the relationship between financial education and financial behavior are still not fully understood (see Kaiser and Lusardi, 2024). Ultimately, just like financial literacy, financial education itself represents a complex and highly heterogeneous concept.²

6 Effects of financial literacy

6.1 Financial behavior

Many studies on financial literacy focus on financial knowledge and its effects on financial behaviors such as making investments or borrowing. As financial behaviors are sometimes included in the term “financial literacy,” e.g. in the OECD/INFE definition, the distinction between components of financial literacy and effects of financial literacy has been considerably blurred. Nevertheless, studies on the connection between financial literacy (in terms of financial knowledge) and the associated financial behaviors can be considered crucial evidence of the causal mechanisms that lead to financial well-being in the long run.

Indeed, there is a growing body of research indicating the importance of financial knowledge and its positive effects on various aspects of personal finance. One of the earliest studies conducted by Hilgert et al. (2003) demonstrated a strong link between having sound financial knowledge and engaging in good financial practices, including managing cash flow, handling credit, saving and investing. Subsequent research has further confirmed this correlation, showing that financial literacy is associated with retirement planning as well as accumulating savings and wealth (Ameriks et al., 2003; Behrman et al., 2012; Hung et al., 2009; Lusardi, 2003; Lusardi and Mitchell, 2007; Van Rooij et al., 2012).

Moreover, a person’s level of financial knowledge appears to be predictive of various investment behaviors, such as participating in the stock market (Christelis et al., 2010; Van Rooij et al., 2011), achieving higher risk-adjusted investment returns (Clark et al., 2017; Van Rooij et al., 2011), selecting low-fee investment portfolios (Hastings et al., 2011) and engaging in better diversification and more frequent stock trading (Graham et al., 2009). People with higher financial knowledge are also less susceptible to exploitation or deception (Andreou and Philip, 2018; Balloch et al., 2015; de Bassa Scheresberg, 2013; Campbell et al., 2011; Deevy et al., 2012; Lusardi and Mitchell, 2011). In contrast, low levels of financial literacy are associated with negative credit behaviors such as accumulating debt (Andreou and Philip, 2018; Lusardi et al., 2020; Lusardi and Tufano, 2009) and resorting to high-cost borrowing (Lusardi and Tufano, 2009).

² Further details on financial education measures and their effectiveness go beyond the scope of this paper but will be explored in other issues of this series.

Overall, financial knowledge appears to be an important predictor of beneficial financial behaviors. However, the results presented above are exclusively based on correlational studies and can thus not serve as causal evidence. Other factors, such as disposable income, could influence both financial knowledge and financial behavior. Financial knowledge may also be the result of behavior itself, where financial knowledge increases through learning-by-doing or other mechanisms.

6.2 Financial resilience and financial vulnerability

According to the OECD (2021b), financial resilience refers to “the ability of individuals or households to resist, cope and recover from negative financial shocks” (p. 35). Salignac et al. (2019) define financial resilience more specifically “as an individual’s ability to access and draw on internal capabilities and appropriate, acceptable and accessible external resources and supports in times of financial adversity” (p. 21). Financial literacy, as an internal capability, may thus help people develop effective budgeting and saving habits and cultivate a mindset of preparedness, eventually enabling them to better cope with events that negatively affect their personal financial situation, such as job loss or financial shocks like the need to repair or replace necessary appliances.

Sometimes, instead of (or in distinction to) financial resilience, the term “financial vulnerability” or similar terms such as “financial fragility” (Lusardi et al., 2011) or “economic insecurity” (Hacker, 2018) are used to describe a person’s *likelihood to fall into financial hardship* (O’Connor et al., 2019). Thus, financial vulnerability does not (necessarily) imply that poverty or a state of financial hardship are already present. High-income earners, for example, may have an expensive lifestyle that prevents them from setting money aside to compensate for future financial shocks. Conversely, people with lower incomes can be very economical and forward looking and therefore able to build the necessary financial buffers. Despite the apparent similarities between financial vulnerability and financial resilience, it remains unclear whether the former can be regarded as the mere opposite of the latter, i.e. whether high vulnerability equals low financial resilience and vice versa, or whether financial resilience, in fact, requires more than the relative absence of financial vulnerability.

Several studies have identified indicators that can be used to quantify current levels of financial resilience or financial vulnerability by considering the potential impact of financial shocks (see e.g. Bialowolski et al., 2022; Mainwaring; 2020; OECD, 2020a; Ratcliff et al., 2022) or the ability to cope with a sudden fall in income or an unexpected expenditure (Financial Resilience Task Force, 2019). On a general level, studies from around the world have found that higher levels of financial knowledge are associated with holding more money in savings and investments and borrowing less (Lusardi and Tufano, 2015; Lyons et al., 2019). Some evidence also indicates that financial literacy might be able to strengthen financial resilience and reduce financial vulnerability (Anderloni et al., 2012; Andreou et al., 2023; Clark et al., 2021; Lusardi et al., 2021).

However, it is important to acknowledge that financial resilience or financial vulnerability are heavily influenced by external factors that are largely beyond a person’s immediate control, including wealth, income, employment opportunities or social support systems (Anderloni et al., 2012; Lusardi et al., 2011; Salignac et al., 2019). Consequently, low financial resilience cannot be attributed solely to a lack of financial literacy without considering prevalent societal and interpersonal disparities concerning income, assets, wealth and social capital.

6.3 Financial well-being

The recent shift from a primary focus on financial literacy to a broader concept of *financial well-being* marks a significant evolution in the field of financial literacy. For many well-known organizations and initiatives active in financial education, such as PISA (OECD, 2012d), OECD/INFE (2020) and the CFPB (CFPB, 2015), financial well-being now represents the overarching goal of their financial education efforts. In parallel, national strategies are shifting their focus toward financial well-being, replacing their previous emphasis on financial literacy (OECD, 2022a).

The new focus on financial well-being acknowledges that people's financial lives are not solely shaped by financial literacy but are instead influenced by a multitude of other determinants (e.g. Warmath, 2022), including contextual and structural factors (Bowman et al., 2017). For instance, a family with strong financial literacy might still face financial challenges due to external factors like high costs of living or local economic conditions. This example highlights the need for a more comprehensive approach to financial literacy and financial well-being.

There is no universally accepted definition of financial well-being (Brüggen et al., 2017; Riitsalu and Van Raaij, 2022). Some studies attempted to define financial well-being from the perspective of consumers or financial professionals. A study by the CFPB (2015) explored the meaning of financial well-being through interviews in the United States, arriving at the definition of financial well-being as “a state of being wherein a person can fully meet current and ongoing financial obligations, can feel secure in their financial future, and is able to make choices that allow enjoyment of life” (p. 18).

Applying a similar approach, Kempson et al. (2017) reanalyzed focus group data previously published by the World Bank (Kempson et al., 2013; Holzmann et al., 2013) and the UK Financial Service Authority (Atkinson et al., 2006; Personal Finance Research Centre, 2005). Based on their analysis, they describe financial well-being as “the extent to which someone is able to meet all their current commitments and needs comfortably, and has the financial resilience to maintain this in the future” (Kempson et al., 2017, p. 19).

Another strand of definitions has emerged from literature reviews, including one proposed by Brüggen et al. (2017), who define financial well-being as an individual's “perception of being able to sustain current and anticipated desired living standards and financial freedom” (p. 229). Financial well-being is also related to various other terms, including “financial satisfaction,” “financial wellness,” “economic well-being” or “financial efficacy,” and is frequently used as a synonym for financial health (Fu, 2020).

An ongoing debate revolves around the question of whether financial well-being should be assessed through objective factors, subjective factors or a combination of both (Brüggen et al., 2017; Riitsalu and Van Raaij, 2022). Objective financial well-being comprises quantifiable aspects such as income, expenses, savings and debt. Historically, financial well-being has often been used as a synonym for income, representing an objective factor (Sorgente and Lanz, 2019).

In contrast, subjective financial well-being reflects a person's perception of their financial situation (Salignac et al., 2019). Here, objective measures such as income may serve as determinants of financial well-being but are not considered part of the concept itself (Warmath, 2022). In the current discourse, the notion that financial well-being is purely objective is rejected. However, it remains a matter of debate whether financial well-being encompasses both objective and subjective dimensions (Kempson et al., 2017; Sorgente and Lanz, 2019) or whether it is solely subjective in nature (Brüggen et al., 2017; Fan and Henager, 2022; Riitsalu and Van Raaij, 2022).

Proponents of a purely subjective understanding of financial well-being consider it to be influenced by personal factors such as social reference groups and personal preferences (Brüggen et al., 2017). Proponents of a combined approach argue that it offers a more balanced perspective and is less susceptible to

distortions (Tenney and Kalenkoski, 2019). Research also consistently highlights that financial well-being encompasses both current and future elements (Fan and Henager, 2022). While most studies combine present and future aspects into one factor, some studies suggest that current and future well-being may be distinct concepts influenced by different determinants (Netemeyer et al., 2018; Riitsalu and Van Raaij, 2022).

Going beyond the concept of financial well-being itself, researchers have made some efforts toward developing a structured understanding of the factors contributing to it. Their studies are mostly empirical and sometimes theoretical in nature (Goyal and Kumar, 2021). The prevailing trend in most studies is to list factors that are empirically linked to financial well-being without delving into an in-depth exploration of the underlying mechanisms. Some recent studies, however, place more emphasis on the theoretical basis of these mechanisms (Bowman et al., 2017; Lučić et al., 2023).

Based on current literature³, determinants of financial well-being can be broadly categorized as:

- *contextual/environmental factors*: economic conditions, income and wealth in particular, cultural norms and policy environments;
- *financial attitudes and behaviors*: individual approaches to finance management, such as spending habits and saving attitudes;
- *financial knowledge and skills*: financial literacy playing a critical role in shaping financial decisions and behaviors;
- *personal traits*: personal characteristics such as resilience and adaptability.

While financial literacy has been consistently shown to have an indirect influence on financial well-being through financial behaviors, contextual factors such as income and wealth are its most influential determinants by far. A recent empirical study (ANZ Bank New Zealand Limited, 2021) found that socioeconomic factors account for 55 % of variations in financial well-being, money management behaviors for 21 % and behavioral traits for 14 %. Financial attitudes and personal traits primarily have an indirect impact on financial well-being through their influence on people's financial behaviors (Çera et al., 2021; Hwang and Park, 2023).

Although financial well-being is today regarded as the overarching goal of financial literacy, the lack of a common understanding of the concepts themselves and of a theoretical basis restricts the ability to conduct meaningful analyses of their interplay. While financial literacy has been demonstrated to have an indirect effect on financial well-being through people's financial behavior, further research is still required to determine what type of financial literacy and financial education interventions effectively contribute to financial well-being or, on a larger scale, to societal financial stability and crisis resilience.

7 Conclusion

This paper aims to provide a suitable starting point for navigating the diverse research and policy landscape of financial literacy. To this end, we explore the concept of financial literacy from six perspectives, ranging from its historical origins and evolution to the promoting organizations, definitions, measurement approaches, determinants and, finally, to its outcomes and effects. We find that financial literacy can be considered the core concept of a research and policy field that is, in essence, concerned with people's ability to improve their financial outcomes. Unsurprisingly, however, we neither find a universal definition

³ ANZ Bank New Zealand Limited (2021); Atkinson and Messy (2012); Brüggen et al. (2017); CFPB (2015); Fan and Henager (2022); Goyal and Kumar (2021); Kempson et al. (2017); Lučić et al. (2023); Xiao et al. (2022).

of financial literacy nor one common methodological approach to measuring it nor conclusive evidence of its causal relevance for financial well-being and similar effects.

We find that the field has progressed substantially in recent years, with evidence-based approaches having emerged just around the turn of the millennium. Despite the leading role of influential (international) organizations in promoting financial literacy, definitions of financial literacy remain heterogeneous across countries, policymaking bodies and research. Based on various popular definitions of financial literacy and closely related concepts, we find five aspects in which definitions differ with regard to fundamental theoretical assumptions, namely: (1) goals, (2) traits considered, (3) inclusion of observable behaviors, (4) thematic focus and (5) inclusion of environmental and context factors. We recommend that researchers, policymakers and educators consider and report on the definitions and fundamental assumptions they adopt to minimize ambiguity and promote comparability of both interventions and research results.

The heterogeneity of underlying assumptions and concepts becomes strikingly apparent in the different approaches to measuring financial literacy. Although many assessments still rely on a few short knowledge questions, others employ much larger sets of questions that cover other components such as behaviors and attitudes. We find that different methodological approaches, such as in sampling or statistical analyses, can considerably influence the results of financial literacy assessments. We thus recommend increasing efforts to reduce the sampling bias in surveys, comprehensive validations of test instruments and the use of advanced statistical frameworks (item response theory in particular) to further advance the measurement of financial literacy and produce useful and comparable measurements.

Finally, the causal relationships between financial education, financial literacy and financial well-being remain ambiguous. Studies report significant differences in financial literacy depending on sociodemographic, socioeconomic and macro-level determinants and show associations both between financial knowledge and financial behaviors and between financial literacy and financial resilience and financial well-being. However, a clear causal link between financial education and financial behaviors, let alone financial well-being or societal financial stability, has yet to be established. More research and evaluation are therefore needed to ascertain what type of financial education and what type of financial literacy affect people's lives in what way.

Given the striking diversity and the interdisciplinary nature of financial literacy-related perspectives and methods, we also argue for establishing some level of consensus on terminology, methodological approaches and reporting standards. While diversity has clearly enriched the discourse, we expect that critically reflecting on the underlying assumptions of concepts and making transparent theoretical and methodological choices can only improve the comparability of interventions and research results. We also call for increased collaboration across institutions and heightened methodological standards in evaluation efforts and large-scale assessments to ensure coherent evidence for effective policymaking.

This paper is restricted in scope and therefore characterized by many omissions. It primarily focuses on quantitative evaluation and research methods and does not discuss qualitative or mixed-method approaches. We mainly explore English-language and international literature and concentrate on definitions and approaches applied by the most prominent (international) organizations in the field, which necessarily means that many authors and organizations have not been considered. We only briefly address concepts beyond the concept of financial literacy, the concept of financial education, in particular. Indeed, our focus on the core concept of financial literacy and its most prominent definitions and approaches aims to provide a structured, accessible overview that explores key aspects of the concept and facilitates informed theoretical and methodological choices in both research and policymaking.

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Part II:

Financial education in practice and in pedagogical theory



Is financial knowledge enough? Reviewing impact assessments of financial education interventions

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This article presents a thematic review of the financial literacy evaluation landscape in OECD countries from 2010 to 2024, analyzing 68 (quasi-)experimental and peer-reviewed studies. We identify key patterns across study contexts, intervention types, outcome measures, results and methodological approaches. Our findings indicate that financial education interventions consistently improve financial knowledge across diverse contexts and target populations. However, evidence on their impact on financial attitudes and behaviors remains mixed and appears to be largely contingent on the intervention and study design. Given the considerable heterogeneity across studies, we propose methodological guidelines to enhance comparability and facilitate more robust conclusions on intervention effectiveness. To better understand the relation between financial education interventions and intended impacts, we recommend that future studies place a stronger focus on the underlying causal mechanisms and devote greater efforts toward assessing these effects in study designs.

The rise of financial literacy and financial education as a policy and research topic can be understood as a reaction to developments of the financial landscape in the late 20th and early 21st century: Increased globalization, digitalization, the proliferation of complex financial products and increased economic and financial uncertainty (including various global crises) have created a financial environment that is increasingly difficult to navigate.

Indeed, consumers' financial literacy appears to be linked to financial stability and well-being on a global scale; the 2008 financial crisis has been attributed, in part, to consumers' lack of financial literacy (see e.g. Kovács and Terták, 2019; Shiller, 2008). This may also explain the sharp increase in policy focus and the exponential growth of research output on financial literacy starting after 2009 (see Goyal and Kumar, 2021).

However, knowledge and skills concerning the management of personal financial resources were certainly relevant before and topics and themes of financial skills do not appear to have changed much in the past century. In 1926, Clason already wrote about topics such as controlling expenditures, saving, compound interest, planning for the future, the risk of investments and home ownership (Clason, 1926). While more complex financial topics are the focus of specialized fields of financial education and financial literacy (for example digital financial literacy, see e.g. Lyons and Kass-Hanna, 2022), the majority of financial education interventions appear to focus on basic issues of personal finance.

Nevertheless, the scientific research field on financial literacy is still very young. The field is not only characterized by a lack of conclusive empirical evidence but also by substantial heterogeneity in definitions and theoretical assumptions. Definitions of financial literacy can vary considerably in terms of the

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assumed goal dimensions, traits, behaviors and content areas.² The prevalent heterogeneity in the field can also be observed in financial education interventions themselves, in target populations, the educational environment, desired outcomes of interventions and study designs.

Evidence from correlational studies indicates that financial literacy (in terms of financial knowledge) is strongly linked to financial behaviors. Several meta-analyses have been conducted to assess the effectiveness of financial education interventions on financial knowledge and behaviors (the most recent being Kaiser et al., 2022). However, meta-analyses showcase how drawing universal conclusions is severely limited by heterogeneity in the design and context of interventions and by insufficient reporting on intervention details. Existing meta-analyses provide little information on underlying causal mechanisms as they rely on a small set of broad metrics that are applicable to the whole study pool.

In contrast to meta-analysis, this paper employs a different approach by thematically reviewing the diverse landscape of financial education studies. First, we review studies to identify patterns across study contexts, intervention types, outcome measures, results and methodological aspects. Where possible, we aim to offer guidance on program characteristics that show promising effects on financial literacy. Second, building on the observations from the literature review, we offer methodological recommendations. These recommendations aim to enhance the quality and comparability of research on financial education interventions and to facilitate more robust meta-analyses and systematic reviews in the future.

For the review we adopted the following methodical approach: We conducted a search³ for peer-reviewed journal articles that report on the effects of financial education interventions on financial literacy (including knowledge, attitudes and behaviors) or financial well-being. The publication period we reviewed was January 2010 to April 2024. We limited our scope to studies conducted in OECD member countries to reduce the overall complexity and heterogeneity stemming from countries' economic conditions and education systems. We further limited our study pool according to the study design, only including (quasi-)experimental studies that (1) compared at least two treatment groups or included a control group and that (2) either used randomized group assignment or a quasi-experimental design with a clear identification strategy. The latter category predominantly used pre- and post-test designs and difference-in-difference estimation.

Our search and selection resulted in 68 studies, which we categorized based on the target population: 29 interventions focused on children and adolescents and 39 on adults. Given the substantial heterogeneity in interventions for adults, we further subdivided them into four groups: (1) interventions targeting vulnerable populations, (2) interventions focusing on retirement savings, (3) interventions applicable to the general adult population and (4) interventions with college or university students.

This article was written for financial education program designers and educators interested in evidence-based financial education results. It also serves researchers and policymakers by providing an overview of the current landscape of financial literacy program evaluations and offering methodological recommendations for advancing this field of research. Additionally, this article is a key component of the OeNB Financial Literacy Evaluation Series⁴, which links the exploration of the concept of financial literacy in

² Zieser et al. (2024) explore the concept of financial literacy in more detail in a previous issue of the OeNB Financial Literacy Evaluation Series.

³ We conducted a search on the Scopus database and Google Scholar and included all studies from recent meta-analyses that fulfilled our criteria (in particular Kaiser and Menkhoff, 2017, 2020, 2022).

⁴ All issues of the OeNB Financial Literacy Evaluation Series can be found under <https://www.oenb.at/en/Publications/financial-literacy/financial-literacy-evaluation-series.html>.

research and policymaking (Zieser et al., 2024) and several practice-oriented issues on evaluation methods (see e.g. Felbermayr, 2024; Lorenz, 2024).

The remainder of this article is organized as follows. In section 1, we introduce the concept of financial education as it is used in empirical research, specifically in relation to financial literacy. Section 2 provides an overview of prior empirical evidence, drawing on insights from meta-analyses. In section 3, we summarize the key characteristics of financial education evaluations for children and adolescents. Section 4 presents our findings for adults, with subsections dedicated to the four adult subgroups discussed earlier. In section 5, we synthesize the findings from the sections on both children and adults to draw conclusions about which types of financial education interventions are effective for different groups. Finally, section 6 offers methodological recommendations for addressing the limitations of existing evidence, providing guidance for future financial education evaluations.

1 Financial education in empirical research on financial literacy

In the international literature surrounding the concept of financial literacy, financial education is generally considered a *determinant* of financial literacy and the primary way to foster financial literacy through active intervention and, by extension, to promote beneficial financial behaviors and financial well-being. The World Bank, for example, follows the understanding that “*financial education is a tool for increasing consumer financial literacy*,” and financial education programs are “*tools for increasing consumers’ financial literacy and capability*” (World Bank, 2014, p. 1).

Financial education can also be considered either the input or the process that leads to financial literacy. The OECD considers it “*the process by which financial consumers/investors improve their understanding of financial products, concepts and risks and, through information, instruction and/or objective advice, develop the skills and confidence to become more aware of financial risks and opportunities, to make informed choices, to know where to go for help, and to take other effective actions to improve their financial well-being*.” (OECD, 2021, p. 21). This definition puts the focus on the active *process* by which people acquire knowledge, skills and confidence. Conversely, financial education could also be defined as the “*input intended to increase a person’s human capital, specifically financial knowledge and/or application (i.e., financial literacy)*” (Huston, 2010, p. 308). Such a definition emphasizes the *input*, i.e. the role of the education intervention.

Depending on the definition of financial literacy, any intervention that addresses financial literacy could thus be considered to be a financial education intervention. In case of the OECD’s definition of financial literacy (OECD, 2020), this could be any intervention addressing any or all of the components “financial knowledge,” “financial attitudes” and “financial behaviors,” with the ultimate goal to improve financial well-being. This appears to be in line with the empirical literature, where approaches that address e.g. saving attitudes or behavior directly, without aiming to improve knowledge, are regularly subsumed under the financial education umbrella. We thus apply this prevalent understanding of financial education for the selection of studies for this review article.

There is debate over whether financial education should focus on economic knowledge applied to decision-making or directly foster prudent financial behaviors (Bartholomae and Fox, 2022). Additionally, some argue that financial education should go beyond personal finance, viewing it as a subset of economic education (Retzmann and Seeber, 2016), while others criticize a narrow “consumer training” approach and advocate for developing critical, reflective decision-making skills (Baumann and Hall, 2012). For the purpose of this review, however, we follow what has been considered financial education in the empirical literature on financial literacy. In subsection 6.1, we discuss causal channels that future research could explore.

The main causal pathway assumed for the majority of financial education interventions appears to be via knowledge, where an intervention improves knowledge, which in turn enables desired behaviors, financial well-being or other desirable outcomes. However, the theoretical underpinnings of this assumed causal chain is rarely clearly stated, let alone explicitly formalized or analyzed.

Nevertheless, empirical research on the isolated link between financial knowledge and financial behavior appears to be abundant. Financial knowledge has been associated with positive financial outcomes (see Gomes et al., 2021), such as financial planning (e.g. Lusardi and Mitchell, 2011a), savings (e.g. Jappelli and Padula, 2013), more sophisticated investment behavior (Gaudecker, 2015; Rooij et al., 2011) or less risky mortgages as an important mechanism in preventing financial crises (e.g. Gerardi et al., 2010; Klapper et al., 2012; Zahirovic-Herbert et al., 2016). Moreover, financial knowledge appears to have a significant relationship with financial well-being that is mediated via financial behaviors (ANZ Bank New Zealand Limited, 2021; Hwang and Park, 2023). However, these results are exclusively based on correlational studies.

Evidence on the causal effects of financial education interventions is less clear. Kaiser and Lusardi (2024) highlight the growing evidence from meta-analyses, from recent natural experiments, particularly those on mandatory financial education in high schools and from a relatively new wave of experimental literature on what the authors label “innovative” financial education measures for adults. They emphasize that factors such as relevance to the situations of participants, accessibility and entertainment value, as well as scalability seem to be important factors for the effectiveness of these initiatives. However, they also conclude that the causal mechanisms driving the effects of financial education remain largely unclear.

The heterogeneity of financial education interventions and related research can be attributed to the concept of financial literacy. Positioned at the center of the assumed causal chain between financial education and financial behavior or financial well-being, financial literacy itself lacks a coherent, accepted definition in the existing literature. Definitions of financial literacy differ, for example, in terms of envisioned content areas and situations in which it is applicable, but also with regard to its specific goal dimensions and the extent to which behavior is included as a component of financial literacy (see Zieser et al., 2024 for an overview). Of course, the same heterogeneity can also be expected in concepts of financial education as the primary form of intervention addressing financial literacy and in financial behaviors and financial well-being as the central downstream outcomes.

Heterogeneity of the financial education literature also stems from empirical studies on interventions. This becomes especially apparent in meta-analysis. Fernandes et al. (2014), for example, provide little information on the specific design of interventions considered. Instead, they state: *“It is striking how many papers do not give enough details about their financial education interventions to be able to code for variables that might plausibly affect their effect sizes.”* (p. 29). They thus recommend that authors report key aspects of the intervention. Similarly, Kaiser et al. (2022) cluster studies by outcome variables and by intervention format (e.g. classroom- or online-based), but do not provide any classification based on intervention content, curriculum or didactic aspects. Moreover, Fernandes et al. (2014) observe a distinct disconnect between theoretical concepts addressed in the studies and how they are measured. They find that financial literacy appears to be frequently defined as a skill or form of expertise. In empirical studies, however, the concept is typically measured merely through a small number of knowledge questions on financial topics.

As opposed to trials testing the effectiveness of clearly defined interventions and outcomes (e.g. drug trials), research on financial education interventions may thus be considered too heterogeneous to draw clear conclusions (see Willis, 2022, for a critical discussion of this heterogeneity). Despite increasingly large study pools analyzed in meta-analysis, it remains unclear what the “active ingredients” of financial

education are and by which causal mechanisms they improve which desired outcomes. Nevertheless, there is mounting evidence from meta-analysis that interventions *typically classified* as financial education exert, on average, a positive influence on financial knowledge and, to a lesser degree, on financial behaviors. The following section summarizes such causal evidence from the most prominent meta-analysis.

2 Prior evidence from meta-analyses

Several meta-analyses on the effects of financial education have been conducted in the past decade. Due to a rapidly growing number of empirical studies over the past two decades, meta-analyses were able to use increasingly strict inclusion criteria with regard to the study design. Overall, conclusions from meta-analyses are fairly consistent: On average, financial education interventions increase financial knowledge. Effects on financial behaviors also appear to be positive, but evidence is considerably weaker.

In one of the first meta-analyses, Fernandes et al. (2014) incorporated both econometric and (quasi-) experimental studies reported in 168 papers in which financial literacy was either measured or manipulated through education interventions. Similarly, Miller et al. (2015) analyzed 188 studies with results from studies on “financial literacy and capability interventions.” Kaiser and Menkhoff (2017) narrowed the methodical scope of studies and included 126 (quasi-)experimental studies. Kaiser and Menkhoff (2020) analyzed 37 (quasi-)experimental studies conducted in schools. Lastly, Kaiser et al. (2022) focused exclusively on 76 randomized experiments.

In meta-analyses, authors typically code studies according to intervention and study characteristics to create variables which are tested for effects on the overall impact of financial education. Fernandes et al. (2014), for example, coded studies according to the intensity of the intervention, the type of intervention (high school, counseling, seminar or workshop, multiple sources, information exposure), the type of behavioral outcome assessed (saving, planning for retirement, absence of debt, stock ownership and investment decisions, cash flow management, activity in retirement plans, financial inertia), the delay in months between intervention and measured behavior and other study and population characteristics. Other meta-studies used similar approaches. It is important to note that none attempt to cluster the thematic focus or hypothesized causal mechanisms of interventions themselves.

Fernandes et al. (2014) find that correlations between measured financial literacy and behavior exceed the comparatively small effects on behavior found in studies where financial literacy was influenced, which suggests that correlations between financial literacy and behaviors cannot serve as suitable indicators for the effectiveness of financial education. While they report stronger overall effects for high-intensity education interventions, they also find a rapid decay of the effects. They also find that seminars or workshops, counseling and financial education in high schools show the strongest effects, mere exposure to information about financial education the weakest. Using an extended study pool of studies assessing financial education interventions, Miller et al. (2015) use a detailed classification of outcome variables, combining results for the following binary outcomes: “savings reported in past period,” “contributes to retirement savings,” “defaulted on a loan” and “keeps financial records/budgets” (p. 234). Overall, their analysis suggests that saving and recordkeeping behavior are impacted most by financial education interventions.

Only considering (quasi-)experimental studies, Kaiser and Menkhoff (2017) find that financial education strongly affects financial literacy and that it has a small but measurable effect on financial behavior. Some outcomes, such as borrowing behavior, appear more difficult to affect than others, such as budgeting or saving. Concerning the interventions themselves, they find that higher-intensity financial education

generally shows stronger effects and that the type of intervention determines effect sizes. Mandatory education, in particular, exhibits weaker effects than voluntary education interventions. Education at a “teachable moment” or “just in time” for relevant decisions provides stronger effects, but no differences are reported with regard to intervention channel and format. Focusing on the school context only, Kaiser and Menkhoff (2020) find results similar to those found in the general population, namely that there are clear and pronounced effects on financial knowledge and less certain effects on financial behavior. More intense treatments produce stronger effects and there appears some decay of the effect on knowledge over time. As school-specific results, the authors find that smaller class sizes appear to be associated with stronger treatment effects.

In the most recent meta-analysis that attempts to identify a generalizable effect of financial education, Kaiser et al. (2022) only consider randomized experiments. In their analysis, they find statistically robust effects of financial education on both financial knowledge and financial behaviors. In contrast to Fernandes et al. (2014), the authors do not find evidence of rapid decay of the effects over time. A small number of studies covering time spans longer than 18 months, however, limit the conclusiveness of this result. Concerning the types of interventions, Kaiser et al. (2022) conclude that, on average, classroom financial education appears to be effective, while less costly and less intensive interventions (such as information “nudges”) are less effective. More costly personalized interventions (such as counseling) appear most effective. In line with previous meta-analyses, the authors find stronger effects on certain outcomes, in particular financial knowledge, budgeting and saving, and smaller effects on credit, insurance and remittance (money transfer) behavior. The authors tie these differences in outcomes to a predominant focus of studies on saving and budgeting behavior and to different causal mechanisms by which financial education can affect different types of behavior.

3 Financial education for school children

Financial education interventions are frequently targeted at children and adolescents (e.g. Mauser et al., 2024). Consequently, it is valuable that many impact evaluations analyze interventions aimed at this group of people. In fact, 43 % of studies in our study pool examine interventions for children and adolescents. Except for one study (Migheli and Coda Moscarola, 2017), all of them take place within formal schooling settings. Therefore, we frequently refer to interventions in this section as “school-based.”

Although financial education programs for children vary in implementation details and pedagogical approaches, they tend to be more homogeneous compared to those targeting adults. This greater consistency is especially apparent in the specific target population (programs designed for specific age groups), the setting (typically school-based), limited variations in program intensity and less diverse contents covered. Additionally, the large share of rigorous school-based evaluation studies contributes to a more reliable assessment of their effectiveness.

Given this comparatively higher consistency, this section first aims to provide an overall summary of the key characteristics of financial literacy evaluations for children and adolescents. The remainder of this section highlights emerging patterns found in interventions’ content and conceptual focus, explores differences between well-established education initiatives with pilot programs, and addresses variations of program development and implementation, outcome measures and effects, as well as teacher training and parental involvement. To better contextualize results and guide their interpretation, we additionally expand on existing literature beyond our study pool.

3.1 Main study characteristics

School-based interventions are typically designed to align with specific educational levels and target students, rather than other influential stakeholders within the school system, such as teachers or parents. We will discuss their role in sections 3.7 and 3.8. Financial education evaluations begin at grade four in primary school, corresponding to children aged approximately 8–11 years in most countries. Research suggests that children make significant advancements in their financial and economic understanding between the ages of six and twelve. This development is largely attributed to cognitive growth, socialization and early personal financial experiences. Even below that age, important cognitive abilities, such as executive functioning, basic numeracy and financial attitudes start to form (see for example Consumer Financial Protection Bureau, 2016b; Van Campenhout, 2015; Webley, 2005). However, our study pool does not include interventions targeting students below the age of eight, likely because financial education programs and their impact evaluations are less common at very young ages (Holden et al., 2009).

Studies of secondary school interventions predominantly focus on grades eight and nine with students approximately aged 13–14, whereas interventions aimed at primary school students are less frequently available. Though less common, some research also explores interventions tailored to specific tracks, such as students in vocational education tracks (see e.g. Amagir et al., 2022), or schools with a high proportion of socioeconomically disadvantaged students (see Gill and Bhattacharya, 2015; Lührmann et al., 2015, 2018).

For the interventions studied, financial literacy is to a large extent not part of the standard curriculum. Nevertheless, school-based financial education interventions are typically taught during regular school hours, either (1) as part of a specific subject, (2) integrated across multiple subjects, or (3) as an extracurricular activity within standard schooling hours. Importantly, all approaches imply that programs are usually mandatory for students. At the primary school level, integration across subjects and extracurricular activities are the most common formats in our sample. In secondary schools, all three formats exist. For detailed information on program intensity for each study see table 1 in the annex. The following programs illustrate the three different implementation approaches:

Amagir et al. (2022) examined a subject-specific financial education program for adolescents with eight weekly lessons focusing on financial decision-making. The program was implemented in either economics or social studies classes and taught by economics teachers. Students in the control group followed the standard economics curriculum, which gave minimal attention to the topics covered in the treatment group. They found positive effects on financial knowledge and financial behaviors but no effect on financial attitudes.

Batty et al. (2020) tested the effect of an experimental non-lecture-based program. Instead of lectures, activities were integrated across the school day in a simulated classroom economy with students receiving artificial money in exchange for individual responsibilities in class (i.e. a “classroom job”) which they could use for rent and other expenses, save or spend on rewards. They report positive effects for financial knowledge, but they did not investigate other financial outcomes.

Moscarola and Kalwij (2021) analyzed the impact of an extracurricular visit to the “museum of savings” in Turin, where primary school students watched videos and played games related to economic and financial topics. They showed increasing financial knowledge for topics that were explicitly taught during the visit.

Given that financial literacy is not widely embedded in curricula, most interventions are short in duration, often spanning only a few hours or lessons over several weeks, with limited variation in program length across studies. Due to the limited variation, we are unable to find patterns in the role of program

intensity in determining effectiveness. Kaiser and Menkhoff (2020) found that smaller class sizes enhance program effectiveness; however, this information was not available for the studies we considered.

Given that disadvantaged children may benefit most from financial education (see OECD, 2024), one key advantage of school-based interventions embedded within regular school hours is that all students participate, given that schools, headmasters or teachers sign up for the program. In contrast, programs with voluntary participation for children offered outside of school hours may be subject to major selection biases, particularly due to parents' and students' self-selection (see subsection 6.9). Among the studies we reviewed, the only intervention conducted outside of school hours was investigated by Bhattacharya, Gill and Stanley (2016), who examined a one-week financial literacy summer camp. Their findings suggest that parents with present bias — those who undervalue the long-term benefits of the program — are less likely to enroll their children.

3.2 Content and conceptual focus

Financial education programs for children and adolescents aim to equip students with essential personal finance skills. These programs typically cover topics that can be categorized as *basic money management*, *saving and investing*, *credit and borrowing*, *banking and financial services*, *economic literacy and financial risk management*. For details on the content focus of each study, see table 1 in the annex.

Many interventions primarily focus on building *basic money management skills*, such as distinguishing between needs and wants, tracking income and expenses (budgeting), setting spending limits and financial goals. These fundamentals are often taught alongside *saving and investing*, where students learn the benefits of regular saving, the role of interest in wealth accumulation, and various saving options, including bank accounts and investment products. Overall, saving and investing is the most commonly taught topic.

Beyond basic money management and saving, approximately one-third of programs introduce students to the topic *credit and borrowing*. These lessons help students understand how loans and credit cards function, including interest rates and repayment terms. They may also explore the potential consequences of accumulating debt and strategies to avoid excessive borrowing. Similarly, about one-third of programs cover the role of *banks and financial institutions*, teaching students about different types of accounts, payment methods, and how financial institutions facilitate saving and borrowing. Some programs cover *financial risk management*, where students learn about insurance as a safeguard against financial loss and how to identify and mitigate financial risks, including fraud and identity theft.

Only a limited number of programs extend beyond personal finance to incorporate economic literacy, some introducing students to fundamental economic concepts relevant for personal money management such as scarcity, trade-offs and market forces, along with supply and demand (Bhattacharya and Gill, 2020; Bhattacharya, Gill and Stanley, 2016). Others provide an introduction into more general economic topics like inflation or the origins of money (Kalwij et al., 2019; Moscarola and Kalwij, 2021; Sconti, 2022). A few studies explore business or entrepreneurship education, providing students with insights into starting and managing businesses (Kalmi and Rahko, 2022; Kalwij et al., 2019). Few studies also incorporate topics such as earning an income (Kalwij et al., 2019; Walstad et al., 2010).

The primary conceptual focus of the reviewed programs is often on teaching financial knowledge rather than fostering positive attitudes or changing financial behaviors, though this is not always explicitly stated. As we will demonstrate in more detail below, school-based financial education programs generally show positive effects on financial knowledge, but their impact on financial behaviors and attitudes is mixed. Given that most studies primarily emphasize financial knowledge, these findings are not particularly

surprising. Studies that *explicitly* focus on financial behaviors or attitudes – particularly those addressing children’s everyday financial experiences – tend to yield more favorable results in these areas (see subsection 6.1 for a discussion on classifying studies by causal mechanisms). Below, we present three studies illustrating this.

For instance, Amagir et al. (2022) integrated personal and group goal setting into the curriculum, encouraging students to establish and work toward their savings objectives. This goal served as a practical framework to apply financial literacy concepts, such as tracking their spending and developing practical saving strategies. It also fostered discussions on financial matters with peers and family through home assignments and promoted reflection on money habits and comparisons using research data. Additionally, students received practical rule-of-thumb advice, such as how small amounts can contribute to achieving savings goals. Amagir et al. (2022) found positive effects on financial behaviors (saving behavior and engagement in financial activities such as budgeting and making ends meet), at least in the short run. Maldonado and De Witte (2021) evaluated a computer game designed to teach students about payment methods and financial security. Their findings indicated that 13–14-year-old students who played the game made better decisions in simulated scenarios involving online fraud, payment safety and digital advertising.

While financial behaviors may be built when children themselves start to act in financial situations, the development of financial attitudes has been demonstrated to already start at the age of primary school education (see Consumer Financial Protection Bureau, 2016b; Van Campenhout, 2015). Regarding attitudes, Migheli and Coda Moscarola (2017) conducted a laboratory experiment with primary school children, where they were asked to draw an item they wished to save for and then discussed with instructors how much money they would need to purchase it. To measure the program’s effect on patience, the researchers conducted a patience exercise both before and after the drawing exercise. While the direct impact of the program on reducing impatience was weak, the combination of the intervention and repeated exposure to patience exercises significantly improved children’s ability to delay gratification.

The relationship between conceptual focus and program outcomes is not always straightforward. Some studies that primarily target financial knowledge still report positive effects on attitudes and behaviors, while others that focus on behavior and attitudes show mixed results. Another challenge in drawing clear conclusions is that while many studies specify the topics they teach, it remains difficult to determine whether they approach these topics through a knowledge-based, attitudinal or behavioral lens.

Another important consideration is the type of financial knowledge being taught. Many of the reviewed studies do not specify whether their programs focus on static, conceptual knowledge – such as factual information about financial concepts – or on equipping students with adaptable skills (procedural knowledge) that prepare them for diverse financial challenges. The limited scope of topics covered and the limited focus on attitudes, behaviors and adaptable skills may be due to the short duration of most programs, which typically last only a few hours.

To sum up, existing financial literacy studies primarily focus on teaching knowledge of personal finance topics, while less emphasis is placed on fostering financial attitudes, behaviors and adaptable skills. Interventions that establish a connection with children’s everyday financial experiences tend to be more likely to have an impact on students’ financial attitudes and behaviors. However, further research and more transparent reporting are necessary to confirm these findings.

3.3 Well-known programs vs. piloting programs

Some studies evaluate the effectiveness of well-established and widely used financial education programs, while others either develop a new program for the underlying study or investigate the impact of a pilot or lesser-known initiative. The widely used programs are typically characterized by distinct branding including a website with information about the program and materials and lesson plans that can be freely downloaded or are available upon registration. The following programs fall into this category:

- Financial Fitness for Life⁵ (FFFL), a program that reaches over 15 million US students each year and has been evaluated by Batty et al. (2015), Bhattacharya, Gill and Stanley (2016) and Gill and Bhattacharya (2015) in the study pool we considered. The program was developed by the Council for Economic Education and teaches broad financial literacy including most of the topics mentioned at the beginning of this section through instructional materials from kindergarten to grade 12.
- FutureSmart⁶, a program developed by the MassMutual Foundation that reached almost 6 million students and was evaluated by Johnson et al. (2021). Through a story-driven narrative and interactive activities, students take on the role of the mayor of a town, where they must navigate real-life financial decisions. They learn skills like saving, budgeting, education and career planning to achieve important financial goals.
- Hands on Banking⁷ (HOB) was part of the program evaluated by Bhattacharya, Gill and Stanley (2016) and is a real-world money management simulation, where young people learn financial skills like budgeting, comparison shopping and evaluating needs, wants and trade-offs. The program was developed by Wells Fargo Bank.
- My classroom economy⁸ (MCE) evaluated by Batty et al. (2020), a program where students earn salaries for completing classroom tasks, manage expenses such as paying rent for their desks, receive bonuses or fines based on behavior and performance and make spending decisions at in-class auctions or stores. The program was designed to integrate into the school day without requiring significant instructional time or specialized teacher training.
- My Finance Coach (discontinued) evaluated by Lührmann et al. (2015, 2018). The My Finance Coach Foundation GmbH provided interested schools and teachers with free educational materials on financial topics. Teachers would receive the materials, including all necessary worksheets in class sets, after they had participated in online teacher training sessions. Main topics include buying, planning, saving, risk management.
- Stock Market Game⁹ (SMG) evaluated by Bhattacharya, Gill and Stanley (2016), Gill and Bhattacharya (2015), Harter and Harter (2010) and developed by the SIFMA Foundation. In the program, students manage a virtual portfolio of 100,000 dollars and invest under real-time stock market conditions. They research, buy, sell and monitor their virtual investments, adjusting their portfolio based on real-world market data. The game concludes after a set period, after which students analyze their performance and reflect on their strategies.

⁵ Council for Economic Education (2025).

⁶ MassMutual Foundation (2025).

⁷ Wells Fargo (2025).

⁸ Vanguard (2025).

⁹ SIFMA Foundation (2025).

All listed programs, except My Finance Coach, originated in the USA. Overall, most well-known programs show positive effects on financial knowledge. However, studies investigating financial behavior found no significant impact on behavior (Batty et al., 2015; Johnson et al., 2021; Lührmann et al., 2015). Findings on financial attitudes are mixed: While Lührmann et al. (2015) and Lührmann et al. (2018) reported positive effects for My Finance Coach, no significant effects were found for the adapted FFFL program (Batty et al., 2015) or FutureSmart (Johnson et al., 2021).

Some interventions combine multiple programs. For example, Harter and Harter (2010) combined SMG with a complementary stock market learning curriculum. Similarly, Gill and Bhattacharya (2015) integrated SMG with their broader stock market learning curriculum and tested it against the FFFL program. In both interventions also elements of HOB were used. Bhattacharya, Gill and Stanley (2016), on the other hand, combined FFFL and elements of a stock market curriculum including SMG in their intervention. Gill and Bhattacharya (2015) found that the broader financial literacy program outperformed the curriculum with special focus on the stock market and investing in financial knowledge. They argue that deeper engagement with learning and applying economic principles creates spillover effects in financial knowledge, ultimately enhancing performance on questions assessing overall financial literacy. Effects on financial attitudes and behaviors were not investigated.

Comparisons of well-known programs and pilot programs reveal no significant patterns in reported effects. Moreover, well-known programs are typically evaluated on a similar scale as pilot initiatives, with most evaluations including 600–2,000¹⁰ student observations, while only a few interventions rely on notably smaller samples of fewer than 200 students (Bhattacharya, Gill and Stanley, 2016; Gill and Bhattacharya, 2015; Migheli and Moscarola, 2017; Özdemir and Kaya Uyanik, 2021). Sample attrition is a major issue in many interventions (see subsection 6.8 for a discussion of sample attrition and possible mitigation strategies).

3.4 Didactic aspects – experimental, active individualized learning

Most of the studies considered involved teaching methods with experiential or active learning elements. According to Burch et al. (2019), experiential learning occurs when “*students actively process an experience that leads to new knowledge, skills, or insight*” (p. 242). This approach provides students with regular opportunities to integrate conceptual, procedural and factual knowledge, supporting deeper understanding and advanced learning (Burch et al., 2019). Examples of experiential financial education include simulated financial scenarios (e.g. stock market games, business management simulations) and real-world financial experiences (e.g. opening and managing a savings account). Active learning, in contrast, prioritizes student engagement, participation and critical thinking over passive information absorption. This approach employs instructional methods such as discussions, problem-solving, case studies, hands-on experiments and group work, requiring students to actively engage in the learning process.

Experiential learning has demonstrated positive effects on students’ overall learning outcomes (Burch et al., 2019). Research further suggests that experiential learning may be particularly effective in enhancing financial literacy compared to traditional teaching approaches (Amagir et al., 2018). Experiential learning may be especially well-suited for improving financial literacy, as it “*(...) has the potential to engage students in topics of interest to them, and it provides opportunities to explore how financial concepts can be applied to*

¹⁰ The studies with the highest numbers of observations are Dare et al. (2020), De Witte et al. (2020), Iterbeke et al. (2020) and Sule and Seda (2018) with more than 2,000 observations each.

real-world situations." (Amagir et al., 2018, p. 75). However, the evidence supporting the impact of active learning on financial literacy is limited and its effect has not been directly addressed in our study pool.

Identification of the causal role of experiential versus active learning is often not possible due to inconclusive descriptions of instructional approaches used. Only some recent studies explicitly integrated experiential learning as a core didactic element (see e.g. Amagir et al., 2022; Batty et al., 2020; Kalmi and Rahko, 2022). Among these studies, only Kalmi and Rahko (2022) directly compared experiential learning to traditional methods, reporting that experiential approaches yielded the highest positive impact. However, the study had some major limitations, including non-random program assignment, an imprecise intervention design and significant sample attrition, with a substantial drop in observations between pre-test and post-test. Several noncomparative studies also indicate that experiential learning enhances financial knowledge, though its effects on financial behavior and attitudes remain mixed.

As mentioned in subsection 3.2, Amagir et al. (2022) studied the effect of a program that incorporates several experimental learning principles (like setting personal saving goals) throughout the program as a framework to apply financial literacy concepts. The program was supplemented with animated videos and students were encouraged to discuss financial matters with peers and family using take-home assignments. The program demonstrated an immediate positive impact on financial knowledge and financial behavior but no effect on financial attitudes. However, nine months after the program, the effects vanished.

Similarly, Johnson et al. (2021) assessed a program incorporating both experiential and active learning elements. Secondary students assumed the role of a town mayor helping citizens make financial decisions. In one module, students advised a local resident on redecorating within a limited budget; in another, they assisted a character in saving for a trip while managing daily expenses. The study indicates positive effects of the assessed program on financial knowledge but no measurable impact on financial behavior or attitudes.

Next to experimental and active learning, a significant share of studies focused on individual student learning approaches, where students progressed through online learning paths at their own pace. Research suggests that tailored financial education programs are more effective (see Van Campenhout, 2015, for a review), and individualized student learning through adaptive learning paths may be an effective way to enhance financial literacy. For example, Compen, De Witte and Schelfhout (2020) assigned students to different learning paths based on their initial financial knowledge scores. De Beckker et al. (2021) examined a structured learning path featuring exercises, summaries and formative assessments. These exercises integrated videos, interactive learning games and case studies tailored to real-life financial experiences.

In summary, the inclusion of active and experiential learning approaches as well as self-paced and individualized learning programs seems to improve financial knowledge. However, they are not a universal solution for enhancing financial attitudes and behaviors. Further research is needed to determine which types of financial education are most effective in fostering positive financial attitudes and behaviors. For doing so, we suggest that researcher provide more information about the didactic focus of the program evaluated, as this aspect has often received limited attention in existing studies (for details see subsection 6.2).

3.5 Program development and implementation

Financial education programs in schools are often provided by external suppliers that implement curricula developed by universities, financial sector firms (such as investment firms and banks), government ministries or non-profit organizations. As shown in subsection 3.3, many well-known programs originate from for-profit companies, but also piloting initiatives are frequently developed by for-profit firms.

Against the background of potential conflicts of interest, it is crucial to critically assess the motivations of these for-profit providers, particularly in the context of children's education. Without careful scrutiny, financial education could inadvertently serve as a tool for corporate influence rather than a neutral, skills-based approach to economic empowerment. For an overview of (private) stakeholders involved in financial education and their influence on the financial literacy landscape see Zieser et al. (2024).

In some cases, financial education programs are not only developed but also delivered by external instructors such as university professors or volunteer employees from firms involved in program design. These instructors typically teach during regular school hours and their involvement is also intended to increase standardized implementation of the program. However, most programs rely on students' regular teachers for implementation. For both teacher- and externally led programs, participation for schools is typically voluntary. Teachers and other school personnel either apply for the program themselves or are recruited by researchers or program developers. Consequently, students' access to financial education programs largely depends on teachers and school administrations, who serve as key gatekeepers.

Moreover, when teachers implement the program, they additionally determine both the content (e.g. which materials they use) and the teaching methods (e.g. instructional approaches), meaning that students experience the intervention indirectly through their teachers' mediation. To minimize the variability introduced by instructors, studies explore ways to standardize financial literacy instruction. Common strategies include: (1) providing standardized teaching materials to ensure consistency, (2) offering teacher training and additional instructional resources to improve delivery, (3) implementing self-directed learning models, where teachers assume the role of a coach or facilitator rather than a direct instructor. Under this approach, students primarily navigate digital materials at their own pace, consulting the teacher only when necessary. For more details on format and delivery channel of each intervention see table 1 in the annex.

As a result, some interventions are more standardized than others, offering predefined lesson plans and materials to ensure uniformity in delivery. Conversely, other programs provide minimal guidance and grant teachers flexibility in how they adapt and implement the content. This variability may lead to differences in program outcomes depending on teachers' approach and expertise. Since teachers – and in particular professional teacher training – play a central role in shaping students' learning experiences, we examine their influence in more detail in subsection 3.7.

3.6 Outcome measures and effects

As shown above, the conceptual focus of most school interventions lies on financial knowledge delivery. The evaluations also primarily assess the impact on financial knowledge. Approximately half of the school studies in our pool additionally also examine self-assessed (hypothetical) financial behaviors. Financial attitudes are studied either alongside with financial knowledge and/or financial behaviors or as the primary focus when the intervention specifically targets attitude. None of the considered studies investigated long-term impacts like financial well-being. In the context of school children, this appears appropriate because financial well-being as defined in the context of financial literacy cannot be considered a concept relevant for the population of children. However, financial well-being may be a relevant outcome measure for future long-term studies that follow children into adulthood.

For financial knowledge, a large proportion of studies report positive effects. In contrast, only a minority of studies find positive impacts on (self-reported or hypothetical) financial behaviors. This aligns with recent findings from a meta-study on the effectiveness of financial literacy in schools by Kaiser and Menkhoff (2020), who report the lowest effect sizes for financial education on financial behaviors for

children. They note that measuring financial behavior is particularly challenging, as young people at this stage have limited engagement in financial activities. In contrast to financial behavior, research indicates that financial attitudes begin to form as early as in primary school. However, the evidence on the effects of financial education on attitudes is mixed, with some studies reporting positive effects while others find no impact.

Only a handful of studies have conducted follow-up investigations, reassessing participants some time after the intervention. These studies generally find that effects persist for several months (Batty et al., 2015; Compen, De Witte and Schelfhout, 2020; Iterbeke et al., 2020). The only study examining long-term effects is Sule and Seda (2018), which assessed children's patience through an intertemporal choice task. Almost three years after the intervention, the effect of the educational program on patience was still present but slightly smaller than in the earlier assessments.

Although we find overall positive effects on financial knowledge and mixed effects on financial attitudes and behaviors, caution is required when interpreting these findings due to the varying definitions of these concepts. As discussed in Zieser et al. (2024), financial literacy and related concepts are defined and operationalized in diverse ways, varying in fundamental aspects of their understanding. This inconsistency poses challenges for impact evaluations, as different researchers may apply varying conceptual frameworks, making comparisons across studies difficult. Therefore, it is insufficient to rely solely on authors' claims that they measured financial knowledge (or any other outcome) – one must consider the individual measurement instruments used to determine what was actually measured.

For example, studies differ in their level of abstraction when measuring financial knowledge. Some apply widely used items such as Lusardi and Mitchell's "Big Three" to try to measure general financial knowledge (see e.g. Lührmann et al., 2018; Sconti, 2022) – while others focus on short-term learning gains with items closely linked to the intervention content (see e.g. Iterbeke et al., 2020). While neither approach is inherently superior, this variability introduces another source of heterogeneity that complicates comparisons across studies. A major limitation to the robustness of studies is that only very few use rigorously validated instruments (see subsection 6.4 for a discussion on validating outcome measures).

Studies measuring short-term financial knowledge gains do not always report positive effects, and the extent to which content-specific knowledge translates into broader financial understanding remains unclear. For example, Moscarola and Kalwij (2021) found that students' scores improved only for topics explicitly covered in their program, with smaller or no effects observed for other topics. Similarly, Bhattacharya, Gill and Stanley (2016) examined the impact of an additional 90-minute financial education session within an existing program and found no statistically significant effects, even for items directly related to the additional instructional content.

Similar challenges arise in defining and measuring financial behavior. Researchers do not use a single, standardized scale to assess financial behavior. Instead, they rely on diverse conceptual frameworks. Some studies measure hypothetical behaviors, such as savings intentions, consumer choices, decisions in fictional financial scenarios (e.g. payment safety, online fraud, advertising) and responses to hypothetical situations (e.g. receiving a suspicious email from a bank). Others assess self-reported, real-life financial behaviors, including actual savings, engagement with financial products, carefulness with different payment methods, financial planning, impulse shopping and making ends meet. As these examples illustrate, financial behavior measures vary widely – not only in distinguishing between hypothetical and real-life behaviors but also in the specific financial topics they cover. This heterogeneity further complicates cross-study comparisons.

Overall, financial education effectively enhances financial knowledge, but its impact on attitudes and behaviors remains mixed. Measurement inconsistencies and varying definitions complicate comparisons across studies, highlighting the need for standardized assessment methods. Long-term effects remain largely unexplored, and more research is needed, particularly for young children, as effects on behavior and attitudes have not been examined at the primary school level, although findings on financial knowledge appear consistent across grade levels.

3.7 Professional teacher training

As discussed in section 3.5, students' teachers are frequently responsible for the implementation of the financial literacy program for children. Training them prior to their teaching in financial education programs is used in several interventions, though the extent and structure of teacher trainings vary. For most studies applying teacher training, the sessions typically last only a few hours.

Teacher training appears to be a critical factor influencing the effectiveness of financial education programs for two main reasons: The literature suggests that many teachers lack confidence in teaching financial topics, or in other words they lack what has been labeled "teacher self-efficacy". Additionally, they often do not possess sufficient financial knowledge to deliver financial education effectively (see e.g. Compen, De Witte and Schelfhout, 2019; Eisenhart, 2024; Sawatzki and Sullivan, 2017). Providing teacher training addresses these issues by familiarizing teachers with program materials, core topics and didactic approaches. Furthermore, if the training is specifically tailored to the materials and teaching methods of the program it is more likely that teachers implement the program as intended (Compen, De Witte and Schelfhout, 2019).

However, based on the training sessions used in the studies we considered, it is hard to draw definite conclusions about their impact on children's outcomes for the following reasons. First, studies provide limited details about the nature of teacher training and its actual implementation. Some report information about the teaching intensity, others only state that teachers have received training (see table 1 in the annex). Second, the training sessions are rarely evaluated. Instead, the relevance of teacher training is often inferred from student performance evaluations. For instance, Batty et al. (2015), Harter and Harter (2010) and Sconti (2022) collect teacher-level data alongside student data, but teacher data are mainly used as a control variables rather than being analyzed in depth. Most studies do not collect teacher-level variables at all. As a result, our understanding of the role of teacher characteristics, particularly their training experience, topical knowledge and self-efficacy, remains limited. This gap is also highlighted by Compen, De Witte and Schelfhout (2019) in a review of the impact of teacher professional development on financial literacy education. Key unresolved questions include whether the length of training, the level of program standardization, or the competencies and self-efficacy gained by teachers are most relevant for program effectiveness.

Still, a small number of recently conducted studies provide initial evidence on the role of teacher training. Two studies conducted in the Flemish region of Belgium between 2017 and 2021 as part of a larger research project examined teacher training alongside financial literacy program evaluations (Compen et al., 2023; Compen et al., 2020). In addition to an investigation of financial education interventions for secondary school students (De Witte et al., 2020), Compen, De Witte and Schelfhout (2020) assessed the effects of a three-hour webinar designed to enhance teachers' financial knowledge and pedagogical skills. The training included instruction on financial concepts (such as saving and investing), program materials and the didactic approach of self-regulated learning – an approach that was used as part of the student intervention. Teachers participated in discussions, collaborative assignments and feedback sessions to

refine their teaching methods. The study measured teachers' self-efficacy (confidence in teaching financial literacy) and teaching behavior before and after the student intervention. For teachers who attended the webinar they find an increase in self-efficacy between students' pre- and post-test, whereas those who did not participate experienced a decline. However, a key limitation of this study is that teachers' self-efficacy was only measured after the webinar, meaning the student intervention itself may have influenced the observed effects, particularly since teachers self-selected into the training.

Several years later, Compen et al. (2023) evaluated a more scalable teacher training approach, where teachers accessed a training module before implementing the financial literacy program. This module included videos, quizzes, discussion forums and other interactive elements, allowing teachers to decide how and to what extent they engaged with the training. While the training did not improve students' financial knowledge, it had a positive impact on their financial behavior. Classroom observations indicated that trained teachers spent more time introducing the program, which may have contributed to these improvements. Consistent with the earlier study, teachers who completed the training experienced increased self-efficacy, whereas those who did not participate showed a decline. Notably, teachers' own financial knowledge did not significantly improve, suggesting that enhanced self-efficacy and instructional modifications were the primary drivers of student improvements.

Another study by Urban et al. (2020) examined the effects of financial education graduation mandates on young people's credit outcomes in the United States using a quasi-experimental synthetic control design. The study compared states with different financial education mandates before 2011, focusing on Texas and Georgia. While both states required financial education courses for high school graduation, Georgia invested more in teacher training and certification, leading to stronger improvements for participants' credit scores and reductions in delinquency. Georgia implemented three key measures: (1) funding expert-led teacher training with ongoing support via webinars and local organizations, (2) linking school funding to teacher certification in economics or social studies to ensure qualified instructors and (3) incorporating personal finance topics into standardized student assessments with structured materials. By contrast, Texas had no formal teacher training requirements, and its curriculum was less structured and loosely enforced. A major limitation of this study is that state-level variation makes it difficult to establish causality, as other policy differences may have contributed to the observed effects.

While these empirical studies suggest that teacher training may positively influence students' financial knowledge and behavior, the mechanisms behind these effects remain unclear. Additionally, the studies have methodological limitations. Compen et al. (2020) only measured teachers after training, raising questions about causality. Compen et al. (2023) had a small sample of teachers, limiting generalization. Urban et al. (2020) relied on state-level comparisons, which introduced uncontrolled variability. To address these gaps, further research on teacher training and its impact on financial literacy education is necessary.

While research on financial education lacks definitive recommendations, reviews of professional financial literacy teacher training – drawing from broader educational disciplines and financial literacy – offer valuable guidance: Compen et al. (2019) and Louis et al. (2024) highlight that effective professional development should strengthen both teachers' subject matter knowledge and pedagogical skills. Training should have significant intensity and is recommended to be supplemented with ongoing support, including follow-up activities, coaching and peer collaboration, such as participation in professional learning communities. Giving teachers a voice in selecting content and structuring training programs enhances engagement and addresses individual knowledge gaps. Additionally, successful programs boost teachers' confidence in teaching financial concepts.

3.8 Parental involvement

Parents play a crucial role in their children's financial socialization – particularly in their early years – which Van Campenhout (2015) summarizes as follows: *“In sum, parents’ financial behavior, norms, and attitudes will be passed down in the financial attitudes and behavior of their children”* (p. 200). Despite parents' significant impact, the role of parental involvement in school-based financial education interventions has been largely understudied. This gap may stem from the common perception that financial education at school and at home are separate and disconnected efforts (Van Campenhout, 2015).

In our review, we identified only two recent studies that examined parental involvement in school-based financial education programs, both conducted in Flanders, Belgium. Maldonado, De Witte and Declercq (2022) investigated the impact of parental involvement in homework assignments by comparing two groups. In one group, students received homework tasks explicitly encouraging parental participation. In the other, homework did not require parental input. While the classroom-based intervention significantly improved students' financial literacy, additional parental involvement in homework did not lead to a statistically significant improvement in financial knowledge. However, for students from low socioeconomic backgrounds and those with initially low family communication, parental involvement in homework had a positive effect on behavioral aspects of financial literacy, such as saving habits.

Similarly, Maldonado and De Witte (2021) examined the effects of providing parents with subject-specific information about financial literacy, delivered through an informational brochure. This intervention aimed to increase parental awareness and engagement by highlighting the importance of financial education and offering practical tips for discussions with their children. Although the study found no significant improvement in students' overall financial literacy or financial knowledge, there was a notable positive effect on students' financial skills, particularly in practical financial decision-making.

These findings provide initial evidence that while parental involvement in financial education may not always enhance knowledge outcomes, raising parents' awareness of the importance of financial education can positively influence financial behaviors. Existing meta-analysis on the general effects of parental involvement suggest that different types of parental involvement can lead to varied student outcomes. For example, parental assistance with homework had either no effect or, in some cases, negative effects, though the underlying reasons remain unclear. In contrast, parental expectations regarding their children's schooling – such as their beliefs and attitudes toward education, teachers and academic subjects – were strongly linked to higher academic achievement. Whether these findings apply universally or only to specific outcomes remains uncertain (see Wilder, 2023, for a synthesis of meta-analysis results).

Assuming that the results of meta-analysis on overall academic achievement are also applicable to financial literacy, the greater effectiveness of interventions aimed at raising parental awareness – compared to homework-based involvement – aligns with this pattern. However, further research is needed to determine which forms of parental involvement are most effective in improving financial literacy. Additionally, the finding that parental involvement influences financial behaviors rather than financial knowledge is consistent with existing literature. Research suggests that while parents predominantly shape their children's financial *behaviors*, schools are more effective at imparting financial *knowledge* (Van Campenhout, 2015). To enhance parental participation, Van Campenhout (2015) recommend that schools and policymakers implement strategies such as raising awareness of parents' critical role, offering financial literacy workshops tailored to parents and addressing practical barriers to engagement.

4 Financial education for adults

Adults face a wide range of financial decisions throughout their lives, from managing day-to-day expenses to planning for long-term financial security, all of which impact their financial well-being and economic mobility. This section examines the effectiveness of financial education interventions across different adult populations.

First, we analyze financial education initiatives designed for vulnerable groups, including low-income individuals and welfare recipients, migrants, people with disabilities and victims of domestic violence. These programs often involve tailored, in-person approaches and aim to enhance financial stability and resilience. Second, we explore financial education interventions targeting retirement planning, which range from employer-based programs and pension fund initiatives to digital tools designed to improve long-term saving behavior. Third, we review financial education interventions addressing general financial topics for the broader adult population, examining various delivery formats such as digital platforms, printed materials and in-person education. Lastly, we examine a specific study population, popular in various research areas, namely university and college students.

For interventions targeted at adults a different review approach was necessary than for children due to the heterogeneity of the population, the limited number of studies sharing a common intervention structure and a larger proportion of studies with major methodological limitations. Consequently, for each population group, we focus on analyzing the studies and identifying underlying patterns. Recommendations on what works are, however, only possible to a very limited extent. Across the four categories, we assess the effectiveness of different educational methods, considering factors such as financial knowledge improvements, behavioral changes and long-term financial outcomes. By comparing intervention designs, target audiences and measured impacts, this section provides insights into practices and the challenges associated with financial education for adults.

4.1 Financial education for vulnerable groups

This section examines financial education interventions for various vulnerable groups. Based on a review of the studies in our pool, we analyze the empirical evidence by categorizing it into three subgroups according to the target populations: (1) low-income individuals and clients of debt and welfare services, (2) migrants and very specific target groups, namely (3) people with disabilities or chronic illnesses and victims of domestic violence. Financial education for vulnerable groups is typically delivered in person and tailored to specific life situations. By analyzing the effectiveness of these targeted interventions, we seek to determine whether they can have a lasting impact on individuals' financial stability and economic mobility.

4.1.1 Low-income individuals and clients of debt and welfare services as target group

Three studies considering financial education for low-income individuals are from the US and involve in-person interventions (either one-on-one or classroom-based). One of these programs is mandatory and designed for clients of financial assistance programs, specifically recipients of housing vouchers (Collins, 2013). Prior research indicates that low levels of financial literacy are linked to poor financial decision-making, which potentially contributed to housing crises in the past – such as choosing high-cost mortgages and managing debt detrimentally (Gale et al., 2012). An interest in financially securing US households has potentially shifted the focus from stand-alone financial education interventions to educational interventions that are integrated with access to relevant financial products and institutions

(Grinstein-Weiss et al., 2015). The other two programs rely on voluntary participation (Modestino et al., 2019; Reich and Berman, 2015). In the Netherlands, a voluntary in-person intervention was implemented for financially vulnerable individuals, primarily clients of debt and welfare services. The sample consisted of 64% of participants whose main income source was social benefits, with over two-thirds receiving professional financial assistance (Bruijn et al., 2022).

Program content generally covers financial planning, budgeting, money management (e.g. bill payment), credit management (e.g. credit reports), savings strategies and debt management. One program also includes counseling, such as personalized budgeting and credit action plans (Modestino et al., 2019). Bruijn et al. (2022) tested two treatment variations: a traditional financial education program focusing on financial knowledge (e.g. income types, expenditures, financial products) and skills (e.g. budgeting, applying for allowances, tracking expenses), and a version enhanced with behavioral insights (BI) designed to foster motivation and practice through a need-driven, adaptive approach. The BI program allowed participants to choose discussion topics (e.g. income, financial administration, short- or long-term budgeting), applied autonomy-supportive teaching and integrated behavioral exercises to encourage financial habit formation.

All interventions were delivered across multiple sessions (at least two, up to six), with the programs requiring between eight and 15 hours in total (Bruijn et al., 2022; Collins, 2013; Reich and Berman, 2015). As far as indicated, courses spanned one to two months. The programs' primary focus was behavioral change – applying financial knowledge to personal financial decisions.

In part, studies used administrative data as outcome measures, including bank balances and credit reports (credit scores and ratings) (Collins, 2013; Modestino et al., 2019). Other measures included financial knowledge tests (Modestino et al., 2019; Reich and Berman, 2015) and self-reported financial behaviors (e.g. saving, timely bill payments), financial attitudes such as self-efficacy (e.g. confidence in financial skills), and financial well-being indicators (e.g. negative life events, future planning, chronic and general financial stress) (Bruijn et al., 2022; Collins, 2013; Modestino et al., 2019; Reich and Berman, 2015). In Bruijn et al. (2022), knowledge was assessed via self-report while the financial behavior measure adapted items from Kempson et al. (2013). Additionally, financial psychological indicators included attitudes toward spending, financial self-assessment and motivation (drawing from Dijk et al., 2022; Lown, 2011; OECD, 2015).

Except for Reich and Berman (2015), who only applied short-term measures, effects were studied over periods ranging from 6–18 months. Reich and Berman (2015) observed short-term improvements in financial knowledge and behaviors, but only for participants who already exhibited positive behaviors before the intervention. No effects were found on self-reported negative life events. Modestino et al. (2019) found that their intervention increased access to credit six months post-intervention and improved credit scores 18 months later. It also positively impacted self-reported financial knowledge, self-efficacy and reduced reliance on alternative financial services. In contrast, Collins (2013) found no significant effects on savings or credit scores after 12 months, though self-reported financial planning behaviors improved. Bruijn et al. (2022) showed that the BI program significantly improved financial behavior – primarily through its positive impact on budgeting – and increased self-efficacy and motivation six months post-intervention. However, no effects were found on subjective financial well-being or the overall financial situation, which the authors attribute to the small sample size. Surprisingly, no significant differences emerged between the BI and traditional programs, possibly due to their similar content and duration.

A common limitation among studies on low-income populations is high attrition due to small sample sizes and participant dropout, leading to potential biases (e.g. hidden characteristics), reduced statistical

power, and weakened internal validity and generalizability (as in e.g. Bruijn et al., 2022; Collins, 2013; Modestino et al., 2019; Reich and Berman, 2015). Additionally, selection biases (e.g. full randomization was not feasible in Bruijn et al. 2022) and the limitations of self-reported measures must be considered (e.g. financial attitude measure had low reliability, and the absence of administrative data limits objectivity in Bruijn et al., 2022). A general discussion of studies' limitations and methodological recommendations can be found in section 6.

4.1.2 Migrants as target group

Interventions targeting migrants are delivered either digitally – via websites providing financial information and assessments (Barcellos et al., 2016) or mobile applications (Blanco et al., 2023) – or in person, through training sessions held at churches, community centers and sports clubs (Gibson et al., 2014). Two interventions are conducted in the US (Barcellos et al., 2016; Blanco et al., 2023), while the third is in New Zealand and Australia (Gibson et al., 2014). All are voluntary, with incentives provided for study participation.

Gibson et al. (2014) implemented a financial education intervention that differs from traditional migrant-focused programs, which typically emphasize integration into the host country's financial system, banking services, budgeting and savings. Instead, their intervention focused on educating migrants about remittance methods and associated costs.¹¹ Participants included three distinct migrant groups (Pacific Islanders, East Asians and Sri Lankans), each varying in remittance frequency, education levels and financial literacy. The intervention was delivered separately for each group, consisting of a one-time, two-hour training session for groups of 30 participants. Originally designed for Pacific Islanders, the curriculum was later adapted for East Asian and Sri Lankan participants. Highly educated trainers were selected from the target populations, ensuring cultural and linguistic appropriateness. The programs materials (and surveys) were available in English, Mandarin and Korean, covering remittance cost components, strategies for reducing costs, cost comparisons and new remittance products.

Barcellos et al. (2016) designed a one-time financial education intervention tailored to first- and second-generation immigrants. They tested the effects using the RAND American Life Panel (a nationally representative online panel), though details on participant incentives were not provided. The intervention by Barcellos et al. (2016) was tested in two versions: (1) general-purpose financial education on banking services, saving and investing, and (2) the same content plus an FAQ section with information relevant to immigrants (e.g. required documents to open a US bank account).

Blanco et al. (2023) implemented a community-based digital intervention among low- to moderate income Hispanics (aged 18+). Their six-month intervention combined weekly financial education materials with nudges (text and email reminders) encouraging digital engagement. Participants received monetary incentives. Blanco et al. (2023) structured their intervention around monthly topics: financial goal setting, financial product selection, bill payments, credit scores, debt and saving. Participants worked at their own pace, with an emphasis on both knowledge acquisition and behavioral change. Activities included income tracking and financial planning. The program was available in English and Spanish.

All three studies focused on financial knowledge and behavior as outcome measures. Barcellos et al. (2016) measured general and immigrant-specific knowledge, as well as hypothetical financial behavior changes (e.g. likelihood of enrolling in retirement plans). Blanco et al. (2023) assessed financial capability

¹¹ Remittances play a crucial role in reducing poverty in migrants' home countries. Yet, many migrants lack knowledge about cost-effective transfer methods.

and self-efficacy through behavioral indicators and measured financial stress (e.g. budget planning, confidence in covering expenses, automatic savings deposits). Gibson et al. (2014) evaluated remittance-specific financial knowledge (three questions), broad financial knowledge (two questions similar to those used in Lusardi and Tufano 2015) and participants' behavior regarding remittance cost comparisons and choices.

Barcellos et al. (2016) found significant short-term knowledge increases, with greater gains for the intervention including immigrant-specific information. However, knowledge effects dissipated after six months, suggesting the need for repeated interventions to ensure retention. No significant behavioral changes were observed, even immediately after the intervention.

Blanco et al. (2023) combined financial education with digital nudges, and examined retention rates, comparing them to those of in-person financial coaching programs. Their program achieved retention rates exceeding 60%, compared to less than 40% in financial coaching programs such as Branches and Financial Clinic (Consumer Financial Protection Bureau, 2016a). However, the study did not isolate the effect of nudging. Regarding financial behavior, participants showed improvements in budgeting and emergency savings, and financial stress was reduced. The authors argue that mobile-based digital interventions can address barriers such as time constraints and inaccessible financial information.

Gibson et al. (2014) found that financial knowledge increased among participants with initially low levels of financial literacy (Pacific Islanders and East Asians). These groups were more likely to compare remittance costs across methods and products. While there were modest effects on avoiding expensive or non-transparent remittance channels, no significant changes were observed in remittance frequency or amount. Knowledge effects slightly declined after six months but remained statistically significant.

Considering study limitations, Barcellos et al. (2016) faced attrition and small sample sizes, preventing subgroup analysis (e.g. differentiation between Hispanic immigrants and other groups, or between first- and second-generation immigrants). Additionally, the behavioral outcome measures relied on hypothetical scenarios rather than real-world actions. Blanco et al. (2023) faced external validity issues due to their small, non-representative sample of the US Hispanic population. The program targeted working age individuals, raising concerns about its applicability to older adults with lower digital literacy. Long-term behavioral persistence is unknown, as the study did not measure outcomes after participants stopped receiving nudges. The intervention also took place during the COVID-19 pandemic, a period of heightened digital engagement, and monetary incentives may have influenced retention rates. Gibson et al. (2014) encountered difficulties comparing the three migrant groups due to high attrition among Sri Lankan participants and irregular remittance patterns among East Asians. The most reliable findings pertain to Pacific Islanders, for whom the intervention was originally developed. Despite a six-month follow-up, the study lacks long-term analysis, particularly of behavioral changes beyond the Christmas remittance period.

4.1.3 People with disabilities and chronic illnesses or victims of domestic violence as target group

Four studies examine financial education interventions for two particularly vulnerable groups: individuals with disabilities or chronic illnesses and survivors of intimate partner violence (IPV). The first group includes US military veterans with psychiatric disabilities (Elbogen et al., 2016) and chronically ill rural women in the US (Haynes et al., 2011). The second group comprises female IPV survivors in the US and Puerto Rico (Hetling et al., 2016; Postmus et al., 2011). Notably, the two IPV studies evaluate the same intervention with the same sample.¹²

¹² Hetling et al. (2016) and Postmus et al. (2011) analyze a program implemented in 14 agencies across seven US states and Puerto Rico, with 457 participants at baseline. While Hetling et al. (2016) report short-term effects immediately after the intervention, Postmus et al. (2011) extend the evaluation to 6 and 12 months.

All interventions were voluntary, with financial incentives provided in two studies (Elbogen et al., 2016; Hetling et al., 2016; Postmus et al., 2011). Data collection involved interviews, conducted in person for veterans (Elbogen et al., 2016) and via phone or face-to-face for IPV survivors (Hetling et al., 2016; Postmus et al., 2011). The interventions differed in format, duration and delivery mode. Two were in-person programs (Elbogen et al., 2016; Hetling et al., 2016; Postmus et al., 2011), while one was an online course (Haynes et al., 2011).

Elbogen et al. (2016) implemented a one-time, one-on-one session (lasting 1–3 hours) designed to help veterans with psychiatric disabilities manage their finances. The curriculum covered budgeting, saving, distinguishing needs from wants, financial fraud and avoiding financial exploitation. It also included information on disability benefits, employment income limits and veteran discounts.

The IPV intervention, examined in Postmus et al. (2011) and Hetling et al. (2016), combined four to eight weekly group workshops (1–2 hours each) with at least one individual financial counseling session (30–60 minutes). The program, available in English and Spanish, focused on financial independence and address disentangling joint financial relationships, repairing credit damaged by abusers, economic safety planning and protecting against financial abuse.

Haynes et al. (2011) studied an online financial education program for chronically ill rural women in the US, integrated into a broader university-led health education project. The intervention was tested in two formats: one with online lessons supplemented by expert-led roundtables and a social support discussion forum, and one with online lessons only, without interactive components. The program lasted 22 weeks, with weekly lessons consisting of a short instructional guide followed by related financial activities. Topics included income tracking, assessing net worth and navigating government assistance programs. Participants were provided with computers and internet access, along with technical support if needed.

Studies measured financial knowledge and behavior. Elbogen et al. (2016) assessed financial knowledge using true/false questions on saving, borrowing and disability benefits, and measured financial behavior with an impulsive-spending scale. Additional self-reported measures included savings, debt, annual income, employment and homelessness status. No significant main effects were found 6 months after the intervention comparing the treatment and control group. Only secondary post-hoc analysis found that participants who reported using learned skills after the intervention increased financial knowledge, improved responsible spending and reduced impulsive spending. The authors suggest that the broad coverage and one-time format may have limited effectiveness, recommending a more targeted, skills-based approach.

Hetling et al. (2016) assessed perceived financial knowledge, self-reported financial behavior, financial attitudes (e.g. views on saving, planning, and recordkeeping) and external factors such as depressive symptoms and IPV experiences with data from 300 participants. Immediately after the intervention, financial knowledge and behavior improved, with positive correlations between financial attitudes and outcomes. Using data from the same study, Postmus et al. (2011) analyzed a longer time frame of 6 and 12 months with data from 195 participants, finding sustained intervention effects. However, both studies report improvements in the control group, possibly due to financial and support services provided by domestic violence agencies.

Haynes et al. (2011) used five multiple-choice questions from a validated personal finance textbook to measure knowledge on wealth, income, money management, saving and assets. While financial knowledge improved overall, only savings knowledge showed a significant increase compared to the control group. There were no significant differences between the two intervention formats, suggesting that the

social networking component did not enhance financial learning. However, participant engagement in discussion forums suggests potential benefits for motivation and retention.

Study limitations include reliance on self-reported measures, which may be biased by overestimation or social desirability effects (Hetling et al., 2016; Postmus et al., 2011). The financial attitude scale in Hetling et al. (2016) was not validated beforehand. The IPV intervention's generalizability is limited due to a non-random sample with a higher proportion of Latina participants. Additionally, control group improvements may reflect external services received from domestic violence organizations. Haynes et al. (2011) faced small sample sizes and recruitment difficulties, which limited statistical power. The unexpectedly high baseline financial knowledge suggests that more advanced materials and a more sophisticated knowledge assessment may have been appropriate. The study did not isolate the effects of discussion forums or expert roundtables, as these components were not tested separately. In Elbogen et al. (2016), intervention effects were detected only through post-hoc analysis, which reduced explanatory power. The single-session format likely limited long-term impact, suggesting the need for more sustained interventions.

4.1.4 Conclusions on financial education for vulnerable groups

Financial education interventions for vulnerable groups are largely delivered in person, unlike those for the general population, which often rely on digital formats (see section 4.3). Many programs are also timed to specific life situations, such as receiving financial assistance, experiencing debt or recovering from financial hardship, ensuring that participants are directly confronted with relevant financial decisions. This context-specific approach may enhance engagement; however, it is important to consider that findings on participants' financial behaviors may be influenced by immediate external pressures.

Findings across the studies show short-term improvements in financial knowledge, financial attitudes (such as self-efficacy) and certain financial behaviors, though long-term impacts on financial stability (i.e. being able to meet financial obligations, manage financial shocks and maintain a sustainable financial position over time) and economic mobility (e.g. income growth, wealth accumulation and movement out of poverty) remain unclear. While some interventions, particularly mandatory ones linked to financial assistance programs, show positive effects on savings and credit outcomes, others fail to translate knowledge gains into lasting behavioral changes. Digital interventions, particularly for migrants and welfare clients, improve engagement and retention, yet their effect on real financial behavior remains uncertain.

Outcome measures vary widely, with some studies relying on self-constructed financial literacy scales and others using a mixture of established measures with additional tailored questions. Objective, administrative financial data, such as bank balances and credit reports, provide more reliable insights but are mostly used in interventions targeting low-income individuals. Self-reported measures dominate but introduce risks of bias, overestimation and social desirability effects, making it difficult to assess actual financial improvements.

A major limitation across studies is low external validity, as interventions often target highly specific populations, such as IPV survivors, individuals with psychiatric disabilities or recipients of social benefits. Small sample sizes, high attrition rates and selection biases further restrict the ability to generalize findings. Additionally, financial incentives for participation, especially in studies involving marginalized groups, may have influenced retention and engagement.

Overall, while financial education for vulnerable groups can enhance financial knowledge and immediate decision-making, evidence on long-term financial security and behavioral change remains mixed. Future

research might want to prioritize longitudinal designs, stronger evaluation methods (e.g. randomized controlled trials) and greater use of objective financial data to better assess the real impact of these interventions. Integrating financial education with practical financial tools and institutional support may further enhance effectiveness, ensuring that knowledge gains lead to sustainable financial behaviors.

4.2 Financial education for retirement planning

This section examines financial education interventions aimed at improving retirement planning. Based on a structured review of studies, we categorize the interventions into three subgroups based on their target populations: (1) members of retirement saving funds, (2) employees and (3) the general adult population. The interventions employ various strategies, including direct financial education, behavioral nudges and decision simplification techniques. By analyzing their effectiveness, we aim to understand how these interventions impact financial knowledge, retirement saving behavior and long-term financial decision-making.

4.2.1 Members of retirement saving funds as target group

One way to engage individuals in financial education for long-term financial planning and retirement preparation is through interventions offered to members of retirement savings funds (Billari et al., 2023; Foster et al., 2015; Ghafoori et al., 2021). These programs are often provided by pension funds themselves (Billari et al., 2023; Ghafoori et al., 2021), raising concerns about economic interests but also leveraging institutional expertise and shared interests with pre-retirees (Ghafoori et al., 2021).

The interventions examined in Italy (Billari et al., 2023) and Australia (Foster et al., 2015; Ghafoori et al., 2021) were all voluntary and one-time, though they varied in delivery. Billari et al. (2023) studied a 25-minute online video using a nudging approach to increase awareness and engagement with pension fund resources, while Foster et al. (2015) tested how modifying the salience of retirement fund information influenced the ability to evaluate fund performance. In contrast, Ghafoori et al. (2021) evaluated a 90-minute, in-person seminar.

Program content typically covered expected lifespan and pensions, retirement savings needs, tracking pension entitlements, investment choices, tax benefits of additional contributions, inflation effects and portfolio diversification (Billari et al., 2023; Ghafoori et al., 2021). Billari et al. (2023) focused on demographic survival literacy¹³, financial knowledge (measured using questions on compound interest, inflation, risk and diversification adapted from Lusardi and Mitchell, 2007b), self-reported behavior and administrative pension fund data. The intervention significantly and persistently improved both demographic and financial knowledge over an approximately nine-month period, with younger participants shifting toward higher-risk, higher-return investments within three months.

While both digital interventions relied on behavioral design, the study by Ghafoori et al. (2021) focused on long-term behavioral changes. Using administrative data over 20 months, the study found a significant increase in voluntary contributions within three months of the seminar, with effects persisting nearly two years later. Participants also improved their portfolio allocation strategies, reducing risk as they neared retirement. Similarly, Foster et al. (2015) found that making key financial information more accessible and

¹³ Defined as the awareness of the need to plan for the long term given life expectancy (Billari et al., 2023).

assessable generally lead to a higher ability to evaluate fund performance and improved financial knowledge¹⁴ among participants with lower financial literacy but not for those highly literate.

A common limitation among these studies is self-selection bias, as all programs were voluntary, limiting generalizability (Billari et al., 2023; Ghafoori et al., 2021). Additionally, Foster et al. (2015) recruited participants from an online panel rather than a broad cross-section of fund members, further restricting applicability to the general population.

4.2.2 Employees as target group

Another approach to providing financial education for retirement planning is to target employees directly, aiming to influence their contributions to employer-sponsored retirement plans. The three interventions analyzed here originate from the US and utilize digital formats, either for direct financial education (Collins and Urban, 2016), behavioral nudging (Clark et al., 2014) or decision simplification strategies using projections and anchoring (Goda et al., 2014).

Intervention designs varied in complexity and participant engagement. In a field experiment at a large financial institution, Clark et al. (2014) tested a low-effort intervention with a subset of newly hired employees – a simple informational flyer distributed primarily via email – highlighting employer matching contributions and the benefits of compound savings (i.e. information nudges). By contrast, Collins and Urban (2016) implemented a more comprehensive approach, offering a ten-module, self-paced online financial education course, which employees at 45 credit unions could complete during work hours. On average, participants spent 8.75 hours completing the program, covering financial planning, investing and retirement saving strategies. Meanwhile, Goda et al. (2014) tested three versions of a four-page mailed brochure that provided varying levels of personalization and income projections, aiming to assess how these features influenced voluntary retirement contributions.

Outcome measures relied on employer administrative data and participant self-assessments. While Clark et al. (2014) found that younger employees were more likely to enroll in a 401(k) workplace retirement plan after receiving the flyer (2 months later), the effect was negative for older employees.¹⁵ Survey responses also suggested that the flyer alone was not enough to influence behavior. In contrast, the more immersive program tested by Collins and Urban (2016) led to a 40 % increase in employer contributions to retirement savings, with no negative impact on other financial priorities such as education savings or bill payments.

Goda et al. (2014) found that employees exposed to any version of the retirement projection intervention increased their annual voluntary contributions by 3.6% on average. However, differences between the versions were too small to determine which features were most effective. The study also revealed that randomly assigned assumptions about retirement age, investment returns and hypothetical contributions significantly impacted behavior, likely due to anchoring effects.

The studies highlight both the potential and the limitations of workplace financial education. The flyer intervention was cost-effective but had minimal impact (Clark et al., 2014), while the education program

¹⁴ The knowledge test used questions adapted to Australian terminology from Lusardi and Mitchell (2011a) and Rooij et al. (2011).

¹⁵ In a footnote, Clark et al. (2014) mention that they actually split the treatment group in two, where one group received the flyer with additional information on the saving behavior of all company employees. This alternate version was supposed to introduce a peer effect. However, no differences between the two treatment groups were observed.

successfully improved saving behavior (Collins and Urban, 2016). The projection-based intervention demonstrated modest effects, but the strong anchoring influence suggests a need for a careful framing of financial information (Goda et al., 2014). In general, the target populations in the reviewed studies have higher financial literacy levels and engage in above-average retirement savings compared to the general US population (Collins and Urban, 2016; Goda et al., 2014).

4.2.3 General adult population as target group

Beyond workplace settings, online financial education interventions on retirement savings have been studied in Australia, the US and Canada (Boyer et al., 2022; Heinberg et al., 2014; Isler et al., 2022).¹⁶ These studies incorporated behavioral insights (Heinberg et al., 2014), educational nudges (Isler et al., 2022) and financial education videos on tax-preferred savings vehicles (Boyer et al., 2022).

Interventions differed in their design and teaching approach. Heinberg et al. (2014) developed a low-cost, replicable financial education program covering five key concepts – compound interest, inflation, risk diversification, tax treatment of retirement savings, and employer matching – delivered via short videos (3 minutes) or written narratives.¹⁷ Isler et al. (2022) tested educative versus default nudges, where participants completed a financial knowledge test before allocating monetary endowments among savings accounts with varying risk and return rates. The educative nudge provided feedback on incorrect answers to improve financial knowledge.¹⁸ The default nudge pre-allocated savings according to financial service recommendations. Boyer et al. (2022) studied the effect of educational videos (the longest lasting only 101 seconds) on Canadians' understanding of tax-advantaged retirement savings vehicles, testing one video on tax features and the same video in combination with one on government benefit clawbacks.

Outcome measures included financial knowledge tests and behavioral indicators. Heinberg et al. (2014) found that both video and narrative formats increased financial knowledge, particularly in areas where prior knowledge was low. Video-based learning tended to enhance self-efficacy more than narratives, but neither format consistently outperformed the other. Focus group insights suggested that written narratives may aid knowledge retention, while videos might be more effective in motivating behavioral change. Knowledge gains diminished after eight months, highlighting the need for reinforcement. Isler et al. (2022) found that default nudges significantly increased savings and reduced financial errors, while educative nudges had no measurable impact. Moreover, individuals with higher cognitive reflection and stronger financial knowledge – particularly in understanding compound interest – managed their retirement accounts more effectively. Boyer et al. (2022) found that participants in the intervention groups scored significantly higher on tax-related knowledge questions and made more tax-efficient savings decisions. However, while the effect was significant for both videos combined, the single-video intervention was only effective for participants with limited prior knowledge.

¹⁶ Isler et al. (2022) used a university-based experimental sample recruited via an online platform for survey participants. Heinberg et al. (2014) employed a probability-based sample of US households from the RAND American Life Panel and Boyer et al. (2022) recruited Canadians aged 35–55, again from an online panel.

¹⁷ Heinberg et al. (2014) applied principles from psychology and marketing, e.g. minimizing complex jargon to reduce cognitive load and ensuring that content was practical and directly applicable to participants' financial lives. The intervention also emphasized short-term benefits – both tangible and intangible (e.g. peace of mind) – to make the content more immediately relevant. The videos aimed to boost self-efficacy through direct visual experience, promoting observational learning.

¹⁸ The test incorporated and modified questions from various existing questionnaires, including the “Big Three” financial literacy test put forth by Lusardi and Mitchell (2011b), the OECD/INFE survey and PISA (OECD, 2018).

Common limitations include self-selection biases, restricting generalizability (Boyer et al., 2022; Heinberg et al., 2014; Isler et al., 2022). Additionally, Heinberg et al. (2014) only measured knowledge and self-efficacy, despite the intervention's goal of influencing behavior. Isler et al. (2022) suggested that the educative nudge may have been too brief to drive behavioral changes. Boyer et al. (2022) relied on hypothetical investment choices, leaving real-world applicability uncertain. All three studies lacked long-term follow-ups, limiting conclusions on the persistence of effects.

4.2.4 Conclusions on financial education for retirement planning

Financial education interventions targeted at retirement planning primarily aim to optimize financial behavior, particularly in preparing for retirement. Unlike other financial education initiatives, these programs frequently integrate alternative methods, such as nudging, alongside traditional education. Online formats are also more prevalent in this domain compared to in-person interventions.

Effectiveness varies by target population and intervention design. Programs for retirement saving fund members show that short educational videos and in-person seminars can improve financial knowledge and influence investment choices. However, self-selection bias and potential conflicts of interest arise when pension funds provide the education. For employees, simplified information and nudges – such as informational flyers or personalized retirement projections – can increase savings participation, particularly among younger employees and those with lower financial literacy. Workplace integration appears to enhance engagement, particularly when employees can complete training during work hours, but overall effects remain modest. Findings on interventions targeting the general adult population suggest that default nudges, which passively guide individuals toward certain financial behaviors, tend to be more effective in influencing savings decisions than educative nudges, which actively provide financial knowledge. Short educational videos have been shown to improve financial knowledge in the short term, but their impact on long-term knowledge retention and behavioral change remains uncertain.

Across all subgroups, outcome measures commonly include financial knowledge tests, self-reported behavior, administrative data on retirement contributions, and experimental decision-making tasks. Many studies report positive short-term effects, but self-selection bias, short observation periods and a lack of long-term follow-ups limit the generalizability of findings. Moreover, while personalized interventions, such as retirement income projections, can enhance engagement, their impact on actual saving behavior remains limited.

Overall, the findings suggest that nudging is more frequently combined with traditional education in retirement planning than in other financial education initiatives. At the same time, online interventions dominate, while in-person education is less common. While financial education interventions can support retirement planning, their effectiveness ultimately depends on delivery format, behavioral strategies and accessibility. Future research might want to prioritize longitudinal designs, stronger evaluation methods and a greater focus on sustained behavioral change to better assess the long-term impact of these interventions.

4.3 Financial education for a more general adult population on general financial topics

In addition to financial education interventions aimed at vulnerable groups and those focused specifically on retirement planning, we also identified several studies targeting a broader population and covering general personal finance topics. Given the diverse nature of these interventions, we take a different approach to our analysis. Rather than categorizing them by target group, as done previously, we examine

them based on their delivery format – digital tools, analog materials and in-person education. By doing so, we seek to identify which approaches are most effective in improving financial literacy and decision-making for a general audience.

4.3.1 Digital and analog interventions

There are various interventions that employ both digital and analog formats to improve adults' financial literacy and decision-making. While digital approaches leverage tools such as smartphone applications, web-based exercises, interactive platforms, educational videos, and narrative formats, analog interventions use printed materials. Despite addressing similar financial topics – including budgeting, saving, debt management, risk diversification and investment decision-making – the studies differ in their delivery design, duration, engagement strategies, outcome measures and target populations.

Digital interventions vary considerably in format and structure. Some studies focus on mobile tools and bundled app interventions. For instance, Angel (2018) evaluated three digital tools among Austrian adolescents and young adults (16–22 years): a 75-minute documentary on over-indebtedness, a mobile budgeting app for tracking income and expenses and an online research exercise. Outcome measures included objective financial knowledge, financial attitudes (e.g. interest in personal finance, saving affinity) and self-reported behavior (e.g. information-seeking, saving habits). Although all tools addressed budgeting, saving and debt management, only the mobile budgeting app led to a statistically significant increase in bank account monitoring; no other significant effects were detected.

In a similar vein, French et al. (2020) tested a bundled intervention over six months among working age adults (16–65 years) in Northern Ireland that were recruited via a credit union. This intervention combined four smartphone applications (covering loan interest comparison, expenditure tracking, a cash calendar and debt management) with additional engagement strategies such as push notifications, a mid-intervention money management workshop and a two-week money skills competition. This multifaceted approach resulted in short-term improvements in financial knowledge, loan confidence and self-reported behaviors like income tracking, although overall digital literacy and financial well-being remained unaffected and engagement diminished over time.¹⁹ Since all four apps were bundled into a single intervention, their individual levels of effectiveness could not be compared.

Other studies have examined how content presentation influences learning outcomes. Kothakota and Kiss (2020) compared a brief (10–15 minute) financial education module – with integrated visual elements – to a text-only format among adults (18–65 years). Administered via SurveyMonkey and including a five-question multiple-choice test on topics such as interest rates and compound interest, the intervention found that visual enhancements significantly improved knowledge retention. Complementing this perspective, Lusardi et al. (2017) recruited US adults from the RAND American Life Panel to test four digital formats focused on risk diversification and investment decision-making: (1) an information brochure (also suitable for print distribution), (2) an interactive visual tool allowing “what-if” analyses, (3) a written narrative and (4) a video narrative. While (1) and (2) emphasized data exploration, (3) and (4) employed storytelling techniques. All formats boosted self-efficacy and confidence in applying financial concepts, with the strongest effects in the video group. Additionally, the video narrative outperformed the written

¹⁹ 32% of participants used the apps infrequently, and 23% did not download them at all. The most common reason for low engagement was forgetting about the app, while 35% of participants indicated they would be more likely to use the apps if the information provided was more relevant to their personal financial situations (French et al., 2020).

narrative in improving objective financial knowledge. The brochure and interactive tool showed no significant differences. Technical issues (no further explanation is provided by the authors) and comparatively lower knowledge gains among women and minority participants highlight potential limitations.

A different digital approach was adopted by Fürstenau and Hommel (2019), who examined an online mortgage loan calculator as an informal learning tool among a small sample of German university students ($n = 45$).²⁰ Designed to enhance both objective and subjective financial competencies related to mortgage decisions, this self-directed tool did not yield significant differences between experimental and control groups. The findings suggest that without structured guidance – or without participants possessing sufficient prior financial knowledge – such digital tools may have limited effectiveness, measured immediately and two weeks after the intervention. Given that participants were final-year economics and business students, this raises questions about the level of required prior knowledge.

Addressing a specific life event, Moulton et al. (2015) implemented a hybrid digital intervention for first-time, low-to-moderate-income homebuyers in the US.²¹ Participants who were engaged in a subsidized homeownership program and had completed a mandatory preparatory course (comprising an online book, a one-hour financial counseling call and an online assessment) were invited to use an interactive online platform for goal setting and action planning, supplemented by quarterly telephone coaching sessions. Outcome measures focused on mortgage delinquency rates (30-, 60- and 90-day), along with self-reported financial behaviors (budgeting, borrowing, saving and planning) and perceived financial well-being. The intervention notably reduced the likelihood of 90-day mortgage delinquency one year later, reinforced saving behavior, lowered revolving credit card balances and doubled the rate of automated mortgage payments, though it had no effect on shorter-term delinquency (30 or 60 days) and was limited by a 50% attrition rate and potential biases in self-reporting.

In contrast to these technology-driven approaches, analog interventions remain a viable alternative. Fan and Chatterjee (2018) investigated the effectiveness of a printed investment information sheet among undergraduate students at a large US public university. Covering topics such as stocks, bonds, interest rates, risk and tax advantages, the material was handed out to the participants with the experimental session lasting no longer than one hour. Participants were randomized into four groups: one received only the printed material, two received the material alongside situational stimuli (one simulating market volatility and the other introducing peer influence), and a control group received unrelated information. The situational stimuli aimed to examine whether external psychological pressures affect financial learning. A ten-question investor knowledge quiz (derived from an existing financial literacy assessment) served as the primary outcome measure. All treatment groups showed significant improvements in investment knowledge relative to the control, with the situational stimuli providing no additional benefit.

Several studies reported common limitations. Digital interventions often suffered from small sample sizes, low retention rates and self-selection biases (Angel, 2018; French et al., 2020; Kothakota and Kiss,

²⁰ In reference to Livingstone (2001) the authors understand informal learning as “any activity involving the pursuit of understanding, knowledge or skill which occurs without the presence of externally imposed curricular criteria” (Fürstenau and Hommel, 2019, p.7).

²¹ Buying a new home places significant financial constraints on households, making them vulnerable to unplanned consumption, which could lead to shifting income toward other expenditures instead of mortgage payments (Moulton et al., 2015). Rather than being an intervention specifically for vulnerable groups per se, we consider this intervention to be targeted at a specific life event – first-time home buying. Therefore, we did not analyze the study within the study pool of financial education interventions for vulnerable individuals but instead included it in this subsection.

2020; Lusardi et al., 2017; Moulton et al., 2015). Additionally, the short duration of many interventions and the reliance on hypothetical or immediate post-intervention testing raise questions about the long-term impact and real-world applicability of these programs. For analog approaches, constraints such as brief exposure, limited sample diversity and the absence of behavioral outcome measures suggest that while effective for short-term knowledge gains, printed materials may require further enhancement to influence sustained financial behavior (Fan and Chatterjee, 2018).

4.3.2 In-person interventions

There are various in-person interventions to improve adults' financial literacy by engaging participants in structured, face-to-face learning environments. While some interventions are offered on a voluntary basis in settings such as financial institutions and workplaces, others are mandatory. The studies differ in delivery methods (from one-time sessions to multi-week courses), duration, content focus and engagement strategies.

Two studies (Bruhn et al. 2014; Horwitz et al. 2021) implemented voluntary interventions in non-military settings, whereas others (Skimmyhorn 2016; Skimmyhorn et al. 2016) focused on mandatory interventions in military contexts. Bruhn et al. (2014) evaluated a large-scale program implemented by a major financial institution in Mexico City, where extensive recruitment efforts (including direct mail, social media ads and street-based surveys) yielded a low attendance rate of only 17.8%. In contrast, the workplace intervention by Horwitz et al. (2021) was implemented in two financial services companies, where employees self-selected into a ten-week program, with one company allowing flexible class timing and the other pre-scheduled sessions. Meanwhile, in the military setting, Skimmyhorn (2016) delivered a mandatory eight-hour personal financial management course, split into two sessions, to newly enlisted US Army soldiers during boot camp, ensuring high participation. Similarly, Skimmyhorn et al. (2016) conducted mandatory financial education at a US military academy, randomly assigning cadets to one of two teaching methods within a semester-long economics course.

The design and delivery formats vary substantially across the studies. Bruhn et al. (2014) describe a one-time, four-hour in-person course covering savings, retirement planning and credit management, developed by banking specialists in collaboration with an economic education organization. To address low take-up, their design included various treatment groups that experimented with monetary rewards, deferred payments, free transportation and promotional testimonials. However, while financial incentives modestly increased attendance, other incentives had little effect. In the workplace intervention, Horwitz et al. (2021) implemented a ten-week course with weekly one-hour sessions covering budgeting, saving, credit, homeownership and retirement planning. Participants paid an upfront fee – refunded upon course completion – as a commitment device. The non-participating comparison group received two financial education booklets covering similar topics, representing a less structured, passive intervention.

In the military boot camp, Skimmyhorn (2016) provided an eight-hour program, split into two sessions, that addressed spending plans, credit, debt management, insurance, consumer awareness, financial ethics and military retirement savings. In the military academy study, Skimmyhorn et al. (2016) compared two pedagogical approaches: (1) a principles-based approach focusing on traditional personal finance instruction and (2) a rules-of-thumb method simplifying decision-making through heuristics. This intervention, integrated into a semester-long economics course, consisted of four two-hour personal finance labs that covered budgeting, consumption, and the time value of money, and employed lectures, slides, videos, exercises and Q&A activities.

The studies employed a mix of objective and subjective measures to assess intervention impacts. Bruhn et al. (2014) used a financial knowledge test based on course content, self-assessments of financial knowledge, self-reported saving and borrowing behaviors, and administrative data on savings account and credit card balances for a subsample of financial institution clients. Horwitz et al. (2021) measured outcomes with a ten-question financial knowledge test adapted from Lusardi and Mitchell (2007a), focusing on knowledge gains while not assessing behavioral changes. In the military boot camp, Skimmyhorn (2016) relied on administrative data to capture credit behavior (including delinquencies, debt balances, legal actions), retirement savings participation and military labor market outcomes such as promotions or involuntary discharge. Skimmyhorn et al. (2016) combined objective and subjective assessments along with self-reports to gauge financial behaviors, including self-efficacy, motivation to learn, likelihood of seeking financial advice, and risk and time preferences.

While all interventions generally improved financial knowledge in the short term, their effects on behavior varied by design and sample. Bruhn et al. (2014), measuring the intervention's impact six months later, found that the course significantly improved both objective and subjective financial knowledge and increased self-reported savings, though no significant effects were observed on borrowing behavior, credit card use or retirement savings; administrative data suggested that any savings effects were short-lived. In the workplace study, participation improved financial knowledge, yet perfect attendance did not outperform partial attendance (missing at least one class), and the study left open whether increased knowledge would translate into improved financial decision-making (Horwitz et al., 2021).

Among the military interventions, the mandatory program during boot camp reduced credit delinquencies and debt balances during the first year and significantly increased military retirement savings participation. The largest effects were recorded among participants with prior credit market activity – although only the retirement savings effect persisted over two years (Skimmyhorn, 2016). In the military academy study, both teaching methods significantly enhanced financial knowledge and self-confidence compared to a control group; however, the principles-based approach yielded greater self-efficacy, while the rules-of-thumb method appeared to reduce the willingness to seek financial advice, potentially fostering overconfidence. Notably, for female cadets, no significant differences emerged between the two methods (Skimmyhorn et al., 2016).

Several limitations affect the statistical power and generalizability of the reviewed studies. Low participation rates were a major challenge in Horwitz et al. (2021) and Bruhn et al. (2014), where voluntary enrollment led to minimal attendance despite incentive strategies, highlighting the difficulty of engaging participants in financial education programs. Short-term effects were common across all studies, with knowledge and behavioral improvements fading over time, suggesting that one-time interventions may not drive sustained financial changes. Self-selection biases were present in Horwitz et al. (2021), as participants voluntarily enrolled, and in Skimmyhorn et al. (2016), as students of military institutions show high professional motivation to learn, potentially skewing results. The findings suggest that while in-person financial education can improve knowledge in the short term, its long-term impact remains uncertain, particularly in voluntary settings.

4.3.3 Conclusions on financial education for a more general adult population and general financial topics

This section examined financial education interventions for the general adult population. Findings suggest that, while financial education can enhance financial literacy and decision-making, effectiveness varies significantly depending on the delivery format, engagement levels and the duration of the intervention's impact.

Digital interventions, including mobile apps, web-based tools, educational videos and interactive platforms, showed mixed results. While videos and visual elements improved financial knowledge retention, mobile budgeting apps encouraged income tracking, though engagement declined over time.

However, many digital tools failed to produce long-term behavioral changes, with studies highlighting low user retention and limited effectiveness in altering financial habits. Self-directed tools, such as mortgage calculators, did not significantly enhance financial competencies, suggesting that structured guidance may be necessary to maximize their benefits.

The effectiveness of printed financial education materials was examined in a study on investment knowledge, which found significant short-term knowledge gains. This indicates that even brief and low-cost interventions can be useful for improving financial literacy. However, the study did not assess behavioral outcomes, making it unclear whether these knowledge improvements would persist over time or would translate into better financial decision-making.

In-person financial education interventions, particularly voluntary ones, improved financial knowledge but faced significant participation challenges. Large-scale programs struggled with low engagement, even when incentives were offered, as many individuals did not perceive the expected benefits as sufficient to justify participation. Workplace programs showed positive knowledge gains, but long-term behavioral effects remained uncertain. By contrast, mandatory financial education in military settings had stronger behavioral effects, particularly in improving saving behavior and reducing financial delinquencies. These findings suggest that linking education to immediate financial decisions, as seen in structured military programs, may enhance its impact.

A key limitation across all intervention types was their short-term nature, with many studies lacking long-term follow-ups to assess the persistence of knowledge and behavior changes. Voluntary participation also introduced self-selection biases, limiting generalizability. Overall, while financial education interventions can be beneficial, sustained engagement and reinforcement mechanisms are necessary to drive lasting financial improvements.

4.4 Financial education for university and college students

As in many other fields of research, university and college students represent a convenient study population due to their availability for recruitment. Universities and colleges could thus be considered one of the testing grounds for more widely applicable financial education interventions for (young) adults. However, university and college students may also be considered a population with specific financial education requirements due to their educational background and life period compared to the general population of (young) adults.

While most interventions considered in this subsection are described in their respective publications as being tailored specifically toward university or college students, only one study actually focused on financial education topics relevant only for university students, namely an intervention focused on financial issues of post-graduate education and on student debt: In a randomized experiment, Salas-Velasco (2022) finds that a short online course on the economic viability of a master's degree and on student loans increases knowledge of the respective issues and the perceived self-efficacy in making relevant decisions.

All other studies covered general personal finance topics that may also be relevant for a general population. The content of these interventions addressed basic personal finance topics (e.g. bank accounts, payment methods, interest rates, inflation, loans), topics related to expenditures (budgeting, responsible consumption, debt management), future-oriented topics (saving, retirement planning and insurance), and more specific investment-related topics (e.g. bonds, stocks, mutual funds, diversification and homeownership).

Gerrans and Heaney (2019) examine the impact of a high-intensity personal finance course conducted at an Australian university. The course consisted of 11 weeks of lectures and tutorials covering, among others, financial planning, loans, taxation, investments, housing and consumer law. In a difference-in-difference comparison, they find that the intervention had positive effects on financial knowledge, self-reported beneficial financial behaviors and behavioral intentions. Using a research design where the control group receives the treatment only after the post-test (waitlist design, see subsection 6.6), Torma et al. (2023) assess the effectiveness of four lectures at a university in Croatia, finding significant effects on financial knowledge and interest, but no differences in a decision task. It is unclear from the publication, however, what topics were included in the lectures. Handy et al. (2021) examine the effect of a general personal finance program in a quasi-experimental setting, finding increased knowledge and increased confidence attributable to the intervention, particularly in minority women.

Concerning the longevity of effects of such personal finance courses, Gerrans (2021) reports results of one of the very few efforts to examine long-term effects through follow-up surveys with the treatment and control group. Analyzing survey responses from the original sample of Gerrans and Heaney (2019) over the subsequent three years after the intervention, the author finds that the intervention appeared to have a stable effect on financial knowledge and some behaviors even three years later.

A third type of studies focuses on the effectiveness of online or social media-based interventions. Agasisti et al. (2023) find that online video lectures have a similar effect on financial knowledge as on-campus lectures. However, the lack of a no-treatment control group and numerous differences in content, format and duration between the online and offline treatment make clear conclusions difficult. Anderson and Card (2015) find that both lectures and a social media-based intervention addressing consumption behavior and credit card use have similar effects on self-reported compulsive buying, despite the very low active participation in the social media group. The authors attribute this result to an awareness-raising effect of the social media-based intervention. In a randomized waitlist design with a fairly small sample at a US university, Choi et al. (2023) find no significant effect of weekly emails and text messages that encourage participating in an online financial education platform on financial self-efficacy or behaviors. Kuntze et al. (2019) tested the effects of a financial education video, finding significant effects on financial knowledge. However, it is not clearly reported whether the post-test-only design used random assignment to the treatment and control group.

Overall, outcome variables assessed in the studies are heterogeneous. This is true for the general type of concept, i.e. if knowledge, attitudes or behaviors are assessed, for the specific concepts addressed and for the specific questions used in questionnaires. Most studies presented here used some type of financial knowledge test as a main outcome variable. The tests used to assess knowledge vary widely: Some studies use published or otherwise independently created test items, in part to avoid an excessive overlap between the content of the intervention and the test questions. Others, conversely, employed specifically designed tests to better capture topics taught in the intervention. Only Handy et al. (2021) explicitly report using a published or validated knowledge test; others appear to use at least partly self-developed collections of knowledge questions without any validation efforts reported. Two studies do not measure financial

knowledge as an outcome but employ published scales to capture attitudes and behaviors (see Anderson and Card, 2015; Choi et al., 2023).

Overall, due to the limited number and methodological rigor of studies, few conclusions can be drawn from the studies with college and university students. First, studies on conventional classroom-based interventions teaching personal finance topics show significant effects on knowledge, while their effects on self-reported behaviors or behavior intentions are less clear. Second, the only study on a student-specific topic (student loans and economic viability of a master's degree) exhibited significant effects on topic-specific knowledge and self-efficacy only, as actual behaviors were not assessed in the short experiment. Any other conclusions are not possible due to methodological limitations of the studies. This concerns the potential of online or social media-based interventions and interventions directly addressing behaviors. Potential study designs that may shed light on the effectiveness of intervention formats are discussed in subsection 6.7.

5 What works for whom

The effectiveness of financial education programs varies across different target groups, reflecting their diverse financial challenges and learning needs. The reviewed literature is highly heterogeneous in terms of study populations, intervention types, study designs and methodological rigor. Consequently, it is exceedingly difficult to determine which financial education interventions are most effective in specific contexts, let alone establish a universally valid impact. Nevertheless, many studies provide valuable insights into the effectiveness of financial education within their specific contexts. Moreover, a subset of studies on knowledge-focused interventions show a high degree of overlap in findings regarding improvements in financial knowledge and, to a lesser extent, financial attitudes and behaviors, allowing for limited general conclusions.

There is consensus that financial education aiming at children should be integrated into school curricula, rather than being left an optional choice, to ensure universal access for all children. These programs should always be tailored to children's cognitive development. Key cognitive developments occur in kindergarten and primary school, becoming more pronounced during secondary education. Yet, further research is needed to determine the most effective methods for teaching very young children. For a review and preliminary recommendations on this topic, see Consumer Financial Protection Bureau (2016b) and Jambunathan et al. (2024).

Evidence starting from secondary schooling level appears to be most robust due to a more homogeneous study pool compared to the adult population. Empirical evidence thus allows the fairly clear conclusion that school-based financial education improves financial knowledge. However, improving short-term knowledge alone is generally not considered the ultimate goal of financial education. The long-term retention of financial knowledge remains uncertain, as does the issue in how far financial knowledge translates into real-world skills and financial well-being, as there are no studies following students into adulthood.

Concerning the most effective teaching methods, research highlights that experiential, active and individualized learning approaches significantly enhance students' financial understanding, although there is no definitive evidence regarding which method is the most effective, partly because methods are often combined and serve different objectives. For example, individualized learning accommodates different learning speeds, whereas active and experiential learning strategies focus on engaging students through practical applications.

Evidence on the impact of financial education on financial behaviors of children is mixed, with most studies finding no significant positive effects. This aligns with previous meta-analyses that report the smallest effects on financial behaviors among children (Fernandes et al., 2014). Preliminary evidence suggests that financial behaviors disconnected from children's current experiences are difficult to influence. However, programs aligned with students' everyday financial experiences show greater promise. Consequently, financial behavior interventions should be gradually adapted to children's financial experiences and interests. If so desired, directly promoting sound financial behaviors among children may require interventions that explicitly emphasize behavioral aspects.

Parents and other influential figures in children's social environments also play a crucial role. Parental beliefs and attitudes toward education strongly correlate with student achievement. However, many parents remain skeptical of financial education programs. While the evidence regarding parental involvement is limited in our study pool and warrants further research, a review by Van Campenhout (2015) recommends strategies such as raising parental awareness, offering tailored financial literacy workshops for parents, and improving communication between schools and families.

Evidence on financial attitudes among children is also mixed, with some studies reporting positive effects and others not. However, research on attitudinal outcomes in children remains scarce, which is unfortunate, given that financial attitudes and cognitive skills such as executive functioning begin forming as early as in kindergarten (see e.g. Consumer Financial Protection Bureau, 2016b; Van Campenhout, 2015; Webley, 2005). Unlike financial knowledge, financial attitudes appear to be directly linked to financial behaviors (Fernandes et al., 2014). This suggests that fostering positive financial attitudes early on can influence financial behaviors later in life.

Programs targeting children are typically implemented in schools. Teachers therefore play a key role in improving children's financial literacy. Regardless of the quality of provided materials, the effectiveness of interventions typically depends on teachers' confidence and willingness to use them. However, many teachers feel unprepared to teach financial topics and thus, professional teacher training is recommended. Reviews suggest that teacher training should enhance both subject matter knowledge and pedagogical skills and be an ongoing process rather than a single event. Strategies such as follow-up activities, best practice sharing among teachers and joint development of teaching materials can enhance training effectiveness.

Concerning adults, drawing conclusions about the effectiveness of financial education is even more challenging due to the heterogeneity of the studied populations and the limited number of evaluation studies for the various intervention types. Indeed, compared to the population of children, the study pool includes a great variety of intervention approaches that focus more on behaviors in various contexts rather than on general knowledge. Interpretations on what works must therefore be made with caution.

Financial education programs targeting vulnerable groups – such as low-income individuals, migrants, people with disabilities or chronic illnesses and individuals who experienced domestic violence – often yield short-term improvements in financial knowledge, attitudes and behaviors. These effects are particularly evident when education is provided “just in time” in situations most relevant to participants' financial lives. Interventions tied to financial assistance programs or debt management for individuals in financial distress show positive effects, especially on savings and credit behavior. However, long-term impacts are uncertain. Digital interventions have increased accessibility and engagement, particularly for migrants and welfare recipients, yet their effectiveness in driving lasting behavioral change remains unclear.

The effectiveness of financial education for retirement planning varies depending on target populations and intervention designs. Educational videos and in-person seminars for retirement fund members improve financial knowledge and influence investment decisions. However, self-selection bias and conflicts of interest due to economic interests of pension funds may affect outcomes. Interventions aimed at employees to increase contributions to employer-sponsored retirement plans are more effective when incorporating behavioral nudges or simplified information, particularly among employees with lower financial literacy. However, overall effects remain modest, and the long-term retention of financial knowledge from these interventions is uncertain.

For the general adult population, interventions can be classified into digital and in-person programs. Digital financial education programs (e.g. videos, interactive tools) typically enhance short-term financial knowledge, but they often fail to sustain behavioral changes due to declining user engagement. In-person financial education programs for adults have yielded improvements in financial knowledge, though participation remains a challenge in voluntary settings, even when incentives are provided. Regarding financial behaviors, some in-person programs also yield positive effects on financial behaviors when linking education provided “just in time” to immediate financial decisions, particularly in improving savings habits and reducing financial delinquencies.

6 Methodological recommendations

We find considerable variation in methodological approaches, methodological rigor and transparency across studies, although the research field has shown substantial improvements in recent years. This section provides an overview of common methodological approaches and limitations within the study pool and offers recommendations for improving the research methodology in future evaluations and impact assessments, aligning them as closely as possible with “gold standard” research practices. We acknowledge that implementing many of these recommendations is challenging in most evaluation and research contexts and may often be impossible due to practical and resource constraints. Some recommendations thus attempt to take typical limitations into account.

6.1 Classifying interventions and outcomes

Most interventions found in the literature are described as “financial education” interventions. At least to some degree, meta-analyses and literature reviews (including this review) treat these interventions as a theoretically coherent group of interventions. However, as described in section 1, financial education interventions by definition largely depend on the intended outcomes of “financial literacy” and its assumed causal mechanisms. In an extreme interpretation of the concept, “financial education” could thus be considered a collection of many distinct types of interventions related only by the content area of “finance.”

It may thus be beneficial to classify financial education interventions further, by the assumed underlying causal mechanisms. Indeed, the World Bank (World Bank, 2013) differentiated between three types of studies that assess interventions with different mechanisms, namely (1) traditional methods of financial education, (2) innovative methods that either use novel approaches to impact financial capability or influence behavior more directly and (3) a combination of methods. Under (2), the World Bank also subsumes tools that improve financial behaviors (e.g. savings) without clear cognitive mechanisms. However, it may be disputed whether all interventions in this category (such as counseling, nudges or information framing) should be labeled as financial education (e.g. Willis, 2022).

Building on the World Bank's classification and the OECD's definition of financial literacy as knowledge, attitudes and behaviors, we propose a classification of common causal channels interventions address and potential outcomes they aim to affect directly. While real-world interventions certainly combine multiple of these channels and outcomes, the categorization illustrates possible key causal channels interventions may use.

It is important to note that any of the suggested channels may be aimed at different ultimate goals, e.g. at individual or – like in PISA's definition of financial literacy (OECD, 2023) – at societal financial well-being. Indeed, Kaczkó (2025) focuses on a similar distinction of overarching goals, defining a spectrum between “individual-oriented” and “socially-oriented” interventions and the related pedagogical notions of “financial training” and “financial education”.

The following suggested categories of causal channels are ordered by the approximate prevalence in currently available impact assessments of financial education interventions:

- (1) **Factual knowledge:** This channel focuses on building an understanding of financial terms, concepts and facts in the general area of (personal) finance and is often the theoretical foundation of higher order procedural skills. Effects on financial behaviors and well-being are assumed to be largely mediated by increased knowledge.
- (2) **Procedural skills:** This channel emphasizes the practical application of financial knowledge through actions and decision-making, often in real-world or simulated environments. Procedural skills are inherently adaptive, enabling learners to apply financial strategies flexibly across diverse and evolving situations. They serve as a critical bridge between the acquisition of knowledge and the development of actual financial behaviors.
- (3) **Instructions:** Through this channel, information and guidance about beneficial behaviors or decisions in specific situations are given. Related interventions do not aim to promote understanding or knowledge but address behaviors directly. Nevertheless, some amount of active deliberation may be necessary on the side of recipients.
- (4) **Nudging:** This channel aims to promote behaviors without providing a significant amount of information and may skip deliberate cognitive processing altogether; it may involve designing decision situations or information framing in a way that guides people toward desired behaviors.
- (5) **Attitudes:** Rather than imparting knowledge, this channel aims to influence participants attitudes toward financial topics and decisions; related interventions may, for example, be aimed at attitudes toward saving and spending, interest in financial topics, perceived financial self-efficacy or social responsibility.
- (6) **Self-regulation:** This channel is aimed at furthering executive functioning and meta-cognitive and emotional skills. Related interventions may aim to foster awareness about one's own thinking, attitudes, behavior, motivations and emotions, for example by training self-control through mental strategies or fostering awareness about one's own psychological biases.
- (7) **Critical thinking:** This channel focuses on promoting reflection on (complex) financial topics, situations and systems. While we assume that such interventions may also aim to foster knowledge, behaviors and well-being, this channel may put a stronger focus on social responsibility and societal well-being as opposed to individual well-being.

All interventions included in this review can be considered combinations of these channels, with most appearing to put the primary focus on a smaller subset of the channels. Indeed, interventions focusing on factual knowledge can be found for all populations. They are most commonly classroom-based and have

been tested most rigorously with school children. Knowledge-focused interventions may also address secondary channels with varying intensity, such as procedural skills, instructions and, more rarely, attitudes, among other potential channels. Some interventions, for instance experimental school-based programs that use simulations of real-life situations, could be considered multi-pronged by nature as they may directly or indirectly address multiple channels, i.e. factual knowledge, procedural skills, attitudes, self-regulation and critical thinking.

Another common type of interventions focuses more strongly on providing instructions to adults to encourage specific behaviors. Such interventions can focus on varying issues and be of varying intensity and format and may also address other channels, particularly factual knowledge. They are often conducted “just in time,” i.e. directly preceding the relevant decisions.

Few studies in our pool directly target attitudes, self-regulation or critical thinking, both with regard to the reported interventions and the study design and outcome measures used. Although these channels may be addressed indirectly, it is largely unclear whether they are affected positively, negatively or not at all. We thus recommend that these missing channels are included in future studies on financial education, both through intervention designs and their assessments.

Moreover, concepts of financial literacy typically imply or explicitly define individual or societal financial well-being as the ultimate goal of financial education. These goals are, however, rarely addressed in studies. We therefore recommend that studies attempt to include these downstream effects in their study designs, in combination with the causal channels that are hypothesized to affect them.

Nevertheless, one of the main obstacles hindering conclusions about the effectiveness and causality of financial education appears to be the heterogeneity in interventions and outcomes. We therefore recommend that studies always clearly and explicitly indicate not only content, method and practical aspects of the intervention under examination but also the presumed causal channels as (for example) listed above and the assumed goal dimensions. One approach to theoretically model the underlying causal channels of the intervention is conducting a theory of change exercise, ideally already at the outset of the planned intervention (NEFE, 2016; Finney, 2020; University of Wisconsin-Madison, 2025). We thus also call for future meta-analyses to integrate a classification of causal mechanisms. Overall, we suggest that the research field at large should move away from treating “financial education” as a single class of interventions and instead focus on clearly identifying causes, mechanisms and (intended) effects.

6.2 Reporting intervention and study details

A substantial proportion of the studies lack detailed descriptions of interventions’ content and didactic aspects. A few studies altogether omit information on delivery format or curriculum. Similarly, some publications leave essential methodological aspects unclear, such as the procedures used to assign participants to treatment and control groups.

Next to our call for a more deliberate focus on underlying causal mechanisms, another important recommendation mirrors one already made by Fernandes et al. (2014), namely that publications should clearly describe all relevant aspects of interventions and studies in sufficient detail to allow meaningful comparisons and conclusions. Descriptions should include details of the intervention (i.e. format, timing, content, didactic aspects), participants and staff (i.e. population context and sociodemographics, teachers’ qualifications, if applicable), study method and procedure (e.g. group assignment method) and all other relevant factors that may improve comparability of results across studies.

6.3 Choosing appropriate statistical models

Concerning the choice of statistical model, studies generally use appropriate difference-in-difference (or similar) estimates with all available data points. However, some studies conducted both pre- and post-tests but only analyzed post-treatment data without further explanation. If non-compliance (i.e. participants do not take part in the intervention) is an issue, some studies use intention-to-treat models, i.e. keep “non-compliant” participants in the analyses to increase internal validity (e.g. Maldonado, De Witte and Declercq, 2022).

Almost all school-based studies use clustered standard errors to account for the clustered structure of the data, e.g. with students clustered in classes and classes clustered in schools. Multilevel models are used by three papers. Two use random intercepts (Amagir et al., 2022; Johnson et al., 2021) and only one (Dare et al., 2020) attempts to model a potential heterogeneity of the treatment effect across clusters as random slopes. As a result, standard errors of treatment effects may be underestimated in many studies using clustered data.

Lastly, some studies use non-significant differences as evidence for the *absence* of a difference or an effect. However, if the absence of differences is of interest, it should be tested using *equivalence testing*. To this end, researchers pre-define upper and lower limits of differences that they consider negligible. If the measured difference falls within these limits with a certain confidence, conclusions about a “significant non-difference” are appropriate.

We therefore recommend that studies use appropriate statistical methods for the collected data. In general, this includes using all available data points and, particularly in a school context, modeling hierarchical data structures using multilevel (mixed) models and taking potential heterogeneous treatment effects across clusters into account using random slopes models.

6.4 Validating outcome measures

We found that studies varied widely in how they defined and measured financial literacy, behavior and well-being. This results in inconsistency and the usage of non-standardized outcome measures. Some used their own independently designed financial literacy tests, while others relied on established measures like Lusardi and Mitchell’s “Big Three” questions (e.g. Isler et al., 2022; Lusardi, Samek, et al., 2017). This lack of standardization makes it difficult to compare results across studies and synthesize broader conclusions.

As Fernandes et al. (2014) observed, many studies appear to draw on skill- or competency-oriented concepts of financial literacy and education. However, many fall back on very few questions of financial knowledge. In addition, some studies appear to avoid excessive “curricular validity” or “teaching to the test” by using independently developed or published test questions, others specifically develop tests to better capture the content taught in the intervention. While both approaches may have merit, their side-by-side application again limits the comparability of studies.

Moreover, a major limitation arises due to the frequent use of self-reported outcome measures, which can be influenced by social desirability bias. Such measures can lead to overestimation or show little correlation with actual behaviors (as in e.g. Postmus et al., 2011). Only a few studies involving adults used objective financial data (e.g. bank balances, credit scores, savings deposits) that provide more reliable indicators (e.g. Skimmyhorn, 2016).

Many studies rely on some kind of knowledge test and many assess attitudes or behaviors using questionnaire items. However, a large proportion of the studies do not report any statistical validation efforts

of the instruments used. This means that no validity or reliability checks (e.g. correlations with convergent or divergent criteria, internal consistency, factor analyses or more advanced analyses) are reported for the questionnaire items. Moreover, some publications appear to give undue weight to significance tests of effects on only single items from composite scales. Only few studies (e.g. Angel, 2018; Batty et al., 2020) used item response theory (see De Ayala, 2013) to validate the knowledge test they use; few report Cronbach's alpha values or results from factor analyses.

We thus recommend that studies validate outcome measures or use instruments where validation results are reported elsewhere. Moreover, measurement instruments should be used in their intended composition. Analyses of single items of composite scales should be framed as exploratory analyses and, in case of multiple significance tests, results should be interpreted cautiously or alpha level corrections should be applied. Additionally, incorporating objective financial indicators, such as administrative data, can help mitigate biases associated with self-reported data and provide a clearer picture of actual improvements in financial behaviors and outcomes.

6.5 Testing long-term effects

While many studies used pre- and post-tests in their design, most measured effects immediately or a few months post-intervention, making it difficult to assess whether financial behavior changes are sustained over time. As an example, Barcellos et al. (2016) found short-term financial knowledge gains among migrants, but these gains disappeared after six months. Few studies conducted long-term follow-ups beyond a year. Gerrans (2021), for example, reports on the effectiveness of an intervention up to three years after the treatment. Ultimately, however, insights into long-term effects on financial knowledge and behavior, let alone financial stability, wealth and long-term financial well-being are limited.

To understand the effectiveness of financial education, future research needs to prioritize longitudinal study designs with follow-ups extending beyond one year, allowing for a better understanding of long-term changes in financial behaviors and financial well-being.

6.6 Using suitable study designs

We only included studies with at least a quasi-experimental design in our review. The majority of studies we considered in our analyses thus used either randomized experiments, randomly allocating participants to the treatment and control groups, or quasi-experimental designs with a non-random group allocation. In general, however, research designs were remarkably simple, with surprisingly few studies comparing different treatment combinations. In the school context, for example, such designs appear useful to test treatments in combination with system-oriented factors such as teacher engagement or parent involvement (e.g. Compen, De Witte and Schelfhout, 2020; Maldonado and De Witte, 2021). In all contexts, however, targeted treatment comparisons would be useful to isolate the effect of intervention aspects, e.g. different combinations of content or format. Such designs, however, need to ensure that the effects of treatment aspects can be isolated (see subsection 6.7).

Some studies, particularly in the school and university context, used some type of waitlist design (e.g. Torma et al., 2023).²² Such designs have the distinct advantage that being allocated to the control group is not associated with missing out on a potential beneficial intervention indefinitely. Instead, the control group receives the treatment after a predefined delay and after completing the post-test. In situations

²² Depending on context and implementation, such research designs may also be called delayed intervention, staggered roll-out or phase-in intervention designs.

with limited resources and ethical or practical concerns about withholding the treatment from study participants, such designs can prove helpful to guarantee internal validity. It is important to note, however, that because in these designs the control group typically receives the intervention not long after the treatment group, it becomes impossible to analyze longer-term differences between the groups.

Nevertheless, we recommend considering waitlist designs in situations where a pure control group would be unfeasible or where assignment to a pure control group could lead to strong unintended effects such as compensatory behavior or resignation. As detailed in the following subsection, if sample sizes are sufficiently large, future studies should attempt more complex research designs in order to compare intervention characteristics and combinations thereof.

6.7 Enabling useful treatment comparisons

Most studies in our pool compare the effects of a financial education program against the alternative of not providing (additional) financial education. However, some interventions take a different approach, comparing two (or more) alternative interventions with each other (Lusardi et al., 2017). For example, some interventions compare the effectiveness of online vs. offline interventions. Studies specifically in the school context compare pedagogical approaches (see e.g. Sconti, 2022) or test the effects of additional teacher trainings (see e.g. Compen et al. 2023).

When rigorously implemented and evaluated, this comparative approach provides deeper insights into the relative impact of specific program components by isolating one aspect while keeping as many other conditions as possible constant across interventions. Such an approach can be extremely valuable in the search for the relevant factors that make financial education interventions effective.

Conversely, comparing different treatments can also severely limit the usefulness of results if these treatments are not carefully implemented. Few studies appear to unsystematically combine several treatment factors into treatment groups, making clear conclusions on the cause of different outcomes impossible. For example, comparing two delivery formats when they are each combined with a different didactic method and course duration makes it impossible to attribute group differences to any one of these factors alone. If such confounded designs do not have a true control group that receives no financial education treatment, possible conclusions are even more limited. As result, such designs only give an indication that one specific combination of treatment factors is more effective than another. The results, however, may be impossible to generalize to other interventions and situations.

We thus recommend that studies always include a true control group that receives either no treatment whatsoever or some type of placebo intervention. Even in the absence of treatment differences, a control group allows interpretations of the overall effects of the interventions. We also recommend that studies make a deliberate effort to isolate the effects of single treatment factors that may influence the effectiveness of financial education. This means that studies could vary specific intervention characteristics, while keeping all other aspects of the intervention constant. Larger sample sizes may also allow more complex designs. For example, a 2x2x2 factorial design could simultaneously test the three factors “type of content,” “level of intensity” and “didactic method” and their interactions. The challenge, however, lies in keeping the other factors constant when varying any of the included factors, which may require considerable sacrifices in the overall design of the intervention.

6.8 Preventing sample attrition

Voluntary participation generally led to low rates of take-up of the interventions. Consequently, many studies, particularly those with (young) adults, suffer from small sample sizes and thus limited statistical

power. Another common issue across all studies with repeated measurements is considerable sample attrition (dropout) over time, which becomes increasingly severe with longer study periods. Attrition is highly problematic as it limits internal validity and statistical power. Even school-based studies with typically short study periods usually lose over 30% of the sample between pre- and post-test. Attrition was particularly severe in some digital and online interventions (e.g. Angel, 2018; French et al., 2020) and in studies on vulnerable groups (e.g. Postmus et al., 2011).

Studies deal with this issue in different ways. Many do not consider attrition at all, others test for systematic attrition and very few use more robust estimation methods, such as weighted regression models (e.g. Maldonado, De Witte and Declercq, 2022).

We highly recommend that studies should put a stronger focus on reducing attrition by proportionally allocating more resources to *retaining* participants through follow-ups and incentives as opposed to only recruiting a very large initial sample. A large initial sample becomes virtually useless if only a small proportion of the sample participates in the post-test. Moreover, attrition may be highly problematic, as it threatens internal validity by introducing an additional source of sampling or self-selection bias that may be associated with the effectiveness of the intervention. Attrition should also be considered in the analyses. At the minimum, this should include testing for non-randomness of dropout and for systematic differences in attrition between the treatment and control groups. Ideally, however, attrition should also be considered in the statistical analyses through (inverse probability) weighting methods.

6.9 Sampling and randomization

Across contexts, most studies are limited by non-random sampling. While experimental or quasi-experimental designs may provide excellent or good internal consistency, results may not be transferable to the general population due to self-selection. For example, participants may consist exclusively of people already interested in actively engaging with financial topics, thus reacting differently to the intervention than the general population would.

Moreover, some studies with adults focus on highly specific populations, making it difficult to apply findings to the general adult population. For instance, studies targeting military personnel (Skimmyhorn, 2016; Skimmyhorn et al., 2016) may not reflect the financial behavior of civilians. Other interventions rely on country-specific financial systems, limiting their relevance elsewhere. For instance, pension fund interventions (e.g. Billari et al., 2023, in Italy; Foster et al., 2015, in Australia) may not be transferable to other countries due to varying retirement savings schemes across countries.

In the school context, most studies are conducted with (likely highly motivated) schools or teachers that actively self-select into the studies. While many studies use randomization mostly at the school level, in some quasi-experimental studies schools or teachers actively self-select into treatment and control groups. While pre- and post-tests allow difference-in-difference estimates, the parallel trends assumption necessary for such models can hardly be deemed fulfilled. Self-selection into treatments (as opposed to random or quasi-random assignment) must thus be considered a severe threat to internal validity.

Concerning non-random sampling from populations, there may be situations of experimental research where non-random sampling does not pose a severe threat to external validity, for example, when university students participate in a short incentivized laboratory experiment without having much information about the content of the experiment beforehand. However, in the case of financial education, participating individuals, schools or teachers often commit to a larger time investment. Allowing self-selection into the study may thus represent a threat to external validity of a larger magnitude.

We therefore recommend that interpretations of impact assessments be limited to the appropriate population, for example the population of interested or motivated schools that have self-selected into the study. Moreover, we recommend that future studies put more emphasis on internal validity and only allow self-selection into the treatment group as a last resort and combined with thorough analyses of group differences and, if possible, pre-treatment trends.

6.10 Exploring mechanisms and processes with mixed methods

Impact evaluations predominantly rely on quantitative study designs. However, methodological limitations such as sampling biases, lack of randomization and other constraints often severely limit their robustness. While quantitative approaches are certainly essential for estimating an overall impact of an intervention, qualitative methods can offer valuable insights into causal mechanisms and processes that should inform future research.

The studies included in this review are almost exclusively quantitative, with only a few incorporating both qualitative and quantitative methods. However, some of them do not fully qualify as mixed-methods research but rather as multi-mode studies (as in e.g. Compen, De Witte and Schelfhout, 2020; Özdemir and Kaya Uyanik, 2021). For instance, Compen, De Witte and Schelfhout (2020) employed classroom observations to examine student-teacher interactions, tracking how students progressed along an adaptive learning path and the extent to which teachers provided content-related assistance.

Complementary qualitative research is particularly useful for understanding underlying processes better (see e.g. Lorenz 2024). We recommend that future research considers incorporating mixed-methods approaches, and indeed, when robust quantitative evaluations are not feasible, qualitative approaches are viable alternatives. For example, more research is needed to assess the impact of financial education on very young children. At the same time, evaluating financial education for this age group presents unique challenges, as traditional assessment methods may not yet be suitable. Consequently, alternative research approaches, such as qualitative studies, can provide valuable insights into early financial education.

Lastly, regardless of the methodological approach, we recommend that future studies generally look into causal mechanisms and processes in greater detail. For example, treatment uptake, knowledge, attitudes and behavior may be considered mediating variables as opposed to outcomes in and of themselves (also see subsection 6.1).

7 Conclusions

In this article, we take a thematic approach to summarizing recent empirical evidence on financial education impact evaluations. We analyze 68 experimental and quasi-experimental peer-reviewed studies published between 2010 and 2024, implemented in OECD countries based on key program characteristics. Where feasible, we highlight program features that show promising effects on the effectiveness of financial education in improving financial literacy – including knowledge, skills, attitudes and behaviors.

Financial education programs targeting children and adolescents are almost exclusively implemented in school settings. However, they are typically not integrated into the mandatory school curriculum but rather offered on a voluntary basis to schools. Program intensity is relatively homogeneous across school-based studies, with programs typically lasting a few hours over several weeks. Most school-based interventions are designed for specific grade levels, predominantly at the secondary school level and delivered through classroom sessions using in-person and/or digital didactic approaches. These programs typically emphasize personal finance topics over a broader understanding of financial literacy, focusing primarily on financial knowledge. By comparison, financial attitudes and behaviors receive less attention.

Financial education interventions for adults exhibit substantial heterogeneity. They range from one-on-one in-person sessions to digital tools and apps incorporating alternative methods (e.g. nudging) that may not be inherently considered as financial education alongside traditional education. Many adult-targeted programs are brief – often a one-time information treatment, a single session or a short course – raising concerns about the depth of learning. Unlike school-based interventions, adult programs tend to focus more on improving financial behavior as opposed to financial knowledge.

Few questions on the effectiveness of financial education can be answered conclusively based on the current state of research. In line with existing literature, our findings confirm predominantly positive effects of knowledge-based interventions on short-term financial knowledge. Although different learning approaches have been tested, particularly in school-based interventions, none have emerged as superior. Early evidence also suggests that parents and teachers are important moderators for the effectiveness of financial education on children.

Studies also yield mixed results regarding financial attitudes and behaviors. Just-in-time interventions tied to a specific decision and interventions directly addressing behaviors as opposed to knowledge appear promising at least with regard to their short-term effects. However, robust evidence is lacking. Financial attitudes are rarely studied independently as the main outcomes of interventions and are often measured alongside financial knowledge or behavior. Long-term outcomes and causal mechanisms are rarely assessed. Additionally, the impact on financial well-being remains almost completely unexplored.

Study results depend not only on the interventions but also to a large degree on research context and methodology. For example, studies involving the adult population that focus on specific behavioral outcome measures typically examine interventions that address these specific behaviors more directly than purely knowledge- or skills-based interventions. Few studies were able to use outcomes beyond self-reports. Those that do use objective measures from administrative data are set in specific contexts, where e.g. retirement savings were increased through pension fund intervention.

Many studies considered are facing major methodological limitations. To strengthen future evaluations and to enhance the robustness of future literature synthesis, we propose explicitly identifying the causal pathways between financial education and various financial outcomes, suggesting seven potential causal channels prevalent in scientific discourse. Additionally, we advocate more detailed descriptions of core program characteristics, particularly in terms of instructional methods and content, and stronger methodological rigor in study design with a focus on causal identification strategies and comparative effectiveness studies that isolate individual program components. We also recommend increasing efforts to mitigate sample attrition and increasingly adopting of mixed-methods approaches to enhance the depth of analysis. Finally, we need more long-term evaluations to assess sustained impacts.

For program and evaluation designers, we recommend developing higher-intensity interventions, as brief programs may lack the depth required for lasting impact. Additionally, programs may consider incorporating emerging topics and broader financial education topics that have gained policy attention but remain currently underrepresented in the programs we considered. Notably, digital financial literacy and sustainability have not been widely integrated into financial education initiatives. While substantial evidence supports the effectiveness of financial education for secondary school students, research on younger children remains limited, highlighting a critical gap that future programs should address. Likewise, financial attitudes, self-regulation and critical thinking are rarely explored as causal channels of financial education. Ultimately, the research field presents ample opportunity to test the effects of both innovative and conventional approaches that go beyond the immediate improvement of financial knowledge.

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Table 1

Overview of financial education assessments for school children

No.	Study	Intervention summary	Country	Target group	Instructor information	Intensity	Content	Format and delivery channel	Study design	Outcome measures
01	Agasisti et al. (2022)	online financial education course for students majoring in languages, sciences and humanities	Italy	secondary school grade 12	students' teachers but with limited role; no prior instructor training	10h over ca. 9 weeks	bank accounts, interest rates, choosing the best installment purchase agreement	online classroom lessons	experimental: random assignment for classes within one school in Milan	financial knowledge
02	Amagir et al. (2022)	real-life financial education initiative with emphasis on financial behaviors	Netherlands	secondary school grade 9 (age: 15 years)	students' teachers; 3h prior instructor training	6.67h over 8 weeks	budgeting, setting savings goals, avoiding unnecessary expenses	in-person classroom lessons	experimental: random assignment for classes that signed up	financial knowledge, financial behavior, financial attitudes
03	Batty et al. (2020)	experiential, non-lecture-based financial education program using an economy simulation	USA	primary school grade 4 (age: ca. 10 years)	students' teachers; no prior instructor training	10 weeks	financial responsibility and decision-making; planning and money management; saving and investing	in-person classroom lessons	experimental: random assignment for schools that signed up; with waitlist design	financial knowledge
04	Batty et al. (2015)	traditional classroom lessons improving financial knowledge and decision-making skills	USA	primary school grade 4 and 5 (age: 9-10 years)	students' teachers or educators from a local credit union; 3h prior instructor training	3.75h over 5 weeks	savings, financial decision-making and money management	in-person classroom lessons	experimental: random assignment for all classes in two school districts; with waitlist design	financial knowledge, financial behavior, financial attitudes
05	Becchetti et al. (2013)	traditional classroom lessons with very limited content information	Italy	secondary school (age: ca. 18-19)	no information available	16h over 13 weeks	no information available	classroom lessons	experimental: random assignment for schools/ or classes from Milan and Rome that already had prior experience in teaching the program (no information how they were selected)	financial knowledge, financial behavior
06	Bhattacharya and Gill (2020)	in-person classroom education featuring two distinct treatments: one centered on money management and the other on financial investment	USA	secondary school grade 11 and 12	instructors from university; no prior instructor training	8h over 8 weeks	budgeting, borrowing, saving, and understanding economic principles relevant to personal finance and focus on money management or financial investment	in-person classroom lessons	quasi-experimental: non-random assignment for classes that signed up to the program, assignment according to predefined logistic constraints	financial attitudes

Source: Authors' compilation.

Table 1 continued

Overview of financial education assessments for school children

No.	Study	Intervention summary	Country	Target group	Instructor information	Intensity	Content	Format and delivery channel	Study design	Outcome measures
07	Bhattacharya et al. (2016)	voluntary summer camp that blends traditional financial education with a savings-based intervention	USA	secondary school grade 8	external trainers from university and a certified financial advisor; no prior instructor training	12.5h over 1 week	economic fundamentals; money management fundamentals; saving and investing; credit and borrowing; financial risk management	in-person financial literacy camp	experimental: random assignment for only a small part of the intervention; among three schools, parents voluntarily signed their children up for the program, random assignment to 90 minutes of extra financial education (+14%) among registered students	financial knowledge
08	Coda Moscarola and Kalwij (2021)	extracurricular financial education program during standard schooling hours at the Museum of Savings in Turin called the Money Learning program	Italy	primary school grade 4 and 5 (age: 8–11)	museum staff, no information about prior instructor training	one museum visit	origins of money, principles of a loan, implications of inflation, barter versus monetary exchange economies, history of the Templars and the concept of interest rates, origin of the euro, interest compounding	museum visit with short videos and interactive intertemporal-choice game	experimental: random assignment for schools that arrived at the museum; waitlist design	financial knowledge
09	Compen et al. (2020)	interactive online teacher training paired with three distinct student treatments, each varying in the level of teacher involvement	Belgium (Flanders)	secondary school and students (age: 13–15) and their teachers	students' teachers with different intensity of teacher role depending on the treatment; 3h of instructor training	3.33h	saving and investing	students: digital learning path; teachers: online webinars	experimental and mixed methods: qualitative teacher class observation; random assignment for schools that signed up except for one treatment; three different treatments with one treatment only included teachers who signed up for teacher training	financial knowledge, financial behavior
10	Compen et al. (2023)	online teacher training module paired with a digital financial literacy game, where student pairs solve problems based on ability matching	Belgium (Flanders)	secondary school and teachers (age: 13–15) and their teachers	students' teachers with limited role; ca. 3 hours of prior instructor training	3.33h	payment methods	students: digital game, classroom lessons; teachers: online platform	experimental and mixed methods: qualitative teacher class observation; random assignment for schools that signed up; two different treatments	financial knowledge, financial behavior

Source: Authors' compilation.

Table 1 continued

Overview of financial education assessments for school children

No.	Study	Intervention summary	Country	Target group	Instructor information	Intensity	Content	Format and delivery channel	Study design	Outcome measures
11	Dare et al. (2020)	large-scale national financial education program involving 124 schools, with two modules: one focused on responsible spending and the other on performing transactions	Netherlands	primary school grade 5 (age: 9–13)	students' teachers; no prior instructor training	sometime between October and March	responsible spending: making budget choices, peer pressure and advertising, estimating prices; performing transactions: cash and digital money, proactive attitude toward money matters, security features of money	digital learning path classroom lessons	quasi-experimental: schools were randomly chosen by the ministry, schools that used the program previously were assigned to treatment and the schools that did not use the program were previously assigned as the control group	financial knowledge
12	De Becker et al. (2021)	digital learning path featuring various exercises (videos, games, case studies), primarily implemented in subsidized private schools, combined with a discrete choice experiment	Belgium	secondary school	students' teachers, but with limited role; no prior instructor training	4h	general concepts in financial literacy education, including saving and investing and deals with budgetary choices in everyday life	digital learning path classroom lessons	experimental: random assignment for schools that signed up	financial knowledge, financial behavior
13	Gill and Bhattacharya (2015)	comparison of two well-known interventions: one focused on personal finance (FFFL program) and the other on investment (SML), implemented in low-income schools	USA	secondary school grade 11 and 12	economics university professor; no prior instructor training	7h	both interventions: principles of economic thinking, banking and financial services; saving and investment, credit and borrowing; FFFL intense: the above topics more intensively; SML intense: the topic of investing more intensively	digital and in-person classroom lessons	quasi-experimental: non-random assignment, randomization is not really addressed in the paper	financial knowledge
14	Harter and Harter (2010)	well-known stock market game (SMG) paired with a comprehensive, complementary stock market learning curriculum	USA	secondary school grade 9–12 (age: 14–18)	students' teachers; prior instructor training without information about intensity	NA	stock market simulation	digital and in-person classroom lessons	quasi-experimental: non-random treatment and control assignment, randomization is not really addressed in the paper	financial knowledge

Note: NA = information not available.

Source: Authors' compilation.

Table 1 continued

Overview of financial education assessments for school children

No.	Study	Intervention summary	Country	Target group	Instructor information	Intensity	Content	Format and delivery channel	Study design	Outcome measures
15	Itebeke et al. (2020)	combinations of ability matching and differentiated student instruction, tailored to their proficiency levels, using an interactive learning game and paper-based materials	Belgium (Flanders)	secondary school grade 8 and 9 (age ca. 13–14)	students' teachers but with limited role; no prior instructor training	3.3h	means of payment, including bank accounts, bank cards, direct debits, standing orders and security issues such as skimming and phishing	digital and in-person classroom lessons	experimental: random assignment for schools that signed up	financial knowledge, financial behavior
16	Johnson et al. (2021)	story-driven narrative with interactive activities where students take over the role of a town's Mayor, navigating real-life financial decisions	USA	secondary school grade 7 and 8	students' teachers, but with limited role; no prior instructor training	4h	financial values and goal setting; budgeting and opportunity cost; saving and investing; payment types; banking, risk versus return; planning for the future	digital and in-person classroom lessons	experimental: random assignment for schools that were recruited and already had prior experience in teaching the program	financial knowledge, financial attitudes, financial behavior
17	Kalmi (2018)	peer learning between higher and lower secondary students, where the former developed digital financial education materials	Finland	secondary school (age: 15)	students from an upper secondary school under the supervision of their teachers; no prior instructor training	1h	no information available	digital classroom lessons	quasi-experimental: non-random assignment according to location of the school; treatment group is located in the area of the sponsoring bank	financial knowledge, financial attitudes, financial behavior
18	Kalmi and Rahko (2022)	comparison of traditional teaching methods versus a game-based approach, exploring how interactive, gamified learning impacts financial understanding and decision-making	Finland	secondary school grade 9 (age: ca. 15)	students' teachers; no prior instructor training	TR1: 4–6h; TR2: 2h; TR3: 5–8h	TR 1: work-life skills, personal finance; TR2: personal finance skills; TR3: teamwork skills, communication skills, decision-making calculations	classroom lessons with digital elements depending on the intervention	quasi-experimental: random assignment for only one out of three treatments for schools that signed up, 3 different treatments	financial knowledge, financial behavior

¹ TR = Treatment group.

Source: Authors' compilation.

Table 1 continued

Overview of financial education assessments for school children

No.	Study	Intervention summary	Country	Target group	Instructor information	Intensity	Content	Format and delivery channel	Study design	Outcome measures
19	Kalwij et al. (2019)	45-minute cash quiz and short video presented during Global Money Week	Netherlands	primary school grade 5 and 6	employees from the financial sector acted as quiz master, no prior instructor training	1h	banks, money, transactions, planning and managing, savings, borrowing, risk and reward, the financial landscape, Dutch central bank tasks, how to earn income, entrepreneurship, happiness and money, costs of a smartphone	in-person classroom quiz	experimental: true RCT with random assignment for all primary schools and a small share of schools that already participated in previous study	financial knowledge, financial behavior
20	Lührmann et al. (2015)	intervention led by an external coach, featuring three modules on shopping, financial planning, and saving/investing, designed for lower-track students	Germany	secondary school (age: 13–15)	employees of sponsoring firms, prior instructor training without information about intensity	4.5h	shopping decisions, financial planning, saving/investment choices	in-person classroom lessons	quasi-experimental: non-random assignment for schools that signed up for the teacher trainings; allocation to treatment and control according to time constraints; waitlist design method	financial knowledge, financial attitudes, financial behavior
21	Lührmann et al. (2018)	intervention led by an external coach, featuring three modules on shopping, financial planning, and saving/investing, designed for lower-track students	Germany	secondary school (age: 13–15)	employees of sponsoring firms, prior instructor training without information about intensity	4.5h	shopping decisions, financial planning, saving/ choices	in-person classroom lessons	experimental: true RCT with post-test only and random assignment for all lower-track high schools	financial knowledge, financial attitudes
22	Maldonado and De Witte (2021)	parental financial education information and parental homework involvement combined with a classroom-based intervention	Belgium (Flanders)	secondary school students grade 8 and 9 (age: 13–14) and parents	students' teachers but with limited role; no prior instructor training	4h	saving and investing	in-person classroom lessons with some digital elements	quasi-experimental: non-random assignment for schools that signed up, assignment according on time preferences for implementation, with waitlist design	financial knowledge, financial skills
23	Maldonado et al. (2022)	parental involvement in homework activities, combined with a classroom-based intervention	Belgium (Flanders)	secondary school students grade 8 and 9 (age: 13–14) and parents	students' teachers but with limited role; no prior instructor training	4h	payment methods	classroom lessons combined with digital homework assignment	experimental: with random assignment for schools that signed up	financial knowledge, financial behavior

Source: Authors' compilation.

Table 1 continued

Overview of financial education assessments for school children

No.	Study	Intervention summary	Country	Target group	Instructor information	Intensity	Content	Format and delivery channel	Study design	Outcome measures
24	Moscarola and Migheli (2017)	children had to draw what they want to save for and discuss with the researcher how long they need to save to achieve that goal	Italy	children (age: 8-9)	researchers	1h	saving; intertemporal choice	laboratory in-person intertemporal choice task	quasi-experimental: random selection of schools but non-random assignment for classes – headmaster decided which class participates	financial attitudes
25	Oezdemir and Kaya (2021)	financial literacy course consisting of eight modules, with lack of further information	Türkiye	secondary school grade 10	no information available	over 8 weeks	financial terms, budgeting, debts, saving, spending, sharing, rational decisions	no information available	experimental (?) and mixed methods: qualitative interviews with students on students' opinions; no information about randomization and level of treatment	financial attitudes, financial behavior
26	Sconti (2022)	digital program (short quizzes on an interactive multimedia whiteboard) versus in-person program (using slides and the rule-of-thumb approach)	Italy	secondary school (age: 16-18)	students' teachers and financial advisor but with limited role of teachers; no prior instructor training	8h over 4 weeks	both treatments cover the same topics: motivational lecture, price and inflation, savings and payment instruments; financial markets; mortgages, insurance and retirement	in-person or digital classroom lessons	experimental: random assignment for classes within one school	financial knowledge
27	Sule and Seda (2018)	teaching the consequences of actions when making intertemporal decisions using mini case studies	Türkiye	primary school grade 3 and 4	students' teachers; instructor training without information about intensity	2h over 8 weeks	temptation goods, smart shopping, coping mechanisms against temptations to meet saving targets	in-person classroom lessons	experimental: random assignment for schools that signed up; version of a waitlist design	financial attitudes
28	Urban et al. (2018) ²	impact of state-level personal finance education requirements on credit outcomes	USA	(age: 18-21)	depends on the state	over one semester	personal finance standards for the treated states of Georgia and Texas	classroom lessons	quasi-experimental synthetic control: Georgia and Texas had financial education mandates already before 2011; calculation of synthetic controls based on state-level macro variables of remaining states	financial behavior

² Note that state-level interventions were only included where sufficient information about the policy intervention was provided.

Source: Authors' compilation.

Table 1 continued

Overview of financial education assessments for school children

No.	Study	Intervention summary	Country	Target group	Instructor information	Intensity	Content	Format and delivery channel	Study design	Outcome measures
29	Walstad et al. (2010)	video-based financial education program accompanied by lesson materials including instructional activities and assignments	USA	secondary school (age: 15–18)	students' teachers; 3–4 hours prior instructor training	6h over 4 weeks	key concepts in personal finance such as saving, wealth building and money management, opportunity costs related to financial decision-making, earning an income; banking and checking practices, credit, debt, annual percentage rate (APR), credit scores, budgeting, saving, investing and risk tolerance	in-person and video classroom lessons	quasi-experimental: non-random assignment for classes that signed up, assignment according to treated teachers' preferences	financial knowledge

Source: Authors' compilation.

Table 2

Overview of financial education assessments for adults

No.	Study	Intervention summary	Country	Target group	Intensity	Content	Format and delivery channel	Study design	Outcome measures
Financial education for vulnerable groups									
01	Barcellos et al. (2016)	financial education for immigrants; focus on general-purpose and immigrant-specific financial information	USA	migrants: first- and second-generation Americans	one-time information	banking services, saving, investing (+ immigrant-specific information in a second version)	online information material (two versions)	experimental: used nationally representative online panel, random assignment, measurement up to six months after intervention	financial knowledge, hypothetical financial behavior
02	Blanco et al. (2023)	community-based digital financial education program including nudges for motivation	USA	Hispanics: low-to-moderate income	6 months (with weekly reminder)	monthly topics: financial goal setting, financial product selection, bill payments, credit scores, debt and saving; activities include income tracking, financial planning; language: English + Spanish	online program + nudges (mobile phone-delivered; motivational text messages)	experimental: waitlist control group design with block randomization, measurement immediately after intervention	retention rate, financial behavior, financial stress
03	Collins (2013)	mandatory financial education which is linked to institutional financial assistance	USA	very low-income families in a subsidized housing program	5 courses over 2 months (total: 12h)	personal finance, budgeting, credit reports, credit management, banking, financial planning, saving, paying bills	in-person, mandatory course	experimental: field experiment with waitlist control group design and random assignment, measurement up to 10 months after intervention	self-reported financial behavior, bank account balance, credit reports
04	De Bruijn et al. (2022)	a behaviorally informed financial education program (need-driven adaptive approach) compared to a traditional program for financially vulnerable individuals	Netherlands	financially vulnerable consumers (debt and welfare service)	5–6 weekly classes (total: 12.5–15h)	traditional: income types, expenditures, financial products, budgeting, applying for allowances, tracking expenses; BI: income, financial administration, making ends meet (now or later)	in-person, course (traditional vs behaviorally-informed program)	quasi-experimental: only partial randomization was possible, two studies: (1) program with behavioral insights (BI) vs. control group and (2) BI vs. traditional educational program	financial skill and knowledge, financial-psycho-logical indicator, financial behavior, financial well-being, financial situation
05	Elbogen et al. (2016)	supporting veterans with psychiatric disabilities in money management (focus on self-directed care taking)	USA	veterans with psychiatric disabilities	one time (1–3h)	money management: needs vs. wants, budgeting, saving for emergencies, and handling disability income (e.g. veteran discounts, how much additional earning without losing disability benefits, avoiding financial exploitation)	in-person (psychoeducational one-on-one intervention)	experimental: random assignment with clinical trial, measurement 6 months after intervention	money-knowl- edge and saving behavior, perceived empowerment, debt, employ- ment, psychiatric symptoms, homelessness
06	Gibson et al. (2014)	financial literacy to help migrants become better informed consumers, shopping for better remittance deals	New Zealand and Australia	migrants	one time (ca. 2h)	components of remittance costs, strategies for reducing these costs, sources of information for comparing costs, remittance products	in-person, course	experimental: random assignment, measurement up to 6 months after intervention	remittance-specific financial knowledge, broader financial knowledge

Source: Authors' compilation.

Table 2 continued

Overview of financial education assessments for adults

No.	Study	Intervention summary	Country	Target group	Intensity	Content	Format and delivery channel	Study design	Outcome measures
07	Haynes et al. (2011)	financial education as part of health education for women to help manage financial challenges, with or without social networking component	USA	chronically ill rural women	22 weekly lessons	recording income and expenses, assessing net worth, navigating government assistance programs; second version includes expert round table + discussion forum	online lessons consisting of 1-to-2-page instructional guide (2 versions)	experiment: random assignment to two treatment (two versions) and one control group	financial knowledge
08	Hetting et al. (2016)	financial literacy curriculum to support financial independence of victims of domestic violence	USA and Puerto Rico	survivors of intimate partner violence	4–8 weekly workshops (1–2h each); individual session (30–60 min)	disentangling joint financial relationships; repairing credit damaged by abusers; economic safety plans; protecting against financial abuse; language: English + Spanish	in-person workshops + individual session	experimental: random assignment, with face-to-face (or telephone) interviews, measurement immediately after intervention	perceived financial knowledge, financial behaviors, financial stress
09	Modestino et al. (2019)	financial literacy and one-on-one coaching with a credit building counselor for low-income young adults	USA	low-income young adults (age 18–29)	one-time workshop (1h); coaching over 1 year (initial session: 1h)	workshop: credit building and access, saving products; coaching: individual's credit report, development of budgeting and credit action plan	in-person workshop + one-on-one coaching	mixed method: (1) quasi-experimental, random assignment among applicants; (2) two focus groups (during + after intervention)	financial knowledge, self-efficacy, financial behavior, credit data
10	Postmus et al. (2015)	financial literacy curriculum to support financial independence of victims of domestic violence	USA and Puerto Rico	survivors of intimate partner violence	4–8 weekly workshops (1–2h each); individual session (30–60min)	disentangling joint financial relationships; repairing credit damaged by abusers; economic safety plans; protecting against financial abuse; language: English + Spanish	in-person workshop + individual session	experimental: random assignment, with face-to-face (or telephone) interviews, measurement immediately, 6 and 12 months after intervention	perceived financial knowledge, financial behaviors, financial stress
11	Reich and Berman (2015)	financial literacy classes targeting low-income individuals	USA	low-income individuals receiving services in nonprofit residential program	4 weekly sessions (ca. 2h each)	budgeting, loan repayment, noncash payment, checking accounts, credit, identity theft, credit report management	in-person course	experimental: with phase-in, random assignment; measurement immediately after the intervention	financial knowledge, self-reported financial behavior and negative financial life events
Financial education for retirement planning									
12	Billari et al. (2023)	financial education provided by a pension fund to motivate proactive financial decision making	Italy	employees enrolled in pension fund	one time (25 minutes)	expected lifetime and pensions, retrieval of information on retirement income, importance of investment choices in pension funds, effects of inflation on investment decisions, concept of portfolio diversification	online video + nudge (encouraging to gain more info)	experimental: stratified random selection of treatment- and control group, measurement 2 weeks, 3 and ca. 9 months after the intervention	demographic survival literacy, financial knowledge, self-reported financial behavior, pension fund data

Source: Authors' compilation.

Table 2 continued

Overview of financial education assessments for adults

No.	Study	Intervention summary	Country	Target group	Intensity	Content	Format and delivery channel	Study design	Outcome measures
13	Boyer et al. (2022)	provision of targeted information to improve individuals' ability to choose the most tax-efficient savings option	Canada	adults (age 35–55)	one time (longer video: 101 seconds)	video 1: tax features of front-loaded (individuals contribute post-tax income) and back-loaded (individuals contribute pre-tax income) savings plans; video 2: tax features + clawbacks on government benefits	online video (2 versions)	experimental: survey experiment via online panel, random assignment, measurement immediately after intervention	hypothetical saving choices
14	Clark et al. (2014)	low-effort intervention (information nudges) to increase employee participation in the employer's 401(k) plan	USA	newly hired employees of a large financial institution	one-time information	information on the employer's 401(k) plan: highlighting employer matching contributions, benefits of compounding savings over time	online e-mail flyers (hard copies for no computer access) including motivational nudges	experimental: field experiment, random assignment, measurement ca. 2 months after intervention	participation in employer's 401(k) pension plan
15	Collins and Urban (2016)	field study on financial education during work hours to improve retirement savings	USA	employees at 45 credit unions	ca. 8.75h	basics of finance, basics of investing, basics of financial planning, basics of retirement, saving strategies, mutual funds, saving for college, understanding investment risks, working with financial advisers, getting started	online course (10 self-paced modules during work time)	experimental: field experiment, random assignment	self-assessed financial knowledge, self-reported financial behavior, data on retirement contributions
16	Foster et al. (2015)	manipulation of information presentation to impact the accessibility and assessability of retirement savings statements	Australia	members of retirement savings funds	one-time information	retirement fund information including administrative costs (manipulation of information location)	online information (different formats)	experimental: survey experiment, random assignment, measurement immediately after the intervention	financial knowledge, self-assessed financial understanding and behavior
17	Ghafoori et al. (2021)	a large-scale financial education intervention focusing on pension planning	Australia	pre-retiree pension fund members (age 53+)	one time (90 minutes)	retirement and preservation ages, determining retirement money needs, monitoring personal retirement savings, age pension mechanics, additional super contributions and tax benefits, asset allocation fundamentals, retirement income streams, financial advice and action planning	in-person seminar	quasi-experimental: fund members who are yet to attend their seminar act as a de-facto control group	voluntary pension contributions

Source: Authors' compilation.

Table 2 continued

Overview of financial education assessments for adults

No.	Study	Intervention summary	Country	Target group	Intensity	Content	Format and delivery channel	Study design	Outcome measures
18	Goda et al. (2014)	variations of retirement income projection information	USA	university employees (below age 65)	one-time information	planning treatment: general saving information for retirement; describing options for signing up for or changing contributions to pension plan; balance treatment: adds customized projection of hypothetical additional contributions; income treatment: adds customized projection of the additional annual retirement income	online, e-mail brochures (with and without customized saving projections)	experimental: random assignment, measurement ca. 7 months after intervention	voluntary pension contributions
19	Heinberg et al. (2014)	short interventions using psychological principles to increase the appeal of retirement saving and motivate behavioral change in retirement saving	USA	adults	one time (5 videos/narratives; each ca. 5 minutes long)	five basic financial planning concepts; compound interest; inflation, risk diversification, tax treatment of retirement savings vehicles and employer matches of defined contribution savings plans	online videos vs. narratives (with and without behavioral insights)	mixed method: (1) two focus groups to feedback intervention, (2) field experiment with online panel, random assignment, measurement immediately and 8 months after intervention	financial knowledge, self-efficacy
20	Isler et al. (2022)	nudging compared to financial literacy training focusing on the self-management of online retirement accounts	Australia	adults	one time	educational nudge: giving feedback on answers to prior financial knowledge test before allocation task; default nudge: automatically distributing savings in the money allocation task	online money allocation task with educational vs. default nudges	experimental: convenience sampling from university-based online platform, random assignment, measurement immediately after intervention	financial knowledge, financial behavior (saving)
Financial education for a more general adult population and general financial topics									
21	Angel (2018)	digital financial education format's influencing personal finance behavior	Austria	adolescents (age 16–22)	movie (75 minutes); app (4–7 weeks); exercise (NA)	documentary movie: over-indebtedness; mobile budgeting app: track income and expenses web research and exercise: budgeting, saving and debt	online movie, smartphone budgeting app and web research exercise	experimental: random assignment, all measures via online survey, time between pre and post measurement: 4–7 weeks	financial knowledge, financial attitudes, self-reported financial behavior
22	Bruhn et al. (2014)	large-scale financial education program, focusing on low take-up rates for voluntary financial training	Mexico	clients of financial institution (mean age 33)	one time (4 hours)	savings, retirement planning, credit management	in-person (classroom setting)	experimental: random assignment, measurement immediately and ca. 6 months after intervention	objective + subjective financial knowledge, financial behavior, for a subsample: saving account and credit card balance

Note: NA = information not available.

Source: Authors' compilation.

Table 2 continued

Overview of financial education assessments for adults

No.	Study	Intervention summary	Country	Target group	Intensity	Content	Format and delivery channel	Study design	Outcome measures
23	Fan and Chatterjee (2018)	information sheet and situational stimuli	USA	undergraduate students from a public university (age 18+)	one time (whole experiment took 1 hour)	stocks, bonds, interest rates, risks and returns, tax-advanced accounts	information sheet (printed copy)	experimental: random assignment, control group and 3 treatment groups: (1) info sheet, (2) + (3): info sheet + situational stimuli of psychological pressures; measurement immediately after intervention	investor knowledge
24	French et al. (2020)	financial education smartphone apps	Northern Ireland	members of a large credit union (age 16–65)	apps available for 6 months	apps: (1) loan interest comparison, (2) expenditure tracking, (3) cash calendar, (4) debt management; money management skills workshop to demonstrate capabilities of app; money skills competition: required app usage	smartphone apps (4 versions) + push notifications + workshop + contest	experimental (?): market research company carried out pre- and post-surveys; apps were evaluated together because sample would have been too small to evaluate apps separately; treatment group (those who complied) + control group (those who would have complied), no further info on participant selection or randomization, measurement immediately after 6 months availability	financial and digital knowledge, self-reported financial behavior, financial well-being
25	Fürstenau and Hommel (2019)	online mortgage loan calculator	Germany	economics and business undergraduate students	one time (NA)	calculator to calculate, e.g. income and expenses, the affordable purchase price for a property, the annuity and costs for renting or buying	online (loan calculating tool)	quasi-experimental: random assignment, very small sample, measurement immediately and 2 weeks after intervention	objective and subjective financial knowledge, self-reported previous investing experience
26	Horowitz et al. (2021)	workplace financial education program	USA	employees of financial services companies	10 weeks (weekly sessions, each 1 hour)	compound interest, risk diversification, stock market basics, budgeting, financial planning	in-person (classroom setting)	quasi-experimental: participants self-selected into treatment, control group received passive educational intervention (booklets with similar content), measurement immediately after intervention	financial knowledge
27	Kothakota and Kiss (2020)	visual representation combined with text-based financial education compared to text-only explanations	USA	adult population (age 18–65)	one time (10–15 minutes)	financial information (not specified)	online (text-only vs. text with visualization)	experimental: survey experiment via SurveyMonkey, control group and 2 treatment groups: (1) text-only presentation and (2) text + visualization; measurement immediately after intervention	financial knowledge

Note: NA = information not available.

Source: Authors' compilation.

Table 2 continued

Overview of financial education assessments for adults

No.	Study	Intervention summary	Country	Target group	Intensity	Content	Format and delivery channel	Study design	Outcome measures
28	Lusardi et al. (2017)	different digital financial literacy information formats	USA	adults (representative of US households)	one time ("short")	risk diversification and investment decision making	online (4 different formats: informational brochure, interactive visual tool, written narrative, video narrative)	experimental: survey experiment via online panel (ALP), random assignment to control group and 4 treatment groups, measurement immediately after intervention	financial knowledge, self-efficacy and confidence
29	Moulton et al. (2015)	financial planning program and coaching for first-time homebuyers	USA	first-time homebuyer (average age: 33)	over 12 months (9–20 contacts during that period)	online platform: financial assessments, changes in certain financial variables (e.g. amount saved each month) affecting future time periods; identify specific goals and set up plan for 1 year; calls: implemented monitoring schedule	online (interactive platform) + phone calls with financial coach	experimental: field experiment (sample consisted of participants of subsidized homeownership program), measured 30-, 60-, 90-day delinquency rates and had 1-year follow up	mortgage delinquency and default, self-reported financial behavior
30	Skimmyhorn (2016)	mandatory in-person financial education interventions in military settings	USA	new enlistees of US military	2 sessions (total: 8 hours)	spending plans, credit, debt management, insurance, consumer awareness, financial ethics, military retirement savings	in-person (mandatory classroom setting)	experimental: staggered rollout across location, measurement 1 and 2 years after the intervention	administrative data of credit behavior, retirement savings, military labor market outcomes
31	Skimmyhorn et al. (2016)	mandatory in-person financial education interventions in military settings	USA	undergraduate university students (military academy)	4 sessions (each: 2 hours) integrated into a semester-long economics course	budgeting, consumption, time value of money	in-person (mandatory classroom setting), 2 versions: (1) principles-based approach (2) rules-of-thumb approach	experimental: random assignment to control group and 2 treatment groups, measurement immediately after intervention	objective and subjective financial knowledge, self-reported behaviors, self-efficacy
Financial education for university and college students									
32	Agasisti et al. (2023)	financial education module embedded in course on business economics; on-campus lectures compared to online video lectures	Italy	undergraduate university students	3 hours (on campus), 55 minutes (online videos)	bank accounts, simple/compound interest, shares and bonds, loans and mortgages	online lectures or in-person lectures as part of experimental design	quasi-experimental; group assignment based on last name; pre- and post-test	financial knowledge test

Source: Authors' compilation.

Table 2 continued

Overview of financial education assessments for adults

No.	Study	Intervention summary	Country	Target group	Intensity	Content	Format and delivery channel	Study design	Outcome measures
33	Anderson and Card (2015)	different formats of financial education on financially responsible behavior	USA	first year undergraduate university students	not reported	responsible consumption behavior and credit card use	in-person lectures or online (social media and website) as part of experimental design	randomized experiment	compulsive buying and degree of irrational credit use
34	Choi et al. (2023)	weekly emails and text messages with different content and linking to financial education online platform	USA	undergraduate and graduate university students	weekly reminders for three months; new curriculum options each week; time spent with material not reported	budgeting, paying for college and debt and saving as topics	emails and text messages leading to online platform	randomized experiment: waitlist design	financial self-efficacy, financial health
35	Gerrans and Heaney (2019); Gerrans, (2021)	semester unit of a "managing personal finances" course	Australia	university students	11 weeks (105 minutes + 45 minutes tutorial per week)	financial planning, savings, debt, personal taxation, investments and diversification, home ownership, insurance, consumer law, behavioral finance, long-term planning, retirement, marriage	on-campus, classroom setting	self-selection quasi-experiment, follow up-surveys up to three years later (reported in Gerrans, 2021)	financial knowledge test, self-assessed financial literacy, decision confidence and satisfaction, self-reported and intended financial behaviors
36	Handy et al. (2021)	personal finance lectures	USA	college students	six sessions	budgeting, credit management, risk and return, mutual funds, retirement planning and risk management (insurance)	on-campus lectures	self-selection quasi-experiment	financial knowledge, confidence in post-college financial future
37	Kuntze et al. (2019)	financial education video module, micro-lectures that could be watched at participants' own convenience	USA	business college students	67 minutes videos in total	not reported	online videos	randomized experiment implied, but not clearly reported	financial knowledge test

Source: Authors' compilation.

Table 2 continued

Overview of financial education assessments for adults

No.	Study	Intervention summary	Country	Target group	Intensity	Content	Format and delivery channel	Study design	Outcome measures
38	Salas-Velasco (2022)	laboratory experiment, intervention focused on debt-financed graduate education decision-making	Spain	final year undergraduate students	"short"	economic viability of investing in a master's degree and financing the master's degree with a student loan; second treatment group with additional information on availability bias	computer-based	randomized experiment	topic-specific knowledge test, self-confidence, self-efficacy
39	Torma et al. (2023)	financial literacy lectures	Croatia	university students	four 90-minute lectures	not reported	on-campus lectures	randomized experiment, waitlist design	financial knowledge test, self-assessed knowledge, financial interest, impulse buying, hypothetical financial decisions and consumption

Source: Authors' compilation.

Conceptions of financial education for “consumers” and “citizens” and their implications for pedagogical action: a discussion

Éva Kaczko¹

This article encourages reflection on desirable objectives and evaluation approaches in financial education. It proposes a novel categorization of financial education conceptions known in the international literature and analyzes the characteristics relevant for pedagogical action. I outline five exemplary groups along a continuum ranging from rather individual- to rather socially-oriented conceptions. Distinguishing between financial education and financial training from an educational-philosophical perspective, I argue that it is possible to place the presented conceptions within both the broader notion of financial education and the narrower notion of financial training. Furthermore, I present teaching and learning foundations from a psychological perspective, hypothesizing that an instructivist pedagogical approach is more likely to be used in rather individual-oriented conceptions and a constructivist pedagogical approach is more likely to be used in rather socially-oriented conceptions of financial education. An illustration assists the reader to select, and reflect on, educational and learning objectives, instructional strategies and the measurement of learning outcomes. The article concludes with six reflection questions for practitioners in financial education, financial training and evaluation research.

In the international literature, we encounter various conceptualizations of personal financial education, each with distinct interpretations that target different educational or learning objectives. In the introduction of the “International Handbook of Financial Literacy,” Aprea et al. (2016) state that different interpretations, and thus pedagogical actions, are linked to values that are ideally presented transparently and in a way that encourages discussion. Similarly, Biesta (2010) stresses that it is not enough to assess data and evidence in the field of educational evaluation, but that we need to deal with values as well. This article aims to follow this path and proposes a classification of conceptions of financial education along a continuum ranging from rather individual-oriented to rather socially-oriented conceptions. In doing so, I also encourage reflection on the values embedded in different notions. This proposal is largely in line with Aprea’s (2014) draft of a conceptualization of financial education grounded in educational theory.

Adopting a pedagogical lens, I will outline how personal financial education may be understood, identify central characteristics and discuss what implications the different interpretations have for pedagogical action. Accordingly, I approach the question of how to evaluate educational or learning outcomes of financial

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education and training efforts from a different angle: I am not asking what educational and training efforts are effective and how we can measure them. Rather, inspired by educational philosopher Gert Biesta (2006, 2010, 2013), I am asking what financial education and training efforts and evaluation objectives are *desirable*. When addressing the question of desirability, I will not provide recommendations or express preferences but let the readers answer the question themselves. Thereby, I aim to raise awareness among practitioners in financial education, training and evaluation research about the broad spectrum of possible educational and learning objectives and corresponding pedagogical actions. Furthermore, I aim to encourage reflection about Biesta's (2010, p. 13) concern about evaluating educational outcomes and measures, i.e. "whether we are indeed measuring what we value, or whether we are just measuring what we can easily measure and thus end up valuing what we (can) measure."

To this end, in the next section, I will distinguish between two positions at the end of the continuum from rather individual- to rather socially-oriented conceptions of personal financial education and training, and also give some examples of conceptions that lie in between. I will identify the characteristics associated with each one and offer some initial pedagogical considerations. In the following section, I will differentiate between "education" and "learning," and thus between financial education and training. In this context, I will highlight some aspects of the purpose of education following Biesta's (2010) thoughts about "good education" and how it is different from "learnification," and also reflect on teaching from an educational-philosophical perspective. This might help the reader to think about what the term "education" means, a term that is widely used in the international discourse but may have different connotations. In the third section, I will give a brief overview of learning theories and corresponding pedagogical approaches, covering some basics of learning and teaching from a learning and instructional psychological perspective. This is central to understanding how knowledge, skills, dispositions and behaviors, or "competences," can be learned and assessed, even though providing a comprehensive overview in a single section is not feasible. Moving on to the fourth section, I will connect my pedagogical considerations to the conceptions of financial education and training and explore further implications for pedagogical action. After pointing out the limitations of this pedagogical discussion, I will conclude with reflection questions for the reader.

1 Conceptions of financial education from a pedagogical perspective

"Education, be it in the form of schooling, workplace learning, vocational training or learning through life, is by its very nature a process with direction and purpose" (Biesta, 2010, p. 2). Accordingly, conceptions of personal financial education as used here are notions that pursue educational goals, such as actively managing life situations characterized by personal financial issues that affect the financial well-being of individuals, the economy or society. As educational efforts are directed at human beings, we might speak of persons who, as consumers, earners or citizens, should have certain knowledge, skills, dispositions, behaviors or "competences," and virtue that help them make informed decisions and critical judgements on issues related to personal finance.

Conceptions of financial education address people in varying roles. I will highlight three primary roles in advance, which will be pertinent to the pedagogical discussion further below. In the financial education literature, individuals are often addressed as (1) *consumers*, (2) *earners* or (3) *citizens* or *economic citizens*. *Consumers* are considered in different roles, depending on the market in which they operate: buyers, savers, investors, debtors and insurance holders (Retzmann and Seeber, 2016). Retzmann and Seeber (2016) also consider individuals in other life situations, such as renters, inheritors and owners. As (2) *earners*, many people see income as a central aspect of their financial decisions, especially regarding financial (in)security.

Consequently, the focus is on income and the life situations associated with career choices, such as employee, freelancer, self-employed entrepreneur, producer, supplier and employer (Retzmann and Seeber, 2016). As members of a community, (3) *citizens* or *economic citizens* actively contribute to social and economic development. Hence, Retzmann and Seeber (2016) refer to roles such as taxpayers, voters and participants in voluntary and political functions. Arthur (2016) advocates politically and financially educated citizens who can develop effective and collective solutions to provide (financial) security for all.

Consequently, I refer to *rather individual-oriented conceptions* of financial education which address people as consumers and earners and to *rather socially-oriented conceptions* which additionally address people's role as citizens. I will begin with the narrower conceptions from an educational perspective and gradually build on these to end with the broadest ones. Considering Aprea's (2014) conceptualization, I organize financial education into *personal financial management* and *critical consumer education* within the realm of rather individual-oriented conceptions. Additionally, I suggest a further subgroup here, i.e., *personal financial management that takes into consideration social and structural conditions*. I put *financial education as part of a more comprehensive economic education* within the realm of rather socially-oriented conceptions. Here, I propose a separation of *financial education as part of a more comprehensive economic and political education*. I wish to emphasize that the conceptualizations presented cannot provide a complete picture but only serve as examples to underline the proposed categorization.

1.1 Rather individual-oriented conceptions

1.1.1 Personal financial management

Personal financial management is concerned with the financial decisions that people make in their personal lives (Aprea, 2014). The OECD², the editors of the Routledge Handbook of Financial Literacy³, Lusardi and Mitchell (2014)⁴, and, apparently, also the European Commission⁵ seem to be among the prominent proponents of this view. It focuses on individuals (human capital, consumers, investors) and the (informed, effective, sound) decisions they make with the aim of participating in economic life and achieving personal financial well-being. Financial well-being is often viewed as the ultimate goal. For instance, Brügggen et al. (2017) describe it as an individual's perception of financial freedom, achieved through maintaining a desirable standard of living. A closer look at the concepts used to describe how to manage personal finances

² First definition of financial education from OECD (2005, p. 26): "Financial education is the process by which financial consumers/ investors improve their understanding of financial products and concepts and, through information, instruction and/or objective advice, develop the skills and confidence to become more aware of financial risks and opportunities, to make informed choices, to know where to go for help, and to take other effective actions to improve their financial well-being." Definition used in the PISA test in 2012: „Financial literacy is knowledge and understanding of financial concepts and risks, and the skills, motivation and confidence to apply such knowledge and understanding in order to make effective decisions across a range of financial contexts, to improve the financial well-being of individuals and society, and to enable participation in economic life." (OECD, 2014, p. 33). Current definition from OECD (2020, p. 3): "Financial literacy: a combination of financial awareness, knowledge, skills, attitudes and behaviors necessary to make sound financial decisions and ultimately achieve individual financial well-being."

³ "We define financial literacy as a component of human capital that includes knowledge, skills, attitudes, and confidence related to financial decision-making" (Cude, 2022, p.5).

⁴ [W]e focus on financial literacy, by which we mean people's ability to process economic information and make informed decisions about financial planning, wealth accumulation, debt, and pensions" (Lusardi and Mitchell, 2014, p. 6).

⁵ "Financial education enables individuals to improve their understanding of financial products and concepts and develop the skills necessary to improve their financial literacy, i.e. to be aware of financial risks and opportunities and to make informed decisions in their choice of financial services." (Commission of the European Communities, 2007, p. 2).

reveals that social aspects are rarely considered.

Individuals are often addressed as *consumers* and partly also as *earners*, which encompasses perspectives people take as, e.g., buyers, savers, investors, debtors, renters and employees. For pedagogical action, it seems relevant to focus on *learning content* that addresses financially shaped life situations from these perspectives. This includes topics such as earning and using money (consumption/saving) and increasing income (Retzmann and Seeber, 2016); planning and money management, saving and investing, credit and debt (Jump\$tart Coalition, 2007); relationships between risk, return and liquidity; present value and discounting (interest); relationships between saving, consumption and investing; opportunity costs of financial decisions; and inflation and the real value of an investment (Pang, 2010). Since the focus is on the financial well-being of the individual, individualized instruction that takes the needs of each student into account appears to be appropriate.

Achieving financial well-being requires not only knowledge of specific financial facts and concepts as described above but also skills (e.g. for decision-making), certain dispositions (e.g. attitudes, motivation, confidence) and corresponding behaviors. It is worth looking more closely at what the terms used in this context might mean as they give teachers and trainers various ideas about selecting and judging teaching and learning activities.

Knowledge and *skills* are inseparable: knowledge relies on skills (e.g. various thinking skills), and skills are necessary to turn knowledge into action (e.g. knowing how to use financial information to make informed financial decision in a certain situation). According to the OECD (2014), financial skills encompass generic cognitive processes such as accessing information, comparing, contrasting, extrapolating and evaluating within a financial context. They also include fundamental mathematical skills, like calculating percentages or converting currencies, and language skills, such as reading and interpreting advertisements and contractual texts.

Dispositions, such as attitudes, motivation and confidence, are needed to put skills into action. They “are not themselves skills, but they represent a readiness to employ such skills – and to employ them in a coordinated and cumulatively reinforcing fashion” (Lipman, 2003, p. 171). Whether and how certain dispositions can be fostered would require a more detailed examination of multifaceted pedagogical approaches such as experiential, service- and inquiry-based learning.

Financial behavior refers to the specific actions individuals take in life situations where they make decisions about, e.g., saving, borrowing and retirement planning. It concerns the extent to which individuals utilize or fail to utilize their knowledge and skills in particular situations. The latter can, for example, stem from a lack of knowledge and skills, from inert knowledge (Whitehead, 1929), certain personal dispositions or various other reasons resulting from situational and personal circumstances.

Decision-making skills have a particular role in personal financial education just as they do in economic education (Loerwald and Stemmann, 2016). Economics explores decision-making from the perspective of rationality and irrationality. While rational decision-making involves selecting the best possible alternative from the available options (Kirchgässner, 2008), behavioral economists question whether this is possible at all and instead concentrate on cognitive biases that can lead to irrational decisions (Loerwald and Stemmann, 2016). Antonietti et al. (2016) adopt a cognitive psychological perspective on effective financial decisions. They approach decision-making from the perspective of individuals’ thinking skills within the context of metacognition. Marchetti et al. (2016) investigated the influence of social norms on the development of individuals’ decision-making skills during childhood from a developmental psychological standpoint. All these different perspectives provide good guidance for pedagogical action, e.g., promoting decision-making skills through instructional strategies such as problem-based learning, case-based lear-

ning or decision science techniques.

1.1.2 Critical consumer education

Critical consumer education is concerned with the financial decisions that people make in their personal lives with a particular focus on their needs and consumption (Aprea, 2014). It encompasses the reflection of consumption decisions, the functions and interests of financial service providers as well as consumers' rights and obligations. Aprea (2014) mentions Atkinson et al. (2007) and Rutledge (2010) as proponents of this position.

This approach has some particular implications for pedagogical considerations. At its center are consumers and their consumption choices, as well as their relationships with financial service providers, who offer, e.g., consumer credit. This makes the selection of learning content a fairly straightforward process. In addition, critical consumer education could also encompass selecting instructional strategies that facilitate reflective decision-making in purchasing situations. Such strategies might draw on findings from the field of behavioral finance and, as has been shown elsewhere (Kaczko and Razen, 2022), embed economic experiments (e.g. on anchoring bias) in an experiential learning approach.

1.1.3 Personal financial management that takes into account social and structural conditions

Personal financial management that takes account of social and structural conditions addresses the financial decisions people make in their personal lives and reflects on access to financial products and services that suit their personal life circumstances. This idea is intertwined with the conception of financial capability according to Sherraden and Ansong (2016), who define financial capability as “an individual and structural idea that combines people’s *ability to act* with their *opportunity to act*.” Together, they “contribute to the freedom to pursue financial well-being and improved life chances” (Sherraden and Ansong, 2016, p. 86). While “the ability to act” is closely linked to the construct of financial literacy as personal financial management (Sherraden and Ansong, 2016), “the opportunity to act” complements it with the reflection about dignified access to appropriate financial products and services. In less developed countries, for example, there are an estimated 2.5 billion people without a bank account (Chaia et al., 2012). Financial and government institutions set boundaries on the opportunities people have, for example for accessing banking services, the presence or absence of government-backed deposit insurance of bank savings and government-backed retirement accounts (Sherraden and Ansong, 2016).

Sherraden and Ansong’s (2016) notion may be a response to Henchoz’s (2016) critique of the term “financial literacy” as defined by the OECD (2005) (see “personal financial management”), which, she argues, reflects a tendency toward neoclassical economics. This definition explains individuals’ socioeconomic situations, among other factors, by their actions aimed at maximizing their own utility and satisfaction (“homo oeconomicus”). Poverty and debt are understood as economic situations, or, more precisely, deficits to be corrected by financial education programs (Henchoz, 2016). Henchoz argues that if financial education programs are to fully empower people, they should not only disseminate information, direct instruction and advice, but also consider the social conditions in which this information is mobilized and translated into practice and well-being. For example, financial classes may even be harmful if they explain that savings accounts are optimal (Sherraden and Ansong, 2016). If people have little money to save and fees absorb their meager savings, this may lead to frustration (Sherraden and Ansong, 2016). In such cases, it would be appropriate to discuss whether there are low-cost savings accounts for more modest savings.

There is also another notion that brings social and structural conditions to the fore: financial vulnerabi-

lity. At first glance, it seems to be the opposite of financial well-being. Voith and Mauser (2024, p. 3) clarify that there is no precise distinction: “[w]hile financial vulnerability and financial well-being both involve meeting financial obligations and achieving sustainable security, the critical distinction lies in financial well-being’s inclusion of the ability to afford *wants*, whereas financial vulnerability focuses solely on sustainably meeting essential *needs*.” They explain vulnerability as the likelihood of experiencing financial hardship which can affect anyone regardless of their living situation. Financial vulnerability depends not only on individuals’ choices but also on structural social inequalities and circumstances beyond their control (Voith and Mauser, 2024).

Considering these characteristics for pedagogical action, we first note that those discussed in the context of personal financial management are also relevant here. Additionally, another aspect comes to the fore: the opportunity to act. Financially capable people need to understand and reflect on the structural opportunities and constraints that affect their finances and be empowered to find personalized solutions. This requires sensitivity from teachers and trainers, particularly an awareness of challenging or disadvantaged life circumstances and the potential personal vulnerabilities that come with them.

1.2 Rather socially-oriented conceptions

1.2.1 Financial education as part of a more comprehensive economic education

According to Aprea (2014), Remmele and Seeber (2016) represent the view that financial education should be considered as part of more comprehensive economic education. They point out that while the concept of financial literacy as defined by the OECD (2014) (see 1.1, “personal financial management”) is designed to help individuals navigate an increasingly complex economic environment, it is a limited view from an educational perspective. They argue that, as schools have the educational mandate to prepare young people for their personal (and, depending on the school type, professional) lives and empower them as citizens to participate in society and social change, financial education should be considered in a broad context, i.e. not only from a personal perspective but also in the context of the economy as well as society as a whole. Accordingly, Retzmann and Seeber (2016, p. 10) state that financial education as part of economic education “should enable students to act responsibly, autonomously and appropriately in [financially] shaped life situations.” Retzmann and Seeber (2016, p. 15) also define a financially educated person as somebody “who is willing and able to judge, decide and act autonomously (self-governing), appropriately and responsibly in accordance with [...] transferable competences in financially shaped life situations.”

Compared to the rather individual-oriented conceptions, Retzmann and Seeber’s (2016) explanation suggests that financial education has a broader focus. First, there is a societal perspective, and individuals are also referred to as *economic citizens* who make decisions, form judgments and act in a self-determined, appropriate and responsible manner. A person who acts responsibly and appropriately can also be considered a person that acts morally and ethically, which is a widely discussed ideal of an educated person. From this point of view, reasons and consequences of personal financial decisions, judgments and actions should also be reflected within the broader context of participation in society as a whole, of which economic life is a part.

Second, this conception addresses — beyond decision-making — the broader concept of *judgment*. Judgment “is the forming of opinions, estimates, or conclusions [...] that] includes such things as solving problems, making decisions, and learning new concepts” (Lipman, 2003, p. 210). Furthermore, judgments can also be the outcomes of deliberations, verdicts of trials, conclusive findings of investigative proceedings, solutions to actual or theoretical problems (Lipman, 2003). They result from inquiry, from “a self-corrective practice in which a subject matter is investigated with the aim of discovering or inventing ways

of dealing with what is problematic" (Lipman, 2003, p. 184). This implies that beyond teaching and learning activities to promote decision-making, other strategies for facilitating judgements need to be considered.

Third, in the educational context, particularly in German-speaking countries, the term *competence* has become established as a comprehensive concept encompassing skills, motivational, volitional and social aspects, as well as responsible action and problem-solving in concrete situations (for a more detailed explanation, see Bader and Müller, 2002; Weinert, 2001). The discourse on competences might be seen as a contemporary understanding of education, at least in German-speaking countries. Consequently, it is not surprising that evaluation efforts, specifically "competence measurement," have also become widespread. Therefore, Retzmann and Seeber's (2016, p. 15) definition of financial competence should be interpreted within this context:

„[T]he sum of an individual's cognitive judgment, decision-making and planning abilities, their practical and technical skills for implementing decisions and plans, including the use of electronic media, and their motivational, volitional and social disposition with regard to [various financial themes] for themselves, as a trustee for other people, and as a social or political representative for the general public, in efficiently and responsibly generating and implementing such assets to achieve the best possible effect on the short, medium and/or long-term well-being of the people concerned.“

Accordingly, financial competence is a network of knowledge, skills, motivation, volition and social dispositions that enables people to responsibly apply their knowledge of and skills in personal finances in real contexts. Financial competences become visible through actions. They also encompass social dispositions and societal aspects, such as the role of individuals as political representatives. Moreover, the ultimate goal of financial competence extends beyond achieving personal well-being to include the well-being of all those affected by actions, decisions and judgments in financially shaped life situations. Accordingly, Retzmann and Seeber (2016) address competences resulting from financial education at three levels: individual judgment and decision-making, responsible relationships with others, and understanding the political order and the financial system as a whole. When considering the selection of appropriate teaching and learning activities, Retzmann and Seeber's (2016) competence model may be used as a guide.

1.2.2 Financial education as part of a more comprehensive economic and political education

Other critics of a narrow understanding of financial literacy view financial education as an integral part of an even wider concept of education, namely economic and political education. For the purpose of this article, I will discuss four perspectives, of which Hippe and Davies are also mentioned in Aprea (2014). Hippe (2012) explores how basic economic and political concepts can be linked in a problem-oriented and integrative way to critically reflect on interdependencies between political and economic order. With regard to personal financial education, Hippe (2010) points out that no "education" is possible without a critical, scientific examination of the financial markets. He calls for critical financial education that encourages young citizens to develop a healthy mistrust of financial product providers.

Davies (2015) criticizes the prevailing individualistic view of financial literacy and proposes a comprehensive educational framework to enable young people to understand the financial world more broadly and to take on an active democratic role. This educational framework also provides good guidance for the selection and judgment of teaching and learning activities. Davies (2015) hypothesizes that if voters have

a sufficient understanding of financial processes and incentives to put pressure on politicians, the latter are more likely to govern in the public interest.

Arthur (2016) argues for financial literacy education as public pedagogy which aims to uncover the constructed nature of financial insecurity. He wishes to educate justice-oriented citizens⁶ who can work together to develop ethical and collective solutions to the unjust distribution of financial insecurity. Therefore, financial education is ideally complemented by political education (Arthur, 2016).

Remmele (2016, p. 53) offers another critical perspective: “Financial literacy education is less about helping to understand but rather about helping to become skeptical and to act in order to change the (social) frames of references and/or the rules of interaction.” He questions whether it is possible to fully comprehend market and money, addressing epistemic and non-epistemic forms of financial incomprehensibility. Consequently, Remmele explores various implications for the (non-)teachability of financial concepts. He emphasizes that financial education should not be limited to the perspective of private households. Instead, it should be considered within the context of democracy, and encourage skeptical and emancipatory thinking that could lead to political action (Remmele, 2006).

The rather socially-oriented conceptions of financial education presented in this subsection address individuals not only as *consumers* and *earners* but also as (economic) citizens such as taxpayers, voters and participants in voluntary and political functions. They may be both financially and politically educated citizens. They possess decision-making skills relevant for individual financial well-being and critical judgment skills regarding financially shaped life situations that also impact the well-being of others. This requires social skills, a caring attitude toward societal issues, the skills and dispositions to think critically and the cultivation of democratic values. These elements provide a solid starting point for selecting *instructional strategies*, such as inquiry-based instruction (e.g. Dewey, 2016; Garrison et al., 2000), community-of-inquiry as a method (e.g. Kaczko and Ostendorf, 2023; Lipman, 2003; Reznitskaya and Wilkinson, 2017) and critical pedagogy (Freire, 2005).

1.3 Conceptions of financial education – first summary

Figure 1 summarizes the key characteristics of the conceptions of financial education relevant to further pedagogical discussion.

Figure 1

Rather individual- and rather socially-oriented conceptions of financial education



Source: own illustration.

⁶ “[I]f participatory citizens are organizing the food drive and personally responsible citizens are donating food, justice orientated citizens are asking why people are hungry and acting on what they discover.” (Westheimer and Kahn, 2004, p. 4)

Rather individual-oriented conceptions tend to focus on individuals as consumers and earners, emphasizing informed and partly reflective financial decisions related to personal financial management. They advocate that individuals should be enabled to participate in economic life and pursue individual financial well-being. In contrast, rather socially-oriented conceptions incorporate the role of (economic) citizens who “judge, decide and act autonomously [...], appropriately and responsibly” (Retzmann and Seeber, 2016, p. 15), so that they can participate in society as a whole (which includes economic life) and pursue well-being for all. To achieve this goal, it is not sufficient to make informed and reflective decisions solely for oneself; individuals should also be able to make critical judgements to navigate an increasingly complex economic, political and social environment. We see that there are no purely individual or social conceptions of financial education; rather, we are dealing with a range of different approaches combining elements from both orientations to varying degrees.

In the next section, I will introduce some pedagogical basics necessary for a further consideration of pedagogical action, starting with a differentiation between “education” and “learning.”

2 Education and learning – an educational-philosophical perspective

Before considering the fundamentals of learning and teaching from a psychological perspective, it is necessary to distinguish between two notions at another level, i.e. between education and learning. Biesta (2006, p. 15) states that

“One of the most remarkable changes that has taken place in the theory and practice of education over the past two decades has been the rise of the concept of “learning” and the subsequent decline of the concept of “education.” Teaching has become redefined as supporting or facilitating learning, just as education is now often described as providing learning opportunities or learning experiences.”

Biesta (2010) emphasizes two aspects of the *language of learning*, namely that learning is an individualistic concept that focuses on the individual's doing, and that it is a process term that focuses on activities, where the direction of the process is not necessarily considered, like in the term “lifelong learning.” Both aspects are inherently different from the concept of education, which implies a relationship between educators and people to be educated as well as the purpose of educational practices (Biesta, 2010).

Biesta (2010) proposes considering three composite functions of education when discussing its aims and ends, drawing on the educational and political tradition of the Enlightenment: qualification, socialization and subjectification. The *qualification function* involves the acquisition of knowledge, skills and dispositions that enable people to do something, such as making sound decisions about money management and borrowing, or something more general, such as an introduction to economic life. Great emphasis is placed on preparing young people and adults for working life, which also has an economic factor, namely securing the future workforce. Yet, the qualification function is also relevant for preparing people for their personal life and their role as citizens.

The *socialization function* of education refers to the way in which people become part of certain existing social, cultural and political systems, with all their desirable and undesirable aspects (Biesta, 2010). One example could be how schools address personal financial education issues – as part of regular classes or during school events – in the context of cultural norms, such as advocating thriftiness. The idea of acting responsibly in financially shaped life situations (see Retzmann and Seeber, 2016) is also related to the

socialization function of education, which places financial education in the role of a producer of certain subjects, such as responsible citizens (cf Biesta, 2010, p. 23). More broadly, financial education can also be seen as a social practice, e.g. by embedding it in a national financial education strategy, i.e. as part of the political order.

In contrast to socialization, the *subjectification function* of education encompasses the process of becoming an autonomously and independently thinking and acting subject (Biesta, 2010). For Biesta (2010), this is an essential element of any “education.” Accordingly, education becomes uneducational when it loses sight of this function and concentrates solely on integrating newcomers into existing socio-cultural and political orders. Thus, teachers and trainers might want to consider how to address the question of human subjectivity, what it means to be human, to exist in a world of plurality and difference (in the Arendtian sense⁷), and what ways exist for a person to be different from existing social, cultural and political orders (Biesta, 2010). This raises the question of “what kind of opportunities for being and becoming the engagement with the field of [financial education] might offer” (Biesta, 2010, p. 25). Arthur (2006) might answer that financial education should shed light on the constructed nature of the concept of financial insecurity. Sherraden and Ansong (2016) might respond by raising issues of fairness and justice, such as enabling people to access appropriate financial products and services with dignity. These ideas might provide opportunities for the development of subjectivity and agency in a broader context of financial education.

Engaging in reflection and discussion on all three functions of education, including their synergies and conflicts, would help to identify educational rationales for financial education in different institutional contexts. This brings me to the final thoughts I would like to share about teaching, before approximating financial education and training from a learning and instructional psychological perspective. For Biesta (2013), it is important that teachers are aware of the need to constantly make situated judgments (in concrete situations with concrete students) about what is educationally desirable in relation to, and by balancing, the three functions of education. These judgments are wise educational judgments that are value judgments, not technical ones. Empirical evidence about what financial education interventions have worked well in the past may inform judgments in present situations, but such evidence cannot replace judgments about what needs to be done (cf Biesta, 2013, p. 131). It implies that teaching is an art rather than a science. Biesta (2013) uses the word “virtuosity” to emphasize that teaching is practical wisdom that demands a certain excellence, and not just an activity in which you apply a set of skills and dispositions or competences. To summarize: “Teaching is not a matter of following recipes but ultimately requires teachers who are able to make wise situated judgments about what is educationally desirable” (Biesta, 2013, p. 140).

Against this background, I suggest using the term *financial training* when financial education efforts do not encompass the educational relationships and purposes described in this section. *Financial training* emphasizes learning and the function of qualification in education, i.e. the acquisition of knowledge, skills, dispositions and behaviors or competences.

⁷ A responsibility for plurality and difference is the condition of human freedom and action (Biesta, 2010). Biesta (2010) proposes a pedagogy of interruption that could open up spaces in which it is possible to engage with the complexity of a world populated by humans who are different, and in which subjects develop not just a representative voice but a unique one. In this sense, a pedagogy of interruption is orientated toward human freedom, which is always linked to the freedom of others.

3 Learning paradigms and pedagogical approaches – a psychological perspective

Learning and teaching are closely related, but they cannot be equated. When a teacher or trainer designs and delivers a lesson on the opportunity costs of financial decisions, it does not mean that students will learn what is taught. Students learn for themselves; teachers and trainers can at most intend to teach (Terhart, 2005). It is also possible that learners acquire new knowledge and skills autonomously, without external support. Therefore, it makes sense to clarify how established theories explain learning and what implications these explanations have for teaching.

Firstly, it is important to emphasize that none of the extant theories comprehensively elucidate the phenomenon of learning; rather, they each address specific facets (Fürstenau, 2016). What they have in common, however, is that they all agree that *learning is a permanent change* in knowledge or behavior based on experience (Mayer, 2008). The change (or potential for change) can be cognitive, affective or psychomotor and may not be immediately apparent. The different theories, which can be grouped into three learning paradigms⁸, offer different explanations for the transition from initial to final behavior: (1) *behaviorism* posits that learning is a change in behavior through reinforcement, (2) *cognitivism* views learning as a mental process involving information processing or the processes of assimilation and accommodation, and (3) *constructivism* interprets learning as a mental process of knowledge construction.

3.1 Learning paradigms

3.1.1 Behaviorism

Behaviorist (or conditioning) theories of learning dominated the first half of the 20th century and gained new prominence with the proliferation of digital media in various learning environments. Well-known names include Pavlov, Watson and Skinner, who, despite differences in their theories, share the view that learning is a “process of forming associations between stimuli and responses” (Schunk, 2009, p. 74). Behaviorists regard the brain as a black box, disregarding mental processes such as meaning, volition and motivation as reasons for action and self-reflection during learning, and instead concentrating on conditioning behavior (Schunk, 2009). Skinner’s operant conditioning involves presenting stimuli that are reinforced – through rewards or punishments – to increase the likelihood of the desired behavior occurring.

Learning through stimulation and response has several implications for teaching. The teacher or instruction designer focuses on stimulation and reinforcement. They formulate learning outcomes (desired behavior and its demonstration), such as learners being able to distinguish between fixed and variable costs in monthly bank statements and allocate their expenditure accordingly. Lessons are organized in small steps; for example, the teacher introduces fixed costs through direct instruction, followed by exercises that learners complete and for which they receive immediate feedback from the teacher (or learning program). When learners complete a sequence, the next step follows similarly until all sequences have been actively completed (“teaching to the test”). Successful steps are rewarded; a typical example from e-learning is the provision of badges that learners can collect for completed tasks.

According to behaviorism, the achievement of learning objectives should be explicitly observable. For instance, knowledge, comprehension and application are tested by asking learners to solve familiar problems, e.g. through multiple-choice questions with objective answers. Behavioral approaches are, for example,

⁸ A learning paradigm is a particular worldview about the explanation of learning shared by a group of scientists.

well suited for novices (e.g. learning the multiplier table) and for the development of procedural skills (e.g. the procedural sequence of taking x-rays of patients) (Larson and Lockee, 2014).

In summary, behavioral learning theories view learners as passive beings who can be controlled and manipulated. Teachers (and learning programs) provide stimuli, break the lesson down into small steps, reinforce learning progress with rewards, and monitor the entire learning process.

3.1.2 Cognitivism

In the late 1960s, cognitivist learning theories emerged, focusing on the mental processes of perception, imagination, thought, judgment and language, i.e. the processes by which an organism acquires knowledge and skills about its environment and/or changes its behavior (Shuell, 1986). In the following, I will highlight some key aspects of various cognitive learning theories.

Cognitive information processing theories regard our brain as an information-processing instrument. "Learning, or encoding, occurs when information is stored in long-term memory" (Shunk, 2009, p. 181). One of the best-known models is the multistore model by Atkinson and Shiffrin (1968), which describes three stages of information processing: the sensory register, working memory, and long-term memory. The more frequently information is repeated in working memory ("rehearsal"), the more likely it is to be stored in long-term memory. Other scientists (e.g. Markovitsch, 1996; Roth, 2001) distinguish between different functions within working and long-term memory.

We change not only when we learn something new but also through biological development. Developmental psychological theories explain the evolution of thinking, acting and feeling throughout a person's life through an interplay of maturation, socialization and learning (Jank and Meyer, 2002). One of the most prominent theories comes from Jean Piaget. He explains learning as the active engagement of the individual with the environment in a given situation, where they actively influence the environment through their actions according to their development level (Jank and Meyer, 2002). Learning is therefore a process of assimilation (the child adapts what is new in its environment to their cognitive schema) and accommodation (the child adapts their mental representations to the demands of reality). Both processes together change the child's cognitive structures, which is understood as learning. They interact with further cognitive processes which together lead to an increasingly complex differentiation of cognitive structures. Cognitive conflicts manifest themselves in the occurrence of errors when learners actively engage with learning content (Jank and Meyer, 2002).

Anderson and Krathwohl's (2001) taxonomy of learning objectives aids in the formulation of cognitivist learning outcomes. In addition to the cognitive dimensions formulated by Bloom et al. (1956), which they reformulate as verbs (remember, understand, apply, analyze, evaluate and create), the authors distinguish between four dimensions of knowledge: factual, conceptual, procedural and metacognitive knowledge. For instance, it is conceptual knowledge when students comprehend the relationships between saving, consumption and investment, and it is procedural knowledge when pupils develop personal strategies for managing money.

In the literature about instructional design, there is abundant practical advice on how different cognitive and knowledge dimensions may be addressed with teaching and learning activities. Here, I will provide some general considerations from a cognitivist point of view. For factual and conceptual knowledge, students may learn from concrete experiences; teachers may induce cognitive dissonance, present new information alternately with familiar content to foster a network of associations, or provide multisensory input linking new information with small exercises and allow it to circulate in working memory (cf Jank and Meyer, 2002). For procedural knowledge, such as developing a strategy, learners may be introduced to concrete

problem-solving and decision-making strategies and tools that enable them to process their solutions using established procedures for managing their finances. Teachers may assess learning outcomes through presentations, assignments or exams, where acquired knowledge must be presented or applied. Similar to behaviorist theories, cognitivist theories suggest that students' performance be evaluated according to objective criteria.

In summary, cognitivist theories of learning regard learners as active, processing beings who actively engage with their environment. Prior knowledge, i.e. cognitive structures, is crucial for learning. Mistakes are considered beneficial to learning and are therefore allowed during the learning process. Teachers are providers of ready-made knowledge, pay attention to the knowledge structure and the active (cognitive) role of the learner, and design learner-centered lessons. They deliberately disturb the cognitive equilibrium of the learners, i.e. they generate curiosity.

3.1.3 Constructivism

Constructivism has been increasingly applied to learning and teaching in recent years. The construct of "competence" is also constructivist in nature, indicating that this theoretical (more precisely, epistemological) approach has become established in educational policy measures, at least in German-speaking countries. Constructivist learning theories are inherently cognitive; however, they stress the "construction" of knowledge and beliefs (Shunk, 2009). They are based on various theoretical contexts, including radical constructivism, the neurobiology of cognition, systems theories and cognitive psychological concepts of learning (Terhart, 2005).

The basic epistemological assumptions come from radical constructivism (e.g. Foerster, 1985; Glasersfeld, 1996). Reality itself is inaccessible; everything we know about reality is an individual construction and is bound to the observer's perspective (Terhart, 2005). This constructed knowledge must be useful or viable in a particular context. Knowledge is not "transferable," has no definite beginning and end, but is a provisional state (Terhart, 2005). Neurobiological studies provide another basic assumption. They show that the sensory organs do not convey a pure image of the outside world. Distorted and incomplete information reaches the brain, which determines this information based on its structure, thereby creating its reality (Terhart, 2005).

Social-constructivist theories of learning, originating from John Dewey's pragmatism and Lev Vygotsky's socio-cultural theory, posit that knowledge is collaboratively developed, exchanged, and negotiated by individuals within the context of social transactions. Consequently, individuals' thoughts and actions can only be understood within a specific social context. Learning is situated, i.e. it is bound to the context and social experiences in the learning situation (Bendorf, 2016). These learning theories are frequently discussed in the literature in relation to promoting critical thinking (cf Volman and Ten Dam, 2015).

In recent decades, a moderate constructivist view has become predominant (Dubs, 2014). Reinmann and Mandl (2006) prefer the term "knowledge-based" constructivism, as it emphasizes the role of a sufficient knowledge base that enables learners to construct meanings for themselves. This practice-oriented position includes instructivist support to help learners develop a knowledge base to deal autonomously and collaboratively with problems, dilemmas, etc. (Kaminski, 2017).

Constructivist theories of learning have particular implications for teaching and learning. They assert, for instance, that it is not possible to transmit ready-made knowledge. Instead, learners acquire knowledge by themselves, thus they are the protagonists of the learning process. Teachers act as learning guides and mentors, engaging in dialog with learners (discussing errors, contradictions, questioning truths), and

encouraging independence and initiative. They design learning environments and create active learning situations where authentic and complex learning activities are offered and the learners' perspectives are considered (Riedl and Schelten, 2013). Additionally, learning content is ideally presented in different contexts and viewed from varying perspectives. There is also an emphasis on the social context, including communication processes and collaborative and cooperative learning opportunities (Riedl and Schelten, 2013).

In this sense, *learning objectives* often target higher stages of Bloom's taxonomy of learning, i.e. analysis, synthesis, evaluation that builds on lower stages, i.e. knowledge, comprehension and application (moderate constructivism). A relevant example from financial education would be for students to understand the reasons for high inflation and to reflect on and discuss its consequences for individuals and society. Constructivist *teaching and learning activities* are linked to self-regulated and collaborative learning. Teachers and trainers may employ various forms of problem-based learning, such as case-based learning, scenario-based learning, project-based learning, simulations (e.g. simulation games) and role-playing. They may also consider different types of dialog, debate and discussion (e.g. discussions of articles) and self-reflection on experiences, e.g. through journals.

According to Reinmann and Mandl (2006), *assessment* has a different weight in constructivist learning environments. Comparative assessments of learning success, predictions of learning outcomes and the creation of rankings among learners are not considered relevant. Formative assessment plays a central role, i.e. providing feedback to learners on the progress of their learning process and indicating necessary corrections, assistance or support measures (Reinmann and Mandl, 2006). Summative constructivist assessment tasks largely do not measure objective learning goals. Based on predetermined criteria, the performance (also in terms of "competences") is evaluated through observing complex activities such as project work, case studies, journals, peer assessments, essays, reflective papers, discussion contributions and presentations. This requires teachers to have appropriate skills and pay particular attention to quality criteria in diagnosis and assessment (Euler and Hahn, 2007).

To summarize, constructivist learning theories view learners as active agents and protagonists in the learning process. Learning is an autonomous and/or collaborative process of exploration where learners acquire knowledge based on prior knowledge through internal inquiry. This kind of learning is context-dependent and influenced by the situations in which it takes place. Teachers design learning environments and act as counsellors and guides. They engage in discussions about errors and contradictions and question what is taken for granted.

3.2 Pedagogical approaches

The notion of a "pedagogical approach" is used in different ways in the literature, often interchangeably with "instructional approach" or "instructional strategy." In this context, I will use the term at a meta level, as Larson and Lockee (2014) do, i.e. as an overarching direction for teaching. I distinguish "pedagogical approach" from "instructional strategy," which Reigeluth and Keller (2009) use to describe macro teaching strategies such as problem-based learning, inquiry-based learning, direct instruction, project-based learning, anchored instruction and role-play.

In scientific discourse, two pedagogical approaches have emerged that, despite being controversially discussed, provide a useful initial orientation: the instructivist pedagogical approach (objectivist paradigm) and the constructivist pedagogical approach (subjectivist paradigm) (Dubs, 2017).

The *instructivist pedagogical approach* encompasses behaviorist and cognitivist perceptions of knowledge, which is objective, assuming that it can be effectively and efficiently transmitted to learners (Dubs, 2014; Larson and Lockee, 2014). Instructors present learning content, monitor and guide learning activities, and provide regular feedback on learners' performance (Larson and Lockee, 2014). Learners are assessed based on their performance against specified outcomes and objectives.

In contrast, the *constructivist pedagogical approach* assumes that knowledge is subjective, individually and socially constructed, and therefore interpreted differently by each person (Dubs, 2014; Larson and Lockee, 2014). Teachers create authentic experiences and set learning objectives aimed at constructing knowledge (or material objects), either individually or in groups. Students benefit from feedback from teachers, peers and even professionals.

At this point, it is also worth emphasizing some of Dubs' considerations on the teaching of economics, which might well align with the positioning of the two pedagogical approaches. Dubs (2014) draws attention to potential indicative objectives in economics education, encompassing design ideas and normative principles that also provide guidance in financial education. He distinguishes between three design ideas in economics: scientific-exemplary, economic-geographical and economic-civic. I will omit the economic-geographical design idea here as it relates to a specific subject area within geography.

According to the *scientific-exemplary design idea*, learners should be introduced to scientific paradigms and working methods of economics to develop a fundamental understanding. Learners study typical concepts in-depth, such as price formation in different markets. This design idea does not require comprehensive orientation knowledge ("exemplary") and remains value-neutral, as it is based on economic model assumptions⁹ (Dubs, 2014). Using a value-neutral idea in teaching typical economic concepts may suggest an instructivist pedagogical approach where knowledge is transmitted and assessed objectively.

In contrast, the *economic-civic design idea* addresses economic issues relevant to individuals as consumers, earners and citizens, and is therefore inherently normative. The primary focus is on fostering a general understanding of economics and society, enabling individuals to navigate the increasingly complex dynamics of the economic and social environment in a self-determined and responsible manner (Dubs, 2014). A normative approach to fostering a general (and critical) understanding of economics and society might suggest a constructivist pedagogical approach where individuals construct their knowledge autonomously and collaboratively, allowing for normative considerations.

It is important to note that there is no best pedagogical approach or instructional strategy. This is one of the few findings from research into teaching methods that can be considered certain (Terhart, 2005). Pure direct instruction, an instructivist strategy, often leads to inert knowledge (Renkl, 1998), but when applied correctly, it can be effective for learning (Pauli, 2006). Similarly, a radical constructivist strategy can lead to little learning without self-directed and collaborative learning skills (Dubs, 2014). Or, as Biesta (2010, p. 4) points out:

⁹ However, I question whether economic models are value-neutral, given the different schools of economic thought (Keynes, Hajek, etc.).

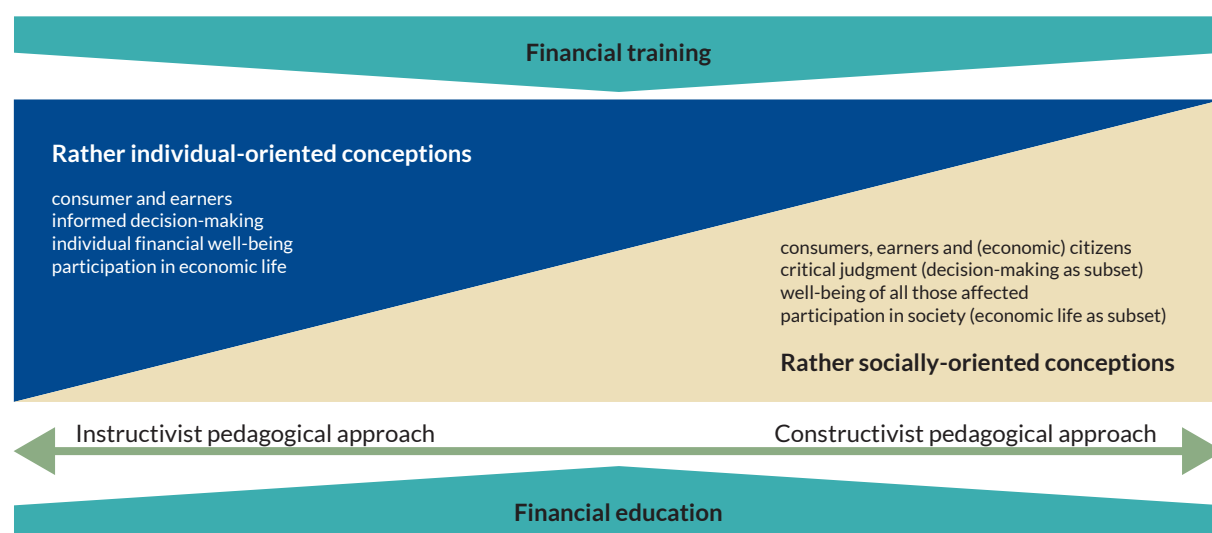
“In some cases students may benefit from interaction, talk, activity, for example if the aim for students is to test their understanding by explaining their views to fellow students. But in other cases, the collaborative classroom would actually hinder learning, for example, when the aim is the mastery of a complex skill – something that may require concentration and perseverance, rather than discussion and collaboration. Whether collaborative forms of student activity are to be preferred therefore entirely depends on the purpose of the activity, that is, on the outcomes that are considered to be educationally desirable. It is only when we are able to say something about the latter question that we can begin to make decisions about how we might want to achieve what is aimed for.”

4 Conceptions of financial education and their implications for pedagogical action

In this chapter, I will conflate the conceptions of financial education as delineated in section 1 with the pedagogical basics I outlined in sections 2 and 3 and summarize them in figure 2. I provide this illustration as a map that may help practitioners select, and reflect on, specific objectives in financial education and training, choose appropriate teaching and learning activities, and to consider how learning outcomes might be measured.

Figure 2

Conceptions of financial education and training from a pedagogical perspective



Source: own illustration.

In figure 2, I put the key characteristics of the conceptions of financial education (shown in figure 1) in relation to financial education and financial training to illustrate the difference between the two notions as outlined in section 2. I argue that it is possible to place both rather individual and rather socially-oriented conceptions within the broader notion of financial education and the narrower notion of financial training. In practice, this is often an unconscious process that also has to do with beliefs and personal values. However, even if teachers focus on qualification, they may still influence (consciously or unconsciously) the socialization and even the subjectification of their students. It is possible that even if teachers consciously prioritize

financial training to teach young people certain skills, they can still achieve educational effects. Similarly, the reverse is also possible. My intention in differentiating is to draw attention to the differences and to enable the reader to consciously choose, judge and transparently address these differences in pedagogical action and in evaluation endeavors.

Furthermore, I have added another continuum of pedagogical approaches. I hypothesize that there is a relationship between first, the type of educational outcome that emerges from financial education and training activities, second, the conscious or unconscious choice of learning theory approaches, third, correspondingly chosen instructional strategies and fourth, certain types of financial education conceptions. Accordingly, an instructivist pedagogical approach is more likely to be used in rather individual-oriented conceptions and a constructivist pedagogical approach is more likely to be used in rather socially-oriented conceptions. This seems to be a rather loose relationship that would be worth exploring in more depth in future empirical work. In the following subsections, I will present examples for each end of this continuum that serve to illustrate my arguments.

4.1 An instructivist pedagogical approach and rather individual-oriented conceptions

Rather individual-oriented conceptions of financial education focus on enabling individuals as consumers and earners to make informed decisions to achieve individual financial well-being. For example, it benefits consumers and earners to comprehend how inflation affects their purchasing power, the value of money in their current and savings accounts, their investments, and their income. Similarly, for savers and investors, it is useful to understand the relationship between return, risk and liquidity in various forms of investment (e.g. bonds, shares, funds, ETFs) to make better choices.

Following an instructivist pedagogical approach, teachers and trainers will focus on learning objectives such as understanding and applying the learning content. They might employ direct instruction, e.g. starting with theoretical input on the development and calculation of the inflation rate, current interest rates on savings and loans. They might combine theoretical input with exercises in individual or partner work so that students learn to correctly apply the concepts presented, such as calculating personal purchasing power in comparison with previous years and calculating real interest rates. Teachers and trainers might use learning materials based on scientifically validated literature and official sources, such as inflation rates calculated by national statistical offices. For the final assessment of learners' performances (e.g. in the form of single- or multiple-choice or true/false questions), instructors prepare assignments with a single correct solution that can be objectively assessed, such as calculating the real value of savings given a savings amount, annual interest, fees and the inflation rate.

Lusardi's (2011) "big five" items for measuring financial literacy include true/false and multiple-choice questions on compound interest, inflation, risk diversification, bond prices and mortgages. To enable people to answer these questions, it is sufficient to appropriately apply instructivist (behaviorist and cognitivist) instructional strategies so learners can thoroughly internalize these concepts.

4.2 A constructivist pedagogical approach and rather socially-oriented conceptions

Rather socially-oriented conceptions of financial education address individuals not only as consumers and earners but also as (economic) citizens who should be enabled to make informed decisions and critical judgements about finances. These decisions aim to achieve personal financial well-being and contribute to the well-being of others, ensuring everyone can participate in society. This approach places the examples described above in a new, i.e. normative, context that points to the need to apply a constructivist pedagogical approach.

In doing so, teachers and trainers may focus on critical judgment about the learning content. Learners might examine the issue of price stability (inflation/deflation) within a broader societal context. Teaching and learning activities might address the various causes of inflation and deflation and their economic and social consequences, such as extreme cases like the hyperinflation after World War I and the deflation during the Great Depression starting in 1929, which led to considerable poverty and suffering among the population and ultimately provided fertile ground for authoritarian regimes like National Socialism. In addition, central banks' and governments' monetary and fiscal policies could be compared across countries and critically evaluated. Promoting deeper understanding, accompanied by critical judgment and ethical considerations, can be beneficial for people's participation in society, e.g. in the role of informed voters.

In the context of saving and investment decisions, aspects such as the Sustainable Development Goals of the United Nations' 2030 Agenda might be considered. Teachers and learners might discuss relevant social, environmental and economic factors that need to be taken into account when making saving and investment decisions, and how these decisions might affect people's personal finances and society as a whole. Unlike with direct instruction, learning takes place in a dialog and might include the development of critical judgment skills that go beyond informed decision-making.

As briefly outlined in section 3.1, the assessment of student performance (e.g. competences) requires complex assessment settings that enable the observation of students' performance. It requires assessment tools that consider quality standards, as assessment criteria are "constructed" and therefore cannot be objective. For instance, the Argumentation Rating Tool by Reznitskaya and Wilkinson (2017) is an example of a scientifically evaluated instrument for assessing argumentation skills in face-to-face dialog settings.

5 Limitations

This article is a theoretical discussion based on selected literature and is limited to my interpretations. It is not an empirical study, such as an analysis of the international discourse on financial education and training. I take an educational perspective when interpreting the differentiation of conceptions rather than an evaluation research or economic perspective. I have endeavored to present my arguments as transparently as possible, offering them for discussion among professionals. To effectively convey the core ideas of the article, I have focused on the basic aspects when explaining the conceptions of financial education and pedagogical foundations. Consequently, some explanations may appear somewhat simplistic.

6 Summary and concluding remarks

In this article, I have attempted to elucidate the distinction between conceptions of financial education based on characteristics pertinent to pedagogical considerations. I proposed situating these conceptions on a continuum from rather individual-oriented to rather socially-oriented conceptions (figure 1). I delineated three conceptions for the former and two for the latter, building on Aprea's (2014) conceptualization grounded in educational theory. After discussing the differences between education and learning and presenting the three learning paradigms and two corresponding pedagogical approaches, I linked them to the conceptions of financial education. I hypothesized that there is a relationship between first, the kinds of educational outcomes of financial education and training activities, second, the conscious or unconscious choice of learning theory approaches, third, correspondingly chosen instructional strategies, and fourth, particular types of financial education conceptions.

Figure 2 summarizes my arguments, providing guidance in selecting, and reflecting on, specific objectives in financial education and training, choosing appropriate teaching and learning activities and considering how learning outcomes may be measured. Returning to the guiding question –what financial education and training endeavors and evaluation goals are *desirable* – I offer six reflection questions to assist readers in finding their own answers:

- **Q1:** What have been your beliefs about personal financial education?
- **Q2:** For those engaged in financial education and/or training: where would you place your activities in figure 2?
- **Q3:** Where would you place a desirable engagement in financial education and/or training in figure 2?
Does your answer differ from the answer to Q2?
- **Q4:** What are the organizational conditions under which you engage in financial education and/or training, and which approaches are conceivable and feasible? And why?
- **Q5:** For those engaged in evaluation research in context of financial education and/or training: what have you measured so far? Why have you focused on this?
- **Q6:** For those engaged in evaluation research: what would be desirable to measure if it were possible?
Does your answer differ from the answer to Q5?

Lucey (2016) highlights that pre-service teachers' beliefs about personal finance – which can be both critical and uncritical of established views in financial education – affect what and how they teach. He calls for more research on the choice of educational content and instructional strategies related to personal finance and advocates more alternative approaches and respect for different beliefs. I hope my discussion will contribute to considering alternative approaches to teaching and evaluating personal finances – not just for pre-service teachers but for all who are engaged in financial education and training.

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Part III:
**Scientific evaluation
methods and guidelines
for financial education**



Introduction to program evaluation and its research cycle

Sandra Mauser¹

Financial education programs aim to improve people's financial well-being. To determine whether this goal and associated outcomes are achieved, it is essential to evaluate how effective these programs are. Program evaluation is thus critical for assessing performance, identifying strengths and weaknesses and guiding improvements. This paper explores the research cycle of program evaluation, covering key phases such as planning, implementation, analysis and communication. It discusses the role of formative and summative evaluation, the importance of stakeholder involvement and the methodological choices required for rigorous assessment. Furthermore, it highlights challenges in evaluation, including biases in interpretation and dissemination. In addition, it underscores the necessity of a structured communication plan to ensure that findings are conveyed effectively to relevant audiences. By understanding the evaluation cycle and its implications, decision-makers can enhance the accountability and impact of financial education programs, thus fostering better financial outcomes for individuals and communities.

Financial education programs are meant to help people strengthen their financial competencies and, as a result, improve their financial lives. A lot of institutional, political, national and even international effort is put into designing and implementing such educational programs. This costs time and money, which raises the question whether the resources are allocated properly. In other words, whether the implemented programs operate as intended and lead to the desired outcomes. This is where program evaluation comes in.² Berry and Sloper (2016, p. 5) use the following analogy to describe the purpose of program evaluation: *"A financial education program without evaluation is similar to an explorer without a compass. Without a compass, an explorer is not able to decide whether he or she is on the right track. Without an evaluation, the educator is not able to decide whether the financial program is producing successful results and meeting the audience's needs."*

1 Program evaluation and its purpose

Program evaluation helps decision-makers set common goals for financial education programs, assess the latter's effectiveness, measure performance and identify challenges and guiding principles. Evaluation results can be used to improve a program's structure, administration or funding. Such results can also help

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² Although we are talking about *financial education programs* in this paper, it should be noted that evaluations are conducted for a variety of interventions (programs as well as policies) in different areas of research. Typically, the interventions of interest are implemented to address social problems, e.g. poverty, inequality, unemployment, racism and problems related to healthcare or education. Often these problems stem from underlying social, economic or political factors and can affect a significant number of people within a society. Social problems require public attention and purposeful, well-organized interventions, which also include thorough evaluation (Rossi et al., 2019).

respond to political pressure regarding the program's eligibility (Berry and Sloper, 2016; Rossi et al., 2019). Priority number one for evaluators when conducting an evaluation is finding out what the purpose of the evaluation is: who wants the evaluation, what exactly do they want and why (Rossi et al., 2019; Chambers et al., 2009). For evaluators, it is not always easy to get answers to these questions; they often need to examine the situation carefully. Sometimes, there are *underlying hidden agendas* to program evaluation. For instance, an evaluation might be conducted to strengthen public relations (PR), promote the program among funders and politicians or to provide rationale for decisions that have already been made. To some extent, (almost) all evaluations involve some political and PR objectives. As long as such objectives do not define the main purpose of the evaluation, evaluation integrity is not compromised (Rossi et al., 2019).

Every financial education program can be regarded as a social framework comprising roles and activities of various groups and individuals who are involved and/or interested in the program. They are connected by a set of social and potentially political relationships. These groups and individuals are called the *stakeholders* of a program. Not only do stakeholders define the underlying social problem to be tackled and the goals of a program, but they also (indirectly) define the purpose of evaluation and influence the way in which an evaluation will be designed (Rossi et al., 2019). Stakeholders broadly fall into three categories, namely (1) people involved in program operations (e.g. program planners and decision-makers, staff, administrators, oversight boards), (2) people served or affected by the program (i.e. recipients) and (3) people intended to use the evaluation findings (e.g. program and evaluation funders, political decision-makers, legislative committees, organizational leads, research community). It can be fruitful for the evaluation process and also for the interpretation of findings to actively include these stakeholder groups' unique and diverse perspectives (Berry and Sloper, 2016).

In a *participatory research approach*, evaluators involve stakeholders directly in the evaluation process (Yoong et al., 2013). Evaluation can be participatory in various ways and at different points in time of the evaluation process. For instance, it is possible to allow program staff, recipients or other stakeholders to provide input for defining both research questions and outcomes to be examined, for interpreting findings and for communicating results to an interested audience. What needs to be carefully considered is how and to which extent evaluators include stakeholders' expertise and experiences in the evaluation. To ensure that the evaluation is not entirely dictated by stakeholders' subjectivity, evaluators are responsible for balancing this input against the objectivity of the process (Yoong et al., 2013).

In this paper, we will focus on two main objectives of program evaluation, namely program improvement and program accountability. These two functional objectives correspond to formative and summative evaluation, respectively (Berry and Sloper, 2016; Rossi et al., 2019). *Formative evaluation* helps determine whether the program's processes and activities are implemented as planned. Identifying the strengths as well as barriers and weaknesses of the program provides information on its quality and guides *program improvement*. Hence, formative evaluation focuses on processes to make adjustments as needed (Berry and Sloper, 2016; Rossi et al., 2019). *Summative evaluation* helps identify outcomes associated with the program and examine whether these outcomes are a result of the program. A financial education program is expected to make a beneficial contribution to the financial lives of individuals. By focusing on the results of the program, summative evaluation can help determine whether these expectations are met. Hence, summative evaluation investigates *program accountability*, providing a summary judgment of the performance of the program (Berry and Sloper, 2016; Rossi et al., 2019).

Formative evaluation

To examine components of program implementation quality, formative evaluation could address the following questions (Berry and Sloper, 2016):

- Has the process of delivering information via the financial education program been implemented as intended?
- Can the entire content be delivered to the recipients as intended?
- Are the program's materials used in an engaging manner?
- Do educators and recipients behave in an interactive way?
- Do the targeted recipients use the offered program?

Summative evaluation

To examine recipients' experiences with the program and the outcomes of participation, summative evaluation could address the following questions (Berry and Sloper, 2016):

- Can intended changes in outcomes be observed for recipients having participated in the financial education program?
- Are there observable differences in outcomes between recipients of the program and nonrecipients?
- Are recipients satisfied with the program as provided?
- Do recipients perceive advantages of having participated in the program?
- Do recipients report long-term benefits after a specific time following the program?

Evaluation objectives are often related to a program's *stage of maturity*. With programs in an early stage of implementation, attention should be paid to program stakeholder needs, invested resources and the key activities of a program (formative evaluation). With programs that have been implemented for a while, evaluation objectives might be to understand the short-, medium- and long-term outcomes of a program (summative evaluation) (Yoong et al., 2013; Berry and Sloper, 2016). Program evaluation does not need to concentrate exclusively on either formative or summative objectives. Consistency and quality of program implementation play an important role especially when program outcomes are investigated. The features of implementation will influence how and to which extent the recipients benefit from the program. Hence, evidence from formative evaluation can provide valuable inputs for exploring program impact during summative evaluation more thoroughly (Berry and Sloper, 2016). This is why the functional objectives of formative and summative evaluation are part of a more comprehensive categorization of evaluation types, namely process and impact evaluation.³ The purpose of process and impact evaluation is to *"help decision-makers understand whether a program delivered what was promised and how well it performed relative to other programs in terms of benefits to participants"* (Yoong et al., 2013, p. 26). Ultimately, the gained information should help decide on future steps regarding the program (Chambers et al., 2009).

³ Examining the literature on evaluation research reveals that the distinction between formative vs. summative evaluation and process vs. impact evaluation is not always clear cut. While these categorizations describe distinct yet complementary roles in understanding programs, they often overlap, and the direction of inclusion remains ambiguous. In this paper, we follow the understanding that formative and summative evaluation objectives define the functional level of evaluation – i.e. which questions to ask – whereas process and impact evaluation reflect the broader purpose of evaluation, placing it in the context of underlying social (and political) goals.

Process evaluation examines the development and implementation of a financial education program (Yoong et al., 2013; Rossi et al., 2019). The goal is to understand how well the program has been executed and how its processes work. It is meant to deliver information on the program's relevance, efficiency and effectiveness. The questions answered by process evaluation can be grouped into three main areas: (1) implementation/operations, (2) outputs and (3) appropriateness/acceptability. Process evaluation focuses on lessons learned and tends to be formative, but it can also include summative objectives (see example questions in box 2). Identifying issues and problems during program implementation is especially important as the program is being rolled out. Conducting process evaluation at an early stage of maturity therefore makes it easier to adjust the program. Stakeholders typically interested in process evaluation are program planners, administrators, oversight boards and funders (Rossi et al., 2019). For them, evaluation insights need to be concrete, timely and immediately applicable.

Impact evaluation examines whether the financial education program is achieving the predefined goals and objectives (Yoong et al., 2013; Rossi et al., 2019). It helps understand which observed outcomes can and cannot be directly and credibly attributed to the program. In other words, impact evaluation measures the *causal effect* of a program on the observed outcomes, ruling out that any other factors could have caused the effects. The desired effects typically refer to behavioral changes in recipients' financial decision-making. They can, however, also include changes in knowledge, skills and attitudes potentially underlying behavioral changes. The investigated effects can be short-, medium- or long-term. Typically, the long-term effects of a program are referred to as its *impact*. For evaluators it is not only desirable to measure the magnitude of impact but also to explore how this impact was achieved by the program. Stakeholders typically interested in impact evaluation are major decision-makers of program oversight, e.g. program funders, political decision-makers, legislative committees and organizational leads (Rossi et al., 2019).

Box 2

Process evaluation

To examine a program's development and implementation, process evaluation could address the following questions (Yoong et al., 2013; Rossi et al., 2019):

- Were the program's goals and objectives defined appropriately for the given context? (formative)
- Was the program implemented as intended? How can implementation be optimized? (formative)
- Did the program reach the intended recipients? If applicable, why are some eligible people not reached? (formative)
- Are recipients satisfied with the program as provided? (summative)
- Of the people in the program, does a sufficient number complete the program? (formative)
- Did the program have any unforeseen effects? (summative)

Impact evaluation

To examine the causal effect of a program, impact evaluation could address the following questions (Rossi et al., 2019):

- Are the program's goals and objectives being achieved? (summative)
- Are there observable tendencies as to the desired outcomes? (summative)
- Does the program have any adverse effects on the recipients? If yes, which? (summative)
- Does the program have different effects on different groups of recipients? (summative)
- Does the program alleviate the problem it is supposed to address? If yes, to what extent? (summative)

Besides program improvement and accountability, program evaluation also contributes – with its findings – to academic research, *generating knowledge* for the community interested in evaluation studies (Rossi et al., 2019). Since the research field of program evaluation is still developing, good evidence for understanding the effects of financial education programs is still sparse (Yoong et al., 2013). Yoong et al. (2013) point out two underlying challenges: (1) standardized and common benchmark measures have yet to be established to allow comparison between different programs and (2) the existing methodologies differ substantially in their quality and nature. Therefore, any additional knowledge on the evaluation process itself and on the causal relationship between a program and its outcomes helps researchers establish more rigorous program evaluation benchmarks, from which also practitioners and policymakers can benefit.⁴

2 Program theory and evaluability assessment

Evaluation can be regarded as an integral part of financial educational programs. To benefit optimally from an evaluation, the evaluation plan should be drawn at the beginning of the program design and implementation. In other words, the evaluation should be *prospective*. Once a program has started, an evaluation can still be conducted *retrospectively*. Yet, without early planning, the information gathered might be limited (Berry and Sloper, 2016; Yoong et al., 2013). Planning an evaluation from the start enhances the quality of evaluation. This ensures that the goals set for the program are appropriate and measurable. Also, that baseline data are collected at the beginning of the program, which can later be compared with the evaluation results. An ongoing, constructive evaluation increases the likelihood of credible results. Additionally, it proves cost efficient by taking advantage of the program infrastructure when planning evaluation activities (Yoong et al., 2013).

Ideally, evaluation plans are based on theory. *Program theory* should guide how evaluation questions are formulated and prioritized, how the research is designed and how the findings are interpreted. The literature lacks a consensus about the description and setup of program theory. Instead, there are various ways that have various names (Rossi et al., 2019). One way is not necessarily better than the other, and to some extent, it will be a question of taste which way and name to choose. In this paper, we present a simple scheme from Rossi et al. (2019), as it provides a good overview of what program theory should

⁴ Addressing the lack of coherent evaluation evidence on financial education, Lorenz et al. (2025b) provide a thematic review of the financial literacy evaluation landscape in OECD countries from 2010 to 2024 that analyzes 68 (quasi-)experimental and peer-reviewed studies. The review identifies key patterns across study contexts, intervention types, outcome measures, results and methodological approaches.

include. The scheme consists of three interrelated components: (1) the program's organizational plan, (2) the program's service utilization plan and (3) the program's theory of change (program impact theory).⁵

The *program's organizational plan* is developed by program managers and outlines the resources (human, financial and physical) as well as administrative and general organizational factors needed for the program. The plan comprises the functions and activities a program is supposed to perform given resources as well as the factors and preconditions needed for the organization to be able to offer the program (e.g. fundraising, acquisition and maintenance of facilities or governmental contacts). In general, the plan can be understood as a set of propositions in the following sense (Rossi et al., 2019, p. 67): *"If the program has such and such resources, facilities, personnel and so on, if it is organized and administered in such and such a manner, and if it engages in such and such activities and functions, then a viable organization will result that can operate the intended service delivery system."*

The *program's service utilization plan* describes how the organization expects to reach its target population and also which service contacts are needed along the program. The plan *"pulls into focus the critical assumptions about how and why the intended recipients of service will actually become engaged with the program and follow through to the point of receiving sufficient services to initiate the change process represented in the program impact theory"* (Rossi et al., 2019, p. 68).

The *program's theory of change* describes the underlying logic model of the program, i.e. the causal theory of how the program is supposed to lead to the intended outcomes. It displays the sequence of causes, i.e. specific activities of the program, and effects that these activities are expected to produce, i.e. the social benefits. In most cases, a program can only indirectly impact the social conditions of interest. It does so by influencing crucial but controllable key aspects of the given situation, which in turn contribute to improvements of the social conditions. The theory of change goes through each step of the chain of events, revealing the assumptions about the change process and the respective dependencies to bring about the desired social benefits (Rossi et al., 2019). Having a clear understanding of a program's theory of change is especially important for an impact evaluation, as its goal is to measure causal effects. The hands-on construction and relevance of a program's theory of change is discussed in Mauser (2025).

If a program is considered ineffective, then the theory of change, together with the organizational plan and the service utilization plan, can potentially explain why the program failed. For instance, by shedding light on whether specific activities were not implemented with the required quality or whether resources were allocated inadequately (Berry and Sloper, 2016). To this end, an exhaustive program theory needs to reveal all critical assumptions and expectations that underlie the program's design. Formulating program theory is not a one-shot job. The plausibility and logic of the proposed program theory need to be reviewed carefully as part of a continuous process, ideally throughout a program's lifetime (Rossi et al., 2019). *"A program whose design is weak or faulty has little prospect for success, even if it adequately implements that design. Thus, if the program theory is not sound, there may be little reason to assess other evaluation*

⁵ Often the evaluation literature does not distinguish between the three components and discusses them together as the theory of change. Given their interrelations, this is a legitimate and practical way to proceed. However, we find it useful to, at least once, look at the components separately in order to understand the power of program theory and the many assumptions a program, and therefore also its evaluation, relies on.

issues such as impact or efficiency” (Rossi et al., 2019, p. 87). Hence, a poorly defined or flawed program theory makes program evaluation meaningless in the first place. For this reason, program theory itself needs to be assessed too.

One approach to determining the quality of program theory is an *evaluability assessment*. A decision-making tool (Craig and Campbell, 2015), it is commonly used to assess whether a program is ready for evaluation and to support evaluation planning as well as program development. At an earlier stage, an evaluability assessment can help identify promising interventions for evaluation (Lam and Skinner, 2021). According to Rossi et al. (2019), evaluability assessment consists of three main parts: (1) describing the program model with a focus on the definition of objectives and goals, (2) assessing to which degree the model is both well-defined and evaluable and (3) identifying stakeholders’ interests in evaluation and usage of evaluation findings. Importantly, evaluability assessments should not be confused with the evaluation per se. Although insights from an evaluability assessment can serve formative purposes (program improvement), they do not serve summative purposes (program accountability) (Lam and Skinner, 2021).

Financial education programs can differ in various ways. Seldom are they entirely built on unique assumptions about how the program will function and how the intended changes are supposed to happen. As a first step of assessing evaluability, literature on social sciences and human services can provide evidence and insights from practical experiences, which can help assess the meaningfulness of the proposed program theory (Rossi et al., 2019). Besides a detailed desk review on information related to the history, design and characteristics of the program (i.e. *secondary data collection*)⁶, Yoong et al. (2013) advise to interview key stakeholders and conduct site visits and observations of program activities. Interviews do not only yield feedback (especially from the program staff) on the program and its operations but also inform about stakeholders’ needs and their commitment to evaluating the program, as well as their potential contribution to evaluation (e.g. time, money, further information). Observing programs in action, additionally, helps check whether the resources, structures and processes necessary to conduct an evaluation are in place.

While conceptualizations of evaluability assessments and their implementation vary (Davies, 2013), the fundamental premise is that they provide a systematic way to judge the desirability and feasibility of evaluation (Yoong et al., 2013).⁷ After all, the *“results of such assessment should be reasonable consensus on a shortlist of potential evaluation questions and a preliminary understanding of the possible scope, methods and design of the evaluation to be deployed”* (Yoong et al., 2013, p. 30). If the evaluability assessment concludes that an evaluation should be conducted, the next steps of the evaluation cycle can start.

⁶ Secondary data collection refers to acquiring data that was not originally gathered for the evaluation at hand. These data come from secondary sources (Rossi et al., 2019).

⁷ Discussing evaluability assessments in details is beyond the scope of this paper. For readers interested in the topic, we recommend having a look at Davies (2013), Lam and Skinner (2021) as well as Craig and Campbell (2015).

3 The cycle of evaluation

The *evaluation cycle* describes the entire evaluation process and consists of four phases: (1) planning, (2) implementation, (3) analysis and (4) communication. Figure 1 gives an overview of these phases and the corresponding steps, which we will discuss in more detail below.

3.1 Planning

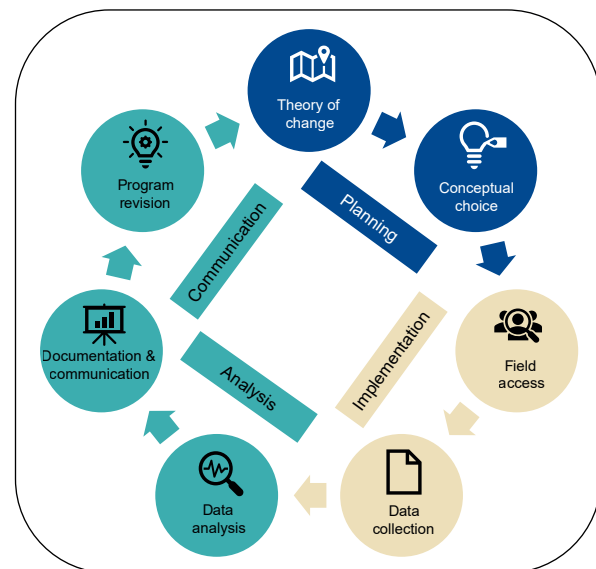
First, an evaluability assessment tells us whether to conduct an evaluation. Second, mapping out program theory, most importantly the theory of change, is already the first step of an evaluation's planning phase. A program's theory of change describes the logic model how certain inputs and activities generating certain outputs are supposed to lead to specific outcomes. It thus informs the decision on an appropriate evaluation concept. Table 1 summarizes the main (measurable) components of a theory of change and gives some examples.

Looking at the examples given in table 1, we see that *inputs*, *activities* and *outputs* are measurable and quantifiable, e.g. the amount of funding, the number of e-learning modules a program includes or the number of participants who attended an offered workshop. To conduct program evaluation, however, it is essential to also identify measurable *outcomes*, which need to be achievable in a reasonable way within a specified period and which require specific resources and activities (Berry and Sloper, 2016). As mentioned in section 1, intended program effects can be short-term, medium-term or long-term.

Often, short-term effects must be established first to create a foundation for more lasting effects. In this sense, outcome measures follow a hierarchical order (Berry and Sloper, 2016): (1) The lowest level of measurable outcomes is the recipients' satisfaction with the provided program. This is not only an important measure in process evaluation (box 2) but also essential for a program to have an impact. (2) Only if the recipients are satisfied with the program can the next outcome level be addressed, namely *learning to*

Figure 1

Evaluation cycle



Source: Theresa Lorenz (OeNB).

Table 1

Theory of change – main components

Theory of change	Inputs	Activities	Outputs	Outcomes
Description	Resources necessary for implementing and operating the program	Specific actions and services that constitute the program	Direct and tangible products of the program	Effects on recipients which can be attributed to the program (short-, medium- and long-term)
Example	• Funding • Staff • Materials	• Workshops • E-learning • Games	• Number of participants • Hours of service • Number of awarded certificates	• Changes in financial knowledge • attitudes • behaviors

Source: Author's compilation based on Berry and Sloper (2016).

change (short-term). This level includes changes in financial knowledge, attitudes and skills and recipients' aspiration toward desirable behaviors. (3) The next level is *taking action to change* (medium-term), which includes desirable changes in recipients' financial behavior and practices, e.g. establishing good routines. (4) The highest level of outcomes are *results of change* (long-term). From the individual perspective, the ultimate result is personal financial well-being, which, in turn, positively impacts on social, economic, societal and environmental aspects.

Based on the theory of change, evaluation questions can be formulated, laying out the evaluation priorities and goals. In box 1 and 2, we presented such evaluation questions, assigning them to formative vs. summative purposes and classifying them as part of process or impact evaluation. Having identified the components and respective measures of the theory of change, we can now assign the evaluation questions accordingly (box 3).

Box 3

Evaluation questions addressing components of the theory of change

Inputs

- Are the materials adequate and relevant for the recipients' needs and interests? (formative, process evaluation)
- Is the recipient-educator ratio appropriate for the workshop? (formative, process evaluation)

Activities

- Can the entire content be delivered to the recipients as intended? (formative, process evaluation)
- Are the program's materials used in an engaging manner? (formative, process evaluation)

Outputs

- Do the targeted recipients use the offered program? (formative, process evaluation)
- Does a sufficient number of the people in the program complete it? (formative, process evaluation)

Outcomes

- Are recipients satisfied with the provided program? (summative, process evaluation)
- Are there observable tendencies as to the desired outcomes (e.g. improved knowledge, greater awareness of financial risks, developing a budget plan)? (summative, impact evaluation)
- Does the program alleviate the problem it is supposed to address? If yes, to what extent? (summative, impact evaluation)

Importantly, evaluation questions should not be confused with survey questions (survey items). While the former outline the evaluation interest in a general sense, the latter are specific items used to examine changes in recipients (Berry and Sloper, 2016). For example, to help answer the evaluation question "Are recipients satisfied with the provided program?", a corresponding survey question addressed to the recipient could be "How satisfied were you with the content of the workshop (response options on a Likert scale from "very satisfied" to "not at all satisfied")?"

"Evaluations should be issues-led not methods-led. And having determined the evaluation questions, the best available method should then be used to answer them" (White, 2010, p. 162). This brings us to the next

step of planning, namely the conceptual choices of the evaluation design and method. Table 2 provides an overview of the key decisions to be made, addressing the following questions: (1) Which type of evaluation (see section 1) should be conducted and (2) how (evaluation design), (3) what type of data should be collected with which method and (4) who should the data be collected from (study sample)?

“A good evaluation design addresses the evaluation questions, is appropriate for the evaluation context (e.g., time, resources), and provides sufficient and critical data” (Berry and Sloper, 2016, p. 16). We differentiate between three types of evaluation design (Berry and Sloper, 2016): (1) *Non-experimental designs* are explorative (descriptive or correlational), looking at relationships among program implementation, participation and outcomes, for instance, using post-test only or examining pre-post changes. Non-experimental designs do not measure causal effects. This means that observed results cannot be argued to have been caused by the program (directionality), nor can they be directly attributed to it. (2) *Experimental designs* allow for measuring attribution and program impact, i.e. causality, by comparing outcomes between treatment and control groups. The treatment group consists of program recipients, i.e. individuals exposed to the program. The control group comprises individuals similar to those in the treatment group; yet, they do not participate in the program. Randomly assigning individuals from the study sample to these groups ensures that they have similar characteristics (except for program participation). This allows for causal conclusions based on observed differences between the groups. (3) *Quasi-experimental designs* also compare outcomes between program recipients and nonrecipients; however, random assignment is not possible. Instead, alternative methods are used to create comparable groups. The advantage of random assignment

Table 2

Key decisions in the conceptual phase

Objective of evaluation	Program improvement	Program accountability
Evaluation type	Process evaluation - Assess recipients' needs, adequacy of resources, quality of provided service - Identify strengths and challenges of program implementation	Impact evaluation - Assess program's effect on recipients - Identify what works for whom
Evaluation design	Non-experimental	- Experimental - Quasi-experimental
Evaluation method	- Needs assessment (quan/qual¹) - Observations (qual/mix¹) - Focus groups (qual) - Surveys (quan/mix) - Document review (qual/mix) - Dosage/attendance (quan)	- Surveys (quan/mix) - Tests or assessments (quan) - Focus groups/interviews (qual)
Study sample	- Program recipients - Program staff/instructors	Program recipients and a comparison group of non-recipients (control group)

Source: Author's compilation based on Berry and Sloper (2016).

¹ quan = quantitative method; qual = qualitative method; mix = mixed method.

is that it eliminates threats to *internal validity* (establishing a causal relationship), whereas alternative methods can only reduce these threats.⁸

Looking at evaluation designs, we need to consider one more question (Berry and Sloper, 2016): When and how often should data be collected? Data can either be collected at one point in time (*cross-sectional design*), or data are collected at least at two points in time (*longitudinal design*). Cross-sectional designs have the disadvantage that they cannot observe changes over time, but they are commonly used to collect data after the program (post-test only) to examine differences between recipients and nonrecipients. Longitudinal designs are more challenging (e.g. time-consuming, costly, program dropouts) but have the advantage that they can explore changes in outcomes over time by comparing outcomes before the program (pre-test) to outcomes after the program (post-test).

Evaluation methods for data collection (mostly *primary data collection*⁹) depend on the nature of the data being gathered (Berry and Sloper, 2016; Rossi et al., 2019): *Quantitative data* involves numerical information, such as test scores, survey ratings and demographic data, which is useful for measuring program outcomes and ensuring accountability. Collecting quantitative data is expected to yield *valid* (accurate) and *reliable* (consistent) results. In contrast, *qualitative data* capture nonnumeric insights, such as written responses and personal narratives. Collecting qualitative data is more flexible and adaptive compared to quantitative data collection, representing experiences of those involved with the program and allowing for a deeper understanding of unintended effects and program dynamics.¹⁰ While both methods have their strengths and limitations, combining them in a *mixed-methods approach* can enhance the value and quality of evaluation. By integrating numerical data with recipient stories, evaluators can provide a more comprehensive picture of a program's impact (or implementation), balancing statistical evidence with human experiences.¹¹ As with the evaluation design, the decision about which data to collect and which methods to employ depends on the evaluation questions as well as on the resources, including staff, funding and time, that are available for conducting the evaluation.

Depending on the evaluation design and method, the final step of the planning phase is to determine the study sample. The actual sampling process then takes place during the implementation phase, as part of accessing the field.¹² If all recipients of a program participate in data collection, a *full census* is achieved, which ensures complete representation. This approach is ideal when the total number of recipients is small and manageable or administrative data are available for outcome measures of interest. However, due to time or resource constraints, it is often not feasible to collect data from all recipients. In such cases, an evaluation sample must be selected (Berry and Sloper, 2016; Rossi et al., 2019). *“The primary goal of sample selection is to provide unbiased and sufficiently precise information on key study measures that adequately represent the target population for the evaluation. A sample is simply the subset of the units that make up the target population for an evaluation”* (Rossi et al., 2019, p. 273).

⁸ Experimental and quasi-experimental designs will be discussed in an upcoming paper of the OeNB Financial Literacy Evaluation Series.

⁹ Primary data refer to information directly gathered by the evaluators conducting the evaluation. However, also secondary data can be valuable for the evaluation, such as information from administrative databases (Rossi et al., 2019).

¹⁰ Qualitative research methods are discussed in Felbermayr (2024).

¹¹ A practical guide for a mixed-methods approach is presented in Lorenz (2024).

¹² Details on survey methodology and sampling in quantitative research can be found in Zieser et al. (2025) and details on sampling in qualitative research in Felbermayr (2024).

There are two main approaches to *sampling* (Berry and Sloper, 2016; Rossi et al., 2019): (1) *Probability samples* are selected randomly, which is to ensure that every member of the population has a known chance of inclusion. This method allows for unbiased, representative findings that can be generalized to the broader population. It also prevents selection bias, which can arise from human discretion in choosing recipients for the sample who portray the program in a particular light. However, probability sampling can be complex and resource-intensive. (2) When resources or time are limited, or when the evaluation is exploratory rather than aiming for broad generalizability, *non-probability sampling* (or *convenience sampling*) is used. This involves selecting recipients based on convenience, specific characteristics or to ensure that diverse perspectives are included. While non-probability samples may not fully represent the population, evaluators aim to include key variations to capture a more comprehensive understanding of the program's impact.

To ensure a well-structured evaluation, careful sampling is essential (Rossi et al., 2019). Evaluators must determine who will provide data, how many recipients are feasible given the budget and timeline and how access to data will be secured. The selected sample should be at least reasonably representative of the target population, since evaluation results can be misled by sample populations which differ in many ways from the target population. Additionally, potentially missing data should be considered and addressed, such as accounting for non-responses by selecting a slightly larger initial sample (*oversampling*). By thoughtfully designing the sampling approach, evaluators can ensure that their findings are as meaningful and reliable as possible (Rossi et al., 2019).

3.2 Implementation

The implementation phase of an evaluation is often the most time-intensive, with *data collection* and management requiring significant planning and effort (Rossi et al., 2019). Even when secondary data sources are used, considerable time must be allocated for securing data access, ensuring ethical compliance, cleaning and integrating datasets and addressing data security concerns. Evaluations that involve primary data collection demand even more logistical coordination, particularly when *fieldwork* involves multiple sites, travel and engagement with various stakeholders (Rossi et al., 2019). Securing permissions, scheduling site visits, recruiting and training data collectors and processing collected data for analysis all contribute to the complexity of this phase. To keep the evaluation on track, a well-structured data collection and management plan is essential to ensuring that resources, staff and timelines align with the evaluation's objectives (Rossi et al., 2019).

"When designing evaluation tools and collecting data, special attention must be paid to protecting the rights of the participants. It is necessary to follow the human subjects governing rules and regulations to ensure that participants' privacy and freedom rights are not violated" (Berry and Sloper, 2016, p. 19). Lorenz and Felbermayr (2024) look at data privacy and research ethics in more detail. For now, we focus on the basic requirements for data collection and fieldwork, which apply to both quantitative and qualitative data collection.¹³ Several critical factors must be considered when designing *evaluation tools* (data collection instruments and processes) to minimize errors, enhance accuracy and ensure meaningful insights: (1) Ensuring clarity and readability of survey questions, (2) addressing sensitive information, (3) minimizing the respondents' burden, (4) ensuring cultural appropriateness and relevance, (5) ensuring natural and unbiased measurement and (6) allowing for unexpected findings. Based on Berry and Sloper (2016) and MacDonald (2013), we summarize these factors in the following.

¹³ For details on qualitative data collection, see Felbermayr (2024).

First, for data collection to yield valid and reliable results, survey questions must be clear, concise and easily understood by respondents. Ambiguous or complex wording can lead to misinterpretation, reducing the accuracy of responses. If a survey includes closed-ended questions (e.g. multiple-choice), all possible response options should be considered or an “other” option should be provided to avoid forcing respondents into inappropriate categories. Open-ended questions should be specific enough to guide meaningful responses while allowing respondents the flexibility to express their thoughts in their own words. For all survey questions, the reading level of the target audience must be considered to ensure that the language used matches the respondents’ comprehension skills. This is especially important when working with children, individuals with low literacy levels or non-native speakers.

Second, when collecting sensitive data such as age, income or financial status, respondents may feel uncomfortable about providing precise information. To improve response rates and maintain comfort, questions should be designed in a way that protects privacy. One effective strategy is using response ranges rather than asking for exact figures. For example, instead of asking for a specific income, a question could present income brackets for respondents to select from. This approach fosters trust and encourages more honest responses.

Third, the process of data collection should be designed to minimize the burden on respondents. This includes considering the time and effort required to, for instance, complete surveys or participate in interviews. Overlapping evaluations within a single setting or population may result in respondents being surveyed multiple times, which may lead to survey fatigue. To avoid this, evaluators should explore opportunities for collaboration between organizations or stakeholders, sharing data collection efforts where possible to optimize resources while reducing redundancy for respondents.

Fourth, data collection methods should also be culturally appropriate to ensure that responses are accurate and meaningful. Survey questions should be sensitive to the values, traditions and norms of the population being studied. This requires engaging stakeholders in the design process to ensure that concepts are relevant and understandable in the given cultural context. Without these considerations, questions may be misunderstood or deemed irrelevant, which would compromise the integrity of the data.

Fifth, to maintain objectivity, survey questions should be written in a neutral manner without presupposing a particular outcome. For instance, instead of stating “increase in knowledge,” a more neutral phrasing would be “level of knowledge” to avoid implying a positive or negative result before data collection is complete. Not implying a preferred direction or value in the formulation of items ensures that the measurement remains unbiased, which allows for a fair and accurate assessment of the program’s impact.

Sixth, while evaluations often focus on predefined outcomes, it is also essential to build in flexibility to capture unanticipated effects of a program. Survey items should be structured in a way that allows for the documentation of unexpected trends or consequences, be it positive or negative. This adaptability enhances the depth of the evaluation by revealing insights beyond initial expectations.

Nightingale and Rossman (2015) point out another pitfall to be aware of when conducting field-based studies, namely the risk of collecting too much information. On the one hand, data collection is time- and resource-intensive, and on the other hand, an excessive amount of data can become overwhelming and disorienting in the subsequent data analysis phase.

To recall the most important principles of data collection, Gugerty and Karlan (2014; 2018) proposed the *CART* principles, where *CART* stands for credible, actionable, responsible and transportable data collection, summarized in box 4.¹⁴

Box 4

CART principles

Credible – Collect high-quality data that can be analyzed accurately

Data should be collected in a consistent way (reliability) and accurately reflect what one seeks to measure (validity). Subsequent credible analysis requires understanding of what can be measured (especially important when attempting estimates of impact).

Actionable – Collect data you can commit to use

Collect only the data that will be used. For every piece of data collected, ask: (1) Is there a specific action that will be taken based on the findings? (2) Are the necessary resources available to implement that action? (3) Does the commitment to take that action exist?

Responsible – Benefits of data collection should outweigh the costs

Data collection should be matched with the available systems and resources for collection, such that benefits exceed costs. Costs include direct costs of data collection but also opportunity costs, i.e. money and time that could have been used elsewhere. Data collection should not exceed the ability of data analysis. Responsible data collection requires transparent processes, protection of individuals' sensitive information and proper documentation.

Transportable – Collected data should generate knowledge

Valuable lessons generated from evaluations should help build more effective programs by applying what has been learned (why a program works) either to the program itself in the future or to other similar programs. Transportability requires transparency, i.e. sharing learnings with others.

3.3 Analysis

Before an evaluator can analyze the collected data, they must first be stored and documented. Thorough *data management* is needed to ensure that the collected data remain secure, accessible and well-documented for both current and future use. A well-structured data system should have sufficient storage capacity, secure backup processes and appropriate hardware and software to support data integrity (Yoong et al., 2013). Regular backups should be stored separately from primary data to prevent loss in case of system failures. When working with sensitive data, especially involving people, strict security measures are needed. Access to individually identifiable information should be restricted and require proper authorization and verification. Where possible, identifiers should be anonymized to protect personal information, and access logs should be maintained to track who interacts with the data (Yoong et al., 2013).

¹⁴ For more details on how to develop effective survey items, we recommend reading chapter 3.3 in Yoong et al. (2013). There, the authors describe how to choose high-quality items according to the SMART principle, short for specific, measurable, attributable, realistic and targeted.

Comprehensive *data documentation* is also critical, as Yoong et al. (2013) note. Final datasets should be accompanied by detailed metadata, including variable definitions, sampling strategies, data collection methodologies and fieldwork details such as nonresponse rates. This ensures transparency, allowing users to accurately interpret, reproduce and apply the data for future evaluations. Proper documentation not only supports the integrity of the current evaluation but also serves as a valuable resource for future research and decision-making.

The *data analysis process* depends on whether the collected data are quantitative or qualitative. On a general level, analyzing and summarizing quantitative data involves calculating percentages and means (or other measures of central tendency). Presenting and visualizing data in a clear and understandable way is also an important part of the data analysis process. This can be done, for instance, by using tables, graphs or bar and pie charts. Analyzing qualitative data typically involves examining responses to open-ended questions, success stories or respondents' observations. A simple way to summarize qualitative data is by identifying common themes and categorizing data accordingly, e.g. by grouping quotes thematically (Berry and Sloper, 2016). See Felbermayr (2024) for details on data analysis techniques for qualitative data and Lorenz et al. (2025a) for quantitative data.

Instead of going into further detail on data analysis processes, we close this section by discussing potential biases and noise that may arise during analysis and when interpreting the findings (Picciotto, 2022, Kahneman et al., 2021):

Bias refers to systematic deviations from objective judgment, often stemming from cognitive tendencies, ideological influences and structural dependencies. One major source of bias is *confirmation bias*, which occurs when evaluators prioritize evidence that supports pre-existing theories or stakeholder expectations while dismissing contradictory data. Another form of bias arises from *motivated reasoning* and *conflicts of interest*, particularly when evaluators are financially or institutionally dependent in their work. In such cases, they may be (indirectly) incentivized to align their findings with relevant stakeholders' expectations rather than conducting independent assessments. *Framing effects* also play a crucial role in shaping conclusions. Even the way survey questions are formulated can influence responses and interpretations, subtly steering judgments in a particular direction. Similarly, *status quo bias*, which is the tendency to favor existing practices and resist change, can prevent evaluators from considering innovative but potentially disruptive alternatives. As mentioned at the beginning of this paper, if political and public objectives compromise evaluation integrity, such biases can be indirectly reinforced.

While bias leads to systematic errors, *noise* represents inconsistent and unpredictable variations in judgment. This occurs when different evaluators, or even the same evaluator at different times, arrive at divergent conclusions based on the same data. One key source of noise is *bounded rationality*, which highlights the limitations in human cognitive capacity when processing complex information. Weighting multiple competing factors can be challenging and lead to inconsistent judgments that are influenced by external pressures, social constraints and information processing costs. Another common issue is *base rate neglect*, where evaluators fail to consider the underlying probability of an event when interpreting results. This can lead to overestimations of a program's success, particularly when positive outcomes are observed in a low-prevalence population. Similarly, the *illusion of validity* skews assessments when evaluators develop strong confidence in their judgments despite insufficient or contradictory evidence, such as when relying on confirmation bias. A further challenge is the influence of *availability cascades*, in which repeated exposure to certain interpretations or perspectives leads to their widespread acceptance,

even if they are flawed. Positive-feedback mechanisms, e.g. through social dynamics and media reinforcement, can amplify these cascades.

Evaluation processes involve assessing a program's *merit* (doing things right), *worth* (doing the right things) and *value* (overall societal benefit) (Scriven, 1991). Each of these dimensions presents challenges that contribute to bias and noise. Merit assessments are often based on predefined standards, but when those standards are set by dominant power structures, they may reinforce existing inequities which can be (socially) counterproductive. Worth assessments address whether a program serves its intended beneficiaries but are inherently subjective, as different stakeholders may have conflicting priorities. Value assessments, which seek to integrate merit and worth into a broader judgment of social impact, are the most contentious, as they require trade-offs between competing goals, such as short-term efficiency versus long-term sustainability.

It is important to note that absolute objectivity in evaluation is unattainable. Evaluators must recognize the limitations of their knowledge and be aware that their judgments are inherently relative and tentative. By acknowledging the inevitable presence of bias and noise, evaluators can adopt strategies to mitigate negative effects, such as diversifying methodological approaches, fostering independence from vested interests and continuously reassessing assumptions. This, in turn, highlights the importance of reviewing program theory, for example, through evaluability assessments (see section 2). Ultimately, while evaluation cannot produce universal truths, it can provide meaningful insights that support informed decision-making for program improvement and accountability and, more broadly, serve the interests of society and the public good.

3.4 Communication

The final phase of the evaluation cycle focuses on documenting and communicating the evaluation findings. The ultimate goal is to use these findings to revise the program.

Like Rossi et al. (2019) point out, a *communication plan* is a crucial component of an evaluation and, strictly speaking, should already be considered during the planning phase. Its purpose is to determine how evaluation findings can effectively guide actions and attitudes in alignment with the evaluation's purpose while maintaining transparency about the process, data, findings and limitations, as well as ensuring evaluator independence. At its core, the communication plan outlines an agreement between the evaluator and key stakeholders (e.g. sponsors) on how evaluation findings will be released. A key consideration is determining who has the right to release the findings, as organizations commissioning evaluations often control the timing and extent of public disclosure. To safeguard evaluation integrity, evaluators may seek to retain intellectual property rights over the data, ensuring that results are communicated in full, regardless of whether the findings are favorable. Establishing clear agreements on information ownership and disclosure conditions during the evaluation's planning phase is essential to avoiding conflicts later, particularly if the results are sensitive or controversial. Without a well-defined communication plan, negotiations about releasing findings can become challenging, especially once the results are known.

Communication of evaluation findings is embedded in a complex environment, shaped by *multiple stakeholders*, who often have diverging, and at times conflicting, interests. These stakeholders, typically those with a direct and visible stake in a program, may hold differing views on both the legitimacy of the evaluation process and the potential consequences of its findings. Given these competing perspectives, evaluators must carefully navigate their relationships with stakeholders, ensuring that their role remains

one of providing empirical evidence rather than making direct judgments. However, the “*distinction between making judgments and providing information on which judgments can be based is useful and clear in the abstract but often difficult to make in practice*” (Rossi et al., 2019, p. 294).

Evaluation involves *political* and *public interests*; therefore, evaluators must recognize that their contributions are just one of many factors influencing stakeholders’ decision-making. Differing political or institutional perspectives can create tensions that evaluators must anticipate and plan for. However, “*the responsibility of the evaluator is not to take one of the many perspectives as the sole legitimate one but, rather, to be clear about the perspective from which a particular evaluation is being undertaken while giving recognition to the other perspectives*” (Rossi et al., 2019, p. 295). Stakeholder reactions to evaluation results can be challenging. Even well-grounded, rigorously conducted evaluations may be perceived as biased or arbitrary, especially if the findings contradict stakeholders’ interests, which leads to resistance or controversy. Even those who commission an evaluation may reject or discredit the results, potentially undermining the evaluator’s professional standing or future opportunities. This puts evaluators in a precarious position, particularly when sponsors or influential stakeholders withdraw support due to unfavorable results.

Another major difficulty arises from *communication barriers*, as evaluation terminology, rooted in social science disciplines, can be difficult for nonexpert audiences to grasp. Ensuring that findings are accessible and clearly communicated is crucial for fostering understanding and encouraging the practical application of results. The diverse backgrounds of stakeholders make this a persistent challenge (Rossi et al., 2019). “*Documenting and communicating your results is the key to ensuring that interested audiences – including program participants and managers, members of your organization, funders and organizations that operate similar programs – can learn from what you found in your evaluation and apply the results*” (Yoong et al., 2013, p. 214). Evaluation results are not only valuable to program stakeholders but also contribute to generating knowledge for academics in evaluation science, as described in section 1. However, “learning from knowledge” can be compromised if not all evaluation findings are documented and communicated. The tendency to publish only results that show large program effects, while neglecting those with weak, negative or no effects, is known as *publication bias* (Yoong et al., 2013). Evaluating financial education programs, particularly their impact, provides valuable insights for future program development. As Yoong et al. (2013) emphasize, it is essential that the evidence base includes all types of findings, not just those showing large effects, to present a balanced and accurate picture of actual program outcomes.

Based on a comprehensive communication plan, writing the report can start once the evaluation findings are available. One way of writing up the findings can be in the form of a *formal evaluation report*. Such a report is often required by program oversight decision-makers and program sponsors. Generally, it is important to approach the report strategically with a view to ensuring it effectively reaches and engages the intended audiences. The structure of a formal report should be clear and purposeful, guiding readers through the key findings, analysis and implications in a way that is both accessible and actionable. We summarize the six main parts of a formal report as described by Yoong et al. (2013) in the following.

First, although mostly written at the end of the report, the *summary* comes at the very beginning of the final report. It should provide an overview of the context, evaluation approach, key findings and recommendations for action. The summary should be brief and self-contained, ensuring that even those who do not go through the full report grasp the essential points.

Second, the *introduction* serves as a roadmap, giving readers an overview of what to expect in the report. It should provide an outline of the program and its purpose, the evaluation’s objectives, the evaluation stakeholders and their relationship to the evaluators and the methodological approach.

Third, the report should contain a detailed *program description*, which presents the program's history, background and development, the program's objectives and goals and also the activities offered and targeted recipients.

Fourth and following this section, the *evaluation design and methods* should be described. This section should clearly state the evaluation questions and the data collection methods used to address these questions. In addition, the analysis methods for each type of data collected should likewise be outlined.

Fifth, presenting *evaluation results* requires synthesizing data into a meaningful narrative. To make it easy for readers to follow, this section should align findings with the evaluation questions (or specific evaluation tasks). In cases where quantitative analysis plays a key role, the writing process may also involve making interpretative decisions to extract clear themes or messages that translate into actionable insights. To improve readability, incorporating visual elements (see section 3.3) is particularly effective. To ensure clarity, any visuals must be accompanied by well-written explanations. Having unclear visual representations or descriptions can prove counterproductive, as they may confuse readers and disrupt the flow of the report. Apart from clarity, sensitivity in presenting findings is critical, especially if the evaluation reveals shortcomings in program performance, staff effectiveness or operational efficiency. The results should be framed objectively and tactfully, with the authors avoiding language by which stakeholders might feel directly criticized. Policy audiences and funders might benefit from a highly condensed version of the result section, which is sometimes referred to as "microcontent." This might include a brief, bulleted summary alongside a compelling visual, designed to fit on a webpage or be used as a standalone document.

Sixth, the *conclusions and recommendations* section is particularly important for audiences expected to take action based on the report. Naturally, this section should build on the results. In other words, authors should ensure that the conclusions are directly supported by the preceding findings (vs. unfounded statements). Additionally, recommendations should not only address current issues but also point to future steps or improvements. Clearly articulating the connections between findings, conclusions and suggested actions supports the credibility and utility of the report.

Once the report is finalized, the final step is ensuring it reaches the right audiences, as underlined by Yoong et al. (2013). This involves selecting the most effective *dissemination channels*, which can include in-person presentations, print distribution and digital platforms. In today's media-rich environment, leveraging multiple modes of communication can be demanding but enhances the report's reach and impact. Overall, careful attention should be given to how and where the findings are shared to maximize engagement and influence decision-making.¹⁵

4 Summary and concluding remarks

Program evaluation plays a vital role in determining the effectiveness of financial education programs. This paper outlines the evaluation cycle, covering the key phases of planning, implementation, analysis and communication. It focuses on the methodological and strategic considerations necessary for a thorough and objective evaluation.

In the planning phase, defining a clear evaluation purpose and setting measurable objectives are essential. The evaluation design should align with the program's stage of development, distinguishing between formative and summative evaluation goals. Formative evaluation focuses on assessing program imple-

¹⁵ A more elaborate description of the communication process is beyond the scope of this paper. Interested readers may refer to chapter 14 in Yoong et al. (2013) and the study by da Costa (2012) for further insights.

mentation and refining processes, while summative evaluation examines program outcomes and impact. As broader categories, process and impact evaluation consider both implementation quality and program effectiveness, aligning with the needs of different decision-makers. Process evaluation is primarily used by program administrators to refine operations, whereas impact evaluation informs policymakers and funders about long-term program success.

During the implementation phase, careful data collection is crucial to ensure reliable and valid findings. The choice of evaluation methods, be it qualitative, quantitative or mixed methods, must be guided by the evaluation questions and available resources. Sampling strategies, ethical considerations and logistical challenges also play a significant role in shaping the evaluation process. A well-structured approach to data collection, aligned with stakeholder needs and practical constraints, enhances the credibility of findings.

The analysis phase involves synthesizing collected data to derive meaningful conclusions. Quantitative analysis may include statistical comparisons of program outcomes, while qualitative analysis provides deeper insights into participants' experiences. Interpreting the findings is not without challenges. Biases, such as confirmation bias, motivated reasoning and framing effects, can influence how results are presented and understood. Additionally, noise in evaluation judgments can lead to inconsistencies. Recognizing these potential pitfalls and applying rigorous analytical frameworks helps ensure that findings accurately reflect program effectiveness.

The final phase, communication, determines how evaluation findings are conveyed to stakeholders. A structured communication plan is crucial to ensure that results reach decision-makers, funders and other interested parties in a clear and actionable manner. A formal evaluation report summarizes the findings. To maintain objectivity and stakeholder engagement, careful attention must be paid to how results are framed, particularly with regard to challenges or unfavorable findings.

In conclusion, program evaluation is an ongoing, iterative process that enhances the effectiveness and accountability of financial education initiatives. How can decision-makers generate meaningful evidence to inform future program development? They can do so by systematically following the evaluation cycle – from planning and implementation to analysis and communication. Moreover, evaluations contribute to the broader research field, thus helping establish best practices and methods for assessing financial education programs.

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Theory of change – understanding the link between a program’s design and its goals

Sandra Mauser¹

This paper explores the role and relevance of a theory of change in the context of designing, implementing and evaluating financial education interventions. Building on the broader framework of program theory, it examines how a theory of change outlines the logical sequence through which an intervention is expected to lead to desired outcomes. The paper provides a practical guide to developing a theory of change, including its key components – such as the initial situation, assumptions, external factors, inputs, activities, outputs, outcomes and impact. Special attention is given to causal pathways and the assumptions underlying each link in the chain. Furthermore, the paper highlights the importance of acknowledging the probabilistic nature of causality, the role of external influences and the distinction between ex ante planning and ex post verification. A well-developed theory of change helps clarify expectations, supports stakeholder alignment and enhances the rigor of impact evaluations by making the underlying logic and assumptions explicit.

The relevance of program theory for evaluation research has already been discussed in Mauser (2025). An essential element of program theory is the theory of change, which describes the mechanisms through which a program or intervention is intended to operate.² In this paper, we examine the definition and purpose of a theory of change, its development and key components as well as how causal pathways are hypothesized.³

1 What is a theory of change? What is its purpose?

“The link between a program’s design and its goals can be understood through the program’s theory of change. Every financial capability program should be based on an underlying theory of change – which simply refers to a logical, sequential argument for how and why an intervention will deliver the desired results, together with any assumptions necessary at each step of the way” (Yoong et al., 2013, p. 36).

A theory of change is a conceptual framework that outlines the structure of an intervention and its intended goal (Yoong et al., 2013; University of Wisconsin-Madison, 2023; Money & Pension Service, 2023): It provides a logical, sequential chain of reasoning, starting with the allocation of resources (inputs).

¹ Oesterreichische Nationalbank, Financial Literacy and Culture Division, sandramauser@hotmail.com. Opinions expressed by the authors of studies do not necessarily reflect the official viewpoint of the Oesterreichische Nationalbank, the Bank of Greece or the Eurosystem. This paper is part of the *OeNB Financial Literacy Evaluation Series*. The series aims to inform researchers, policymakers and educators about the current state of research on financial literacy and education and to provide guidelines for designing and implementing comprehensive evaluation studies. The series was developed by the OeNB in collaboration with the Bank of Greece. For details and further publications of the series, see <https://www.oenb.at/en/Publications/financial-literacy/financial-literacy-evaluation-series.html>

² The terms *program* and *intervention* are often used interchangeably in evaluation literature. In this paper, we follow a practical approach for clarity in writing: program will be used, for instance, when we refer to program theory, the program’s theory of change, program managers or program design, while intervention will be used otherwise.

³ The literature offers a variety of names for the theory of change (e.g. logic model, results chain, outcome pathway, action theory, implementation theory, impact theory or even program theory). However, no general agreement on terms or definitions exists (Mayne, 2015; Rossi et al., 2019). In this paper, we use the term *theory of change* and refer to a specific element within the broader framework of program theory.

These resources are used in specific activities, which in turn produce outputs. The outputs are expected to lead to desired outcomes, which ultimately contribute to the overarching goal (impact) of the intervention. A theory of change provides a structured overview of the intervention, enabling reflection on its individual components and fostering a better understanding of the relationships between them. Each component offers a clear point of reference for identifying underlying assumptions and potential risks, since it is crucial to embed the intervention within an appropriate context: (1) What is the initial problem or situation that the intervention aims to address? (2) What underlying assumptions inform the intervention? (3) What external factors could influence the intervention's success and what risks must be considered? In essence, a theory of change answers the key questions: What? How? Why?

Having developed a thorough and explicit theory of change is beneficial at each stage of an intervention – design, implementation, monitoring and evaluation (Dhillon and Vaca, 2018). A theory of change helps strengthen the design of an intervention by ensuring internal coherence and connecting strategies and activities with expected outcomes. It enhances understanding among those involved, aligning their perspectives and clarifying key assumptions that need to be considered. Additionally, a theory of change provides insights into how the intervention is expected to function effectively. Beyond the organization, it also improves communication with external stakeholders by clarifying their roles and level of involvement. All these beneficial steps also contribute to the implementation of the intervention (Dhillon and Vaca, 2018).

Further, a theory of change illustrates where monitoring efforts can be implemented (Yoong et al., 2013). *Monitoring* refers to the process of collecting and analyzing information on an intervention regularly to assess its progress against the original plan and defined milestones. It is a descriptive process that covers indicators for inputs, activities, outputs and sometimes outcomes. A program's theory of change can help determine which elements can be quantified and which indicators should be used for monitoring⁴ (Yoong et al., 2013).

As part of the program theory, a theory of change serves as a critical foundation for evaluation planning (Yoong et al., 2013; Dhillon and Vaca, 2018): Before evaluating an intervention, it is crucial to have a clear understanding of how the intervention is expected to generate change within the target group. A theory of change provides a detailed description of an intervention's activities and their expected effects on recipients' knowledge, attitudes, behaviors and decision-making processes. In other words, it answers the question: What changes are intended to result from the intervention's activities? By breaking down the intervention into its components, a theory of change allows for the formulation of hypotheses regarding causal relationships and expected effects. Therefore, evaluators will not only be able to answer whether the intended outcomes and goals were achieved but also why or why not, which is an important part of impact evaluation.

Overall, the program's theory of change is important to:

- gain a deeper understanding of the goals and mechanisms of the intervention;
- ensure that all stakeholders involved in the design, implementation, monitoring and evaluation processes share a common understanding of the intervention;

⁴ Monitoring provides regular data to program decision-makers and stakeholders, helping them make timely adjustments to the implementation of the intervention. Therefore, monitoring is closely related to process evaluation, often relying on the same infrastructure and data sources (Yoong et al., 2013). For details on monitoring, see chapter 4 in Yoong et al. (2013).

- identify existing evidence that links the intervention's activities to its defined goals and determine what additional data collection and measurement tools are needed;
- identify intended outputs and outcomes that should be considered in the monitoring and evaluation processes;
- formulate hypotheses regarding the effectiveness of the intervention, which can then be tested through evaluation.

The benefits of a clear and well-formulated theory of change serve diverse needs and interests of different stakeholders (The Annie E. Casey Foundation, 2022): (1) As described above, for evaluators the theory of change offers a foundation for evaluation planning and continuous improvement through learning, also serving the research community. (2) For people involved in program operations (e.g. program planners and decision-makers, staff, administrators), the theory of change functions as a strategic plan and a road-map to maintain alignment with the set goals. Additionally, it is a documentation and communication tool. (3) For major decision-makers of program oversight (e.g. program funders, political decision-makers, legislative committees and organizational leads), the theory of change provides a basis for external communication. It allows to clearly state what is expected to change when and how, and which priorities are set. (4) For people served or affected by the program (i.e. recipients and communities), a theory of change provides transparency, which helps build trust. Furthermore, communities can hold organizational leads accountable for the outcomes of interventions.

2 Developing a theory of change

In this section, we provide a stepwise practical guideline for a theory of change by describing the key components of a basic generic model (Yoong et al., 2013; University of Wisconsin-Madison, 2023; Money & Pension Service, 2023), as depicted in figure 1. Further, we discuss general considerations in the development of a theory of change.

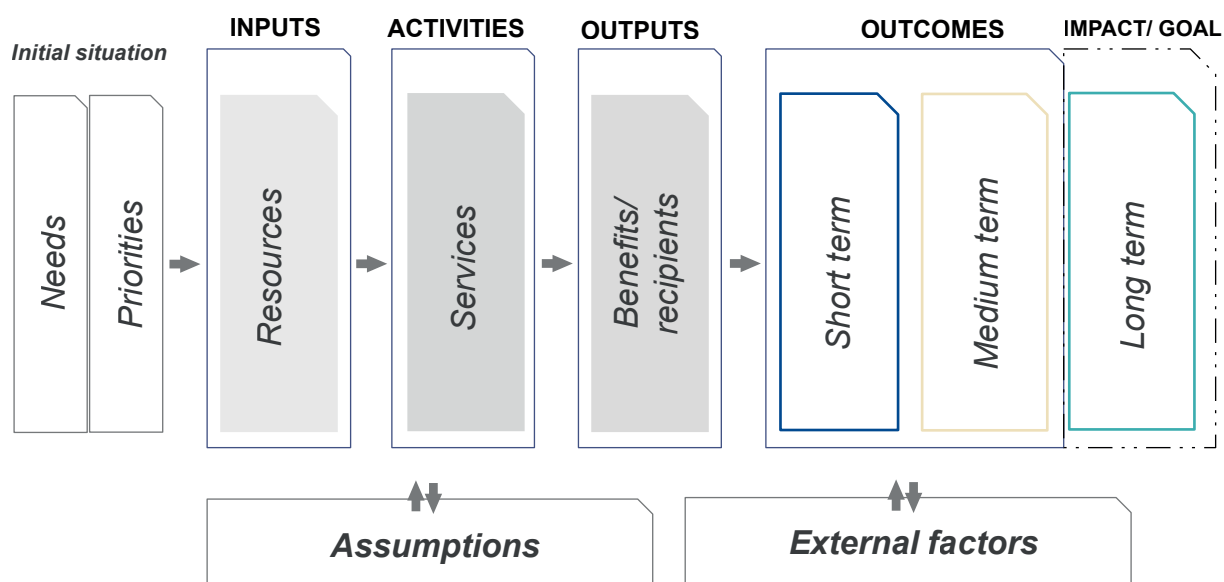
2.1 Practical guideline – key components of a theory of change

As a checklist, any theory of change – independent of the visualization or complexity of the used model – should address the following key questions:

- Who is the target audience of the intervention?
- What is the overarching goal of the intervention?
- What outcomes must be achieved to reach this goal?
- What activities (tasks, material development, etc.) will be carried out to generate the intended outputs?
- What intended outcomes can be achieved through the planned activities?
- What assumptions are made regarding how the intervention contributes to the desired outcomes? (This is crucial for hypothesis formulation in the evaluation.)

Figure 1

Basic generic theory of change



Source: Author's compilation.

Initial situation

The initial situation (or problem statement) describes the underlying problem, i.e. the reason why action is needed, as well as the associated priorities that guide the intervention's implementation. The following questions should be addressed (box 1).

Box 1

Causes and symptoms

- Why does the problem exist, and what does it look like?

Demand and need

- Who is affected by the problem? (Individuals, households, groups, communities, society at large)
- Why is the intervention needed?

Stakeholder engagement

- Who is involved in addressing the problem?

Experiences and data

- Which insights does the implementing institution already have about the problem and the target group?

Priorities and vision

- What is the institutional vision/mission?
- What expertise, resources and experience are available to set priorities?
- What criteria determine the priorities, and who contributes to this process?

Assumptions

Assumptions are the underlying factors that shape the theory of change. They include theories, beliefs, experiences and research (e.g. secondary literature) that inform how the intervention is expected to work. Assumptions influence decision-making and intervention design, and they should address the following elements (box 2).

Box 2

- Problem itself
- Available resources
- Way the intervention operates
- Expected outcomes
- Existing knowledge base
- Internal and external environment
- Target group, including their learning behavior, motivation and response to the intervention

External factors (including risks)

Interventions do not exist in isolation. They are embedded in an environment involving various external factors, including potential risks. External factors may relate to cultural context, economic structures, living circumstances and demographic characteristics of the target group, political frameworks and media influence (e.g. social networks). Importantly, an intervention may also impact the external factors, creating a potential reciprocal effect. The following aspects should be considered (box 3).

Box 3

- Resource allocation
- Implementation and design of the intervention
- Target group
- Speed and effectiveness of achieving outcomes

Potential risks

- Limited accessibility of the target group
- Resource constraints

Inputs

Inputs are the material and immaterial resources allocated to the intervention. They are under the control of the implementing institution and are measurable as well as quantifiable. See box 4 for typical resources.

Box 4

- Personnel
- Time
- Budget
- Equipment
- Materials
- Technology
- Volunteers or partner organizations

Activities⁵

Activities are the concrete actions carried out within the intervention, i.e. the answer to the question: what does the institution do and offer? Activities are controlled by the implementing institution and are both observable and measurable. See box 5 for examples of activities.

Box 5

- Workshops
- Services
- Conferences
- Funding opportunities
- Advisory sessions
- E-learning programs

Outputs

Outputs are the direct, immediate results of the activities. They are the tangible deliverables produced by the intervention, either related to the benefits or recipients of the intervention. Like inputs and activities, outputs are under the organization's control and are typically quantifiable and measurable. See box 6 for examples of outputs.

Box 6

- Number of workshops conducted
- Number of certificates issued
- Number and characteristics of recipients

Outcomes

Outcomes describe the actual changes (effects) the intervention aims to achieve. Unlike outputs, outcomes cannot be directly controlled by the implementing institution but can be influenced through the intervention's activities and outputs. Outcomes are categorized into short-term, medium-term and long-term. We discuss long-term outcomes separately in the next step. See box 7 for various outcomes, with specific examples of effects that financial education interventions may have on recipients.

Box 7

Short-term outcomes can be changes in

- knowledge (of basic financial concepts such as inflation or interest rates)
- awareness (of financial risks such as scams)
- attitudes (reduced anxiety related to managing personal finances)
- skills (learning how to create and manage a budget)

⁵ Not all theories of change distinguish between activities and outputs. Often, activities are considered part of the outputs, as the distinction is not always clear-cut. However, we have chosen to separate them, defining outputs as the tangible deliverables produced as a result of the activities.

- opinions (changing perception of value of financial education in daily life)
- motivation (feeling encouraged to start investing)
- aspirations (setting financial goals such as building an emergency fund)

Medium-term outcomes can be changes in

- actions (opening a savings account with regular saving amounts)
- behaviors (comparing prices and avoiding impulsive purchases)
- applications (using mobile banking tools to manage finances)
- decision-making (choosing to save for retirement)
- strategies (developing a plan to reduce debt and build financial security)
- social interactions (discuss financial matters openly with family and friends)

Impact/Goal

Impact refers to the long-term overarching goal of the intervention. It describes not only changes of individual well-being but also the broader societal, economic or environmental changes that the intervention seeks to achieve. Box 8 includes some examples, specifically potential impacts of financial education interventions.

Box 8

Social improvements

- increased financial well-being (reduced financial stress)
- enhanced financial inclusion

Economic advancements

- Reduced debt levels
- Increased entrepreneurship

Societal transformations

- Reduction in poverty
- Improved consumer protection

Environmental sustainability

- Reduced waste and overconsumption
- Ethical and green investing

2.2 General considerations for developing a theory of change

Developing a theory of change includes various general challenges and important considerations:

(1) **Terminology** (The Annie E. Casey Foundation, 2022): In any theory of change, language and terminology should be clear and consistent – especially when distinguishing between actions (things being done) and changes (the results of those actions). *Inputs* and *activities* refer to actions, while *outputs*, *outcomes*, *results*, *goals*, *impacts* and *indicators* describe changes. Some terms – such as *objectives*, *accomplishments*, *measures* and *metrics* – can refer to either actions or changes, depending on context. Therefore, it's important to clarify exactly what is meant.

(2) **Changes over time** (Mark, 2023): A common pitfall in working with theories of change is treating them as fixed blueprints. In reality, a theory of change should be understood as a living framework,

subject to revision as circumstances change. Financial education interventions, like other social interventions, operate within dynamic environments, and both contextual factors and program implementation itself are likely to evolve over time. The challenge lies in the unpredictability of such changes – stakeholders cannot always anticipate what will change or when. Ongoing dialog with the stakeholders directly involved in the intervention can help identify significant shifts since the theory of change was first developed. Particular attention should be paid to changes that could influence the assumptions or causal pathways within the theory, as these may directly affect the program's effectiveness. In this way, regularly revisiting and updating the theory of change becomes essential to ensure its usability and relevance.

(3) **Heterogeneity** (Mark, 2023): The development of a theory of change often implies a single, unified pathway from actions to changes. However, such a perspective may overlook the diversity of experiences, contexts and responses among recipients of the intervention. Recognizing heterogeneity means acknowledging that outcomes may differ significantly across recipients and settings. These differences might stem from variations in how the intervention is implemented, the contextual environments in which it operates or how recipients engage with it. Failing to address this complexity can lead to overly simplistic models that neglect important nuances. By actively exploring heterogeneity, one can uncover multiple pathways to change, which can improve the explanatory power of a theory of change and also allows for more targeted and adaptive strategies in program design.

(4) **Theory knitting** (Mark, 2023): Program design often draws on a range of ideas and strategies to address complex problems, and this diversity should be reflected in the underlying theory of change. Rather than relying on a single theoretical perspective, the process of theory knitting involves weaving together concepts from multiple theories to build a more comprehensive framework. For instance, efforts to reach the target group might be informed by one theoretical approach, while behavior change strategies draw from another theory. This integrative process allows the theory of change to better capture the multifaceted nature of real-world interventions. By thoughtfully combining theories, one can construct a more robust and context-sensitive understanding of how and why change is expected to occur.

(5) **Validity** (Mark, 2023): Developing a theory of change is not only about articulating a logical sequence of events. It is also about ensuring that the conclusions drawn from that model are accurate and well-founded. Internal validity refers to the credibility of claims that the intervention caused observed changes. To protect against misleading conclusions, one must actively consider alternative explanations for observed outcomes and be open to evidence that challenges the initially proposed causal pathways. Doing so can help avoid *confirmation biases*, i.e. the tendency to favor information that supports existing beliefs. A well-developed theory of change should allow critical questions and promote ongoing inquiry into whether the intended pathways hold true under real-world conditions.

(6) **Frustration** (The Annie E. Casey Foundation, 2022): The development process itself can be time-consuming, complex and even frustrating for those involved. Especially when inclusive⁶, the process may involve divergent thinking, ambiguity and differing perspectives. This can create pressure to shortcut the

⁶ An *inclusive* theory of change promotes *equity* in the development process by encouraging the participation and influence of all the people directly affected by the intervention – especially those who are often excluded from contributing. The Annie E. Casey Foundation (2022, p. 8) describes equity as “Conditions that ensure those who have historically been excluded or oppressed have clear roles and authority in seeking change that affects them. ‘Advancing equity’ can include changes related to access to resources, quality of services and public or organizational policies that result in fair treatment, opportunity and outcomes.” Discussing inclusivity and equity within theory of change research and practice is beyond the scope of this paper. For readers interested in the topic, we recommend having a look at The Annie E. Casey Foundation (2022), Gates et al. (2024) and Lemire (2024).

process, which introduces risks of its own. For this reason, it is important to emphasize the value of a well-developed theory of change for all stakeholders, and to structure the development process in a clear, efficient and collaborative way.

(7) **Visualization** (The Annie E. Casey Foundation, 2022): Challenges in developing a theory of change also lead to limitations in its visualization. First, visualizing a theory of change can be misleading, as expected causal pathways may appear overly predictable. It is important to remember that a theory of change is not a predictive model of exact causal pathways; rather, it is based on underlying assumptions and influenced by external and risk factors. Ideally, the theory of change helps stakeholders identify critical questions for regularly measuring the progress of an intervention – and use those learnings to revise the theory as needed. Second, causal pathways are often depicted as straight lines, but in reality, change may follow a different or more complex path. Financial education interventions, like other social interventions, take place in dynamic social environments where progress can unfold in unexpected ways. Forward movement is not always linear – it may include setbacks, revisions or new directions. Capturing these nuances visually can be challenging. Therefore, it is essential to remember that a theory of change represents expectations about how positive outcomes might be achieved, not a definitive roadmap.

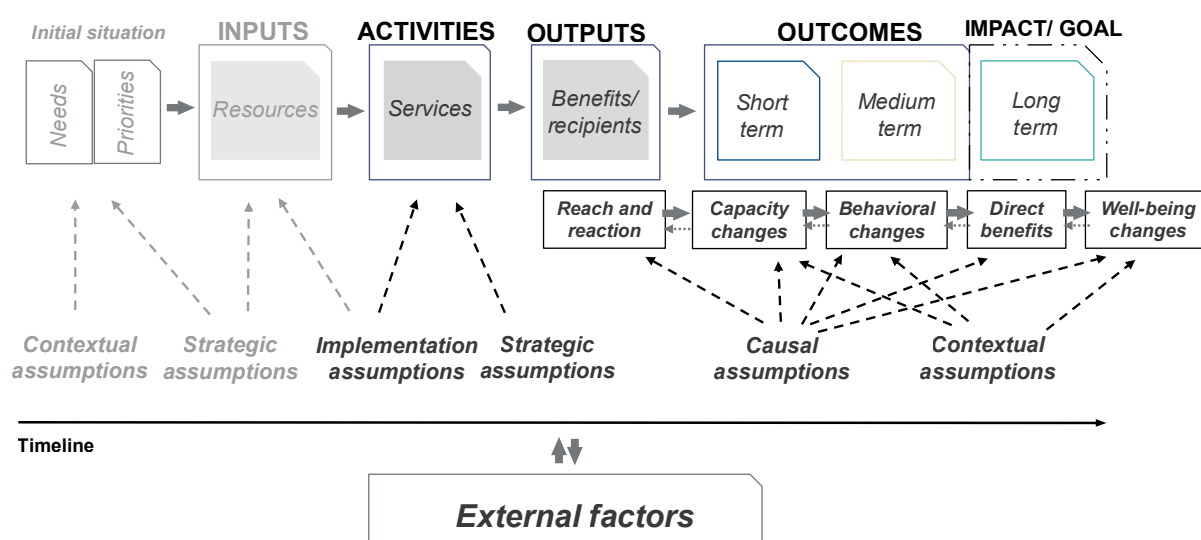
3 Causal changes

“To understand how and if an intervention is working, we need to understand how the activities of the intervention are expected to lead to the desired results – both (a) the causal pathway from activities to outputs to a sequence of outcomes to impacts and (b) the causal assumptions showing why and under what conditions the various links in the causal pathway are expected to work” (Mayne, 2015, p. 121). Figure 2 illustrates the causal pathway and its corresponding assumptions (causal and others), which are discussed in detail below.

In impact evaluation, the primary goal is to identify the actual changes that result from an intervention. However, evaluators may mistakenly assume changes have occurred when they have not, or attribute changes to the intervention when, in fact, the true causes lie elsewhere – or only partially within the

Figure 2

Causal pathways and assumptions



Source: Author's compilation.

intervention. Change is continually occurring, and it is crucial to recognize that some influencing factors may remain unknown or outside our awareness. Moreover, our understanding and assumptions of how change happens is inevitably shaped by our own perspectives and values. These limitations should be openly acknowledged and carefully considered when developing a theory of change (Dhillon and Vaca, 2018). Therefore, a thorough theory of change should not only depict the expected causal pathway but also make explicit the (causal) assumptions on which this pathway relies.

In general, *assumptions* are the often unspoken beliefs or expectations about how the world works and how change is expected to occur (van Es et al., 2015). In the context of a theory of change, they reflect our understanding of what needs to be in place for an intervention to lead to desired outcomes. These assumptions may stem from personal or collective belief systems, shaped by social, cultural or ideological backgrounds. Because they are so embedded in how we view the world, we may not always be aware of them – let alone question them. Yet, they influence how we interpret evidence, design strategies and judge the feasibility or relevance of interventions (van Es et al., 2015). For instance, we may assume that providing financial training will lead to improved financial knowledge and skills and that improved financial skills will translate into financial behavioral change. These are assumptions, not facts, and they require scrutiny. Making assumptions explicit is therefore a critical step in developing a theory of change, which helps us surface the underlying logic we take for granted.

Assumptions form the backbone of causal reasoning, i.e. *causal pathways*, in a theory of change (van Es et al., 2015; Mayne, 2015). They underpin the logic that connects activities to outcomes and ultimately to impact. If these assumptions do not hold true, even the most well-designed interventions may fail to produce the expected change. For example, if a strategy assumes that older people will participate in a digital financial education intervention specifically designed for them, but they do not, the entire pathway may break down. Recognizing and articulating assumptions allows for critical reflection, facilitates evidence-based planning and highlights where further research, stakeholder engagement or monitoring might be needed. Moreover, some assumptions carry more weight than others. Identifying which ones are most critical helps prioritize risk analysis and adaptive planning (van Es et al., 2015). This is particularly important in complex or dynamic environments where change processes are influenced by many interacting factors.

In practice, assumptions in a theory of change can be categorized into several types, which can appear at different points along the causal pathway (van Es et al., 2015; Mayne, 2015):

- **Causal assumptions:** These are the assumptions that explain why and under what conditions each step in the pathway is expected to lead to the next. For example, if the intervention is a workshop on basic concepts of finance for low-income households, we may assume that recipients who gain knowledge about budgeting and saving will not only understand the concepts but will also have the confidence, opportunity and social support to apply them in their daily lives. This assumption could be broken down into several layers, namely (1) cognitive capacities (recipients understand and retain the financial concepts taught), (2) motivation (recipients feel motivated and confident enough to change their financial behavior), (3) social/cultural acceptance (saving money is not perceived negatively or mistrusted within the household) and also (4) contextual/structural assumptions, which are discussed in the next paragraph.
- **Contextual assumptions:** These relate to the broader environment in which the intervention takes place. They include assumptions about the political, social or economic context, the behavior and motivations of the target group (and stakeholders) or even the absence of disruptive events like natural

disasters or regime changes. While some may seem obvious or outside our control, others are critical to success and warrant close attention. In the example of the workshop, we assume that the local economic environment is stable enough to allow recipients to act on the knowledge gained – meaning that they have at least some disposable income, regular sources of income and access to basic financial services, which enable them to budget and save. This type of assumption relates to external factors that shape whether the intervention has a realistic chance of success. The success of the intervention could fall short if, for instance, inflation is so high that saving is not feasible or financial services are unavailable in the recipients' region.

- **Implementation assumptions:** These relate to the operational aspects of the intervention – for instance, that staff will be trained, that activities will be implemented on time or that funding will be secured. While essential for project management, these assumptions are generally less central to the causal logic of change and are often made implicitly. However, if an impact evaluation fails to identify positive causal changes, it can be relevant to examine whether key implementation assumptions were not met.
- **Strategic assumptions:** These underpin the selection of particular approaches or strategies. For example, we might assume that public awareness campaigns are an effective tool for shifting attitudes or that participatory governance mechanisms can lead to greater accountability. These reflect our beliefs about what works and should be critically examined in light of existing evidence or stakeholder perspectives. For example, we assume that providing financial education through community-based workshops is an effective strategy to improve budgeting and saving among low-income households. This assumption reflects a belief about what kind of approach is likely to work best in the given context. It is not about a specific causal step but rather about the choice of intervention strategy itself. It could be based on prior experience, stakeholder input or theoretical reasoning – but it still needs to be questioned.

As described in section 2, a well-constructed theory of change outlines a sequence of changes, from activities to long-term outcomes (impact).⁷ Each link in this *causal pathway* represents a different type of change, each dependent on distinct assumptions (Mayne, 2015):

- **Activities:** These are the planned actions undertaken as part of the intervention (e.g. workshops). While they are under the direct control of implementers, assumptions still apply. For example, the assumptions are that the activities are designed based on sound evidence, are appropriate for the target group and are delivered with sufficient quality and consistency. These assumptions are often implicit but crucial, as poorly designed or executed activities can undermine the entire causal pathway.
- **Outputs** (related to benefits): These are the immediate change resulting from the activities of an intervention (e.g. number of workshops held or materials distributed). The assumption is that these outputs will reach the target group.
- **Reach and reaction** (outputs related to recipients): This refers to who receives and engages with the outputs. Key assumptions here include that the right target group is reached and that they view the outputs as relevant and valuable.

⁷ Typically, the initial situation – defining needs and setting priorities – and the required inputs are not considered part of the causal pathway. However, assumptions still play a role at these early stages of a theory of change, as depicted in figure 2.

- **Capacity changes** (short-term): These involve shifts in knowledge, attitudes, skills or opportunities among the recipients. Assumptions here relate to whether the outputs (benefits) are comprehensible, culturally appropriate and aligned with prior experiences or values.
- **Behavioral changes**⁸ (medium-term): These are actual shifts in practices or decision-making. Crucial assumptions include social acceptability, access to resources and the perceived utility of adopting new behaviors.
- **Direct benefits** (long-term): These represent tangible improvements in people's lives – such as increased income – stemming from behavioral changes. Assumptions here often relate to whether new practices lead to real gains under current conditions.
- **Well-being changes** (long-term): These are the long-term impacts, such as improved quality of life or greater empowerment. They depend on a chain of prior changes holding true and often on broader structural or systemic factors remaining favorable.

Table 1 summarizes the linkage between the pathway and corresponding assumptions.

Table 1

Causal pathway and corresponding assumptions

Causal pathway	Type of change	Assumptions
Initial situation	Problem definition, needs and priorities	• Contextual assumptions (e.g. accurate understanding of needs, political/social context) • Strategic assumptions (e.g. choosing the right focus area)
Inputs	Resources allocated (staff, funding, materials)	• Implementation assumptions (e.g. sufficient and timely resources) • Strategic assumptions (e.g. selecting suitable partners or tools)
Activities	Actions taken in the intervention	• Implementation assumptions (e.g. staff trained, activities happen as planned) • Strategic assumptions (e.g. chosen approach is effective)
Outputs	Services and goods delivered	(see reach and reaction)
Reach and reaction	People receiving/using the outputs and their reactions	• Causal assumptions (e.g. outputs reach and engage the recipients)
Capacity changes	Changes in knowledge, attitudes, skills, opportunities	• Causal assumptions (e.g. recipients have sufficient cognitive capacities and motivation) • Contextual assumptions (e.g. relevance, cultural appropriateness of outputs)
Behavioral changes	Actual changes in practice or behavior	• Causal assumptions (e.g. recipients recognize utility of adopting new behaviors) • Contextual assumptions (e.g. social norms, access to resources)
Direct benefits	Tangible improvements in recipients' lives	• Causal assumptions (e.g. behaviors actually result in direct benefits)
Well-being changes	Long-term cumulative impacts	• Causal assumptions (e.g. direct benefits improving quality of life) • Contextual assumptions (e.g. enabling environment remains stable over time)

Source: Author's compilation.

To close this section, we shortly discuss the probabilistic nature of causality and its implication for a theory of change (Mayne, 2015):

⁸ Behavioral changes are not easily achieved. Extensive research has explored the challenges and underlying theories. For readers interested in this topic, we recommend referring to Darnton (2008).

Understanding causality within a theory of change framework involves acknowledging that change rarely results from a single factor acting in isolation. Instead, causality is often probabilistic, meaning that an intervention increases the likelihood of a particular outcome occurring, but does not guarantee it. Therefore, a cause can be understood as something that raises the probability of an outcome, rather than something that invariably produces it. In this light, it becomes useful to think in terms of *probabilistically necessary* and *probabilistically sufficient* causes. A *likely necessary assumption* refers to a condition or factor that almost always needs to be in place for a specific change to occur – without it, the causal link is unlikely to hold. A *likely sufficient assumption* refers to a condition that, on its own or as part of a package, is often enough to produce the expected change. Within a theory of change, many causal assumptions fall into the category of likely necessary assumptions, because they describe conditions that are critical for each step of the causal pathway to function as intended.

It is also important to, once again, recall the role of *external factors* in shaping outcomes. A theory of change models the contribution of an intervention to an outcome, rather than claiming full causality – since in most real-world settings, other factors are also at play. These external factors can either enhance or undermine the intended effects and may account for part (or even all) of the observed outcome. Significant external risks that could affect outcomes are often also embedded in the causal assumptions, which highlights the need to account for them.

The distinction between *ex ante* and *ex post* thinking is also critical when working with theories of change. *Ex ante*, the theory of change serves as a prediction of effectiveness – a forward-looking hypothesis based on the assumption that, if the intervention is implemented as planned and the assumptions hold, it will contribute to the intended outcomes. It sets out a reasoned expectation of how change will unfold. *Ex post*, after implementation, the theory of change becomes a tool for causal testing. At this stage, evidence is used to verify whether the expected changes occurred and whether the key assumptions were realized. If no major external influences are identified and the results align with the expected causal pathway, it may be reasonable to conclude that the intervention was not only contributory but likely sufficient to explain the observed changes.

4 Summary and concluding remarks

A theory of change is a critical tool for understanding the link between a program's design and its intended goals. It outlines the pathway through which resources and activities are expected to bring about changes, offering a logical structure that supports both planning and evaluation of interventions. In this paper, we examined what constitutes a theory of change, its core components and how it can be developed and applied effectively – particularly in the field of financial education.

The development process begins with clearly articulating the initial situation, including the underlying problem, needs and institutional priorities. From there, assumptions, external factors (including risks) are identified to contextualize how the intervention is expected to work. Inputs and activities describe what the intervention does, while outputs, outcomes and impacts represent the changes resulting from the intervention. Throughout this pathway, assumptions – especially causal ones – are essential for explaining why one step is expected to lead to the next. These assumptions must be made explicit and continually revisited to ensure the logic holds in practice.

We further explored the complexity of causal change, emphasizing that interventions typically contribute to outcomes rather than cause them in isolation. Causality in a theory of change is often probabilistic, meaning that interventions raise the likelihood of certain outcomes under specific conditions. This

reinforces the importance of identifying both likely necessary and likely sufficient assumptions. Additionally, recognizing the influence of external factors helps clarify the limits of attribution and the scope of program impact.

Finally, we highlighted the importance of distinguishing between ex ante and ex post perspectives. Ex ante, a theory of change serves as a prediction of how change is expected to occur, while ex post, it becomes a tool for testing that causal hypothesis. A robust theory of change not only strengthens internal alignment and transparency but also supports a more rigorous and meaningful evaluation process. As such, it is an indispensable element of evidence-based intervention design and continuous learning.

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Data privacy and ethics in financial literacy evaluation research

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Like all research involving individuals, evaluation research presents specific ethical and data protection challenges that must be considered throughout the entire evaluation process, from the evaluation design to the final publication. In this introductory article, we provide an overview of the key legal and ethical aspects that are typically relevant for the evaluation of financial literacy interventions. To do so, we present important legal terminology and regulations alongside fundamental ethical principles. Furthermore, we suggest potential strategies for addressing ethical and data protection challenges.

Since evaluations of financial literacy interventions typically involve individuals, researchers must meet certain *legal* and *scientific* requirements. In the EU, the legal requirements primarily pertain to data protection, governed by both national and supranational laws, with the General Data Protection Regulation (GDPR) serving as the cornerstone. In contrast, in the US, there is no single equivalent to the GDPR. Instead, data protection is addressed through a sectoral approach and state-level regulations. Globally, countries maintain their own data protection frameworks, and even within the EU, member states incorporate additional national provisions, including specific derogations or margins of appreciation allowed under the GDPR.² Given the complexity and diversity of these legal frameworks, this article focuses primarily on the EU's GDPR while referencing relevant national provisions of Austria where applicable.

The scientific requirements mirror the central principles of research ethics (e.g. the principle of voluntary participation), which should guide researchers' actions. While the GDPR is grounded in ethical principles such as privacy and transparency, it transforms these principles into binding legal obligations that researchers have to meet. Therefore, data protection and research ethics overlap, but are not necessarily identical. For example, even if a researcher's actions are adequate from a data protection perspective, they may still be questionable from a research ethics perspective (Unger, 2018). In contrast to data privacy laws, the field of research ethics is less standardized, with guidelines not being universally established at the national or supranational level. Instead, ethical standards – outlined in codes of ethics – can vary significantly. This variation arises because ethical considerations differ greatly based on who is being researched, the methods used and the specific research contexts.

In the following article, we provide an overview of data protection law and research ethics principles that are relevant for researchers working on financial literacy evaluation projects. The laws and principles presented are not exclusive to financial literacy evaluation research but rather apply broadly, to all research.

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² In Austria, for example, researchers also need to comply with the Austrian Data Protection Act (Datenschutzgesetz – DSG) and the Austrian Research Organization Act (Forschungsgesetz – FOG).

Our contribution is adapting these general ethical standards and legal obligations to the context of financial literacy evaluation, highlighting the aspects most pertinent to evaluators in this field. Furthermore, ethical and legal considerations often receive limited attention in research in general (Felbermayr, 2023) and in existing financial literacy evaluation handbooks in particular.

This paper is structured as follows: Section 1 examines key aspects of GDPR compliance, addressing two crucial legal questions: When does the GDPR apply and what actions are required to ensure compliance? Section 2 explores core research ethics principles, highlights ethical challenges specific to financial literacy evaluations and offers strategies to address these challenges effectively.

1 Data privacy and the General Data Protection Regulation

In the context of evaluation research, it is often necessary to collect personal data. That means researchers need to pay special attention to both data protection regulations and ethical considerations when carrying out evaluations. To ensure adherence to data protection laws, particularly the GDPR, it is advisable to take the following steps before and throughout the evaluation process:

- **Understand the fundamentals of the GDPR:** The EU's GDPR is laid out in Regulation (EU) 2016/679 of the European Parliament and of the Council of the European Union. An essential first step involves getting familiar with key legal terminology as well as key principles and obligations for researchers. A concise overview of GDPR fundamentals will be provided below.
- **Conduct a data mapping exercise:** For researchers, it is critical to document the types of personal data they are planning to collect, the purposes they will use them for, the data recipients (if any), storage locations and retention periods, and anyone who will have access to these data (Article 30 GDPR). This mapping exercise is vital for identifying potential compliance issues early on. It is also helpful for planning the technical and organizational safeguards necessary to protect personal data effectively and in compliance with the GDPR.
- **Consult data protection officers or GDPR experts:** Given the complexity of data protection regulations, especially in projects involving multiple institutions or sensitive data, it is advisable to seek expertise from data protection officers (DPOs) or experts in data protection law. Many institutions offer such legal expertise. Expert guidance can be invaluable in navigating the nuances of GDPR compliance and of national law, as well as in implementing appropriate safeguards. Finally, if there is a DPO, they must be consulted (Article 38 para 1 GDPR).
- **Incorporate data protection by design and by default:** From the outset of your project, integrate GDPR principles and compliance with data subjects' rights (see below) into the planning and execution phases. The GDPR emphasizes the accountability principle, requiring organizations to take a proactive stance on data protection. This means designing your research project in a way that privacy and data protection measures are embedded at every stage, ensuring compliance as a fundamental aspect of an evaluation project.
- **Assess the risk for data subjects:** Certain types of data processing, particularly those that rely on new technologies, are likely to result in a high risk to the rights of the data subjects. If a research project involves such data processing methods, a data protection impact assessment (DPIA) is required prior to processing. This assessment comprises a detailed analysis of all the risks and threats that could lead to a breach of any obligation laid down by the GDPR.

1.1 GDPR basics

The GDPR aims to protect *personal data* and applies to both private and public entities in the EU that *process* personal data. Personal data are any pieces of information relating to an identified or identifiable *natural* person. From this follows that data that does *not* qualify as personal data does *not* fall within the scope of the GDPR (i.e. anonymous or anonymized data or data relating to legal persons). In the table below, we define key GDPR terminology (Articles 4 to 10 GDPR; Mondschein and Monda, 2019; Wilms, 2019).

Table 1

Key GDPR Terms

Term	Definition
Processing	Any operation performed on personal data, including accessing, collecting, saving, changing, analyzing, using, sharing, deleting, publishing, archiving data for follow-up research, etc.
Secondary Processing	Processing of personal data that were originally collected for a different purpose, such as data from another research project.
Personal data	Any information related to an identified or identifiable natural person (individual), also called data subject.
Identifiable Person	A person who can be identified with a serious possibility, either directly through direct identifiers or indirectly through indirect identifiers.
Direct identifiers	The International Organization for Standardization (ISO) interprets direct identifiers as "data that can be used to identify a person without additional information or with cross-linking through other information that is in the public domain" (ISO 25237, 2017, section 3.21). Information in the public domain includes information that can be easily found on the internet. Examples for direct identifiers are typically names, addresses, social security numbers, personalized email addresses or phone numbers (Polonetsky et al., 2016).
Indirect identifiers	Indirect identifiers are data points that can reveal an individual's identity when combined with other information. These typically include sociodemographic details like date of birth, age, gender, income, geographic location (e.g. postal codes, census areas) or other characteristics that, when aggregated, may lead to the identification of a person (Polonetsky et al., 2016). However, this does not necessarily imply that variables such as income, age and gender always function as indirect identifiers. A person's identifiability depends on the specific context and the size of the population involved in the evaluation project.
Sensitive personal data	Sensitive personal data involve a greater level of risk to the data subjects and thus require an additional legal basis (Article 9 GDPR) and supplemental privacy measures if processed (e.g. a data protection impact assessment). Sensitive data categories include racial or ethnic origin, political opinions, religious or philosophical beliefs, trade union membership, genetic data, biometric data for the purpose of uniquely identifying a person, health data and information about a person's sex life or sexual orientation, among others.

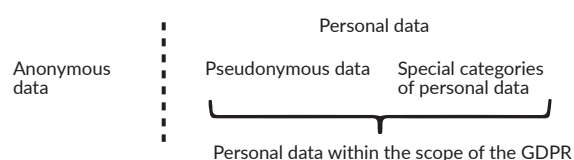
Source: Articles 4 to 10 GDPR; Mondschein and Monda, 2019; Wilms, 2019.

1.2 When does the GDPR apply?

The distinction between pseudonymous and anonymous data is decisive for whether the GDPR applies. While pseudonymous data still fall within the scope of the GDPR, anonymous data do not. However, even when a research project exclusively involves anonymous data, researchers must still adhere to established principles of research ethics (for more details on research ethics, see section 2).

Figure 1

Hierarchy of personal data types under the GDPR



Source: Mondschein and Monda (2019).

The terms pseudonymous data and anonymous data can be summarized as follows:

- **Pseudonymous data:** Data are defined as pseudonymous if all information that can identify an individual is removed (de-identification), but the individual can be re-identified with the help of additional information. This additional information is often referred to as a key. The GDPR still applies to pseudonymous data because re-identification remains possible as long as the key is accessible. Therefore, when handling pseudonymous data researchers must comply with the GDPR but may be subject to more lenient conditions due to reduced identifiability.
- **Anonymous data:** Data are defined as anonymous if de-identification makes re-identification *impossible*, i.e. if it is impossible to link the data back to an individual, either directly or indirectly. Anonymization can occur after data collection (e.g. if the key to the data subject is destroyed). However, until the anonymization process is complete, the information is treated as personal data under the GDPR (Wilms, 2019).

Achieving true anonymization, particularly in evaluation research with individuals, is challenging and sometimes not even desirable. “[P]seudonymization under the GDPR is considerably far-ranging and circumventing compliance obligations under the GDPR by virtue of utilizing anonymous data is rather unlikely, as the usefulness of data for research purposes stands in contrast to the stringent criteria of anonymization under the GDPR” (Mondschein and Monda, 2019, p. 60).

For example, to achieve anonymization, direct identifiers must be removed, and indirect identifiers must either be removed or generalized to prevent identification of specific individuals. However, in many cases, indirect identifiers like income, age or gender may contain critical information that is necessary for evaluation research (Polonetsky et al., 2016). We discuss possible approaches to addressing this challenge from an ethical standpoint in section 2.3.

1.3 What to do when the GDPR applies?

If the GDPR applies, researchers need to establish a legal basis for data processing (Article 6, 9 or 10 GDPR) and follow the principles of processing (Article 5 GDPR). Both steps will be described below.

Establish a legal basis for data processing: If the GDPR applies, one of the six legal bases defined in Article 6 GDPR for processing personal data must be fulfilled. In financial literacy research contexts, obtaining consent from participants tends to be the most relevant legal basis. This involves obtaining consent from individuals before processing personal data relating to them. Their consent must be (1) freely given, (2) specific, (3) informed, (4) unambiguous (Article 7 GDPR) and (5) the age of consent must be fulfilled (Article 8 GDPR). Below, we provide further details on the criteria for consent.

- (1) **Freely given:** Participants must have a genuine choice and be able to refuse or withdraw consent without facing negative consequences. This implies providing them with enough information to understand risks and alternative choices.
- (2) **Specific:** Researchers must clearly articulate the purposes of data processing.
- (3) **Informed:** Participants must be informed about the purposes and means of processing they are consenting to. This includes providing them with information on the right to withdraw their consent at any time and without any justification and negative consequences. Participants must also be made aware that, if they withdraw consent after data collection, their data will be deleted. Furthermore, they have a right to get information about the categories and the nature of the data being processed (e.g. sensitive personal data), details on who processes them (data controller) and on any

third party the data will be shared with (data recipients, if any). Finally, researchers need to ensure that participants know about the data retention period and about their rights as data subjects (Chapter III GDPR).

- (4) **Unambiguous:** Participants must give their consent through clear, affirmative action, such as signing a consent declaration, ticking a box or clicking a button. Silence, pre-ticked boxes, or inactivity cannot count as consent. The absence of negative consequences for participants withholding consent must be communicated.

Box 1

Dealing with the consent of children and young people

The GDPR allows member states to set the minimum age of consent for data processing between 13 and 16 years (Article 8 GDPR).

Addressing the needs of children and vulnerable individuals: While not legally mandatory, ethically, it is advised to also seek and obtain consent from individuals with a legal guardian and children participating in research. This is based on the idea that “[c]hildren who are capable of forming their own views should be granted the right to express their views freely in all matters affecting them, commensurate with their age and maturity” (Wilms, 2019, p. 16).

Oral consent: The GDPR places a strong emphasis on the need to be able to demonstrate that the data subject has given their consent (principle of accountability). Thus, if oral consent rather than written consent is collected this must be documented somehow (e.g. recordings) (Wilms, 2019).

Box 2

Evaluation research in schools and other institutional settings

When conducting evaluation research in schools, it is advisable to obtain approval for the study from both the school administration and the relevant educational authorities (e.g. ministry of education, local authority in charge of education, school principals, etc.). This step is also recommended when teachers are surveyed in their professional capacity – even if this happens outside of school hours. In some cases, national regulations may mandate such approvals as a legal requirement. However, legal obligations for research in institutional settings can vary significantly. For instance, it may be necessary to additionally obtain parental consent for students’ participation, even if the children have reached the minimum age of consent. Therefore, it is prudent to familiarize oneself with the specific legal requirements applicable to research in schools or any other institutional setting in question; or to seek legal advice before starting the evaluation.

Follow the principles of processing: When processing personal data, researchers must stick to the following GDPR principles: (1) lawfulness, fairness and transparency, (2) purpose limitation, (3) data minimization, (4) accuracy, (5) storage limitation, (6) integrity and confidentiality as well as (7) accountability. Below, we provide further details about the principles of processing.

- (1) **Lawfulness, fairness and transparency:** Individuals must be informed in detail about the processing of their personal data, including the purposes, legal basis and rights available to them. Article 13 and Article 14 GDPR include a list of information that the data subjects need to receive before data relating to them are processed.
- (2) **Purpose limitation:** Clearly define and communicate the purposes for which personal data are collected and processed to participants; ensure that their personal data are not used for any other purposes without a further legal basis (e.g. additional consent).
- (3) **Data minimization:** Collect and process only the personal data that are necessary for the intended purposes. Avoid collecting excessive or irrelevant data. If anonymous or pseudonymous data are sufficient for the research purposes, such data must be used.
- (4) **Accuracy:** Implement mechanisms to ensure the accuracy and currency of personal data throughout their lifecycle. Provide individuals with the opportunity to rectify inaccurate data.
- (5) **Storage limitation:** Specify the period for which the data are needed to achieve the previously clearly defined purpose (retention period). Data must either be deleted or anonymized as soon as they no longer serve the purposes for which they were gathered. If the personal data are still needed for definite future scientific analysis, once again, this requires a specific legal basis (for further details, see the information on legal bases above). In the context of research and statistics in the public interest, Article 89 GDPR provides certain exemptions from this principle (for further details, see the information below).
- (6) **Integrity and confidentiality:** In order to comply with the GDPR, it is advisable to follow the principle “data protection by design and by default” (Article 25 GDPR). Proactively implement appropriate technical and organizational measures to comply with all the GDPR principles, to uphold data subjects’ rights and to protect personal data against unauthorized access, disclosure, alteration or destruction. This includes having adequate data management plans as well as access rules for researchers and user authentication. Special measures are required if cloud computing services are used to store or process data, to safe electronic communication and for the use of certain other IT control instruments (e.g. backups). Pseudonymization techniques may also be necessary. More details on how to comply with confidentiality have been summarized by Wilms (2019).
- (7) **Accountability:** As outlined in Article 5 para 2 GDPR, the researcher(s) or the organization they work for are responsible for ensuring and demonstrating compliance with all GDPR provisions.

Therefore, if researchers claim that they have obtained consent, they need to be able to prove it.

Exemptions and safeguards: There are some exemptions and safeguards concerning personal data processed in the context of research and statistics in the public interest. These are outlined in Article 89 GDPR and for Austria additionally in: Section 7 Data Protection Act (Datenschutzgesetz – DSG) and the Research Organization Act (Forschungsorganisationsgesetz – FOG). Exemptions require appropriate safeguards for the rights and freedoms of data subjects, which may include data minimization, technical and organizational measures, privacy by design and by default as well as pseudonymization. In line with these principles, it is crucial to maintain comprehensive records of processing activities, including the nature, purposes and duration of data processing. Only then will the GDPR’s requirement for documentation and demonstrating compliance be fulfilled (Article 30 GDPR).

Box 3

Applying pseudonymization and data segregation to comply with integrity and confidentiality

It is advisable to implement a pseudonymization strategy to demonstrate increased compliance with GDPR. Researchers can enhance data privacy by segregating identifiable personal information from other, non-personal data and creating two discrete datasets – one for identifiers and one for analysis. According to an advanced approach to data governance, researchers or the institution they work for should store, analyze and transmit these datasets independently (=institutional separation).

If, for example a research institution evaluates a program, the entity responsible for the program rollout may retain a dataset with identifiers while passing another, pseudonymous dataset on to the research institution in charge of evaluating the program. Spreading datasets across different institutions significantly reduces the risk of both datasets being compromised.

Table 2

Raw dataset

Name	Student ID	Date of birth	Grade	School	Attendance
					%
Müller	987-654-001	April 12, 2005	8.2	Gymnasium Einstein	95
Luca Rossi	987-654-002	August 22, 2004	7.5	Istituto Leonardo	98

Source: Authors' compilation according to best practices for ensuring data security in research in O'Toole et al. (2018).

Table 3

Identifiers dataset

Name	Student ID
Alice Müller	987-654-001
Luca Rossi	987-654-002

Source: Authors' compilation according to best practices for ensuring data security in research in O'Toole et al. (2018).

Table 4

Pseudonymous dataset for analysis

Student ID	Grade	School	Attendance
			%
987-654-001	8.2	Gymnasium Einstein	95
987-654-002	7.5	Istituto Leonardo	98

Source: Authors' compilation according to best practices for ensuring data security in research in O'Toole et al. (2018).

Through this introduction to GDPR, we aim to provide financial literacy evaluation researchers with a foundation for appropriately protecting personal information. However, focusing solely on data privacy law will likely be insufficient. Researchers are also expected to adhere to ethical standards in their work. In the following section, we outline what constitutes ethical behavior in the context of social science research, where the relationship between researchers and participants is fundamental. Once again, we emphasize ethical considerations that are particularly important for the evaluation of financial literacy initiatives.

2 Research ethics

Research ethics are a central component of all research. In many disciplines, the ethical requirements researchers have to meet are summarized in a code of ethics. One example are the Ethical Guidelines for Educational Research³ published by the British Educational Research Association (BERA, 2011). A code of ethics comprises the ethical principles and rules of a discipline in the form of recommendations. This also has to do with the fact that research methods and contexts can vary greatly, which, in turn, makes it impossible to implement a uniform set of ethical standards. It is the researchers' responsibility to adapt research ethics guidelines to their specific needs and the study in question (Unger, 2018). Compared to interviews with adults, for example, research involving children gives rise to different ethical questions and challenges (e.g. when contacting them or obtaining their informed consent). The Compass for Research Ethics⁴ provides a helpful overview when engaging in research with children and adolescents.

The intensity of the research ethics discourse varies from one country to another. Accordingly, the ethical requirements researchers must meet also vary strongly. Contrary to the USA or Canada, for example, in Austria or Germany research proposals in the social sciences do not have to be reviewed and approved by an ethics committee (Felbermayr, 2023; Kremsner, 2017). Leaving legal obligations to undergo a review aside, we think it is part of every researcher's duties to reflect on their actions with a view to the central principles of research ethics.

2.1 What are the central principles of research ethics?

The following principles of research ethics should guide researchers' actions (Felbermayr, 2023; Griffin and Balandin, 2004; Hatry et al., 2015; Unger, 2018):

- **Voluntary participation:** Participation in evaluation research must be voluntary. Participants must be able to withdraw, or take a break, from participation at any point without fear of negative consequences.
- **Anonymity and confidentiality:** Researchers must anonymize personal data by changing or removing all information that would allow others to link any participant to their data. In addition, researchers must maintain absolute confidentiality. This duty of confidentiality continues to apply even after the end of a research project. Personal data must never be passed on without the express consent of participants.
- **Potential for risk and harm:** Participation in evaluation research must not lead to any harm to, or negative consequences for, the participants or their environment. When conducting e.g. qualitative, narrative interviews, there is a risk of retraumatizing participants when asking them to talk about past traumatic experiences. Researchers must assess all potential (long-term) consequences as best as they can. It might prove impossible, however, to realistically anticipate all risks of harm, such as harm to participants' mental health. If necessary, researchers should refer participants on to contact points offering professional support.
- **Transcripts provided by third parties and publication:** Researchers must inform participants about what happens to their data. This applies, for example, when reproducing anonymized quotations in scientific publications or when contracting third parties (e.g. external agencies) to transcribe audio files. Informed consent means a participant's verbal or written consent to participate in evaluation research. Written consent means having a consent form that is signed by the participant and that contains essential information about research ethics and applicable data protection law (for further details on data privacy, see also section 1.3).

³ <https://www.bera.ac.uk/publication/ethical-guidelines-for-educational-research-2011>

⁴ <https://core-evidence.eu/compass-for-research-ethics>

“Obtaining active, written consent is the most transparent way of holding an evaluation accountable. But an alternative is passive, verbal consent. In this format, research administrators would develop a description of the research effort that includes all the elements from the above bullets and would read it to potential participants exactly as written so that there is no variation depending on which member(s) of the research staff may be recruiting the participant. Once research administrators have finished reading the script, participants may then ask any questions they might have and are given the opportunity to verbally refuse participation, without a consent form. In some cases, verbal consent is more appropriate for literacy-related reasons: Those who have low literacy may not be able to read or fully understand the consent form, nor may they be able to meaningfully provide written consent” (Yoong et al., 2013, p. 204).

Participants can thus give verbal or written consent, but their decision to consent always has to be an informed one. To be able to make an informed decision, potential participants must receive detailed information in advance (Lewis and Porter, 2004). However, there is no consensus in research on how much information participants should receive. At one end of the spectrum, it is considered necessary to fully inform participants about the project, including its research questions, whereas, at the other end, it is considered sufficient to communicate some general aspects of the project (Kruse, 2015). In the end, it is up to each research team to decide how much information they wish to share and to be transparent about this decision in their research report. In this context, we would like to mention an evaluation study we conducted on the implementation of new financial literacy interventions in kindergartens. Before the start of the study, we provided participants (teachers, parents) with information materials and spoke to them about the background of the study, the research design and the planned surveys. Providing this kind of information meant fully educating participants about the project, which has proven to be a good basis for a trusting relationship with them.

Box 4

Note

Obtaining informed consent for conducting a study can be a lengthy and time-consuming process. When evaluating e.g. financial literacy interventions in institutional settings, the school principals and authorities in charge may also have to give their consent. “Evaluators may have to consider alternatives such as requesting data without any personal identifiers or requesting group data only” (Hatry et al., 2015, p. 819).

2.2 What are the ethical challenges involved in qualitative and quantitative evaluations?

Challenges regarding research ethics are as varied as the research settings and the research methods applied. In the context of evaluations, ethical challenges arise more often than in other research contexts “due to the applied and political nature of the work, along with the existence of diverse sources of guidance that do not suggest a single ‘correct’ response to a given ethical dilemma” (Hatry et al., 2015, p. 819). Particularly when it comes to qualitative research, questions concerning research ethics tend to be difficult to answer, given the diversity of qualitative research approaches and individual research questions (Hopf, 2012; Unger, 2018). Quantitative evaluation research, by contrast, may come with fewer ethical challenges due to often limited personal contact with participants. However, this, too, depends entirely on the respective research setting and design. The following list contains a number of ethical challenges that may arise in both qualitative and quantitative research.

- **Conflicts of interest:** When conducting evaluation studies, the various parties involved (e.g. funding bodies, implementers, researchers, political agents) may enter into conflict. Maintaining objectivity and independence can be challenging, for example, when researchers are supposed to independently evaluate a financial literacy intervention while, at the same time, being paid by the body funding the intervention (Yoong et al., 2013, p. 210). Yoong et al. (2013) recommend the following basic rules of cooperation.

“Specifying roles of funders and the implementing agency in the evaluation. Establishing clear lines of communication between evaluators, implementers, funders, and other relevant stakeholders. Articulating and committing to the evaluation’s standards for transparency and accountability. Describing any known or potential conflicts of interest” (ibid., pp. 210–211).

- **Ensuring complete anonymity:** In the context of evaluations, collecting personal data is often unavoidable. However, a key principle of research ethics is to guarantee anonymity during data analysis and publication. This means that researchers must remove or change any information that could lead to an individual being identified (e.g. name, region, institution, profession) in qualitative interview transcripts and publications. Doing so comes with certain challenges for qualitative studies: On the one hand, researchers must ensure the promised anonymity by removing all personal identifiers from interviews. On the other hand, they must not reduce the information contained in the interviews to such an extent that the specific context of an interview is lost (Felbermayr, 2023). It is debatable whether complete anonymity can ever be obtained, especially in this day and age, where the internet allows us to identify participants based on a combination of known personality traits (Lewis and Porter, 2004; Unger, 2018). The way a person speaks or the (technical) terms they use may also allow us to draw conclusions about them. Despite such difficulties, researchers must not forgo anonymity. They must take promises of anonymity and confidentiality seriously and protect participants as best as they can. When addressing anonymity, researchers must also comply with all applicable data protection law. In line with this, researchers must keep personal data only for as long as is necessary for the purposes of a given research project. They must also have an appropriate legal basis for processing personal data, such as informed consent (for further details, see section 1).
- **Benefit or added value for participants:** In the context of research, participants’ motives for, and benefits of, participating in studies also play an important role. Imbalances may arise in particular for research conducted to obtain an academic degree. In these cases, the added value for the researchers (e.g. obtaining an academic title) is higher than for the participants. It is therefore important for researchers to think about the added value for participants and how to thank them for their participation. This also raises the question of whether or not incentives should be offered to participants for participation. The academic debate about monetary incentives is controversial. According to some, a greater willingness to participate would speak in favor of such incentives. Others claim, however, that participants should not take part in research for monetary reasons, but primarily out of personal interest.

- **Ensuring informed consent:** Another important question is how to ensure that potential participants in a research project understand the project-related information they receive in order to be able to give their informed consent. This question is particularly relevant for research involving children, people with special needs or people with low literacy. “Whether informed consent is written or oral, it is critical to realize that in dealing with people with low literacy, even basic financial terms – such as ‘interest’ or ‘budget’ may not be understood and should be defined in the consent process. It is well-documented that access to financial services in the developing world is extremely low, so basic financial terms are likely to be unknown to many participants” (Yoong et al., 2013, p. 204). The use of supporting materials (such as pictures) can minimize the risk of people agreeing without understanding the information provided (Griffin and Balandin, 2004). Documents in plain language can also be helpful, as they present complex content in a simplified form.
- **Research involving vulnerable groups:** Researchers need to be particularly considerate when working with members of certain groups, namely “vulnerable and underserved populations – such as the poor, children, the elderly, those with little education, and those with mental or physical illness or disabilities [...]. [...] These populations may also be more likely to experience negative repercussions from participating in certain types of research and evaluation, including risks to their personal safety, social ostracism, or exclusion from a particular program” (Yoong et al., 2013, p. 201).
- **Building relationships:** Qualitative research in particular relies on research methods that usually allow researchers to enter into direct contact with participants (e.g. through interviews or focus groups). This gives rise to questions about the nature of the relationship between researchers and participants, which is less of an issue in quantitative research. As researchers and participants usually do not know each other before the start of a project, establishing a trusting working relationship is essential for the success of (qualitative) research. The challenge for researchers lies in maintaining a good relationship with those involved in their projects, while also making sure that participants do not confuse this relationship with friendship. This is particularly true for longitudinal studies, where the working relationship with participants may grow over time, given their participation in several surveys (Detamore, 2010; Felbermayr, 2023; Thomson and Holland, 2003).
- **Discussing sensitive topics:** Qualitative research in particular may also involve (planned or spontaneous) personal conversations about sensitive topics. This may lead to the retraumatization of participants and may also cause psychological stress for them (Kelle and Erzberger, 2006). Researchers are responsible for those who participate in their projects and must not leave participants in a state of agitation following an interview. In social research in particular, it is advisable to prepare a list with the contact details of psychosocial support services and to hand out this list during or after interviews, if necessary (Felbermayr, 2023). Topics related to finance, such as debt, may also be a sensitive subject area for some people. Researchers conducting evaluation research should be aware of this, too.

In this section, we provide an overview of relevant ethical questions that researchers should consider for evaluation research. We hope to have clarified that – contrary to legal regulations – the standards of research ethics can only ever serve as a point of orientation for researchers’ actions. It is every researcher’s responsibility to take, and be transparent about, their own decisions concerning research ethics. These decisions, of course, will always depend on the context of their research (research method, target group, etc.).

3 Summary and concluding remarks

In this paper, we outlined key data privacy and ethical considerations for evaluation researchers given that processing personal data is often inevitable in evaluations. We discussed essential concepts from the EU's General Data Protection Regulation (GDPR) that are necessary for understanding the legal framework. Moreover, we provided fundamental legal guidelines to ensure GDPR compliance.

While formal legal frameworks like the GDPR mandate accountability in data processing, research ethics offer broader principles to guide responsible research. Ethical concerns often overlap with data privacy issues, as both frameworks emphasize the protection of participants' personal information. This includes obtaining their informed consent and ensuring that personal data are used only as long as necessary. At the same time, decisions regarding data privacy and research ethics are often complex, requiring careful judgment and the weighing of multiple factors.

For example, financial literacy interventions often target children and adolescents (Mauser et al., 2024). This raises significant research interest in conducting evaluation research with children and young people. Ethically, children and young people are considered "social actors in their own right and should have a voice in research" (Vogl et al., 2023, p. 2). At the same time, research involving vulnerable groups, such as children, demands stricter data protection measures and ethical considerations. Researchers must work closely with data privacy and ethics experts to carefully balance their objectives with evolving privacy regulations and ethical standards. It is important to say that this should not discourage researchers from conducting evaluations involving vulnerable but highly relevant groups.

Examples like this only offer a small glimpse into the challenges and considerations that are involved in data privacy and research ethics. Since this article can only be an introduction to data privacy and research ethics we highly recommend consulting experts in both areas. In the context of the GDPR many evaluation projects may present specific challenges, with national variations and continuous updates of the regulation going far beyond the scope of this article. Moreover, if a research institution has appointed a data protection officer (DPO), it is mandatory to involve the DPO in any evaluation project concerning data privacy matters. The contexts of and the resulting ethical challenges in both qualitative and quantitative evaluations are diverse. Therefore, we also recommend engaging with the ethical stakes involved in such research beyond this introductory article. Researchers are encouraged to reflect on potential ethical challenges with their research fellows and delve deeper into the relevant literature to gain a more comprehensive understanding of these complex issues.

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Qualitative research evaluation – how to get from first ideas to a final paper

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This report provides an overview of the key steps involved in evaluating qualitative research. It is aimed at readers who have little experience in applying qualitative methods and evaluating qualitative research. Readers will learn how to follow an evaluation cycle: it begins by choosing a suitable qualitative inquiry method and ends by analyzing the collected data. Note that I am not going to discuss qualitative methods and methodological approaches in depth here. My goal is to show readers how qualitative evaluation works and to point out its added value. To dive deeper into the topic, I recommend that readers get hold of one of the many qualitative method handbooks around.

1 Introduction: characteristics of qualitative research evaluation

The evaluation process consists of four phases: *planning, implementation, analysis and communication*. In each phase, researchers must go through a series of methodological steps, making informed decisions. I am not going to explain the phases and the methodological steps at length here. This will be covered in one of the publications of the OeNB Financial Literacy Evaluation Series that discusses the evaluation cycle. But first things first: what is an evaluation? In what ways does an evaluation differ from basic research? Hirschauer (2006, p. 405) explains that evaluation research usually refers to applied social research that is characterized by a triangular relationship: (1) A client (2) commissions evaluation research (3) to assess an area of practice. By examining the quality of programs and measures, i.e. their effectiveness, efficiency, acceptance, etc., the research is meant to inform decisions and make decisions more rationally productive.

Researchers must decide for a qualitative evaluation study in the planning phase. Then, they must also decide which type of evaluation they are going to pursue (impact or process evaluation). *Impact evaluation* refers to “a type of evaluation research focused on assessing the effects, outcomes or impacts of a program, intervention or policy. It aims to determine the extent to which desired changes have occurred and the attribution of these changes to the program or intervention” (Lorenz, 2024, p. 17). In contrast, *process evaluation* refers to a type of evaluation “focused on understanding the implementation, delivery and mechanisms of a program, intervention or policy. It examines how and why the program works (or not), the fidelity of implementation, and the contextual factors influencing outcomes” (ibid.). Qualitative methods are mainly used in the context of *process evaluations*. This has to do with the questions that are of interest in the context of a process evaluation, “such as those about the public acceptability of the intervention and participants’ experiences” (Yoong et al., 2013, p. 63). Such questions are best answered by using qualitative methods such as interviews or focus groups. After all, qualitative methods “tell the *program’s story* by capturing

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and communicating the *participants' stories*. [...] They tell what happened when, to whom, and with what consequences" (Patton, 2015, p. 18). Qualitative methods can thus be used to determine whether and in what way or through the actions of which actors interventions have an effect (Kelle and Erzberger, 2006, pp. 291–299). This already addresses a first characteristic of qualitative research. Other central characteristics of qualitative research relate to topics such as context, knowledge generation, openness/diversity of methods and are presented in more detail below.

Context: Qualitative study results cannot be considered in isolation from the individual context and are usually based on a small sample. Although this allows for in-depth and detailed findings (micro-analytical view), it is not possible to make statements that are valid generally. Statements are only ever possible with regard to specific persons, a specific context, etc., and therefore result in a theory with limited scope. Qualitative research and the interpretative methods used do not aim at generalizations. Instead, the aim is to capture the diversity of perspectives and particularities of the respective voices (deeper understanding).

Deeper understanding: The use of qualitative methods provides valuable information about the actors involved and their context. The added value of qualitative data in evaluation and monitoring is that they complement quantitative data. Qualitative data provide "a depth of contextual understanding and a level of detail that one can't get with quantitative data alone" (Yoong et al., 2013, p. 113). Especially in the context of a process evaluation, it is important that qualitative methods help gain greater process-specific and detailed knowledge (Kuckartz et al., 2008, pp. 74–75). This allows for arriving at a deeper understanding of the object under investigation. During the implementation of a new measure, qualitative methods as part of a process evaluation can provide valuable information on how the measure is experienced by the actors or what the reasons are why measures are effective or not (Goodrick and Rogers, 2015).

Exploring the inner perspective: Qualitative research is interested in individuals' inner perspective and subjective opinions and aims to understand these amid various systems or contexts that influence each other. Qualitative methods are therefore particularly suitable for surveying a person's experience or perception in a specific context. Especially with regard to vulnerable target groups (e.g. people with disabilities), qualitative research can also reflect the diversity and, above all, complexity of experiences (Coons and Watson, 2013). This makes it possible to give vulnerable target groups a voice.

Flexibility and openness of methods: Given its flexible nature, qualitative research allows for making methodological adjustments in advance, e.g. with regard to the interview guide or the interview situation. Moreover, adaptations are possible during the research process, e.g. by compiling new data to follow up on a new hunch (Charmaz, 2014; Felbermayr, 2023). Qualitative research thus offers the methodological flexibility that is necessary to justifiably adapt the research to individual needs.

Discovery of new phenomena: Qualitative research is considered a discovering science (Flick et al., 2012, p. 25). This means it aims at discovering phenomena or detecting topics that have so far stayed under the radar with a view to generating new knowledge. Due to their inherent flexibility, qualitative methods are particularly well suited for investigating fields of research that have been little researched to date. This leads to the discovery of new knowledge that is often not present in the researchers' prior knowledge and is therefore not taken into account in the construction of standardized (quantitative) instruments (Kelle and Erzberger, 2006).

Interaction and reflection: Qualitative research is characterized by a variety of data collection and evaluation strategies that are applied based on different methodological principles. Despite this diversity, qualitative researchers are united by "the self-image of understanding research as an interaction between the researcher or researchers and the research subjects" (Mensching, 2006, p. 339). Related to this is the

question of data independence. While in quantitative research the researcher's independence from the object under investigation is key, in qualitative research the methodically controlled subjective perception of the researcher is an essential component of gaining knowledge (Flick et al., 2012, pp. 2–25). The researcher therefore consciously adopts a reflexive stance. Qualitative researchers are always part of the reality they are researching and are not neutral, value-free experts. This makes it all the more important to adopt a reflexive stance to be aware of one's own role in the research process, such as social (power) position, origin, gender or cultural affiliation, and to constantly reflect on this (Charmaz, 2014). All of these aspects can have an influence on data collection, the quality of the data and therefore on the result. This calls for permanent self-reflection in the research process, which is supported, for example, by writing memos (Felbermayr, 2023).

2 Choosing the appropriate qualitative survey method

Deciding on the appropriate qualitative survey method is an important step in planning a qualitative evaluation. The choice must be made in accordance with the research questions and the objectives of the evaluation. According to Patton (2015, p. 248), "getting clear about purpose" is one of the central steps at the beginning of the evaluation. Another thing to be taken into account is the underlying research design. If, for example, an evaluation is meant to find out whether a financial education measure leads to changes in the behavior of students, an observation seems more appropriate than a qualitative interview. However, there is no magic formula for decisions like these. To choose a suitable method, researchers must be familiar with the various qualitative methods, their advantages and disadvantages, fields of application, etc. Here, we therefore first look at ways of using qualitative methods. Next, you will get an overview of the many different qualitative methods available. The following methods will be highlighted in more detail: interviews, focus groups, observation and desk review of documents.

2.1 Ways of using qualitative methods in research evaluation

The World Bank toolkit outlines three fields of application for the use of qualitative methods in evaluation (Yoong et al., 2013). First, qualitative data are used to enhance quantitative research material, such as survey questions. This concerns both the content of such questionnaires and reformulating questions. "For example, evaluators of a financial capability program may conduct a small number of focus groups (qualitative data) to examine how potential program beneficiaries talk about the issues the program addresses (such as savings, budgeting, and so forth), which can then help construct the surveys (quantitative data) using the most appropriate terminology" (Yoong et al., 2013, p. 113). Second, qualitative data can provide a deeper understanding of the development of financial education programs as part of a formative process evaluation in the implementation phase. To this end, for instance interviews are conducted with the people involved. The qualitative data and information collected are meant to help improve the design and implementation of the program. "In particular, qualitative data can shed light on program implementation and operational issues, including questions about the most appropriate mode of delivery, the identification of the target population, and so forth" (ibid., pp. 113–114). Third, qualitative data can provide a valuable input for summative research. This includes both process and impact evaluation. "Although qualitative data can't by themselves establish causality between the evaluated financial capability program and observed outcomes, they can help produce a richer and more informative picture of the program being evaluated and provide insights that can't be fully captured through surveys and other quantitative data" (ibid., p. 114). Qualitative methods can, in particular, close gaps in interpretation that can arise through the exclusive use of quantitative methods.

In the various fields of application, qualitative methods can be used in different ways depending on the research interest. The following list provides some key examples of how qualitative methods can be used.

Using qualitative methods as...

- single method
- in a mixed method design
- in a (qualitative) triangulation design
- in a participatory approach
- in a case study
- in a cross-section or longitudinal-study design

The areas of application differ in terms of whether qualitative methods are used as a single method, in combination with a second qualitative method (triangulation) or in combination with quantitative methods (mixed method). Participatory approach and case study refer to further research approaches that allow using different qualitative methods. In a participatory approach, the research participants should be given an active role in the research process and play an active part in shaping it. At the center of a case study is a case. A case can be defined in different ways, e.g. an institution, a family or several students of a year group, and is examined by applying different methods to gain a better understanding. In addition, qualitative methods can be used in a cross section (one point in time data collection) or longitudinal study design (longitudinal data collection). The individual fields of application are discussed in more detail in separate publications of the OeNB Financial Literacy Evaluation Series.

2.2 Overview of qualitative methods

Numerous methodological handbooks provide a good and in-depth insight into the variety of qualitative research methods (e.g. Denzin and Lincoln, 2018; Monique, 2020; Patton, 2015). However, no specific research methods are used for qualitative evaluation research. Instead, researchers rely on the general canon of methods of qualitative social research (Mensching, 2006, p. 340). Approaches to qualitative research differ in terms of the “type of depth” (Charmaz, 2011, p. 103) that can be achieved with the data collection/evaluation methods. What varies depending on the qualitative research approach are the theoretical point of reference, the understanding of the subject matter and the methodological focus. As a researcher, it is important to select the appropriate qualitative method from the extensive canon of methods (single method). However, there is no universally valid scheme for categorizing the variety of qualitative methods. The methods are structured differently depending on the focus. Patton (2015, p. 14) structures the qualitative methods with reference to the type of qualitative data. Qualitative results are therefore based on three types of qualitative data: (1) interviews, (2) observations and fieldwork and

Table 1

Types of qualitative data

1. Interviews	Open-ended questions and probes yield in-depth responses about people’s experiences, perceptions, opinions, feelings, and knowledge. Data consist of verbatim quotations with sufficient context to be interpretable.
2. Observations and fieldwork	Fieldwork descriptions of activities, behaviors, actions, conversations, interpersonal interactions, organizational or community processes, or any other aspect of observable human experience are documented. Data consist of field notes: rich, detailed descriptions, including the context within which the observations were made.
3. Documents	Written materials and documents from organizational, clinical, or program records; social media postings of all kinds; memoranda and correspondence; official publications and reports; personal diaries, letters, artistic works, photographs, and memorabilia; and written responses to open-ended surveys are collected. Data consist of excerpts from documents captured in a way that records and preserves the context.

Source: Patton (2015, p. 14).

(3) documents, as summarized in table 1. In my opinion, this classification is particularly well suited to categorizing and describing qualitative methods and the different types of collected data. This report presents four qualitative methods in more detail. Each of these methods can be assigned to one of the three categories or types of qualitative data according to Patton (2015, p. 14): *Interviews* and *focus group* (interviews), *observations* (observations and fieldwork) and *desk review of documents* (documents).

What the various qualitative methods have in common is that qualitative data are collected, which in turn form the basis for the analysis. But what is the difference between qualitative and quantitative data? The OECD²/INFE³ (2010a, p. 6) states the following in its Guide to Evaluating Financial Education Programmes:

“Data such as written or spoken thoughts and conversations, photographs or drawings is very useful for understanding the experiences of people and exploring questions such as why or how something happened. It is called Qualitative data and is beneficial when you want to describe the variety of experiences, rather than the proportions of people experiencing certain things.

Data that provides you with numbers for analysis is useful when you want to answer questions like how many or how much and is called Quantitative data. However, this type of data does not reveal reasons for not achieving or exceeding programme objectives. You will need to use qualitative data to find reasons, to find strengths and weaknesses of the programme” (OECD/INFE, 2010a, p. 6).

As outlined in the introduction, qualitative methods make it possible to gain a deeper understanding and capture diverse perspectives. Why and how questions are predestined to generate this knowledge, e.g. by means of interviews.

2.2.1 Interviews

Interviews are a popular qualitative method in evaluation and research. An interviewer asks questions on predefined topics to obtain information from the interviewee. Interviews usually take place as individual interviews, i.e. in a one-on-one setting (1 interviewer and 1 interviewee). Conducting pair interviews or group interviews is also possible and common. Individual interviews are preferable for sensitive topics, e.g. debts. Pair or group interviews are suitable should a one-on-one setting be perceived as too stressful or too intimate (Michael, 2022).

Interviews can be conducted in various ways. The following variants can be distinguished (see OECD/INFE, 2010b, p. 11; Michael, 2022):

- **Face-to-face interviews:** The interviewer and interviewee(s) are present in the same room at the same time and usually sit opposite each other. The conversation therefore takes place face-to-face. This is the most common way of collecting qualitative data using interviews. Advantage: In addition to verbal information, the interviewer also gains a deeper insight into the nonverbal behavior of the interviewee. Recommended for discussing sensitive, in-depth topics. Disadvantage: Takes more time to plan and implement.
- **Phone or video interviews:** The conversation takes place via telephone or video. The people are therefore not present in the same room. This form of interviewing has become increasingly important,

² OECD = Organisation for Economic Co-operation and Development.

³ INFE = International Network on Financial Education.

not least due to the coronavirus pandemic. Advantage: Calls can be arranged more flexibly and are easier to conduct over longer distances and across national borders. Disadvantage: Both parties must have the technical prerequisites and know-how. Conversation and dynamics differ from face-to-face interviews.

- **E-mail interviews:** In this form, the interviewee receives written questions by e-mail. The questions are answered and sent back. This form of survey is only suitable for a structured guideline. Advantage: The interviewees can decide when they want to answer questions. Disadvantage: It is not possible to ask ad hoc questions.

A wide range of qualitative methods can be used in evaluation and research. The same is true for interview forms. Choosing the appropriate interview form depends on the research interest and research questions. The interview form then determines the structure of the guidelines, the degree of openness and structuring as well as the role of the interviewer. Two common interview forms are key informant interviews and in-depth interviews.

Key informant interviews are primarily conversations with experts, which is why this form of interview is often referred to as expert interviews. Key informant interviews are used to gather in-depth knowledge from experts. The focus is on knowledge about a specific topic and not on the interviewee's biography or person as such. Discussions with experts can contribute to a better understanding of complex topics. Such interviews are often conducted as structured interviews, which is often due to experts' limited time for interviews (Flick, 2006b; OECD/INFE, 2010a).

In-depth interviews focus on interviewees' expectations, experiences, opinions and feelings, which are recorded in detail. This requires an interview format that gives the interviewees enough time and space to reflect and share their insights. The interviews are often conducted in an unstructured manner, which allows for discussing individual topics in greater depth. In-depth interviews are therefore particularly suitable for discussing sensitive topics that people would not address in a group setting (focus group) (Yoong et al., 2013, pp. 114–115). The interviewer's skills and experience are also of central importance when conducting in-depth interviews:

"In qualitative research, the nature of the interaction with the respondent is critically important. Because in-depth interviews call for a high level of skill it is important that those facilitating them have substantial interviewing experience, either in a research setting or some other context involving nondirective interviews (i.e., interviews that are allowed to follow the course the interviewer may set). It is very important that the interviewers do not influence what the respondent says and, above all, that they allow and encourage the respondent to speak at length on the topics to be covered. Interviewers should have well-developed listening skills and be familiar with techniques to probe replies and encourage the respondent to elaborate, such as using neutral prompts like "Why do you say that?" and "Can you tell me more about that?" A good in-depth interviewer will allow respondents to stray from the order of the topics in the topic guide if that is how the respondent wants to tell the story" (ibid., p. 162).

Depending on how the interview guide is structured, a distinction can also be made between (1) structured interviews, (2) unstructured interviews and (3) semi-structured interviews (Michael, 2022).

- **Structured interviews:** In a structured interview, the interviewer follows an interview guide with predefined topics and interview questions. Throughout the interview, the interviewer keeps to the sequence of topics and questions as well as the predetermined structure. Therefore, the conversation leaves little room for flexibility. However, structured interviews are more easily comparable in the analysis. Structured interviews are ideal for obtaining relevant data on a specific topic or for answering a specific research question. This form of structured inquiry is therefore particularly suitable for impact evaluations. (Michael, 2022; Yoong et al., 2013, pp. 121–122).
- **Unstructured interviews:** Unstructured interviews lack a list of interview questions that are asked in a specific order. Here, the interviewer only relies on topics that are of interest and that are used as a conversation starter. Based on the interviewee's statements, the interviewer asks follow-up questions and questions of interest. This form of interviewing corresponds most closely to having a natural dialogue and allows for conducting interviews in a highly flexible way. Unstructured interviews are suitable during exploratory research aimed at collecting new knowledge. However, evaluating and comparing unstructured qualitative data is usually much more complicated than with structured interviews (Michael 2022; Yoong et al., 2013, pp. 121–122).
- **Semi-structured interviews:** Semi-structured interviews are often conducted with a view to evaluating programs. Such interviews comprise some predefined topics and interview questions. In contrast to structured interviews, however, the interviewer does not have to rigidly follow the sequence of questions. This makes it possible to gather new, previously unknown knowledge. In addition, interviewees can address new topics in the interview, which enables deeper insights into their perspectives (Yoong et al., 2013, p. 115). "This allows interviewers to plan questions that are specifically geared towards their research questions and data that they need while also allowing them to be flexible and reflexive" (Michael, 2022). According to Adams (2015, p. 493), conducting structured interviews requires a special "interviewer sophistication": "Interviewers need to be smart, sensitive, poised, and nimble, as well as knowledgeable about the relevant substantive issues."

A decision must also be made about how to record interviews. There is no right or wrong here, but there are advantages and disadvantages that need to be weighed up (see Adams, 2015; Helfferich, 2011).

Audio: In most cases, interviews are recorded with a recording device with the consent of the interviewees. A small recorder with external microphones is recommended for working in the research field, as these provide better sound quality and are often better at filtering out ambient noise. This is particularly necessary when conversations cannot be held in a quiet, enclosed room. Sufficient storage capacity must also be ensured. Recording with a cell phone and storing audio files in a cloud should be viewed critically from a research and data protection perspective. The advantage of tape recording is that it can be listened to repeatedly and the interviewer can concentrate better on the conversation and asking the next questions. At the same time, being recorded can make interviewees feel self-conscious or ill at ease and thus influence the interview. It is advisable to place the recording device in the middle of the table before starting the recording and only start recording the conversation after a warm-up phase.

Video: Recording with video is another option that is particularly popular with focus groups. In addition to the spoken word, people's behavior is recorded on video, which allows for analyzing nonverbal behavior

or group dynamics. Due to their size, video cameras are more likely to be noticed by interviewees than recorders and are not forgotten as quickly. The presence of a camera can therefore influence interviewees' behavior, which must be reflected in the evaluation.

Paper and pencil: Instead of technical recording options, written notes can also be taken during or after the interview. Taking notes during the interview might bother the interviewees and give the impression that the interviewer is not really listening. At the same time, this presents interviewers with the challenge of having to take notes awhile also keeping the conversation going by asking questions. Taking notes only after the interview requires a high level of concentration and a good memory on the part of the interviewer in order not to forget important points. In both cases, the note-taking is subject to the interviewer's selective perception. What is noted down is what the interviewer remembers and feels was important. "But notes could not sufficiently preserve the participants' tone and tempo, silences and statements, and the form and flow of questions and responses" (Charmaz, 2014, p. 91). But when is it advisable to use the paper and pencil technique? Note-taking is useful if the participants refuse audio or video recordings or for particularly sensitive topics.

2.2.2 Focus groups

A focus group is "a moderated discourse procedure in which a small group is encouraged to discuss a specific topic by means of information input or (focused) questions" (Krueger and Casey, 2015; Schulz, 2012). Similar to interviews, there are also different forms of focus groups. According to Morgan (1998, p. 29), the various forms of focus groups have three things in common: "They are a research method for collecting qualitative data, they are focused efforts at data gathering, and they generate data through group discussions." Focus groups are intended to reflect the diversity of (divergent and controversial) perspectives on a topic. Achieving group consensus is not the goal of focus groups, nor is making decisions (focus groups are not decision-making forums) or promoting disputes among participants. "Focus groups are conducted to gather the range of opinions and experiences" (Krueger and Casey, 2015, p. 509).

A special feature of the focus groups is the dual communication: Communication takes place between the participants, but also between the participants and the moderator. Focus groups can be used to gather both individual opinions and group opinions on a topic or issue. Focus groups make use of the dynamics of collectives (groups) when collecting data, which proves to be particularly beneficial for collecting attitudes, opinions or taboo topics (Flick, 2009). Just hearing other points of view can be thought-provoking. One advantage of focus groups is therefore that the participants stimulate each other, which leads to open answers and minimizes, to a certain extent, mechanisms such as saying yes or social desirability (Cropley, 2002, p. 110–11). In addition to the verbal statements, one may analyze the interaction between the participants, e.g. by using the method of observation. This also highlights the difference to group interviews, where a question from the moderator is answered by each participant, but there is no discussion among the participants (Yoong et al., 2013, p. 118).

- **Knowledge generation:** Focus groups are particularly well suited to generating (new) knowledge from a group (multiple perspectives). In other words, the focus is on generating knowledge and not on testing hypotheses (Bürki, 2000, p. 101 cited in Schulz, 2012, p. 12).
- **Efficiency:** The opinions of several people can be collected within a short time (a lot of information in a short time, with fewer resources).
- **Naturalness:** Group discussions correspond to natural communication behavior. The atmosphere can contribute to people expressing themselves more spontaneously and freely.

- **Validation:** Opinions or views expressed by individuals are “validated” by the other group members and may have to be justified or revised.
- **Stimulus:** Listening to other opinions from the group can contribute to the formation of new ideas/opinions in the individual (Lamnek, 2005; OECD/INFE, 2010b).

When are interviews preferable to focus groups?

- When sensitive topics are addressed that are difficult to discuss in front of a group, e.g. experiences of violence.
- When individuals are expected to report on their experiences in more detail. In other words, when it is about gaining an in-depth understanding of individuals and less about gauging the diversity of perspectives on a topic or the group opinion. While in an individual interview, the interviewer attempts to evoke all possible aspects, arguments and value judgments of the interviewee on a given topic, in a focus group the moderator presents a few stimuli and these are then discussed in turn by the group participants (Zwick and Schröter, 2012, p. 27).
- When people are asked to talk about a topic at greater length, e.g. their own biographical journey through life in a biographical interview.

What needs to be considered when implementing focus group?

Focus groups take place under controlled and planned conditions; they would not occur in real life in this way (Lamnek, 2005). When conducting focus groups, we distinguish four phases. In the (1) *opening phase*, the focus group begins. The moderator refers to important formal aspects, such as anonymity, voluntariness, audio and/or video recording (Bohnsack and Schäffer, 2001). In the (2) *introduction round*, the participants introduce themselves. People decide for themselves how much information they want to disclose about themselves, e.g. first name only or also surname, profession, etc. The discussion begins with the (3) *stimulus*. The stimulus can be varied (e.g. movie, picture or provocative statement) and should stimulate the conversation or discussion about a topic. This is followed by the (4) *guided discussion*. The moderator uses the guidelines for the discussion as a structure or orientation framework (Lamnek, 2005).

- **Number of participants:** The literature diverges on the recommended sample size for focus groups, which ranges from 4 to 10 people (Yoong et al., 2013, p. 118). The World Bank (2013, p. 145) names 8 people as the ideal number to ensure “a full discussion.” What needs to be taken into account is the size of the basic sample. In the case of vulnerable target groups or a very specific topic, four people may be the maximum group size that can possibly be reached. The composition of the sample must also be considered. Too much diversity among the participants (heterogeneous group) can lead to individual participants dominating the discussion. Too little diversity (homogeneous group), on the other hand, can hinder a discussion – in the sense that topics are not debated due to similar opinions (Yoong et al., 2013, p. 145).
- **Seating arrangement:** Ideally, the seating arrangement corresponds to an “egalitarian structure,” so that all participants experience equal treatment in their subjective perception (Lamnek, 2005, p. 120). A large, round table is best suited for this; with no one “chairing” or sitting at the head of square table formats. The seating arrangement must also take into account participants’ individual needs. Sign language interpreters must sit opposite deaf people or in the deaf person’s field of vision. Visually impaired people should not sit opposite windows due to the incidence of light, for example, as it is difficult for them to see people sitting in front of windows.

- **Moderator:** The environment in which the focus group is conducted should be comfortable and not frightening. The role of the moderator should not be underestimated. The moderator should be perceived as a friendly person who is open to all perspectives (Krueger and Casey, 2015). “Moderating focus groups requires considerable skill, because a number of issues may arise. It calls for all the skills of a good in-depth interviewer, plus the ability to manage the group dynamic and ensure that everyone contributes more or less equally” (Yoong et al., 2013, p. 150). The challenge as a moderator is to lead the group and, for example, to stop frequent speakers and encourage those who are silent to talk. This is because both types of participants can under certain circumstances make a meaningful discussion impossible (Lamnek, 2005, p. 161). Krueger and Casey (2015, p. 511) put it in a nutshell: “Skillful moderators make facilitation look easy. They are friendly, open, and engage with participants before the group starts, making people feel welcome and comfortable. [...] A focus group is working well when participants begin to build on each other’s comments rather than continually responding directly to the moderator”.

It should be noted at this point that focus groups – as well as interviews and observations – can also be used to generate some quantitative results data. “For instance, focus group leaders can count the number of people who agreed or disagreed with a particular statement, or they can conduct a “ranking” exercise whereby participants rank program elements in a certain way” (Yoong et al., 2013, p. 151).

2.2.3 Observation

The interview and focus group are also explained as scientific methods in the World Bank toolkit (Yoong et al., 2013). One method that is completely lacking in the World Bank toolkit is scientific observation. In social research, observation means directly observing human actions, linguistic utterances, nonverbal reactions (facial expressions, gestures, body language) and other social characteristics (e.g. clothing, symbols, customs, forms of living) (Dieckmann, 2017, p. 548). Observation therefore aims to standardize, document and make observations intersubjectively comparable (Bortz and Döring, 2009, p. 262). As a researcher, the aim is to describe descriptively what you observe and write it down accordingly, e.g. a person raises and lowers their head. The observer’s interpretation follows only in a second step and must take into account the (cultural) context. In German-speaking cultures, for example, raising and lowering the head can be interpreted as nodding. However, it should be borne in mind that other explanations are also possible. A person could try to release tension through movement. Scientific observations therefore require qualified researchers who are trained in applying the method of observation and can observe descriptively. According to Patton (2015, p. 331), the path to a skilled observer comprises the following six points:

- (1) “Learning to pay attention: Seeing what there is to see, and hearing what there is to hear
- (2) Writing descriptively
- (3) Acquiring expertise and discipline in recording field notes
- (4) Knowing how to separate detail from trivia in order to achieve the former without being overwhelmed by the latter
- (5) Using systematic methods to validate and triangulate observations
- (6) Reporting the strengths and limitations of one’s own perspective, which requires both self-knowledge and self-disclosure”

Appropriate training is essential for implementing both qualitative and quantitative methods. In the case of qualitative methods, such as interviewing or observation, it is often assumed that one already “naturally” masters the activity, which is why training is often neglected. “Training to become a skilled observer is a no less rigorous process than the training necessary to become a skilled survey researcher or statistician. People don’t “naturally” know how to do systematic research observations. All forms of scientific inquiry require training and practice” (Patton, 2015, p. 330). An important point in the training is to learn how to concentrate during the observation. It is important not to lose the focus of the observation and not to be distracted by other observation stimuli.

An observation is carried out purposefully and with a clear focus that results from the research interest. At the same time, an observation is always selective, i.e. it is never possible to observe and record the entire abundance. Neither is this possible with a camera, as it only ever has a certain angle of view or section of the action in the picture (Bortz and Döring, 2009). Naturalistic observations take place in the field. The field can be understood differently depending on the research approach. For an organizational researcher, the field is an organization, for ethnographers it is a cultural setting, and for evaluators, the field is the program being evaluated (Patton, 2015). As a researcher, it is important to adopt a neutral position when observing. Especially during longer ethnographic field observations, the tension between closeness and distance to the research participants can be a challenge for researchers.

The following table is intended to provide a better overview of the characteristics of scientific observation. The characteristics are presented using various pairs of opposites (see: Boer and Reh, 2012; Bortz and Döring, 2009; Lüders, 2012; Pauli, 2012) in table 2.

Table 2

Characteristics of an observation

Event sampling The observed events are not structured in time, i.e. the observation is decoupled from temporal information. The focus is on frequencies, i.e. whether or how often the observed events occur (Bortz and Döring, 2009, p. 270).	Time sampling The observed event is divided into fixed time periods. The division of the time interval depends on the subject of examination (e.g. interval of 5 seconds) and requires a high level of concentration. Some 30 minutes at one go are recommended, to be followed by a break (Bortz and Döring, 2009).
Qualitative The focus is on an interpretative approach to observation	Quantitative Quantitative data are produced through observation.
Open The participants know that they are being observed.	Covert The participants do not know that they are being observed (to be reflected on in terms of research ethics!).
Participatory The researcher is part of what is being observed. The role of observer is obvious to everyone.	Non-participatory The researcher is not part of the discussion, conducting the observation as an outsider.
Structured (standardized) The observation is carried out using an observation grid (the degree of structuring may vary).	Unstructured The observation is carried out without using an observation grid. Anything that stands out with regard to the research interest is generally noted down.
With technology The observation is recorded with audio and/or video. Technology, especially audio, can also influence the observation.	Without technology The observation is carried out with paper and pencil, i.e. without any other technical aids.

Source: Boer and Reh (2012); Bortz and Döring (2009); Lüders (2012); Pauli (2012).

We distinguish four types of observation depending on whether the observer participates in the discussion and whether the participants know that they are being observed. In other words, whether the observation is carried out openly or covertly (secretly). Any covert observation must be critically considered from a research ethics perspective and must be carried out in a well-thought-out manner (see the OeNB Financial Literacy Evaluation Series publication on data privacy and research ethics). In agreement with Patton (2015, p. 342), I advocate full disclosure to the participants. This means informing people in advance about the planned observation and obtaining their permission. “Trying to run a ruse or scam is too risky and adds to evaluators stress while holding the possibility of undermining the evaluation if (and usually when) the ruse becomes known” (Patton 2015, p. 342). To better understand the different variants, see the following examples from Bortz and Döring (2009, p. 267) in table 3.

Table 3

Variants of observation

a) Participatory-open: A company psychologist participates openly in employee appraisals to explore group problems.	b) Participatory-covert An official of the authority for the protection of the constitution covertly participates in a demonstration while observing demonstrators' behavior.
c) Non-participatory-open: A soccer coach observes the players' performance on the sidelines.	d) Non-participatory-covert: A developmental psychologist observes an argument between two children behind a one-way mirror.

Source: Bortz and Döring (2009, p. 267).

Example

We may confirm the added value of the qualitative method of observation based on an evaluation study we conducted. The aim of the study was for teachers to evaluate a newly developed educational game for students aged 10 to 14 years as part of a training course. An observation is a good way of gaining a deeper insight into teachers' perspectives when testing the didactic method (educational game). The data collected were anonymous, written observation notes using a structured observation grid. The focus of the observation was on verbal statements and not on nonverbal aspects of communication, e.g. facial expressions or gestures. The participatory open observation took place on two consecutive days. Some 40 teachers were present in each observation setting, who, under the guidance of a moderator, tried out the didactic method in small groups of 6 to 8 teachers. Qualitative observations are used to collect data directly in the situation (in situ) and not retrospectively. This makes it possible to make statements about the teachers' experiences directly when trying out the method, which might not have been discussed in a subsequent interview. The qualitative observation provided valuable information on how the game needs to be changed from the teachers' point of view, e.g. clear and plain language, less text to read.

2.2.4 Desk review of documents and materials

A systematic document review is a central method that is often used in the evaluation of programs. This method is described in more detail in the World Bank toolkit (see Yoong et al., 2013). The aim is to review documents, records (including archive records) and data in general that are related to and relevant to the program to be evaluated. The World Bank toolkit distinguishes between the following categories of potentially informative information:

- “Official documents and materials describing the program's aims, structure, and so forth (including, perhaps, a program's website)

- Program materials not intended for public circulation (such as meeting minutes, internal progress reports, internal communications about the program, etc.)
- Data gathered in the course of implementing a program (for instance, demographic information about the program beneficiaries, results of specific activities, logs of program activities, etc.)
- Photographs and audio and video recordings
- Nonprogram data (such as financial transactions, school enrollment records, and so forth)" (ibid., p. 122)

Added value of a desk review

- **Source of information:** Documents, records and data of all kinds are a valuable source of information to learn more about the goals, the design, the planned implementation, the process or the stakeholders (donors, employees, customers, etc.).
- **Changes:** In many cases, changes that have occurred during a program can be mapped in the documents. These changes can be significant for the evaluation of programs.
- **Program quality:** Working with documents can make a direct contribution to the evaluation of program quality. "For example, the quality of program materials (inputs) can be reviewed and assessed by skilled peer reviewers, as can an audio recording of the program being delivered (outputs)" (Yoong et al., 2013, p. 122).

2.2.5 Summary

Table 4

Selective qualitative methods for carrying out an evaluation

Qualitative research method	Description	Benefits	Limitations
Key informant interview	One-on-one setting: an expert is interviewed on a specific topic	In-depth expert knowledge is gathered	Access to the field
	Mostly structured interview	Can help better understand complex topics	Limited informative value Can be costly if travel is involved Interviewer bias: requires reflection
In-depth interview	One-on-one setting: one person is asked more in-depth questions on topics	Insight into individual opinions, feelings, expectations	Time-consuming procedure
	Mostly unstructured interview	Person can communicate their own opinion in more detail Suitable for discussing sensitive topics	
Focus groups	A moderated group discussion	Information from several people in a short time	Not suitable for discussing sensitive topics
	A moderator addresses questions to a small group of people who react to the answers	Group dynamics: People's behavior becomes visible/observable	Group dynamics: individuals dominating the discussion Individual opinion can be influenced by others Moderation bias: Requires reflection
Observation	Persons' behavior and actions are described by observation	Valuable information to better understand people's behavior	Observer bias: separate direct observation from interpretation of what is seen
Desk review of documents	Systematic review of documents, records and data in general	Usually more direct and faster access to the field	Gives no insight into the individual perspectives

Source: Author's compilation based on OECD/INFE (2010a); Yoong et al. (2013).

3 Designing the instrument

Let us assume researchers have both a clear research interest and clear objectives and questions in mind for an evaluation project. They have already selected research design and method in accordance with the objectives. Now, the next step is developing survey instruments.

No single qualitative survey instrument fits all research contexts. Guides for interviews, focus groups and observations must be adapted to the requirements of the research project at hand. Qualitative research offers the methodological flexibility necessary to conduct research in ways that cater to the individual needs of specific target groups. We can make methodological adjustments to the interview guide both *before* the surveying stage and *during* the research process (Charmaz, 2014; Felbermayr, 2023). The time it takes to develop guides should not be underestimated (Flick, 2006b). Therefore, it is important to allow sufficient time for creating, validating and revising guides for evaluation projects. The following explanations can only provide an initial insight into the development of guides. For further details, see the handbook by Patton (2005).

3.1 Developing interview guides and questions

In evaluation research, researchers often work in teams. In qualitative research, this can mean that several people conduct interviews based on an interview guide they developed together. In these cases, it is important to make sure in advance that all researchers understand the interview questions in the same way. They should thus clarify: What exactly does each question mean? What is the aim of asking each question? How should the guide be applied? Otherwise, researchers might interpret individual questions differently and emphasize different things during their interviews. Several factors must be taken into account when developing interview guides: the specific interview method and, depending on this, how structured the interview should be (structured, unstructured, semi-structured) as well as the type of questions. For the design of a structured interview guide, a list of topics is usually drawn up. Questions on the respective topics are written down and prioritized subsequently. In other words, keeping in mind what would be considered an appropriate interview duration, researchers must critically reflect on which questions from the list they should definitely cover (Patton, 2015, p. 256). In a next step, they need to determine a sensible order in which to work through the different topics and questions (Yoong et al., 2013, p. 115). Inefficiencies in the order should be avoided as they have a negative impact on the interview itself and, thereby, on the quality of the survey data (ibid., p. 150). If there are interview questions regarding some sort of evolution over time, for example, it is advisable not to jump back and forth between different points in time. People might get confused if they first have to answer questions about how they are currently doing in implementing a specific measure to promote financial education (present), then about their wishes for the future (future) and finally about prior implementation experiences (past). Instead, it would make sense to cover the different subjects following a chronological order (past, present, future). Asking questions in an effective and logical order can also help interviewers establish rapport with their interviewees by making them feel more at ease during the interview (ibid., p. 150). In structured interviews, the interviewer will stick to a predefined order of questions. Guides for unstructured interviews, on the other hand, can be seen as a point of orientation that supports the interviewer's memory. Interviewers can also deviate from the guide and spontaneously react to new topics raised by interviewees.

An important part of creating interview guides is coming up with questions that are appropriate for the target group. In general, a distinction is made between open-ended and closed questions. How much interviewees reveal varies depending on the type of question.

- *Open-ended questions* are asked to encourage people to narrate, reflect on or make an elaborate statement about an issue. These include questions starting with “why,” “what,” “how,” “when,” “to what extent,” etc. The question “Why are you taking part in the training?” is, for example, meant to make the interviewees give the reasons for their participation. Different people may answer it differently. In this sense, the answer is open, which also explains the term open-ended question. In-depth interviews in particular are characterized by open-ended questions.
- *Closed questions* (yes/no questions), on the other hand, have a limiting character that allows researchers to gather knowledge (quickly) (Michael, 2022). The question “Do you take part in the training?” can only be answered with “Yes” or “No” and results in more superficial knowledge about the situation.

Table 5 uses various topics to illustrate very clearly how both open-ended and closed questions are asked on the same topic, thereby pursuing different research interests and collecting different information.

Table 5

Open-ended and closed questions

Focus of the inquiry	Open-ended inquiry question	Closed question inquiry framing (to be avoided)
Immigration experiences	What are the processes that immigrants experience during immigration? What are the implications of these processes for how they engage where they have immigrated?	Do immigrants' experiences during immigration affect how they engage in the community after immigration?
Program evaluation	What works for whom in what ways with what results and in what contexts?	Does the program work?
Homeless youth	What are the experiences of homeless youth? How do they perceive and talk about their experience of homelessness?	Are there patterns in the experiences of homeless youth?
Ecology and climate change	How, if at all, is the ecological system of the Great Lakes changing? What factors are contributing to those changes? What are the implications of those changes for the future health of the ecosystem?	Is the social ecological system of the Great Lakes changing? Is climate change causing the ecological system to change? Can the implications for the future be identified?

Source: Patton (2015, p. 253).

What needs to be considered when creating an interview guide?

- **Adequacy for the target group:** The wording of the questions must be adapted to the (linguistic) skills/needs of the respective target group. Possible target groups are experts, parents, young people, children – with/without disabilities, migrant background, etc. The various needs of the target group must be considered when developing a guide as well as later, when conducting the interviews. For example, the wording in an interview should generally be easily comprehensible, not only when interviewing people with an intellectual disability. Researchers should also consider the extent to which specialist or technical vocabulary is necessary (Buchner, 2008). According to Charmaz (2014, p. 96), it is advisable to speak the language of the interviewees: “Following threads in our participants' everyday language and discourse helps us to form questions from their terms and learn about their lives.”
- **Balance regarding complexity:** It is important to challenge, yet not overwhelm people. Respondents should not feel underestimated by banal wording. At the same time, abstract or complex wording can also be overwhelming (Flick, 2006b). For example, using too many technical terms and specialist vocabulary creates too much of a challenge for children as interviewees.

- **Balance regarding the number of questions:** Participants' time is precious. Therefore, researchers should only ask as many questions as necessary in order to find out what they need to know. They should avoid questioning people further out of personal interest.
- **One thought – one question:** The guiding principle is that every question should express one thought. Do not ask several questions about different aspects at once, e.g. "How did you use the cash provided to you through the program, and did you get any other type of assistance?" (Yoong et al., 2013, p. 150). Respondents will not know which question they should answer first then – if they even remember all the questions, which is often not the case.
- **Appreciative and neutral wording:** Researchers should avoid wording that can be interpreted as expressing a negative opinion or criticism. An example would be "Why did you do such a terrible thing?" It is also important not to influence respondents with specific wording and ask neutral questions instead. For example, the question "Was the instruction provided by the program effective and suitable?" suggests that the program was effective and suitable. In contrast, open phrases (e.g. "What was your impression of the instruction provided by the program?") leave more room to respondents for both positive and negative reactions (Yoong et al. 2013, pp. 149–150).

Example: Interview guide

There is no single set of universally applicable standards for the design of interview guides. However, it has become common practice to group questions by topic. Some researchers make lists of topics and corresponding questions below each topic. The following is an example of a slightly different approach: The topics are listed in the left column, with the corresponding questions to the right (see table 6). There are both open-ended questions and combinations of closed questions with open-ended follow-up questions. Open-ended questions are intended to encourage respondents to talk about a given topic, thereby generating a story (Helfferich, 2011, pp. 102–103). Therefore, they are also referred to as story-generating questions. In addition, there are also so-called elaboration probes. The example does not contain any of these. Their aim is not to generate new stories, but rather to keep a story going. Elaboration probes thus do not have any content per se, in the sense that they provide no or as few presuppositions or content-related impulses as possible (ibid., p. 104). An example would be the question "And what happened next?"

Table 6

Example of an interview guide

Topic	Questions
Experience	What is your experience with the new didactic method? (open-ended question) Could you please describe your own role as part of the project? (open-ended question)
Application of the method	Have you exchanged ideas with other colleagues about the application of the method in lessons? If yes/no, why? (closed question followed by open-ended question)
Challenges	Where do you see the biggest challenges? (open-ended question) Have you tried to overcome the challenges? If so, how? (closed question followed by open-ended question) If you could change one thing about the method, what would it be? (open-ended question)

Source: Author's compilation.

3.2 Developing interview guides and questions for focus groups

When developing guides for focus groups, researchers can follow the four phases that characterize a focus group according to Lamnek (2005): Opening phase, round of introductions, setting a stimulus and guided discussion to conclude. Focus groups are meant to discuss a given topic. The structure and questions in the guide facilitate the discussion process. After the opening phase and introductions, researchers use a stimulus (e.g. picture, video clip, provocative statement) to kick off the discussion about the topic at hand. Ideally, the stimulus introduces the topic and encourages a lively debate. The intro is then followed by the actual discussion based on the guide for focus groups. Similarly to creating an interview guide, it is a good idea to write down a list of topics with questions for focus groups. Here, researchers may choose between open-ended and closed questions. When researchers develop guides for focus groups, they must consider the same aspects that apply to interviews: balancing complexity and the number of questions, target group adequacy, etc.

Example: Guide for a focus group

The following guide was developed for a focus group with students and teachers of geography and economics. The guide is based on the four phases according to Lamnek (left column in table 7), to which topics and corresponding questions have been assigned. In my experience, this provides a good framework for researchers working with focus groups.

Table 7

Example of a focus group guide

Phase	Topic and questions
Opening phase	Welcoming the participants, discussing the consent form, etc.
Round of introductions	Could you briefly introduce yourselves, please, by telling us your name, the school you teach at and how much professional experience you have got?
Stimulus	(Different terms are printed out and put on a table, such as financial education, economics education, economic education, financial literacy, etc.). Understanding financial education Please take a look at these terms: To what extent do they mean the same thing to you? Which term is missing? Which term do you use in class and why? What do you think of the following statement: Terms are subject to trends, change over time, but basically always mean the same thing?
Guided discussion	Providing financial education Which topics do you think are particularly difficult to teach in economics/financial education classes? What makes teaching them so difficult? How do you feel about talking to students about money/pocket money? And which topics are particularly easy to teach in class?

Source: Author's compilation.

3.3 Developing an observation grid

Before developing such a grid, researchers need to decide how to go about observing. Here, it is particularly important to distinguish between structured and unstructured observations. Let us assume the goal is to observe how new teaching material on finance/economics is used in a classroom. Observing how often students raise their hands or teachers call on girls would count as *structured* observations, for example. An *unstructured* observation would be a descriptive record of how girls react to a teacher's response.

For both types of observations, it is important to define categories in advance, i.e. what is to be observed (topic). If observations are carried out by several researchers, it is important to ensure they have a common understanding of the topic. It is advisable to explain the individual subtopics in more detail in a separate document that also includes examples. In the observation grid, researchers usually leave enough space next to the topics to write down observations. Ideally, it also contains an additional column for notes. This should help observers separate the descriptive part from their interpretation of what they see. Writing down that a person raises and lowers their head would be an example of a descriptive observation. Understanding the head movement as consent would be an interpretation. It is possible, for example, that the person has a sore neck and therefore raises and lowers their head. Interpretations must therefore always be contextualized and recognizable as such.

Example: Observation grid

During a training course, two participant researchers observed around 80 teachers while they were trying out a new didactic game. The researchers carried out their observations on two consecutive days. Some 40 teachers were present in each observation setting, who, under the guidance of a moderator, tried out the didactic method in small groups of 6 to 8 teachers. Using a structured observation grid, the researchers collected anonymous data in written observation notes. The observations focused on various aspects of the teachers trying out the new game. Table 8 shows excerpts of an observation grid.

Table 8

Example of an observation guide

Topic	What is being observed?	Notes (interpretation)
Reading the game manual		
Cooperation among participants		

Source: Author's compilation.

3.4 Pretesting

Guides should be tested before using them to collect data in the field. This applies to interviews, focus groups and observations. In pretesting, also known as pilot testing, one or several individuals test the guide. Ideally, they belong to the target group the guide was developed for. The added value of pretesting a guide is often underestimated, and this phase tends to be skipped due to time constraints. However, it is during pretesting that potential weaknesses in a guide are revealed.

How does pretesting work? Let us assume that researchers want to ask teachers about their teaching experience. During pretesting, a researcher will print out and test the guide with a teacher, who will not be interviewed afterward. All the questions in the guide are read out to this person word by word. The researcher encourages the teacher to point out any ambiguities, anomalies, etc. and writes down all the feedback. For example, the interviewed teacher might criticize the wording of certain questions (comprehensibility), the length of the guide or the order in which the questions are presented. Feedback helps improve the guide and ultimately also the quality of future survey data. Furthermore, problems that might otherwise arise in the field can be avoided (Yoong et al., 2013, p. 147). So, before gathering data, at least one person should test the guide.

4 Sampling and access to the field

Sampling refers to “the process of selecting units, such as individuals or organizations, from a larger population” (Yoong et al., 2013, p. 139). This is important as not everyone can take part in an evaluation due to financial, structural or time constraints. Sampling is an essential step in data collection, and researchers need to be very careful when choosing the sample. Getting the sample wrong may make it difficult to answer the research questions.

Qualitative and quantitative research approaches use different sampling strategies. These are explained in more detail below. According to Patton (2015, p. 264), researchers apply different techniques following a logic that belongs to the category of either qualitative or quantitative methods. For *qualitative research*, they select participants through *purposeful sampling*. This means that participants are selected according to predefined criteria. Qualitative research primarily aims at obtaining more in-depth knowledge. Therefore, the sample is rather small. Even a sample size of 1 ($n=1$) is possible. After all, the goal is not to make empirical generalizations or representative statements. Instead, qualitative research aims at generating detailed, information-rich cases which contribute to a better or in-depth understanding of the object of investigation (ibid.). In contrast, *quantitative research* usually relies on a large sample that is randomly selected and intended to provide information about a certain population as a whole. This sampling strategy is referred to as *random sampling*. The logical structure and power of quantitative methods are grounded in statistical probability theory. “A random and statistically representative sample permits confident generalization from a sample to a larger population. Random sampling also controls for selection bias. The purpose of probability-based random sampling is generalization from the sample to a population and control of selectivity errors” (ibid.).

Sampling and access to the field are often regarded as incidental (Lau and Wolff, 1983, p. 417). However, it is worthwhile exploring questions of field access, not least because the way in which researchers manage to access a field is indicative of certain central field characteristics (Lüders, 2012, p. 392) and opens up another source of potential knowledge (Wolff, 2012, p. 336). Field access is also an interesting subject as there is no one-size-fits-all approach to looking for and getting access to a field (ibid.). Therefore, at the beginning of each research project, researchers have to think about how (best) to access the field in question and to select their sample (Felbermayr, 2023, p. 66).

How large should the sample be?

The first step is to define the target group, the selection criteria and the size of the sample. Contract-based research projects that receive external funding tend to be subject to predefined sampling criteria (Kuckartz, 2006, p. 277). For example, if an institution wants to evaluate whether teachers use its financial literacy program, the target group is implicitly defined by the question. Research interests are usually decisive for choices regarding the fine-tuning of the sample (e.g. teaching subject, teachers' age or gender). There are no widely accepted rules for determining the adequate sample size. It varies from project to project and is largely determined by the research interests, the qualitative method of choice, the size of the target population and the desired heterogeneity. Compared to quantitative research, the sample size is significantly smaller because the aim is to generate in-depth, contextual knowledge and not to be able to make statements about the general population (Yoong et al., 2013, pp. 143 and 145). Instead of a concrete sample size in the form of a number, qualitative research aims at a state called theoretical saturation, a

term coined by Grounded Theory⁴. Theoretical saturation is reached when additional surveys do not reveal, or cannot be expected to reveal, any additional knowledge (Charmaz, 2014). This point marks the ideal sample size (Yoong et al., 2014).

Which sampling strategy should be pursued?

The first question to clarify is whether the participants for qualitative surveys should be recruited from a larger quantitative sample or whether the sample should be selected directly from the population (Yoong et al., 2013, p. 144). Yoong et al. (2013) distinguish between different strategies for qualitative sampling.

- **Purposeful sampling:** Evaluations pursue a clear objective or research interest. In most cases, the selection of participants is purposeful in order to obtain the information necessary to answer the research questions. The aim of this type of sampling is to increase the credibility and depth, but not the representativeness of the results (Yoong et al. 2013, p. 143). There are many different types of purposeful sampling. Patton (2015, pp. 264–272) provides a comprehensive list of 40 purposeful sampling strategies including detailed descriptions.
- **Maximum variation sampling:** Here, the goal is to capture a maximum of heterogeneity in the sample. Research can aim at diversity in accordance with predefined criteria (such as educational qualifications, migration background or age) in order to take different perspectives and experiences into account (ibid.). In this context, Kelle and Erzberger (2006, p. 293) point out that capturing and describing heterogeneity or the greatest possible diversity often requires a larger number of cases to carry out a contrasting analysis. Researchers must take this into account when planning evaluations in terms of the resources they require. After all, a larger number of cases also means more time for analysis.
- **Outlier sampling:** During sampling, it is also possible to specifically select outliers, i.e. cases that are unusual or outside the norm in one respect or another (Yoong et al., 2013, p. 143). If, for example, researchers test students at a school for their competencies in certain areas of economics, outlier sampling would mean to select the students with the best and worst results (outliers on both ends of the spectrum).
- **Snowball sampling:** Snowball sampling is a very popular sampling strategy. It means starting with a small group of people and recruiting new participants from their contacts or networks (ibid., p. 143). This strategy can be particularly effective for sensitive topics or target groups that are difficult to reach. If you are interested in what young people know about debt or how they feel about it, for example, you would first discuss the subject with a few of them. In a next step, you could then ask them to tell their friends about the study and the opportunity to participate. The idea is that motivated participants will hopefully be able to inspire others.
- **Theoretical sampling:** I would like to add to the list by Yoong et al. (2013) strategies that researchers rely on in combination with specific methods or research designs. Theoretical sampling is a well-known approach to sampling for recruiting participants for Grounded Theory studies. *Theoretical sampling* differs from conventional sampling strategies in that the first steps in accessing the research field have to be seen as provisional. Based on an initial, analytical knowledge base and first theoretical insights, additional participants have to be recruited throughout a research project. This allows for

⁴ Grounded Theory refers to a theory that is developed directly from the qualitative data collected (Charmaz, 2014, p. 6), i.e. the theory is grounded in the subject matter. There are specific methods for this, such as initial sampling or intensive interviewing for data collection, as well as initial and focused coding for data analysis. The foundations of Grounded Theory were established in the 1960s by the sociologists Barney G. Glaser and Anselm L. Strauss.

participants to join at a later stage. The open character of theoretical sampling is in line with Wolff's (2012, p. 336) claim that researchers should understand paving the way into the field as an ongoing (and active) task. Recruiting additional participants along the way is difficult to achieve for longitudinal studies where researchers should interview the same people several times over an extended period of time (Felbermayr, 2023, p. 67). This also applies to evaluations, which are often subject to time pressure, which makes it impossible to add participants later on.

How to get in touch with your sample?

Identifying and overcoming gatekeepers is a key step in accessing the research field. Gatekeepers are usually in key positions and can enable or prevent researchers' access to the field (Helfferich, 2011). As intermediaries, Lewis and Porter (2004, p. 192) consider researchers to be in a position of power because at the interface they are "shaping what is researched and whose voices are heard," which significantly influences our perception of the field. Therefore, it is important to consider in advance *who* the relevant gatekeepers are and *how* one can get past them. When interviewing teachers, for example, the school principals and authorities in charge usually also have to give their permission. If minors are involved, their parents' consent will always be required. Obtaining the necessary permits and getting past gatekeepers can cause delays. The evaluation design should thus allow enough time for these steps.

There are various ways of contacting the sample. Let us assume researchers are planning to evaluate financial education activities in some schools. To this end, they will run focus groups with pupils. One option is to make contact in person on site, i.e. the researchers go to these schools and present their project in class with the schools' consent. This has the advantage that there is no selection by third parties. A second variant would be making contact via a third party. Researchers entering schools are always subject to institutional restrictions. Teachers, acting as gatekeepers, are likely to specifically address certain students more so than others. Gatekeepers' preselection, assumptions and prejudices would then largely predetermine the sample (Felbermayr, 2023, p. 69). Preselection could, for example, exclude people with intellectual disabilities from an interview if gatekeepers considered their verbal skills not advanced enough to express themselves adequately (Buchner, 2008, p. 518). Following Lau and Wolff (1983, p. 418), researchers need to ask themselves: What can gatekeepers do? Who influences them? Do they receive instructions? And consequently, whom are they allowed to let past? A third option is to contact potential participants online. Here too, researchers can either contact them directly (if, for example, they know their e-mail addresses) or gatekeepers can forward e-mails to them.

How can people express interest in participation?

To gain access to the field, it is essential to produce information material that is appropriate for the target group. Information about the project (process, objectives, handling of data, etc.) can be provided in the form of handouts or folders, for example. In addition, researchers need to prepare consent forms and ideally send them to participants before they start gathering data.

Information material should always contain the researchers' contact details, so that interested people can get in touch with them directly. From a research ethics perspective, forwarding contact information to other scientists is a touchy issue. Griffin und Balandin (2004, p. 73) alternatively refer to "information drop" as a more ethical practice. In this case, potential participants receive the researchers' contact information and may decide whether they want to get in touch with them or not. This, in turn, can result in a rather specific sample of highly motivated and committed participants, which must be taken into account when analyzing the data.

5 Collecting data

Collecting too much data is a major risk as it might be impossible to analyze the data adequately then, due to a lack of time or human resources. This risk applies to the collection of both qualitative and quantitative data. People tend to think that collected data do not need to be analyzed later. “However, once collected, data pile up, with a pyramiding effect in terms of data processing and analysis effort (as well as adding to the cost of data collection)” (Hatry and Newcomer, 2015, p. 710). In this section, I am providing methodological advice for the collection of qualitative data. The following remarks will always remain incomplete, as researchers can come across countless different challenges, given the diversity of research contexts, methods or target groups they may work with.

Let us assume that, in accordance with their research interests, researchers have already selected an appropriate method and developed, validated and revised a survey guide. Once access to the research field has been gained and the sample has been determined, the actual data can be collected. For qualitative evaluation research projects, tensions arise due to the different demands of qualitative research and evaluation. Qualitative research is characterized by a methodological openness in the research process, which is hardly given when conducting evaluations and externally funded, contract-based research projects. In qualitative basic research, data are often collected in an iterative process. Researchers thus still get to revise their research questions or add new participants to the sample while already collecting data. This openness does not apply to research that is funded by third parties and evaluations, where clients have a specific research question in mind. As a result, the circularity of the research process that characterizes qualitative research is also lost (Kuckartz, 2006, p. 276).

Conducting the interview: Establishing rapport plays an important role in qualitative research, where data are often collected through direct contact with people. It is important to reflect on one’s own role as a researcher as well as one’s relationship with the participants and to ensure, for example, that participants do not confuse a professional relationship with friendship (for such ethical challenges, see Felbermayr and Lorenz, 2024). In addition to getting the relationship dynamics right, the actual conduct of interviews and focus groups is also demanding for interviewers. “It is very important that the interviewers do not influence what the respondent says and, above all, that they allow and encourage the respondent to speak at length on the topics to be covered. Interviewers should have well-developed listening skills and be familiar with techniques to probe replies and encourage the respondent to elaborate [...]” (Yoong et al., 2013, p. 148). Regarding wording choices during the interview, the same aspects are relevant as when creating guides. Researchers need to find the right balance in terms of complexity (not too challenging, but challenging enough), ask an appropriate number of questions, and choose terminology that is adequate for the target group as well as appreciative and neutral.

Technical equipment: Researchers often make audio or video recordings of interviews and focus groups. They should check how their equipment works ahead of conducting the interview, in order to solve potential problems in advance. Before recording, they should also make sure that there is enough memory left on their recording devices and bring spare memory cards as well as batteries on the day. If, unexpectedly, the recording does not work or a person does not agree to a recording, researchers should write down what was said during the interview right after. However, analyzing interview content as remembered by an interviewer is not the same as analyzing a transcript. After all, the former depends on the interviewer’s memory capacity, which can result in (non)arbitrary omissions or subjective distortions (Kelle and Erzberger, 2006, p. 295).

5.1 Notes

Notes should be written immediately after data collection. This documentation should contain information about both interviewer and interviewee as well as the interview frame (such as location, duration, etc.) (Helfferich, 2011, p. 193). The notes must be treated confidentially. Researchers use them to document the research process. According to Helfferich (2011, p. 193), research notes consist of shorthand notes about the interview atmosphere and specific aspects of the interviewer’s personal relationship with the interviewee. This can help prompt the interviewer’s memory of a specific interview, particularly after a longer period of time. Notes thus serve to reconstruct the interview situation as well as to self-reflect (Charmaz, 2014).

Example of notes

There are no strict rules for writing notes. The notes should be adapted to the method used, the context in which data were collected and the needs of the researcher. In my own work, for example, I have used Helfferich’s (2011, p. 201) template and included additional aspects, such as the weather, the time of day and seating arrangements (Felbermayr, 2023, p. 82).

Table 9

Example of notes

Background information	Interview scenario
Interview code:	Location:
Name of the interviewer:	Before the interview:
Date and time:	Seating arrangements:
Weather on the day of observation:	Particular incidents during the survey:
Attendants:	Perceived atmosphere:
	Interaction/difficult parts:
	After the conversation:

Source: Helfferich (2011, p. 201); Felbermayr (2023, p. 82).

6 Analysis

Once the data have been collected using an appropriate method, they must be analyzed. The logic and steps involved in analyzing qualitative data differ from those involved in analyzing quantitative data. What both have in common, though, is that, metaphorically speaking at least, data collection never takes place in a vacuum. This means that data are always collected with a specific research interest in mind and should help answer a specific question (Yoong et al., 2013, p. 65). When analyzing their data, researchers should not lose sight of the goal of their evaluation. Furthermore, the analysis of both qualitative and quantitative data must be rigorous, systematic and intersubjectively comprehensible (ibid., p. 172).

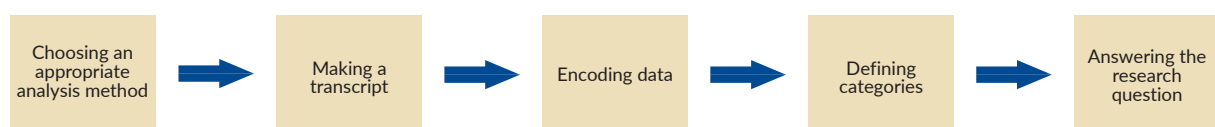
Just as making the right methodological choices at the beginning of the evaluation cycle, researchers must also select an appropriate analysis method (*choosing an appropriate analysis method*). Ideally, they already pick an analysis method at the beginning of the research process as the methods they use for data collection and analysis must be a good fit. For example, relying on a method that focuses on group dynamics (as is the case with focus groups) would hardly work when analyzing an individual interview. After interviews, focus groups and observations, audio recordings or observation notes are used to make

transcripts (*making a transcript*). These transcripts will then be subject to qualitative analysis. The different strategies for analyzing qualitative data follow a similar structure: The *encoding of data* precedes the *definition of (more abstract) categories*. The methodological background of the selected analysis method determines how the individual coding steps are designed and how the categories are defined. At the end of the analysis process, researchers must *answer the research question* and write a final report.

Figure 1 illustrates the steps described above, which underly many methods used to analyze qualitative data. Once they have selected a specific method (e.g. content analysis according to Mayring, 2018; Grounded Theory, Charmaz 2014), researchers should still thoroughly examine the literature on the selected method. For example, many qualitative data analysis methods require researchers to encode the data, but the coding procedure differs depending on the underlying methodological approach. Coding is not a uniform process. It can vary greatly. Furthermore, figure 1 suggests that analysis is a linear process. However, in many cases this does not correspond to the iterative and procedural character of qualitative research in reality.

Figure 1

Main steps of qualitative data analysis



Source: Author's compilation.

6.1 Choosing an appropriate analysis method

A wide range of methods for qualitative data analysis corresponds to the diversity of qualitative methods for gathering data. In many cases, more than one method can be used to analyze qualitative data. For example, no standardized procedure applies to analyzing focus groups. Which analysis method to choose depends on one's research interests. If the focus is primarily on content (What is being said?), researchers are more likely to choose a method for content analysis. If researchers are more interested in developing subjects or in learning about the opinions of a group (What do people say and how do they say it?), they will choose methods that allow them to focus on the group in their analysis. As a point of orientation, Goodrick and Rogers (2015, p. 566) have created a useful heuristic to classify the various qualitative analysis methods. They distinguish between four groups of methods; each group has a different primary focus and objective: enumerative, descriptive, hermeneutic and explanatory. Table 10 provides an overview of these methods. A more detailed description follows below.

Table 10

Four groups of methods for analyzing qualitative data

Primary purpose	Description	Examples of methods
Enumerative	Summarizing data in terms of discrete and often a priori categories that can be displayed and analyzed quantitatively	Classical content analysis (Krippendorff, 2013) Word count Cultural domain analysis (pile sorts, free lists) (Spradley, 1980) Ethnographic decision models (Gladwin, 1989)
Descriptive	Describing how concepts and issues are related	Matrix displays (Miles et al., 2014) Timelines Concept maps/mind maps (Trochim, 1989) Template/framework analysis (Crabtree and Miller, 1999; Ritchie and Spencer, 1994)
Hermeneutic	Identifying or eliciting meanings, patterns and themes	Thematic analysis (Boyatzis, 1998) Constant comparative method (Strauss and Corbin, 1998) Thematic narrative analysis (Riessman, 2008) Discourse analysis (Wetherell et al., 2001) Qualitative content analysis (Schreier, 2012)
Explanatory	Generating and testing causal explanations	Qualitative comparative analysis (Ragin, 1987) Process tracing (Collier, 2011)

Source: Goodrick and Rogers (2015, p. 567).

- **Enumerative methods:** As the name suggests, enumerative methods convert qualitative data into numbers. Qualitative data are prepared in such a way that they can later be analyzed quantitatively. This is done “by sorting the data into coding frameworks, tallying the data, and then developing categories” (Goodrick and Rogers, 2015, p. 566). Enumerative methods can be used to analyze open-ended interview questions or to depict patterns in data, such as frequencies. An enumerative method is not suitable for analyzing rich data (e.g. in-depth interviews) “as it [the method] tends to be overly reductionist and can decontextualize or distort the meaning” (ibid., p. 567).
- **Descriptive methods:** Descriptive methods are used to illustrate how different entities (e.g. ideas, stakeholders, aspects of a certain problem) are related to each other. For this purpose, the data are usually presented in tables or diagrams to allow for comparisons and contrasting (ibid., p. 566). “While enumerative methods are descriptive, our classification of these methods as descriptive indicates that the intended focus and product is not enumeration” (ibid., p. 572).
- **Hermeneutic methods:** Hermeneutic methods are characterized by (a) an iterative process between the collection and analysis of qualitative data and (b) the recognition of the role of researchers in producing data (ibid., p. 576). The focus is on identifying or determining manifest and latent meanings in the data (ibid., p. 566). “Manifest meanings are visible, descriptive labels that appear in the data. Latent meanings are underlying meanings, gleaned from an iterative process of examining the material, looking for similarities and differences, and identifying themes” (ibid., p. 576). Hermeneutic methods are suitable for analyzing rich data and provide in-depth insights. However, hermeneutic analyses are significantly more time-consuming than enumerative or descriptive analyses (ibid., p. 579–580).

- **Explanatory methods:** Here, the focus is on testing causal hypotheses. To do so, researchers establish and analyze theories about the relationship between a cause and its effect. Explanatory methods are suitable for evaluations when, for example, clients want to know whether a newly developed measure works and how it has helped achieve the desired results (ibid., pp. 566 and 580).

Qualitative methods for analyzing data are often characterized by a high level of accuracy, which is associated with being very time-consuming. This often conflicts with the limited time resources available for conducting evaluations (Flick, 2006a, p. 21). This is particularly the case with time-consuming qualitative methods (e.g. Grounded Theory), which do not fit into the tight time constraints of evaluation designs and often require a pragmatic approach (Kuckartz, 2006, pp. 278–279). One possible way out of this dilemma is to use so-called shortcut strategies. What is meant here are pragmatic ways of analyzing data (Kuckartz, 2006, p. 278) or justified deviations from maximum requirements for completeness and accuracy (Flick 2006a, p. 21). In line with the requirement for intersubjective comprehensibility, it is important for researchers to be transparent about how they apply shortcut strategies, as these must also meet scientific quality standards (Kelle and Erzberger, 2006, p. 285; Kuckartz, 2006, p. 280). Examples of shortcut strategies:

- **Transcript:** An appropriate strategy could be to transcribe only relevant parts of an interview, as opposed to the entire conversation, which is usually the state of the art in qualitative basic research (Kelle and Erzberger, 2006, p. 294). However, the parts of the interview that are not transcribed should then be paraphrased accordingly (Flick, 2006a, p. 21).
- **Coding:** Similarly, it is possible not to encode an entire transcript. The focus can be on segments that are relevant for answering the research question.

6.2 Making a transcript

Making a transcript, i.e. writing down what was said in the course of an audio recording, is an essential step in the research process. Here, it is important to keep in mind that a transcript is already a first interpretation. After all, the researchers decide, for example, whether an expression is transcribed as emphasized or not (Felbermayr, 2023). Researchers can make their own transcripts or use contract agencies. Agencies must be provided with a detailed set of transcription rules. Aspects of data protection, such as ensuring secure data transfer, must also be taken into account. Today, a wide variety of free and paid software is available to produce transcripts. Software programs have the advantage that certain keyboard shortcuts can be used to transcribe much faster. However, such programs must be used carefully, and researchers should consider questions like: Where are the (original) data uploaded to? Where are the data stored (e.g. in a cloud)? Who has access to the data? The same approach applies to using AI-supported transcription software.

Depending on the purpose of a research project, different rules for transcribing apply, which leads to varying levels of depth and detail. A rough distinction can be made between simple and in-depth transcription rules. When deciding which to apply, it is important that researchers consider the objective of their analysis and the time available. An in-depth transcript serves to reconstruct the interview situation in the best way possible. Making such a transcript is very time-consuming. Therefore, it should only be made if the research question requires a more in-depth analysis of the data; or if the in-depth transcript adds analytical value (Kuckartz, 2006, p. 274). Analyzing a biographical interview using reconstructive methods may require a more in-depth transcript. If the way of speaking does not need to be reproduced in detail in order to answer the research question, it is preferable to opt for a transcript that is based on

simple transcription rules and focuses on the content. Applying simple transcription rules means getting a transcript of an audio file in a relatively simple form. The audio file gets transcribed word by word while spelling and syntax errors do not get corrected. However, pauses, for example, are only indicated approximately (duration as felt) by a hyphen in the transcript, e.g. “-” for a short pause. In an in-depth transcript, however, pauses would be indicated in brackets, e.g. “(1 sec.)” to mark a pause of one second. Table 11 contains further examples of simple and in-depth transcripts.

While transcribing, it is also necessary to determine whether and, if so, how expressions in a specific dialect are to be rendered. To ensure anonymity, it may be appropriate to translate local dialect into standard language. Otherwise, dialect, as a personal characteristic, might help identify the speaker. From a research ethics and data protection perspective, this aspect must be taken into account, particularly for research involving small and vulnerable groups (Felbermayr, 2023, pp. 81–83).

Table 11

Examples of simple vs. in-depth transcription rules

Examples of simple transcription rules	Examples of in-depth transcription rules
Indicate pauses approximately, e.g. “-” = for a pause of medium length.	Specify pauses in the transcript, e.g. “(1)” = for a pause lasting one second.
Write down filler words (e.g. hm, ah).	Indicate a switch of speakers using empty lines as well as exact time stamps.
Indicate a switch of speakers using empty lines, but no time stamps.	Write down or indicate when speakers talk at the same time.
Underline words that speakers emphasize, e.g. I rode the bike.	Include comments about expressions regarding the speaker’s emotional state (e.g. laughing) or about non-verbal expressions (e.g. coughing, clearing their throat).
	Reproduce unfinished words, e.g. Good mor_.
	Reproduce phonetic extensions, e.g. Go=od morning.

Source: Author’s compilation.

6.3 Encoding data

Coding is “a central part of all qualitative data analysis” (Yoong, 2013, p. 174). A code is “a descriptive word or phrase that is intended to describe a fragment of data” (Goodrick and Rogers, 2015, p. 564). The type of coding depends on a researcher’s method of choice, how findings shall be obtained and the role ascribed to (theoretical) assumptions. First, a *deductive* approach to coding requires that codes and categories are developed in advance that are deemed relevant for answering the research question. Once obtained, data are then sorted according to the previously developed codes and categories. During the analysis, new codes and categories can and should always be added. Second, an *inductive* approach requires that codes are developed based on the collected data, as opposed to developing codes and categories before collecting data. Third, an *abductive* approach refers to a creative, process-based type of analysis that is typical of constructivist Grounded Theory (Charmaz, 2014).

The extent of coding depends on various factors, such as type of data, level of abstraction, aim of the study or stage of the research process (Charmaz, 2014, p. 128). Three popular forms of coding are word-by-word, line-by-line and incident-with-incident coding (ibid., 133). A researcher must be transparent about and justify their choice of one way of encoding over another, because “[t]he size of the unit of data to code matters” (ibid., p. 124). The way of coding can be changed while analyzing data. However, for transparency reasons, changes must also be disclosed. Word-by-word coding means that there is a code

for each word, while line-by-line coding focuses on a larger data unit. Line does not mean a complete and grammatically correct sentence, but one line in the transcript. Accordingly, researchers analyze the transcript line by line (Charmaz, 2014, p. 124; Felbermayr, 2023, p. 84). The two types of coding described (word, line) are probably too detailed and time-consuming for analyses in the context of evaluations. Incident-by-incident coding is recommended, where entire paragraphs or events in the transcript are assigned a code. The codes can operate at different levels. In a first step, they are mostly descriptive. As the analysis moves forward, the codes become increasingly analytical and abstract (Yoong et al., 2013, p. 174). So-called in vivo codes are a way of directly representing the perspective of participants. These are codes taken word by word from a transcript. As in vivo codes represent the language of the interviewees, they serve as “symbolic markers of participants’ speech and meaning” (Charmaz, 2014, p. 134).

Researchers can use software for coding or do it manually (paper-pencil technique). Various software programs support the process of encoding and analyzing qualitative data. Goodrick and Rogers (2015, p. 585) divide these into the following four groups:

- (1) “Word processing and spreadsheet software, such as Word and Excel, which can be used for simple coding and analysis;
- (2) specialized qualitative data analysis software, such as NVivo, HyperResearch, MaxQDA, and Atlas-TI;
- (3) machine learning software, such as MonkeyLearn;
- (4) visual analysis software that produces word cloud or network diagrams of text.”

Using software for data analysis can be helpful in the research process. Programs for analysis “merely enable evaluators to store, organize, search, categorize, and group large volumes of mostly text-based data” (Yoong et al., 2013, p. 175). However, coding is still done by the researchers, which is why the software itself, despite all its advantages, is not an analytic solution (Birks and Mills, 2011, p. 101). There is also a risk that software will generate too much code and thus an unmanageable amount of data (Goodrick and Rogers, 2015, p. 586). At the same time, computer programs can help (better) organize and analyze the amount of data involved in large research projects with several researchers. However, sometimes researchers still prefer manual data analysis, as it gives them a more holistic view of the data. This may well be the case for in-depth analyses of individual interviews (Yoong, 2013, p. 175).

6.4 Defining categories

Coding means labeling text segments that can be smaller or larger. In a next step, the individual codes are grouped into categories. A category thus comprises a varying number of codes on a similar topic, problem, etc. in the data (Goodrick and Rogers, 2015, p. 564). For example, if researchers want to evaluate the implementation of a new financial education program, they can group codes that deal with teachers’ experiences. This group of codes then needs to be further analyzed and categorized, e.g. by creating a category with negative and a category with positive experiences (ibid.). In the course of coding, researchers must find a way to systematically abstract away from or summarize the abundance of codes obtained (Nightingale et al., 2015, p. 470).

6.5 Answering the research question

Answering the research question is the final step in the research process. Usually, a final report summarizes the main results of the evaluation through the lens of the research questions. The requirements for such a final report must be clarified in advance and often depend on client expectations, the target group, disciplinary requirements, etc. According to Mensching (2006, p. 340), clients expect not only a descriptive presentation of the results, but also practical evaluations and recommendations for action. There is no standardized way of presenting results. Researchers can visualize qualitative results in different ways, also using a variety of software programs. In many cases, the visualization will depend on the evaluation method and the research question. If a research project investigates the relationship between different categories (how aspects are connected), the visualization should reflect this.

In a scientific context, research projects are assessed based on quality criteria. According to Mayring (2018, p. 21), quantitative quality standards such as objectivity, reliability and validity can be a point of reference for qualitative research but are not sufficient on their own. They must therefore be supplemented with a view to assessing or evaluating the quality of qualitative research processes. Qualitative evaluations must adhere to the quality standards of qualitative research in general. These include, for example, intersubjective comprehensibility, reflective subjectivity or empirical anchoring (Mensching, 2006, p. 340). In this context, the literature discusses the extent to which there can be uniform standards at all, given the diversity of approaches and methods for qualitative research projects (Flick, 2006c). Flick (*ibid.*, p. 439) wonders, for example, how to measure how much knowledge was actually gained in an exploratory study or how to evaluate the appropriateness of the methods used for a specific research field and question. Assessing the originality of a researcher's access to the field or their methodological approach is equally challenging (*ibid.*). The same holds true for assessing creativity in dealing with data as well as the relationship between the single steps involved in the research process and the process as a whole (*ibid.*).

Leaving these questions aside, stakeholders agree that there need to be criteria for assessing research projects. In addition to standards for qualitative and quantitative research, specific criteria for evaluation studies can also be used, such as those of the Joint Committee on Standards (JCS) of the American Evaluation Association (AEA), the SEVAL standards by the Swiss evaluation society SEVAL or the standards of the German evaluation society DeGeVAL (Caspari, 2006).

7 Summary and concluding remarks

How can qualitative evaluations be carried out? Which are the methodological steps that need to be considered? Which qualitative methods can researchers use to collect and evaluate data? What are the advantages and disadvantages of each method? In which contexts is qualitative evaluation actually appropriate?

These are the questions this report is meant to provide answers to. It gives an overview of the main steps in the different phases of qualitative evaluations. These include planning, implementation, analysis and communication. In section 1, I introduced the characteristics of qualitative research. An important step in the planning phase is selecting adequate methods for collecting and analyzing data. Advantages and disadvantages of common qualitative surveying methods – interviews, focus groups, observations and desk reviews of documents – are mentioned in section 2. I discussed the development of appropriate survey instruments for each method in more detail in section 3. Next, section 4 dealt with a topic that is often strongly underrepresented in the literature, although it is essential for implementing evaluations or studies successfully: sampling and access to the research field. In section 5, I discussed how to apply the different methods, while section 6 focused on the analysis of qualitative data.

Note that I only provided an overview of the individual methodological steps in a qualitative evaluation cycle. It is advisable to explore these methods further, e.g. when making decisions about methods for gathering and analyzing data. An essential step here is adapting methods to the respective target group. Developing a guide to interview children, for example, poses different challenges for researchers than developing a guide for adults. To ensure that research in the field of inclusion and disability is not research about, but research with the target population, people with disabilities must take part in relevant research projects, regardless of their disabilities, and researchers must make methodological adaptations catering to their individual needs (Coons and Watson, 2013; Felbermayr, 2023; Unger, 2014). Researchers can face numerous challenges throughout evaluations. Hatry and Newcomer (2015) provide a list of 27 pitfalls to be avoided during the evaluation process (from planning to implementation and the dissemination of results). I recommend reading their article in the planning phase of an evaluation project to avoid methodological pitfalls. It is also important to consider challenges regarding research ethics and data protection regulations, which are covered in another publication of the OeNB Financial Literacy Evaluation Series.

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The do's and don'ts of qualitative interviewing

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What needs to be considered when conducting qualitative interviews in evaluation research? Following up on the article “Qualitative research evaluation – how to get from first ideas to a final paper” (Felbermayr, 2024) published in the OeNB Financial Literacy Evaluation Series, this paper takes a closer look at the do's and don'ts of conducting qualitative interviews. It addresses interviewers who have little experience in conducting qualitative interviews and are interested in learning about frequent sources of errors in interviewing. Its focus is on the subject areas of reflection, language and communication, interview settings and technology use. Moreover, it also discusses the additional considerations and methodological adjustments necessary for interviews with special target groups, pointing out, by way of example, the challenges that arise when interviewing children and people with disabilities.

Interviews are a popular qualitative method in evaluation and research. Schäfers and Schachler (2022, p. 229) even consider interviews the “supreme way of empirical social research”² when it comes to collecting data on subjective experiences, perspectives and evaluations. There is a wide range of qualitative methods available in both evaluation and research (Hopf, 2004). “The same is true for interview forms. Choosing the appropriate interview form depends on the research interest and research questions. The interview form then determines the structure of the guidelines, the degree of openness and structuring as well as the role of the interviewer” (Felbermayr, 2024, p. 6). What all forms of interviews have in common is that participants answer directly and immediately to the questions asked by the interviewer. Nevertheless, qualitative interviews cannot be compared with conversations on talk shows or political programs (Patton, 2015). While on the one hand, qualitative interviews must meet methodological requirements, on the other, they primarily serve as a data collection tool that helps us answer underlying research questions (McGrath et al., 2019, p. 1002).

“Particularly for beginners who dare to take on this medium, the activity of the interviewer may be seen as a sequence of tasks that can bring about uncertainties in the interviewer” (Hermanns, 2004, p. 209). With a view to avoiding errors, the literature provides a wide range of tips on how to conduct qualitative interviews (e.g. Patton, 2015; McGrath et al, 2019). Dieckmann (2017, p. 447) distinguishes three categories of error sources that frequently occur in interview situations: (1) respondent characteristics, (2) question characteristics, (3) interviewer characteristics and interview situation characteristics. This paper addresses interviewers who have little experience in conducting qualitative interviews and points out frequent

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² All passages quoted from German-language sources have been translated by the OeNB.

sources of error. It is essential that interviewers deal with possible sources of error, or with the do's and don'ts of interviewing, given that the quality of interviews has a "direct influence on data quality" (Emde and Fuchs, 2012, p. 342–343).

This paper is structured as follows: Section 1 provides useful information for the actual conduct of interviews. It focuses on interview situations and issues related to reflection, language and communication, interview settings and technology use. Given that interviews with special target groups require additional considerations and methodological adjustments, section 2 deals with conducting interviews with (1) children and (2) people with disabilities. By way of summary, section 3 provides an overview of important points to consider as an interviewer.

1 Guidance for conducting interviews

Interviews can be roughly divided into three stages: preparation, conduct and follow-up. For details on the methodological steps that must be taken before an interview can actually be conducted (e.g. preparing consent forms or interview guidelines), see the article "Qualitative research evaluation – how to get from first ideas to a final paper" (Felbermayr, 2024), published in the OeNB Financial Literacy Evaluation Series. The present paper focuses on interviewing as an activity and on the challenges related to it. What is essential for both drafting interview guidelines and conducting interviews is that interviewers familiarize themselves with the subject matter the interview will cover (McGrath et al., 2019). The challenges that might materialize in an interview situation can be assigned to the following subject areas:

- reflection
- language and communication
- interview settings
- technology use

1.1 Reflection

To ensure quality assurance, interviewers should reflect on the following aspects while collecting or evaluating data:

1.1.1 Proximity and distance

Interviewers should establish proximity with interview participants by empathizing with what is being said and putting themselves in their place to understand how they experience and interpret the world. At the same time, it is important to maintain distance to participants' statements by constantly questioning the horizons of meaning of terms used in the interview (Hermanns, 2004). Keeping distance to what is being said can be more difficult when interviewing people who have had the same experience as the interviewer. This is the case, for example, when a prospective teacher interviews other teachers. If participants have too much background experience in common with the interviewer, they may not elaborate on certain content because they assume that the interviewer will know this content anyway (Dieckmann, 2017). Interviewers, on the other hand, run the risk of failing to ask detailed questions about terms the participants use or processes they describe. Interviewers will ask fewer questions because they believe they are familiar with the content and that, therefore, they do not need more detailed descriptions. According to Dieckmann (2017, p. 300), "unfamiliarity can thus promote explication." If it is unavoidable that interviewers have a similar or even the same background as participants, this must be taken into account at the latest when analyzing the data – which is ideally done in interpretation groups.

On an emotional level, it is important for the interviewer to establish proximity and familiarity with the participants. Particularly in longitudinal research projects, where people are interviewed several times, there is a risk that participants will interpret the working relationship with the interviewer as friendship and that they will therefore experience the end of the survey as abrupt (Felbermayr, 2023). This is why it is important for interviewers to always reflect on their own proximity-distance relationship with the participants and to inform them in good time about the approaching end of their working relationship. For Przyborski and Wohlrab-Sahr (2014, p. 8), interviewers and participants are like travelers on the same train: strangers unlikely to meet again after the ride (in this case, the interview). This is why it is essential in qualitative research to establish a trusting working relationship between researchers and research subjects, who are usually strangers to one another. And this is also why a trusting working relationship is a core element of many definitions of research ethics (for details on data privacy and ethics, see Lorenz and Felbermayr, 2024).

1.1.2 Culture and power relations

As cultural beings, people have different expectations of an interview situation. An interview with deaf persons or with people with a migrant background may require the services of an interpreter – “someone who is culturally sensitive to the interviewee’s situation” (McGrath et al., 2019, p. 1003). The response behavior of participants can also be influenced by their cultural background. The following example from Dieckmann (2017, p. 441) illustrates that the type of questions or the way questions are asked must always be adapted to the respective cultural context:

“At least in Western cultures, most people have internalized one norm to such a degree that they will try to give a ‘true’ answer to questions from strangers as long as the cost of sincerity does not exceed a certain threshold that may vary individually. This, however, also means that distorted answers are to be expected for unpleasant or sensitive questions. In other cultures, by contrast, politeness norms play a predominant role and, consequently, priority is given to the idea of not offending the person who asked the question rather than to the truthfulness of the answer. Answering ‘no’ to a question, for instance, is frowned upon in some cultures.”

From a research ethics perspective, it is important for researchers to reflect on their own role and the power that comes with it. In an interview, the interviewer’s power is evident in the fact that they determine the interview topic or specify the content of the guideline. It is the interviewers who are (more or less) in control of the process. Then, there is a power relationship between the interviewer and the participants, and also a power relationship among the participants – especially if underage children and teenagers are part of the sample. McCarry (2012, p. 60) draws attention to the powerless position of the latter, which is evident, for example, in the fact that interviewing them requires a declaration of consent. Underage children and teenagers can only participate in research if their parents or legal guardians give their consent. This can lead to situations where underage children or teenagers would like to participate in a study, but their parents or guardians refuse to give their consent.

1.1.3 Protective behavior

Interviewers may tend to try to protect participants from unpleasant, problematic situations and thus to adopt protective behavior during interviews. Hermanns (2004, p. 211) gives two reasons for this: the “*fear of embarrassments* in the course of the conversation” and the “*fear of injuring intimacy or of personality crises.*”

Often, however, such protective behavior does not protect the participants but rather the interviewers themselves, e.g. from having to deal with traumatic episodes they do not know how to react to. This protective behavior is particularly evident when interviewers interview their own peers (ibid.). Protective behavior should not be confused with a responsible approach to unpleasant situations, e.g. when participants begin to cry during an interview or when they report traumatic experiences. Generally, the guiding principle of Charmaz (2014, p. 71) applies: “[Don’t] Leave when the participant seems distressed.” Interviewers have a responsibility toward participants and must not leave them in an agitated state after interviews. It is advisable to compile a list of contact addresses for psychosocial services, etc., and share them if necessary. Also, it is recommendable to ensure that participants are not left alone after a disturbing interview, but that a family member or some other trusted person is present.

1.1.4 Co-construction

In qualitative research, the prevailing understanding is that interviewers create the data together with the participants (McGrath et al., 2019, p. 1004). The interview situation is a joint construction of the interviewer and the study participants, especially as understood by constructivist grounded theory. Interviewers should therefore reflect both on their own behavior in the survey situation and on that of the participants, e.g. with regard to their choice of words, emotions and body language signals (Charmaz, 2014, p. 57–58). Other key influencing factors are aspects such as gender, race, age and power structures (ibid., p. 74). The interview setting must always be interpreted with regard to the individual context. This is because interviews always take place at a specific historical point in time and in a specific social context (ibid., p. 57–58). Certain gestures or facial expressions, for instance, can have different meanings in different cultural contexts (Felbermayr, 2023). “Consequently, the interviewer needs to be reflexive, conscious, and aware about how his or her role might impact the conversation between the interviewer and interviewee” (McGrath et al., 2019, p. 1004).

1.2 Language and communication

“Practice makes perfect,” as the saying goes, and this certainly applies to interviewing. People conducting interviews for the first time are often overwhelmed by the task. This is because during an interview, they must carry out many activities simultaneously: listen to participants and respond to their content; reflect on the guideline and prioritize questions if necessary; pay attention to participants’ nonverbal signals; reflect on their own nonverbal signals and communication, etc.

To ensure quality assurance, interviewers should consider a number of aspects that have to do with language and communication. Table 1 lists the most important do’s and don’ts in interviewing.

Table 1

Do’s and don’ts in conducting qualitative interviews: language and communication

Do’s	Don’ts
Ask short questions that are easy to understand	Don’t imitate the language of participants
Ask if there is something you don’t understand	Don’t ask too many questions at once
Allow for breaks, even if they feel long	Don’t rush when speaking or asking questions
Talk less and listen more	Don’t use too many leading questions and don’t evaluate statements
Think carefully before interrupting	Try not to provoke distorted responses from participants
Adopt a neutral stance	Try not to refer to the interview guideline too often

Source: Author’s compilation.

1.2.1 Do's

- **Ask short questions that are easy to understand:** During an interview, interviewers should ask short, easy-to-understand questions. They should not use sentences with complicated structures, string together several questions or use foreign words and long sentences. They should not use research questions as these are unsuitable in interviews. They are too complicated and usually contain theoretical categories or abstract terms the participants may not be familiar with. Interviewers should ensure that they phrase questions in a way that fits in with participants' life-world (Hermanns, 2004).
- **Ask if there is something you don't understand:** Interviewers should not hesitate to ask questions to ensure that they understand what the participants meant to say. People doing qualitative research want to understand the world from the participants' perspective. So it is essential that they ask participants to explain terms, processes and situations in detail and that they ask questions if there is something they do not understand. "Try, through the interview, to understand your counterpart's life-world so well that you could write a film-script for scenes from this world and could direct a production." (ibid., p. 213). Detailed questions should also encourage participants to talk more about the research topic. Questions such as "Why do you say that?" or "Can you tell me more about that?" are suitable for this purpose (Yoong et al., 2013, p. 162).
- **Allow for breaks, even if they feel long:** Asking detailed questions is important in interviews. Nevertheless, interviewers should make sure that they give participants enough time to think about the questions they ask. Beginners, in particular, might easily begin to feel insecure if participants are silent for what they feel is too long. They might interpret prolonged silences as a mistake in their question and, consequently, begin to correct their question, add to it or rephrase it. This, in turn, can make participants feel insecure. Of course, it is possible that participants do not understand a question. Based on my experience in conducting qualitative interviews with various target groups, however, I can say that participants will always clearly communicate that there is something they do not understand. In many cases, their silence can therefore be interpreted as a pause for reflection. Such pauses can help participants sort out their thoughts (Dieckmann, 2017; Hermanns, 2004). Dieckmann (2017, p. 297) recommends the following rule of thumb: "Every time the participant stops speaking, interviewers should pause for three seconds before resuming communication."
- **Talk less and listen more:** This request to interviewers by McGrath et al. (2019, p. 1004) sounds simpler than it actually is. Beginners, in particular, may have "Problems with the passive-receptive aspects of interviewing: difficulties and lack of patience in listening and in picking up stimuli for supplementary questions." (Hopf, 2004, p. 208). They want to do their job particularly well and are therefore overly active during interviews (McGrath et al., 2019). However, the interviewer's task is precisely to listen in order to learn from the participants. Interviews are not conversations with friends, with everyone ideally having an equal share in the conversation (Felbermayr, 2023). Interviewers have to keep a low profile. Not talking does not mean doing nothing. Active listening is more strenuous than it may sound. "Actively listening to the interviewees means respecting silence and identifying such silent moments as an opportunity for ongoing reflection" (McGrath et al., 2019, p. 1004).
- **Think carefully before interrupting:** For many interviewers, the question is whether and, if so, when participants may be interrupted. Interruptions should not be ruled out in principle, but they require careful consideration. Participants should not be given the feeling that they are being cut off or that their comments are of no interest to the interviewer (Dieckmann, 2017). As an interviewer, you

should generally “intervene as little as possible in ways that specifically influence participants’ narrative flow or explications” (ibid., p. 308). Neither should they interrupt long, digressive narratives that feel like they might not relate to the interview topic. Often, the relevance of such narratives to the research interest only becomes apparent when the respective text passages are analyzed. In addition, interviewers are often pressed for time or focus on collecting information on a specific topic, which is why they cannot let conversations run indefinitely. As a compromise in dealing with digressions, Dieckmann (2017, p. 312) recommends the following approach: “Interviewers [should] let the communication process run for another three to four minutes as soon as they have the feeling that the participant is digressing. If, after this time, no idea has yet emerged as to what it is all about, it may actually have nothing to do with the research question.”

- **Adopt a neutral stance:** Interviewers should adopt a neutral stance in interviews, i.e. they should neither agree nor disagree with what is being said. It is not the interviewer’s job to evaluate or sanction statements made by the participants – neither verbally nor nonverbally. The requirement of neutrality is another feature that distinguishes interviews from everyday conversations. Unlike in interview situations, in everyday conversations we react directly to what the other person says. It is this behavior in particular that is not desired in interviews (ibid.).

1.2.2 Don’ts

- **Don’t imitate the language of participants:** Taking into account participants’ life-world should not be confused with imitating their language. If participants use certain terms or names, the interviewer should use these as well. However, interviewers should not imitate the language of the participants’ milieu, e.g. by using youth language when interviewing students (Hermanns, 2004). Instead, it is important to speak to participants as a layperson and to use sentences that are easy to understand (McGrath et al., 2019).
- **Don’t ask too many questions at once:** Asking questions is part of the interviewer’s core business and is essential for understanding. Nevertheless, questions should be asked carefully and, in general, care should be taken not to ask too many questions at once. This might be the case when interviewers e.g. ask three questions in a row, shedding light on a different, new aspect with each question. Participants may get confused by too many questions. Moreover, their answers may be influenced by cognitive psychological factors (e.g. primacy effects) (Dieckmann, 2017).
- **Don’t rush when speaking or asking questions:** Beginners often tend to go much too fast in interviews. On the one hand, their own speaking rate may be too high. On the other, they may tend to mask their own uncertainty by asking too many questions at a time, without giving the participants time to think (ibid.). As the speaking rate usually increases when speakers are nervous, interviewers should pay conscious attention to their speed of speaking, aiming to create a pleasant, unrushed atmosphere. Moreover, high speaking rates may require increased concentration from participants, causing (earlier) symptoms of fatigue. If participants show signs of fatigue or exhaustion, the interviewer might want to discontinue the interview. In such situations, it may make more sense to make a follow-up appointment.
- **Don’t use too many leading questions and don’t evaluate statements:** According to Hopf (2004, p. 208), interviewers should avoid a “domineering communication style.” This includes using a suggestive communication style (e.g. by frequently asking leading questions, using suggestive specifications,

interpreting what is being said) or evaluating or commenting the participant's statements as such behavior can have a disruptive or distracting effect on the course of the interview.

- **Try not to provoke distorted responses from participants:** In general, interviewers should make sure that they do not influence participants' responses. Although interview guidelines are validated before use and interview questions are tested for suitability, communication issues may occur in interviews, e.g. when participants ask questions back, when they do not understand interview questions, ask for further explanations or begin to answer prematurely. In cases like these, it is the interviewer's job to solve these issues in a nondirective manner. In any case, interviewers should take care to avoid getting distorted responses from respondents (Schäfers and Schachler, 2022).
- **Try not to refer to the interview guideline too often:** Beginners in particular tend to refer to the guideline frequently when conducting interviews, e.g. by repeatedly mentioning it to participants (e.g. "Now these were the questions of Part 1.") or using it as a disciplinary tool (e.g. "Please keep your answers short so there will be time for all my questions."). Lack of practice coupled with fear and uncertainty often leads to an "inflexibility in dealing with an interview-guide" (Hopf, 2004, p. 208). This means e.g. that interviewers do not bring forward a question from the guideline when suitable but rigidly adhere to the sequence of questions in the guideline.

1.3 Interview settings

In addition to aspects of language and communication, interviewers must consider a number of issues regarding the setting of an interview (for further details, see Felbermayr, 2024). Specifically, they should avoid making unclear agreements with the participants. The place, time and duration of an interview should be agreed upon precisely. Otherwise, this "(...) leads to a set of unfavorable conditions, because, for example, the subjects are short of time and the interviewer comes under pressure of time in the course of the conversation" (Hermanns, 2004, p. 210). Allocating too little time for an interview is one of the most common planning mistakes. Interviewers should make sure to avoid it (Hopf, 2004). They should also be on site in good time before the start of an interview, allowing for a time buffer for travel. If interviews take place on the premises of an institution, the interviewers may try to make the interview room comfortable in advance to ensure that the interview does not take place in a sterile atmosphere (Williams, 2015).

1.3.1 Location of interview

It is not always possible for the interviewer to determine the location of an interview (e.g. when interviewing students at school). In any case, the needs of the participants should take priority. Ideally, they should be free to choose the interview location. This requires flexibility from interviewers and can lead to interviews taking place in different, possibly distant locations. On the other hand, this offers the advantage that the interviews are conducted in a place where the participants feel comfortable and safe (Buchner, 2008; Felbermayr, 2023).

Some people like to conduct conversations in their own homes, while others see this as an invasion of privacy. Interviewers should be able to offer participants an alternative interview location if private premises are not an option. Ideally, this should be a professional setting (e.g. in a meeting room at the institution involved). Conversations in public spaces, such as cafés or parks, require careful consideration. There may be loud background noises, which may affect the sound quality of audio recordings. Also, in public spaces the interviewer cannot guarantee that what is being said will not be overheard by others,

which might give rise to concerns of data privacy and research ethics. In a café, moreover, interviews may be interrupted by waiting staff.

In general, the location of an interview can influence interview dynamics. This is particularly obvious in an educational context. Students behave differently depending on whether they are interviewed at their school or in a more private setting, e.g. at a youth center. School as an institution is associated with certain feelings, behaviors, sanctions, etc. All of this can have an influence on students' answering behavior in an interview. This does not mean that conducting interviews at schools should be avoided. However, as with every interview, the interviewer will have to reflect on the possible influence the interview location may have on analyzing the answers given.

1.3.2 Seating arrangements

Interviewers should take care to create a pleasant atmosphere for interviews. "In the first few minutes the interviewer has to create a situation that is so relaxed and open that the people in it can lay bare, without fear, a great variety of aspects of their personality and their life-world" (Hermanns, 2004, p. 210). In addition to the choice and design of the interview location, seating arrangements can make a significant contribution to creating a pleasant atmosphere. Comfortable seating is the preferable choice (McGrath et al., 2019). Whether the interviewer chooses a table with wooden chairs, armchairs or other seating arrangements depends on the target group and interview format. Tables create a barrier between people, but at the same time they can also provide a kind of protection for interviewers. Such an arrangement can be more effective for expert interviews. For biographical interviews with young people, comfortable armchairs or couches may be more appropriate. "Ideally, the seating arrangement corresponds to an 'egalitarian structure,' so that all participants experience equal treatment in their subjective perception" (Lamnek, 2005, p. 120). "A large, round table is best suited for this; with no one 'chairing' or sitting at the head of square table formats. The seating arrangement must also take into account participants' individual needs. Sign language interpreters must sit opposite deaf people or in the deaf person's field of vision. Visually impaired people should not sit opposite windows due to the incidence of light, for example, as it is difficult for them to see people sitting in front of windows" (Felbermayr, 2024, p. 10).

1.4 Technology use

Before conducting an interview, the interviewer must decide whether the interview should be technically recorded and, if so, whether by tape and/or video (on the advantages and disadvantages of various recording options, see Felbermayr, 2024). A popular method is audio recording with a dictation machine. Before the interview, interviewers should familiarize themselves with the technology they are going to use (McGrath et al., 2019) to make sure they feel safe and relaxed when operating it. This, in turn, can have a positive effect on the atmosphere of the interview. Many interviewers find tape recording unsettling. They feel uncertain because of possible technical complications and because they feel uncomfortable having to listen to their own voice and possible mistakes later on (Hermanns, 2004). In most cases, however, interviewers are not the only participants to be excited during an interview. For many participants, it may be the first time they are being interviewed, and they might be nervous and uncomfortable. According to Hermanns (2004, p. 210), it is the interviewer's duty "to accept responsibility (...) for *managing the feeling of 'recorder-discomfort'*, by demonstrating – with the machine running – that, irrespective of the fact that a recording is being made, it is possible to speak in a relaxed and open way (...)"

There are several options for dealing with the technology used. In my experience, it is not advisable to

place the audio recorder on the table and start recording without comment. Such an approach is to be rejected from a research ethics and data privacy perspective as well. Participants must always be told when a recording begins and how long it will take. For me, it has worked well to place the audio recorder on the table after preliminary talks and the warm-up phase and to comment on it accordingly, e.g. “I am now placing the audio recorder on the table and will start recording.” It is okay for interviewers to communicate their own excitement about this to participants, e.g. “It’s always exciting for me to start the tape. Will it work?” Whether or not this is advisable must be decided on an individual basis, however. Not all interviewers feel comfortable sharing their own insecurities with the participants.

2 Interviewing special target groups

The points discussed in section 1 also apply when conducting interviews with special target groups. In addition, specific aspects must be considered. In the following, we will examine two target groups (children, people with disabilities) and their special needs. Before conducting interviews with members of these target groups, interviewers will be well advised to thoroughly study the relevant methodological literature. Moreover, it is advisable that they ask the prospective participants or – if necessary – their parents or guardians about participants’ individual requirements.

2.1 How to interview children

In research involving children, according to Kuhn (2019, p. 237), “we are faced with the fundamental problem that adults cannot ‘think’ and ‘feel’ children’s thoughts and feelings.” This means that being adults, they face the challenge of adequately reflecting children’s views. While interviewers have experienced their own childhood, they have long outgrown it and must now try to reapproach it (Punsch, 2002). Against this background, the question of how interviewers can ensure that they understand a child’s perspective has been widely discussed in qualitative research (Hartnack, 2019; Heinzl, 2012). Interviewers are faced with the fundamental challenge that they have to “totally understand the world from a child’s point of view” (Punch, 2022, p. 324–325). A wide range of qualitative research methods provide helpful tools for approaching the world of children.

It was not always a matter of course that children were interviewed for research purposes at all. For a long time, to gain information about children’s lives, research interviews were conducted with children’s primary caregivers only (Emde and Fuchs, 2012). When adopting a participatory approach, however, research should be conducted *with* and not *about* the group of persons concerned (Unger, 2014). This general requirement is in line with the current maxim for the conduct of childhood research to include children as actors in research and to collect data on their views (Mey and Schwentesius, 2019). Children are “experts on their own life-world” (Hunger et al., 2019, p. 169) and should be interviewed as such.

“Generally, qualitative methods are considered more appropriate for researching children owing to the methods’ greater orientation towards participants’ needs.” (Vogl et al., 2023). Various qualitative methods can be used to reflect the voices of children, such as photo evaluation (e.g. Feichter, 2015), thematic drawings (e.g. Schultheis, 2019; Kuhn, 2019) or puppet interviews (e.g. Weise, 2021). Another popular method is conducting interviews. However, the methods of empirical social research, and especially different forms of interviews, have been designed primarily for adults (Vogl, 2021, p. 142). When researching children, therefore, the interview method needs to be adapted, which, according to Vogl (2021, p. 142), requires a “basic understanding of age-specific abilities and developmental challenges.” At the same time, it is obvious from methodological literature that research involving children requires particular care (Hartnack, 2019;

Heinzel, 2012; Vogl, 2021; Vogl et al., 2023). This refers to ethical research considerations as well as methodological challenges in adapting survey instruments to children's individual needs. What definitely needs to be taken into account in researching children are "development-related aspects" (Vogl, 2021, p. 154) such as their linguistic, cognitive or interactive skills. When collecting data from young children, in particular, it is recommended to combine linguistic (e.g. interview) and nonlinguistic, symbolic forms of expression (e.g. photo evaluation) (Schultheis, 2019). This also means having conversations with the children. If children only produce drawings, for example, interviewers will have to focus on interpretation rather than on children's knowledge, as the drawings come without any further explanations.

2.1.1 What is most important when interviewing children?

"What you have to expect is short attention spans in young children, a literal understanding of language vs. unspecific language use, children's uncertainty and shyness, problems in the temporal localization of events (see Poole and Lamb, 1998) and in specifying frequencies and quantities" (Vogl, 2021, p. 147).

Age plays an important role, not only in children's social and communicative development (how can children express themselves in an interview?) but also in determining when children can be considered responsible and able to decide for themselves whether to participate in an interview or not. The age and *developmental stage* of children must always be taken into account. Abilities and developmental challenges can vary greatly with children's age (ibid.).

As far as children's development is concerned, according to Vogl (2021, p. 145), children at the age of five already have *language skills* that are sufficiently developed for them to take part in qualitative interviews. However, the question at which age children can definitely be "questioned by social scientists cannot be answered per se; rather, this depends on the research interest and the methods and techniques used" (ibid., p. 148). At any rate, interviewers should adapt their communication to the linguistic abilities of the children they interview. They should avoid questions that are too complex or questions that are divided into several individual questions (Emde and Fuchs, 2012). Moreover, most children lack experience with interview situations. Interviewers need to take this into account when designing interviews with children, as younger children in particular can be overwhelmed by complex interactional situations (Vogl, 2021). In any case, children must never be overburdened.

2.2 How to interview people with disabilities

It has been frequently the case that people with disabilities have not been trusted to participate in qualitative interviews or research as such because – as Buchner (2008, p. 518) points out – in the opinion of some people they "do not have sufficient verbal skills to express themselves in line with the method's requirements [...]" To ensure that research in the areas of inclusion and disability is in fact research *in cooperation with* the people concerned, regardless of their disability, it is important to actively involve them in the research and give them a voice (Coons and Watson, 2013; Unger, 2014).

The needs of people with disabilities can vary greatly depending on the type of disability. As this paper can only outline a few aspects and examples, interviewers are advised, at any rate, to read the research literature relevant for the respective target group before designing or conducting interviews (e.g. Buchner, 2008; Hermanns, 2004; Scott-Barrett et al., 2018). When interviewing deaf persons, for instance, different questions arise than when interviewing people with intellectual disabilities. Interviewing deaf persons

requires an interpreter, which means an additional person will be present at the interview. If interviewers lack experience with such communication situations, the setting should be discussed with the participants in advance: Do the deaf persons choose their interpreter themselves? Who pays the costs? Should interviewers look back and forth between the deaf person and the interpreter? Where does the interpreter sit during the conversation?, etc. For people with intellectual disabilities, in turn, it may be more important to adapt the wording of questions to ensure comprehension.

2.2.1 What is most important when interviewing people with disabilities?

As a place of cooperation and communication, an interview depends on the eloquence of its participants, which “may mean making high demands on persons with poor communication skills [...]” (Dimbath, 2012, p. 311). It is the interviewers’ task to adapt the interview situation and interview guidelines to the individual needs of the participants. For some people, for example, free narratives can be too demanding, which is why interviewers should reflect on the suitable interview form and the guideline (Kremsner, 2006). It may be necessary to avoid foreign words or complicated sentence structures. Interviewers should conduct their interviews in an *easy-to-understand and comprehensible language*, which is by no means to be understood as “an oversimplified use of language” (Buchner, 2008, p. 522). According to Charmaz (2014, p. 96), it is advisable to speak the language of the participants: “Following threads in our participants’ everyday language and discourse helps us to form questions from their terms and learn about their lives.”

“Especially in interviews with people with cognitive impairments, however, there will be deviations from the ideal interview process. This has to do with comprehension problems on the part of the respondents, e.g. problems understanding abstract content, time estimates, questions with linguistic modifications and negations, which can lead to frequent clarifying questions, inadequate answers, lower response rates or systematic response tendencies” (Schäfers and Schachler, 2022, p. 230).

In a study, Schäfers and Schachler (2022) examined interviewer behavior in personal interviews with people with so-called cognitive impairments, analyzing the resulting problematic and unproblematic sequences of the interviews observed. They found, for example, that *repeated questioning* by interviewers was problematic for the flow of the interview. This is in line with the general requirement to avoid asking too many interview questions at once (see section 1.2.2), which applies to all target groups.

The research literature (Lewis and Porter, 2004) also discusses the use of visual methods, such as images, photographs or videos, as aids in interviews. Such aids should help participants capture the content of the interview in the best possible way. For participants with visual impairments, visual materials can be printed in a larger format and in stronger, contrasting colors. When interviewing people with learning-disabilities, pictures can be used in support. However, interviewers should carefully consider the use of images or visual aids as such. In the best case, they will reflect on their selection of images with other interviewers. There is always the risk that images symbolize something different than intended, are understood in a different way and thus lead to confusion rather than clarity. When an interview is recorded on audio tape, interviewers must bear in mind that the act of pointing at images will not be recorded. This means that, with this information missing, it may be difficult or impossible to perfectly reconstruct the meaning of individual text passages in the transcript. To eliminate any ambiguities, in such cases it is particularly important to take notes (for further details, see Felbermayr, 2024).

3 Conducting interviews – a summary

This paper highlights the do's and don'ts of conducting interviews in qualitative research, regardless of the specific interview method. It focuses on the subject areas of reflection, language and communication, interview settings and technology use. In this context, conducting interviews is only one part of the overall process under review; what is equally important is preparing for the interview (e.g. drafting interview guidelines and consent forms) and follow-up (e.g. writing notes). Qualitative interviews are very popular in the social sciences and are used in a wide variety of research contexts with many different target groups. Interviewing special target groups such as children and people with disabilities requires specific methodological adjustments. Interviewers working with these groups need sufficient time to carefully plan and conduct interviews and to collect and analyze data while being aware of the related challenges.

It is important to bear in mind that conducting interviews requires experience and knowledge. Just as expertise is required for creating quantitative survey instruments, a basic understanding of the methodological challenges and the necessary know-how are required for the careful conduct of high-grade qualitative interviews. Experienced interviewers know “when it is appropriate, in terms of content, to depart from the question guidelines, when it is essential to ask more intensive follow-up questions and when it is of particular significance, for the research interests of the project, to ask only very unspecific questions and to arrange for interviewees to have broad opportunities for self-expression” (Hopf, 2004, p. 207). Interviewers who lack practical or theoretical experience will be able to conduct qualitative interviews provided they undergo precise preparations such as reading the corresponding methodological literature (e.g. Felbermayr, 2024).

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Conducting representative surveys and impact assessments: an overview of survey methodology

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Yielding findings from samples that are applicable to broader populations is a fundamental goal in financial literacy research. Generating valid and representative findings is methodologically challenging, however. It requires the careful consideration of sampling strategies, survey modes, participant recruitment and weighting procedures – methods that are often summarized under the umbrella term of survey methodology. This paper offers an accessible introduction to the core principles of survey methodology, with a particular focus on how these principles can support robust, representative insights gained from financial literacy research. It aims to introduce financial literacy program designers to these fundamental methodological considerations and encourage researchers to conduct more robust and methodologically transparent studies and evaluations.

Quantitative research usually aims for results that are applicable to a wider population. In the context of financial literacy, this may concern e.g. nation-wide surveys on the levels of financial literacy and similar properties of a country's adult population or the effects of a financial education intervention on the saving behavior of a country's secondary level students. Obtaining results that can be generalized to the target populations is thus a fundamental aspect of research on financial literacy and financial education.

Imagine a research project designed to assess the level of financial literacy of a country's adult population. To accomplish this, researchers could develop a survey that measures key dimensions such as financial knowledge, attitudes and behaviors. To ensure that their findings are not merely reflective of those who responded to the survey but representative of the broader population, the survey must be grounded in robust principles of *survey methodology*.

Survey methodology is a collection of methodological concepts and approaches concerned with enabling valid and accurate statistical inferences from samples conducted through surveys. Often, however, the foundations of survey methodology, such as sampling, take a back seat in the design, analysis, interpretation and reporting of studies, despite the undeniable importance of survey methodology to the overall quality of most quantitative research in the field.

This paper explores the foundational concepts of survey methodology. We focus on the aspects of sampling and touch on the closely related issues of participant recruitment, survey modes and weighting. The paper presents conceptual foundations and methodological approaches in an accessible way, aiming to highlight their importance for research in the context of financial literacy and education and to provide financial literacy researchers with the fundamental insights necessary to understand and apply survey methodology.

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Because real-world research and survey designs are sometimes complex and strongly depend on the specific research context, this paper is limited to the conceptual foundations. Comprehensive introductory textbooks such as Groves et al. (2004) and Lohr (2021) provide additional details and cover a much wider range of concepts and methods. Methodological reports of large-scale survey or assessment projects (e.g. Albacete et al., 2023; European Social Survey, 2025b; OECD, 2024) provide concrete and detailed examples of complex, rigorous methodological approaches in different research contexts.

In the following sections, we explore basic conceptual foundations of survey methodology: In section 1, we outline the logic behind sampling, including the concepts of populations, samples, representativeness, bias, error and variance. In section 2, we describe common probability and nonprobability sampling approaches. In section 3, we highlight the importance of recruitment strategies and survey modes in field-work. In section 4, we briefly outline the logic behind different weighting approaches.

1 Sampling for statistical inference

Random sampling is the foundation of *statistical inference*, i.e. of the process of drawing conclusions about the broader population from a sample (see Lorenz et al., 2025, for a basic introduction to quantitative data analysis including statistical inference). This section covers key concepts of statistical inference closely connected to sampling, including populations, parameters, statistics, error, biases, representativeness, (external and internal) validity and generalizability. It explores how and why drawing random samples is essential for statistical *inferences* in any quantitative research context, whether it is used to compute a simple average, to measure complex constructs (e.g. financial literacy), to assess associations between variables or to conduct randomized experiments.

1.1 Population parameters and sample statistics

The population is the entire group of entities – typically individuals – researchers wish to research and learn about. It is specified by the researchers for their respective research project. Its size can range from very large, e.g. every resident in a country, to very small, e.g. all participants in an education initiative.

Quantitative researchers are often tasked with accurately measuring specific characteristics of an entire population, ranging from simple physical attributes (such as height) to abstract concepts (such as financial knowledge) to relationships between groups or variables, such as differences or correlations. In statistical inference, a numerical value that describes a characteristic of an entire population (e.g. the average age of all adults in a country) is called a *population parameter*. Since in most cases it would be very costly, impractical or even impossible to examine every single entity of a population, researchers often must rely on data from a subset of the population – a sample – to estimate the population parameter. For this purpose, they calculate a *sample statistic*.

For example, researchers may want to calculate the mean (average) from of a financial knowledge test conducted with 1,000 people living in a country (the sample statistic) that reflects the *true* average financial knowledge of all adults living in that country (the population parameter). Moreover, they may also be interested in *differences* in financial knowledge between population groups defined by sociodemographic characteristics such as age or gender. They thus seek difference estimates that reflect the *true* differences in the population. In general terms, any estimated statistic (e.g. a mean, difference or correlation) should accurately reflect the true population parameter.

Key terms of statistical inference

Target population: the entire group of entities (typically individuals) that researchers aim to study and learn about. The target population is defined by the researchers and can range in size from small groups (e.g. financial literacy program participants) to large groups (e.g. all residents of a country).

Population parameter: a true characteristic or measure of the entire target population, such as the mean financial knowledge score of all adults in a country. The population parameter is what researchers seek to estimate using data.

Sample: a subset of the target population. A sample is used when studying the entire population is impractical.

Sample statistic: a numerical estimate derived from a sample that is used to approximate a population parameter. Examples of sample statistics are e.g. means, differences and correlations.

1.2 Sampling principles

To perfectly calculate the population parameter of the mean financial knowledge, researchers would need to obtain answers from every single person of the population. As surveying an entire population is often not feasible, it is necessary to select a sample.

The goal of sampling is often (somewhat vaguely) termed “representativeness.” However, the concept of representativeness lacks a single, universal definition. For example, it is often linked to how well a sample reflects the population with regard to the distribution of certain population characteristics, e.g. age and gender (see section 2.2.4).

In a general statistical sense, though, a sample can be considered representative of a population if *any* relevant statistics calculated from it closely reflect the population parameters, i.e. the statistics that would be obtained if the *entire* population participated in the respective survey. In other words, a sample can be considered representative when insights derived from it can be generalized to the entire population. This means that there is low *error* that stems from the selection of participants, i.e. the differences between population parameters and statistics are minimized both in terms of *random sampling error* and in terms of systematic *sampling bias*, two concepts that are explored in subsection 1.4. We employ this statistical understanding of representativeness for the remainder of this paper.

The conceptually simplest way to achieve a representative sample is to draw a large, perfectly *random* sample from the whole population. While not every entity of the population is actually represented in the sample, by using such a *simple random sample*, each entity has the *same probability* of being selected into the sample (see subsection 2.1.1).

If every entity that is sampled participates in the survey, increasing the number of randomly sampled entities (i.e. the sample size) improves the chances that any population characteristic is well-represented in the data. Consequently, any sample statistic tends to approach the true population parameter as the sample size grows. Random sampling thus also enables *unbiased* statistical inferences about the target population, with confidence increasing with sample size. In a simple random sample, the sampling bias is (by definition) zero because each participant has an equal chance of being sampled (for an explanation of sampling bias, see subsection 1.3).

All statistics such as means, medians, difference estimates, correlations, regression coefficients and their associated significance tests or confidence intervals thus rely on sampling in the sense outlined above. With a random sample, the only limiting factor for the accuracy of an estimate is sample size. In other words, a sufficiently large simple random sample allows for deriving statistics that reflect any population parameters with high confidence.

Box 2

Representativeness in relation to generalizability, external validity and internal validity

Statistical representativeness is related to other terms that are frequently used in statistics, particularly the concepts of generalizability and external and internal validity. While statistical representativeness refers to the accuracy of a measurement resulting from the sampling process, i.e. to its generalizability to a specific target population, the concept of generalizability alone refers more broadly to the applicability of results, e.g. to other populations or contexts. External validity focuses on the findings of a study and whether they are applicable to contexts outside the research context. A study may have low external validity even if it provides a precise estimate of a population parameter, especially when the research setting – e.g. a highly controlled laboratory experiment – is overly artificial and does not reflect real-world conditions. In such cases, internal validity may still be high if the study accurately captures the causal relationship within that specific context. Internal validity is often ensured through randomization. The relation of randomization to sampling is outlined in subsection 1.5.

In practice, obtaining a simple random sample of a population is often anything but simple. First, sampling requires reliable data on the target population to serve as the *sampling frame*, e.g. a complete address register. From the sampling frame, a random sample can be drawn, which results in the *gross sample*. Second, the sampled entities in the gross sample, e.g. households, must be contacted, invited and willing to participate in the survey. Only then do they become part of the *net sample*, i.e. the final set of entities of which data are included in the analysis. The process of first selecting the sampling frame, then the gross sample and ultimately arriving at the desired net sample comes with many practical obstacles and thus several potential sources of bias (see section 1.5).

There are generalizations of the simple random sample that aim to mitigate some of the practical challenges of sampling from large populations. These sampling methods are no longer “simple” but are considered complex *probability sampling* methods, as they assign different probabilities of selection to different entities (see section 2.1). All sampling approaches that assign known and nonzero probabilities of selection to each unit in the population are referred to as probability samples or as (complex) random samples. They allow for the same valid inferences about the population as simple random samples, provided that proper design weights and variance correction techniques are used (see section 4). However, they come with many of the same challenges in achieving an unbiased gross and net sample as simple random sampling.

Nonprobability sampling, in contrast to probability sampling, typically does not rely on complete sampling frames of the population nor on systematic selection or invitation approaches (see subsection 2.2). It thus constitutes a fundamentally different class of sampling approaches that often severely limit the ability to

identify sources of bias or correct bias through weighting and thus to make statistical inferences. In practice, the principles and methods presented in this paper are primarily applicable to probability sampling approaches, where a complete sampling frame exists and the gross and net samples are clearly identified.

Box 3

Key terms of sampling

Statistical representativeness: in the context of statistical inference, the ability of sample statistics to reflect population parameters with low error. The theoretically simplest way to achieve a representative sample is drawing a large simple random sample.

Simple random sample: a sample drawn in such a way that each entity in the population has the same probability of being selected. If all sampled entities participate, this method guarantees that the sampling bias is zero and that the sample statistics will approximate the true population parameters better the larger the sample is.

Sampling frame: a register of entities that reflects the target population and can be used as the basis for sampling, e.g. the list from which individuals are randomly drawn

Gross sample: the sample that results from selecting entities from the sampling frame, e.g. the list of sampled individuals that are invited to participate in the survey

Net sample: the final set of entities from which data has actually been collected, e.g. individuals whose responses can be used for data analysis

Probability sampling: a generalization of simple random sampling in which each unit in the population has a known (but not necessarily equal) and nonzero probability of selection. Probability sampling allows researchers to account for selection probabilities and make the same population inferences as with a simple random sample.

Nonprobability sampling: a family of sampling approaches that do not rely on known and nonzero sampling probabilities, but rather on implicit or subjective methods of selecting participants.

1.3 Random sampling error and sampling bias

Error in a statistical estimate, i.e. the difference between the true population parameter and the sample statistic, can stem from any stage of conducting a survey, particularly from measurement (e.g. through questionnaires), the sampling frame, the sampling method itself and from nonresponse. The goal of survey researchers is to minimize all sources of error in the final estimates. This perspective on the quality of a survey is sometimes called the *total survey error framework* (see Groves et al., 2004). It is important to note that in this framework, the concepts of random sampling error (or sampling variance) and sampling bias relate specifically to the sampling method used to select entities from the sampling frame into the gross sample. However, these concepts can also be considered foundational concepts of statistical inference.

The sampling error can be regarded as consisting of two parts: 1) the *random sampling error* stemming from *sampling variance*, which is the component that may over- or undershoot the population parameter for a single sample but is zero *on average* over many (hypothetical) repeated samples; the random sampling error thus shrinks with larger samples; and 2) the *sampling bias*, which is a *systematic* deviation that causes

over- or underestimation even in large samples or on average over many (hypothetical) samples. We will describe the concepts of random sampling error and sampling bias in more detail below.

Selecting a (simple) random sample always leads to some degree of random difference between the sample statistic and the true population parameter. The extent of this difference is known as random sampling error. In classical (frequentist) statistical inference, a sample statistic is viewed as one possible outcome among many hypothetical repetitions of the same sampling process. To estimate how far the sample statistic might deviate from the true population parameter, statistical inference examines the distribution of these hypothetical outcomes, quantifying the sampling variance and thus the *expected* random sampling error.

The random sampling error is directly related to the sample size. If only few entities are randomly chosen as a sample, there is little chance that they will represent the population on average, i.e. any statistic estimated from this sample may be quite different from the population parameter it intends to capture. However, sampling more entities reduces the influence of random variations on the estimated summary statistics. It is possible to estimate the random sampling error from the values of a sample themselves, e.g. as the *standard error* of the sample mean, which indicates how much the sample statistic would vary across repeated samples from the same population. The standard error thus allows for quantifying the uncertainty of our estimates and enables tests of statistical *significance*.

The sampling bias is the more problematic part of the sampling error. It stems from a *systematically* (i.e. nonrandomly) unbalanced selection of entities from the population. For example, through their sampling process, researchers could be selecting a disproportionate number of young people even though they are interested in the entire population of a country. If the age of participants (or any other characteristic that may be misrepresented) is associated with the parameters researchers want to estimate, the estimates suffer from sampling bias.

Sampling bias cannot be prevented by choosing larger samples. Even if more and more entities are sampled in the same way, sampling bias cannot be expected to shrink. It is also impossible to directly recognize it from the collected values (e.g. the measurements of financial knowledge) themselves. In practice, there are different sources of bias that stem from different steps in the overall process of selecting entities from the population, as outlined in subsection 1.4.

Box 4

Key terms of error and bias

Random sampling error: the error in estimates that occurs due to random variation from sample statistics. This error decreases as the sample size increases.

Sampling bias: the systematic error of a statistic that is attributable to the sampling process. Unlike with the random sampling error, increasing sample size does not reduce the expected sampling bias.

1.4 Sources of bias

The overall bias from selecting survey participants can be differentiated into biases from different stages or from different processes in the overall sampling method. The following distinctions of biases are particularly relevant for probability sampling (see subsection 2.1), where samples are drawn in a systematic way. In nonprobability samples (see subsection 2.2), these distinctions may be artificial, for instance, if

neither a clearly defined sampling frame nor a sampling procedure exist. However, this by no means implies that such biases are absent in nonprobability samples. Similar types of bias still occur, often even as a result of the chosen nonprobability sampling method. Moreover, the bias may be exceedingly more difficult to identify, categorize or correct due to the unstructured and opaque nature of nonprobability sampling.

First, it is relevant whether a reliable register of the population – a sampling frame – exists. Discrepancies between the target population and the sampling frame may lead to *coverage bias*. For example, the telephone book may be an inappropriate sampling frame for the target population of a city's residents and relying on it when sampling may lead to a large coverage bias. An official address register may be more suitable. However, it may lack residents who are not officially registered. Coverage can sometimes be improved by manually removing wrongly included respondents from the sample or sampling frame or by making an extra effort to include initially excluded respondents.

Second, the sampling method itself is important, i.e. the process of selecting entities from the sampling frame into the gross sample, e.g. the method of selecting individuals to be invited to participate. In probability sampling, the sampling process may be a simple random sample, which does not introduce *sampling bias*. Other, more complex approaches of probability sampling, however, may sample clusters of entities based on certain characteristics (cluster sampling, see subsection 2.1.3) or use adjusted sampling probabilities for certain groups of entities (stratified sampling, see subsection 2.1.2). In principle, these processes result in sampling bias through the sample design itself. Because in probability sampling, the sampling probabilities are known, this type of bias can be corrected through design weights (see subsection 4.1).

Nonresponse bias (also called participation bias) occurs when some of the contacted entities in the gross sample do not respond, which potentially leads to systematic differences between respondents and nonrespondents, i.e. between the gross and net samples. Nonresponse bias can be further differentiated: *Noncontact bias* may occur when people have been sampled but cannot be reached through the chosen contact method. *Self-selection bias* arises when contact is successful, but people choose not to participate. For example, it is possible that people interested in a survey's topic are more likely to participate (self-select) or that some people invited via e-mail do not regularly check their e-mail inbox (noncontact) – both of these scenarios can lead to nonresponse bias.

Box 5

Key terms of sources of bias

Coverage bias: bias stemming from discrepancies between the population and the sampling frame

Nonresponse bias (participation bias): a form of bias that arises when sampled entities in the gross sample do not respond to a survey. This may lead to systematic differences between respondents and nonrespondents, i.e. between the gross and net samples, and affect the outcome.

Noncontact bias: a subtype of nonresponse bias that occurs when entities cannot be reached with the chosen contact method

Self-selection bias: a subtype of nonresponse bias that occurs when entities opt in or out of a survey based on characteristics that are associated with the estimated outcome

1.5 Sampling for experimental impact assessments

To evaluate the impact of a financial education intervention, e.g. on the financial knowledge of school children, researchers may want to know what would happen if the population of children received the intervention in question. To this end, they need a control group, i.e. a sample from the population of school children that do not receive the intervention, and a treatment group, i.e. a sample from the (hypothetical) population of children that do receive the intervention. They may achieve this by drawing two random samples from the population of school children and providing financial education to only one of them.

This approach is largely equivalent to drawing a random sample from the population and subsequently randomly assigning individuals to treatment and control groups. Such random assignment is called *randomization*. It guarantees that the assignment process is independent of the outcome or any other sample characteristics. In principle, randomization allows clear causal interpretations of the treatment effects and guarantees internal validity.

However, randomization can also occur after an initial *nonrandom*, biased sampling of participants. For example, researchers could first draw a nonprobability sample of school students, e.g. using a process where schools self-select into the education program, and only then randomly allocate students to treatment and control groups. Indeed, such a process is often used in practice (see Lorenz et al., 2025). While it guarantees *internal* validity, it sacrifices external validity and generalizability. In other words: The difference between treatment and control groups can be clearly attributed to the education intervention itself. However, the treatment effect cannot be generalized to the whole population of school students unless one can assume homogenous treatment effects across the sampled and unsampled population.

Indeed, although many impact assessments of financial education interventions are based on randomized experimental designs, studies often appear to give little consideration to the sampling bias of treatment effects, even though it appears likely that these effects would differ between the sampled and nonsampled parts of the population, e.g. in cases when schools self-select into studies on financial education interventions (see Lorenz et al., 2025).

2 Sampling methods

Although the simple random sample can be regarded as the foundation of quantitative research and statistics, the different goals and practical limitations of studies necessitate a range of other (nonrandom) sampling approaches. Sampling methods can be broadly categorized into probability and nonprobability sampling techniques.

Probability sampling techniques include all generalizations of simple random sampling that still employ random selection processes, even if some groups may have larger or smaller probabilities of being sampled than they would have in a simple random sample. In probability sampling, every entity of the population frame has a *known* and *nonzero* probability of being selected. Using weighting methods, the sampling probabilities can be counterbalanced for statistical analyses (see subsection 4.1).

In a nonprobability sample, nonrandom and often subjective methods are used to collect the sample, which means that the probability of selecting entities of the population is *unknown* or *zero*. While nonprobability sampling approaches can be valuable in the appropriate research contexts, they differ considerably from probability sampling and are typically severely limited in their ability to generalize findings to the entire population.

Understanding common sampling approaches may not only help in choosing a sampling approach for collecting data, but also in interpreting their results and the relevance they hold for policymakers and practitioners in the field of financial education. For instance, different sampling approaches may at least in part explain potential discrepancies in the results of different surveys on the same issues. The following subsections aim to provide a general overview of the key types of probability sampling, including a general outline of more complex approaches such as multistage stratified sampling.

2.1 Probability sampling

Probability sampling describes selection processes based on objective probabilities rather than subjective decisions. The simplest form of probability sampling is a simple random sample (as outlined in subsection 1.4). In a simple random sample, every entity in the population has the same independent chance of being selected. Other probability samples are generalized forms of the simple random sample, where each element in the target population has a *known* and *nonzero* chance of being selected into the sample.

Under ideal circumstances (i.e. a perfect sampling frame and no nonresponse), probability samples combined with proper design weights (see subsection 4.1) enable statistical estimates without sampling bias. Although probability sampling allows bias-free estimates in principle, in practice all sampling approaches tend to introduce bias. While weighting may help to reduce bias, it rarely eliminates it entirely (see section 4). Nevertheless, probability samples must be considered superior to nonprobability samples (Cornesse et al., 2020).

Probability sampling of large populations tends to be costly. It requires a suitable sampling frame, an appropriate method of contacting and incentivizing participants (see subsection 3.1) and considerable organizational effort and methodological expertise. Different methodological approaches can reduce the organizational effort and cost of probability sampling, as with cluster sampling (see section 2.1.4) for instance, but increase the complexity of the sampling and weighting process.

2.1.1 Simple random sampling

As outlined in section 1, simple random sampling is the most fundamental form of probability sampling and serves as the basis for statistical inference. In simple random sampling, every unit of the target population has an equal probability of being selected. In practice, a computer randomly selects the desired number of units from the sampling frame.

A nuance in (simple) random sampling is the distinction between sampling with replacement and sampling without replacement. Sampling with replacement would mean that any element of the population is replaced into the sampling frame after being selected and can be sampled more than once. As it is typically not desired that the same individuals participate in one and the same survey several times, sampling without replacement is used in practice (Groves et al. 2004).

2.1.2 Stratified sampling

Sometimes researchers want to guarantee representativeness with regard to some known characteristics of the population, e.g. residency in a geographical region. If the relevant characteristic is known and available to be included in the sampling frame, the sampling frame can be divided into subgroups based on this characteristic, and random samples are drawn from each subgroup. This ensures that the subgroups of interest are all well represented in the sample. The subgroups are called *strata* and the sampling approach is called *stratified sampling*.

Stratification protects against obtaining a sample that does not match known distributions of characteristics. Moreover, stratification can guarantee precise estimates for the specific strata. For example, in a national survey, researchers may want to ensure that both urban and rural populations from different regions of the country in question are represented adequately. A simple random sample might make it difficult to obtain precise estimates for some rural areas. By stratifying the sample on the basis of urban and rural regions, researchers can guarantee that both groups are properly covered and that accurate comparisons between them will be possible.

Sample sizes within the strata can be adjusted to match the intentions of the researchers. For instance, with proportional sampling, the sample size from each stratum proportionally reflects the size of the respective subpopulations. Using disproportionate sampling, the sample size in each stratum could be adjusted to achieve the required precisions of estimates. To achieve this purpose, there are techniques for the optimal allocation of sample sizes to strata. Based on prior estimates of variability within strata, these techniques can maximize statistical efficiency (e.g. the Neyman Allocation, Neyman, 1934; see also Groves et al., 2004).

2.1.3 Cluster sampling

In this section, we have assumed, so far, that researchers can obtain a suitable sampling frame that contains all relevant population members and that there are no practical limitations to the sampling process and the field work (i.e. recruitment and survey modes). In reality, however, both the sampling process and the fieldwork may be limited due to data availability and financial or time resources. To facilitate the sampling process and field work, probability sampling can be based on sampling groups (so-called *clusters*) of entities rather than on the entities themselves.

For example, researchers may want to obtain a probability sample from the population of all school students of a certain age to test their financial literacy (e.g. as in the OECD's PISA assessments, see OECD, 2024). However, it may be unfeasible or undesirable to 1) obtain a sampling frame of all eligible students, 2) directly contact students or 3) organize tests for students from too many schools. Researchers may thus decide, instead, to sample groups of students that are naturally organized in *clusters*.

For instance, researchers could draw a simple random sample of *schools* as clusters of students, instead of directly drawing a simple random sample of students from all schools. In this way, a probability sample is still possible, because 1) all students have a nonzero probability of being sampled via their school cluster and 2) the sampling probability of each sampled student can be calculated from the number of students per sampled school. Sampling a subset of schools can also facilitate the recruitment and testing procedures, because researchers may be able to rely on organizational support by the sampled schools' administrations and teachers.

Clusters may also be geographical regions. For instance, instead of sampling from the population of all individuals in a country to assess financial literacy levels, researchers may draw a sample of geographic administrative districts. Geographic clusters can considerably reduce organizational and travel costs of in-person interviews, because the sampled clusters of participants are located closely together and interviewers can visit them more easily and quickly.

Importantly, clusters may be populated by too many entities to interview or test all within a sampled cluster. This means a second sampling stage within clusters may be necessary, as outlined in subsection 2.1.4.

2.1.4 Multi-stage sampling

Multi-stage sampling involves selecting samples through a series of stages, moving from larger groups to progressively smaller ones. In practice, this often involves clustered sampling, where the population is first divided into clusters and sampling is then carried out at different levels of aggregation.

In the example of sampled regional clusters given in subsection 2.1.3, we noted that clusters may contain too many students to allow for interviewing all of them within each sampled cluster. Researchers may thus turn to another sampling stage at the level of individuals within clusters, which results in a two-staged clustered sampling design. For example, in stage one, ten schools (clusters) may be randomly selected. In stage two, within each selected school, twenty students are randomly sampled to participate in the survey. Indeed, the OECD's PISA assessments use a second sampling stage at the student level (OECD, 2024).

In the case of regional clusters, researchers may sample a certain number of individuals from each sampled cluster, e.g. a fixed number of participants or a number that is proportional to the subpopulation size of the regional cluster. To obtain a true probability sample, this second stage requires a complete sampling frame of individuals within clusters to enable sampling.

Multi-stage sampling can also be combined with stratified sampling. For instance, stratification may occur at the level of large geographical regions to ensure that these regions are well represented in the sample, while clustering happens at the level of smaller geographical regions for practical reasons and to reduce costs and final sampling at the level of individuals to limit the number of total participants. This design would be referred to as a stratified, multistage sampling design with clustering at the second stage.

In practice, the final sampling stage is sometimes quasi-probabilistic. For example, a survey could rely on *random route patterns*, i.e. interviewers following a pre-defined randomized path, starting from a randomly selected point within a given area. While such sampling methods may be necessary, for instance, if suitable sampling frames are unavailable, they are not considered probability sampling and may lead to bias (see e.g. Bauer, 2014). In other approaches, households are sampled first, and then individuals within households are selected by using probability or quasi-probability approaches, e.g. the Kish method or the birthday method, which may influence bias and interview refusals in different ways (see e.g. Jabkowski, 2017).

In survey research, sampling levels are typically numbered from the most aggregate level downward: starting with primary sampling units (PSUs) that are followed by secondary sampling units (SSUs), tertiary sampling units (TSUs) etc., down to the ultimate sampling units (USUs), which typically represent the entities of interest as defined by the target population.

As long as the sampling probabilities from the different sampling stages are known for the USUs (e.g. the individuals), design weights can be calculated that offset the different probabilities of individuals being selected into the sample. We give a short overview of weighting in section 4.

2.2 Nonprobability sampling

Nonprobability sampling is key to *qualitative* empirical studies that do not aim at the generalization of findings in a statistical sense but rather at the in-depth exploration of a phenomenon. However, non-probability sampling is also frequently used with quantitative surveys for the reasons outlined below.

First, nonprobability sampling typically costs less than probability sampling. Probability samples are more expensive as they require more rigorous planning, complete contact data and the contacting of

entities (often multiple times). The economic downside of probability sampling has become more pronounced recently as these methods often face low response rates (Elliott and Valliant, 2017). In contrast, nonprobability sampling typically relies, for instance, on volunteers, people living in geographical proximity, personal acquaintances, colleagues or members of an institution such as university students, which makes it easier to gather a substantial number of responses. While it may be economically viable, it is important to recognize that nonprobability sampling differs fundamentally from probability sampling in terms of the ability to make inferences about the target population.

Second, experimental studies such as randomized controlled trials often rely on nonprobability sampling. Although randomization ensures internal validity, generalizing results to a broader population rests on the assumption that the treatment effects are *homogeneous* across all units within the relevant population – an assumption that may not always hold. Randomization in the assignment of participants to treatment and control groups must therefore be distinguished from random or probability sampling of study participants (see subsection 1.5).

Third, certain research interests focus on populations for which comprehensive registers or lists are unavailable, i.e. where the population itself is not perfectly known. In financial education research, such populations may include groups like migrant communities, the unemployed or individuals at risk of poverty. For such populations, probability sampling may not be feasible and statistical representativeness may not be the primary research concern. Instead, the goal is to gain insights into specific phenomena or key issues affecting certain groups (Statistics Canada, 2021). In the following, we briefly present examples of nonprobability sampling techniques (for more details, see e.g. Statistics Canada, 2021; Vehovar et al., 2016).

2.2.1 Convenience sampling

Convenience sampling is a nonprobability sampling technique where participants are selected on the basis of their availability and proximity to the researcher, without any further selection criteria being applied. Typically, this method involves limited planning and preparation, which makes it a convenient approach to data collection. Participants may, for example, be recruited through personal contacts, through institutions or by approaching individuals in public (online) spaces. Convenience sampling is useful for gathering sizable data within a short period; the results it yields, however, are prone to severe sampling and nonresponse bias.

2.2.2 Volunteer sampling

Volunteer sampling relies on nonpersonalized invitations, thus allowing entities from the target population to actively opt into the sample. This approach emphasizes the proactive effort required by participants to self-select into the sample. Examples of volunteer sampling include invitations to a large audience (e.g. on TV, the radio or social media) or the use of printed invitations (e.g. posters or leaflets). An issue that is particularly pronounced with volunteer sampling is the problem of self-selection bias (see subsection 1.4). Those who actively volunteer to participate in the survey may differ significantly from those who do not (see Heckman, 1990).

2.2.3 Judgment sampling

Judgment sampling uses specific criteria or expert insights to actively select participants considered relevant to the research topic, aiming for a context-specific form of “representativeness” rather than statistical generalizability. Judgment sampling is useful when the composition of the target population is unknown or when it is defined by an abstract concept, such as “people who are financially vulnerable.” In such cases, researchers establish suitable selection criteria and actively recruit participants accordingly. Judgment sampling can be susceptible to subjective biases if the researcher has an inaccurate understanding of the target population. Therefore, the theoretical basis of the sample selection should be made transparent.

2.2.4 Quota sampling

Quota sampling is often used in convenience or volunteer samples or in web panels. It aims to align sample characteristics with the distribution of certain sociodemographic characteristics found in the target population, often including e.g. age, gender and education. Researchers thus select participants in a way that the resulting sample fulfills the quotas derived from census data. In practice, this may be achieved through the recruitment process, e.g. by distributing invitations through channels relevant for certain groups, or by simply rejecting people from participating in a survey if a quota is already fulfilled.

Although quota sampling utilizes information about the target population and may superficially resemble stratified sampling (see subsection chapter 2.1.3), it remains a nonprobability sampling method. As such, despite its ability to match certain characteristics of the target population, it likely suffers from (unquantifiable) sampling bias. In market research and opinion polling, quota samples may sometimes be labeled “representative,” as they match demographic quotas of the population (e.g. all age groups may be well “represented”). However, they cannot be called statistically representative, as they do not meet the criteria for statistical inference.

2.2.5 Access panels

Online access panels (which are sometimes called “online panels” or “web panels”) are a pool of registered participants who expressed a willingness to participate in future surveys. In response to declining participation rates in traditional surveys, nonprobability access panels have become widely used as cost-effective alternatives (Elliott and Valliant, 2017). While they are widely utilized not only in market research and opinion polling but also increasingly in scientific studies, access panels typically rely on volunteer sampling and financial incentives to achieve high participation rates. They should thus not be viewed as a replacement for probability sampling in studies where statistical representativeness is critical.

With access panels, polling firms maintain a pool of registered respondents who self-select into various surveys. Within these panels, quota sampling is often employed to align the sociodemographic characteristics of the sample with those of the target population (see subsection 2.2.4). Access panels must not be confused with probability-based panel designs such as rotating panels, which aim to collect longitudinal data from probability samples over multiple points in time.

3 Fieldwork

After selecting the sample, the next steps are known as fieldwork. Fieldwork includes recruiting participants and collecting data through the chosen survey modes. Fieldwork may be closely linked to bias, particularly with regard to nonresponse bias, as people's willingness to participate in a survey when contacted can be associated with both the survey mode and the recruitment strategy. Below, we provide an overview of recruitment strategies and survey modes.

3.1 Recruitment strategies

The goal of recruitment is to reach as many sampled entities as possible to maximize participation and minimize survey nonresponses. However, recruitment strategies depend largely on the available contact information for potential survey participants and vice versa. For instance, the available sampling frame may only include postal or e-mail addresses, which limits the potential contact channels.

When researchers can choose among different contact channels, they should carefully consider how certain contact channels are perceived by potential participants and how they affect response rates. For example, e-mails may often go unread or raise concerns about their trustworthiness. In contrast, personal visits before the interview or invitation letters tend to be more effective, yet perhaps costlier, alternatives (Daikeler et al., 2020).

Prenotification is a method to provide advance notice to respondents before recruitment. It has been studied as a potential strategy to increase participation. However, research suggests that prenotification is only effective when traditional communication channels such as letters or phone calls are used. In web-based surveys, prenotification efforts may have limited impact, as e-mails are more likely to be ignored (Daikeler et al., 2020).

Providing *incentives* is another effective strategy used to reduce nonresponses, with monetary incentives being the most effective one (Abdelazeem et al., 2023). Monetary incentives can be unconditional (paid before survey participation) or conditional (paid after survey completion). Unconditional incentives have been shown to be more effective, as participants may feel a sense of obligation to complete the survey in exchange for the monetary reward (see Abdelazeem et al., 2023, for a meta-study on the effectiveness of monetary incentives to increase response rates).

For some nonprobability sampling methods, sampling and recruitment occur simultaneously, as sampling is directly tied to participant recruitment. For example, in convenience sampling via a social media announcement, individuals are both selected and invited to participate at the same time. In such cases, the mode of data collection (discussed in detail below) may be directly linked to recruitment: When participants are recruited online, web-based surveys are likely to be much more convenient than arranging phone or face-to-face interviews.

Although probability sampling typically follows distinct procedures that can be considered independent, practical synergies between sampling, recruitment and survey mode can be used. For instance, if a personal interview at the respondent's home is planned, sending an advance prenotification letter to a sampled address to announce the interviewer's visit may increase participation. The actual contact attempt by an interviewer should then lead to the interview, if possible. Similarly, for telephone interviews, immediate recruitment via telephone may be the most practical approach.

3.2 Survey modes

Once participants agree to take part in a survey, their responses are collected using different survey modes (also referred to as interview modes). These modes can be categorized based on two key factors: The first factor is the data collection *channel*, which determines whether the survey is conducted face-to-face, by telephone, online or through a paper-based format. The second factor is the type of *administration*, which differentiates between interviewer-administered and self-administered surveys.

In interviewer-administered surveys, an interviewer poses the questions and records the responses. In contrast, self-administered surveys require respondents to complete the questionnaire independently. Face-to-face and telephone surveys are typically interviewer-administered, while web-based and paper-based surveys are self-administered.

A growing trend in survey methodology is the use of mixed-mode surveys, where multiple modes are combined. This approach may help mitigate biases associated with self-selection or the undercoverage of certain population groups, as it allows respondents to participate through different means while it potentially also reduces costs (Revilla, 2010). For example, older populations with limited internet access may be underrepresented in web-based surveys, while younger individuals may be less inclined to engage in face-to-face or telephone interviews. By integrating multiple modes or allowing respondents to select their preferred method, researchers may be able to reduce certain forms of bias. Additionally, mixed-mode surveys improve accessibility when some segments of the population require assistance in completing the survey.

Most survey modes utilize digital tools to facilitate data collection. Three common digital-assisted methods are *computer-assisted personal interviewing* (CAPI), *computer-assisted telephone interviewing* (CATI) and *computer-assisted web interviewing* (CAWI). In CAPI, the interviewer conducts the survey in person while entering responses into a digital device. In CATI, the interviewer administers the survey over the telephone while simultaneously recording responses digitally. Different types of online surveys are generally labeled CAWI.

The choice of survey mode depends on two primary considerations: *mode effects* and *resource constraints*. Mode effects refer to differences in *how* respondents respond and *which individuals participate* in the survey given the survey mode. These differences can arise from how questions are presented, how comfortable respondents feel providing answers or from the extent of interviewer involvement (Groves et al., 2004). Mode effects, therefore, play a crucial role in shaping data quality.

Self-administered surveys tend to reduce *social desirability bias*, which may make them particularly suitable for sensitive topics. However, they also tend to have higher *item non-response rates* or implausible responses as respondents may skip certain questions or may rush through the survey. An interviewer, on the other hand, can help clarify ambiguous questions, encourage participation and increase response rates. However, interviewer presence can also influence participants' responses, a phenomenon known as *interviewer effect*. This effect is especially pronounced when there is a significant power or authority gap between the interviewer and the respondent, such as an adult interviewing a child (Vogl, 2021).

Another key aspect of mode effects is whether survey questions are presented *verbally* or *visually*. Groves et al. (2004) explore the effects of some survey question presentation effects and conclude that verbal surveys, such as those conducted by an interviewer, often lead to *recency effects*, where respondents are more likely to choose the last response option they hear. In contrast, Groves et al. (2004) note that visually presented survey questions, such as those in web-based or paper-based formats, are more susceptible to *primacy effects*, where respondents tend to select the first option they see. The way questions

are displayed also influences response patterns and can lead to specific biases. For example, *straight-lining bias* occurs when respondents repeatedly select the same answer category; *central tendency bias* refers to the tendency to avoid extreme responses and consistently select middle answer categories (see Groves et al., 2004).

Different survey modes are associated with varying response rates. Meta-studies indicate that face-to-face interviews consistently yield the highest response rates, while web-based surveys experience the lowest (e.g. Daikeler et al., 2020). These differences are largely attributed to variations in recruitment strategies. When respondents are approached by an interviewer they perceive as credible they may be more likely to participate in the survey. In contrast, web-based surveys often rely on e-mail invitations, which may be disregarded (see subsection 3.1). The COVID-19 pandemic also appears to have had a lasting impact on survey participation. Since the pandemic, response rates – in particular in face-to-face interviews – appear to have declined significantly – a fact that has contributed to an increase in mixed-mode surveys (Charman, 2024; European Social Survey, 2025a).

For instance, the recent Austrian wave of the OECD's International Survey of Adult Financial Literacy (see Voith and Zieser, 2024a, 2024b) combined around 30 % of CAWI with 70 % of CAPI. Respondents were randomly assigned to either mode. Prenotification letters were sent to participants in both modes. While CAPI participants did not receive incentives, CAWI respondents received a EUR 5 unconditional incentive included in the invitation letter, followed by an additional EUR 10 upon survey completion. Despite these incentives for CAWI respondents, response rates remained notably lower, with CAPI achieving an average response rate of approximately 40%, compared to only approximately 20% for CAWI.

While the literature provides only a limited number of precise quantifications of economic costs across survey modes, web-based interviews are generally associated with lower financial costs, whereas face-to-face interviews are the most financially demanding due to their time- and labor-intensive nature. The costs of in-person interviews can be particularly high when experienced interviewers are required to travel long distances, make multiple contact attempts or conduct follow-up visits.

Choosing the most appropriate survey mode can be complex as it involves balancing mode effects, cost structures and survey objectives. Each mode presents unique advantages and disadvantages, making no single method universally superior. Researchers must carefully weigh trade-offs based on their study's priorities.

To minimize bias, face-to-face interviews – which can be supplemented by web-based interviews – continue to be recommended (see Daikeler et al., 2020). Due to their high cost, they are typically used for large-scale scientific surveys by public authorities that demand high standards. For highly sensitive topics, self-administered surveys can yield better data quality as they minimize social desirability bias. Particularly when surveying children, interviewer-administered modes may introduce authority bias, making self-administered surveys more suitable. However, a trade-off between sensitivity and complexity often needs to be managed: for example, actual financial outcomes or behaviors are likely to be highly sensitive, while their composition can be complex and difficult for non-experts to follow. In such cases, the presence of trained interviewers can help improve data quality by assisting respondents in navigating complex questions.

For studies involving repeated cross-sectional or panel data, it is crucial to maintain mode consistency over time. Frequent changes in survey mode can introduce measurement inconsistencies, making longitudinal comparisons unreliable. Thus, stability in the chosen survey mode is essential to ensure data comparability across survey waves.

4 Weighting

Weighting aims to reduce the sampling bias of estimated statistics that stems from unequal selection probabilities in complex probability sampling approaches by applying *design weights* and nonresponse bias by applying *nonresponse weights*. Moreover, under certain conditions, *post-stratification* or *calibration weights* can further reduce random sampling error and nonresponse bias by adjusting sample weights so that the distribution of some of the sample's characteristics match known distributions of these characteristics in the population. In practice, different types of weighting approaches can be combined.

It is important to note that weights typically increase the *variance* and *standard error* of the estimated statistics. Variance can increase because weighting gives more influence to entities from groups that are underrepresented in the sample compared to the population. Moreover, responses from sampling units within clusters are often correlated. For example, students sampled within a subset of schools may share similar characteristics. This *intracluster correlation* reduces the effective sample size because the information gained from each additional entities within a cluster adds less information than the information gained from entities that were sampled independently.

While weights can mitigate different sources of bias, complex sampling designs can reduce the precision of estimates compared to cases in which a simple random sample of the same size is employed. For accurate variance estimation and statistical inference, the sampling design and data structure must be incorporated into data analysis to obtain valid variance estimates, standard errors, confidence intervals and significance tests. Methods of variance estimation are beyond the scope of this paper but can be found e.g. in Groves et al. (2004), Lohr (2021) or ECB (2023).

4.1 Design weights

Design weights directly offset differences in sampling probabilities among entities of the sampling frame that result from complex probability sampling methods such as stratified or clustered sampling. For instance, if a fixed number of individuals is sampled from regional strata (see subsection 2.1.3), individuals from a more populated region have a smaller probability of being sampled than individuals from a less populated region. Without accounting for this circumstance, the individuals from the more populated region would be underrepresented in the sample. Similarly, if in a two-stage design individuals are sampled from households at the second stage, individuals from one-person households have a larger probability of being sampled than individuals from multi-person households.

A way to counteract these unequal probabilities of being selected into the gross sample is to calculate the different sampling probabilities within strata and clusters. Using the reciprocals of the sampling probabilities as weights for the sampled units of analysis (e.g. individuals) corrects for unequal selection probabilities and offsets the corresponding bias in the statistics.

Let us suppose, for example, that we use a stratified two-stage sampling approach where we sample 100 households for each of ten regions as strata. This is a disproportionate stratified sampling approach because the strata are populated by different numbers of households. From each household, we sample one individual, regardless of the size of the household. The sampling probability for one sampled individual from a two-person household in a region with 100,000 households may thus be calculated as follows: 1) The individual was sampled from a two-person household, i.e. as one from two possible individuals, with a probability of $1/2$. 2) The household was sampled as one of 100 from a region with 100,000 households with the probability $100/100,000$. The overall sampling probability of the individual is the product of these two probabilities, namely $1/2,000$, and its weight is 2,000. Weights calculated in this fashion can be applied to the calculation of statistics (e.g. point estimates such as means or proportions).

4.2 Nonresponse weights

Nonresponse weights aim to reduce nonresponse bias, i.e. bias that stems from an underrepresentation of participants that were sampled in principle (i.e. who were included in the gross sample) but could not be contacted or refused to participate (i.e. who were not included in the net sample). If nonresponse is associated with characteristics that are known for both sampled participants and sampled nonparticipants (i.e. for the entire gross sample), the probability of participation can be predicted for each entity on the basis of these characteristics.

The characteristics used as predictors may not only stem from information about the individuals themselves, but also from *paradata* such as interviewer characteristics, neighborhood observations or the documentation of contact attempts noted by the interviewer (Olson, 2013). The reciprocals of these participation probabilities are then the basis for the nonresponse weights that are applied each of the participants.

Suppose, for instance, that researchers have information about the gender, age and zip code of all sampled individuals. After sampling and data collection, and now knowing which sampled individuals have actually participated, they can use statistical models to predict the probability that an individual participates based on their gender, age and zip code (and any combinations thereof). The inverse of this probability can serve as the basis for the nonresponse weight for those that actually did participate.

For instance, young men living in large cities may, on average, be comparatively hard to contact or likely to refuse participation in an interview. Those who *do* participate are assigned low response probabilities based on their characteristics, and thus high nonresponse weights. Data from young men living in cities is thus upweighted so that they are better represented in the population estimates and nonresponse bias is reduced. This logic is of course applied to all participants and combinations of characteristics in this hypothetical example.

4.3 Post-stratification and calibration weights

Post-stratification is usually the last step in the weighting procedure. It aims to weight a sample so that its characteristics match the known distributions of these characteristics in the target population. It is thus similar to stratified sampling (see subsection 2.1.2) in its purpose, only that post-stratification is implemented after sampling and data collection.

For example, suppose researchers know (from reliable census data or other sources) that 70% of the target population live in region A and 30% in region B. However, in their sample, 50% of participants live in region A and 50% in region B. Decreasing the weights for all participants living in region A and increasing them for participants living in region B so that the sample's regional distribution matches the known population distribution ensures that both regions are properly represented in estimates and may help reduce bias.

If the characteristics used are correlated with the outcomes of interest (e.g. the region in question with the respective digital financial literacy scores), post-stratification can help reduce both nonresponse bias and random sampling error. For example, individuals from urban regions may be slightly more digitally financially literate than individuals from rural regions. If urban respondents are underrepresented, e.g. due to self-selection, the overall estimate of the average digital financial literacy of the population may be slightly underestimated. This bias can be reduced through post-stratification by region.

There are extensions of post-stratification (e.g. regression-based post-stratification methods and *calibration* methods) where the distribution of (sometimes complex combinations of) multiple variables is

used to minimize the overall deviation between sample characteristics and known population distributions. If the necessary variables are collected in the survey, such approaches can be used e.g. with distributions of age, gender, geographic region and other information available from national censuses or other reliable sources.

Post-stratification or any of its extensions thus hinge not only on the accuracy and availability of the benchmark distributions but also on the strength of the association between the characteristics used and the outcomes of interest. Even when many reliable benchmark distributions are used, post-stratification cannot correct for all sources of bias if relevant predictors of the outcome are omitted.

5 Conclusion

This paper provides an accessible overview of key components of survey methodology, with a particular focus on their relevance for achieving findings in research on financial literacy and financial education that can be generalized to the intended populations. Outlining fundamental methodological principles in an applied research context, it seeks to support both researchers and program designers in making informed decisions throughout the research process. It also aims to raise awareness of how methodological design influences the generalizability and the interpretability of findings.

Section 1 lays the theoretical foundation of the paper by introducing central concepts such as representativeness, target population, sampling frame, sampling error and biases. These concepts provide a rationale for why sampling is critical when it comes to statistical inference and to generalizing results from impact assessments and large-scale national surveys to broader populations. The subsequent sections relate these conceptual foundations to practical methods of sampling, fieldwork and weighting.

Section 2 emphasizes the importance of choosing appropriate sampling strategies to ensure that study results meaningfully reflect broader populations. It distinguishes between probability and nonprobability approaches and outlines widely used sampling strategies such as stratified, clustered and multi-stage sampling.

Section 3 focuses on fieldwork, showing different recruitment strategies and survey modes. It discusses how the choice of contact methods, incentives and data collection modes (e.g. self-administered vs. interviewer-administered; single-mode vs. mixed-mode) can significantly influence response rates and introduce or mitigate bias.

Section 4 introduces weighting procedures as tools to correct for various biases from the sampling method and from nonresponse. It explains the rationale behind design weights, replicate weights and post-stratification.

This paper is limited in scope and provides only a brief conceptual foundation to a vast and increasingly complex field of survey methodology, focusing on the most fundamental available methods and simplifying certain complexities. For instance, it does not address advanced sampling designs for longitudinal surveys, such as rotating panel designs (see Lynn, 2009), and it does not explore item-level nonresponse or imputation methods (see Little and Rubin, 2020). Also, the development of survey instruments themselves – such as selecting question types, testing for item validity and reliability, the importance of interviewer training and pretesting – goes beyond the scope of this paper.

Nonetheless, by condensing and simplifying complex issues, this paper helps newcomers and non-specialists deal with surveys in a financial literacy context. We also hope to contribute to a deeper understanding of the genesis of survey data by highlighting the importance of survey methodology as a sometimes neglected aspect that is essential to ensuring the quality of research in financial literacy and education.

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Quantitative data analysis in financial literacy evaluation research – visualization and statistical inference

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To evaluate financial literacy programs, researchers often work with quantitative data. To derive meaningful insights from raw data requires data analysis. In this article, we introduce two types of quantitative data analysis: descriptive statistics and statistical inference. Data visualization is a powerful way of descriptive data analysis but is frequently overlooked in evaluation handbooks. This article therefore emphasizes effective methods of data visualization. We show different types of visualizations that are effective for plotting time series, proportions, relationships and distributions. Additionally, we provide general guidelines to ensure an accurate representation of data and facilitate the effective interpretation of the intended message. For impact evaluations, however, descriptive statistics alone may be insufficient. Methods of statistical inference enable us to gain insights into populations from sample data and quantify the uncertainty of these insights. We thus provide an introduction to statistical inference by focusing on the principles behind hypothesis and significance testing.

Financial literacy evaluation research aims to assess the impact and underlying processes of financial literacy interventions. To this end, researchers must collect data, often in the form of structured quantitative information, e.g. from questionnaires or knowledge tests. However, raw data typically must be summarized to a high degree to enable general statements about the effectiveness of the intervention in question or of other relevant processes or associations. Data collected for this purpose can be qualitative or quantitative. Generally, each data type requires and allows for very different analysis techniques. Quantitative data (i.e. numbers in some sort of structured format) are analyzed using *statistical methods*. In contrast, qualitative data rely on textual information and a very different set of analysis techniques. In this paper, we provide techniques for quantitative data analysis, while qualitative and mixed-method approaches are discussed in other parts of the Financial Literacy Evaluation Series (see Felbermayr, 2024, and Lorenz, 2024).

The choice of quantitative analysis techniques depends on several factors: The first thing to consider is the goal of the analysis. Methods for quantitative data analysis can be assigned to either *descriptive statistics* or *statistical inference*. The goal of descriptive statistics is to summarize, visualize or identify patterns in the collected data. However, if the aim is to quantify the uncertainty of conclusions about a broader population (i.e. the target population of a financial literacy intervention) or test hypotheses, statistical

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inference is required. Because inferential statistics *infer* generalizations about the target population from a sample statistic, the related techniques must be based on a representative sample of that population to ensure validity. For impact evaluations, statistical inference is typically combined with descriptive statistics to reveal patterns in the data (e.g. outliers or missing data) and provide a basis for statistical inference analysis. Moreover, the same measures – like the mean – can be used in both descriptive statistics and statistical inference, but their purpose and interpretation differ. In descriptive statistics, the mean summarizes the sample data. In statistical inference, the sample mean is used to estimate or make conclusions about the population mean.

Another critical factor is understanding the nature of the collected data, particularly its scale of measurement (e.g. nominal, ordinal, interval or ratio). The scale of each variable collected defines what type of descriptive statistics and what sort of inferential statistics is applicable. For example, categorical data might require the visualization of proportions (descriptive statistics) or chi-square tests (statistical inference), while continuous data can be represented by distributions, measures of variability (descriptive statistics) or t-tests (statistical inference).

Finally, the target audience is another crucial factor when choosing the type of data analysis as it will significantly influence decisions regarding the choice of representation, such as tables or charts, the level of domain-specific language and how visually appealing an analysis needs to be. If the main purpose of an analysis is for the research team to better understand the data, the visual presentation and unambiguous interpretation will be less important than when the data are supposed to be communicated to a wider audience or used in publication.

This article is structured as follows: In section 1, we explore descriptive statistics, including fundamental summary statistics like mean and variance, which are essential for understanding data. The section also emphasizes the importance of data visualization, a powerful yet often underutilized tool in evaluation research. Moreover, it discusses guidelines and best practices for effectively visualizing relationships, time series, proportions and distributions. Section 2 focuses on statistical inference, covering key concepts such as hypothesis testing, statistical significance, p-values and regression analysis. As these methods are used in the majority of evaluation studies, a basic understanding of these inferential techniques is indispensable when it comes to interpreting findings and assessing evidence on the effectiveness of financial literacy interventions. Section 3 concludes.

Presenting statistical methods in a structured way and in the context of financial literacy research, this introductory article addresses evaluators, program designers and educators interested in descriptive statistics and statistical inference. It aims to refresh researchers' understanding of quantitative (inferential) analysis and advance their visualization skills. Moreover, it can be used by program designers or educators who wish to better understand the methods used in quantitative data analysis and derive meaningful insights from evaluations.

1 Descriptive statistics

1.1 Summary statistics

In quantitative data analysis, the concept of a *variable* is fundamental. A variable is an attribute that can vary over the unit of analysis. In the context of financial literacy interventions, the unit of analysis often covers *program participants* or *observations* across points in time. Variables can encompass all sorts of sociodemographic characteristics, typically age, gender, level of education or income. For instance, each

participant (the unit of analysis) of an intervention has a value for their variable age. Sociodemographic variables are often used to give context or to provide results for certain subsamples.² Moreover, financial literacy evaluations often include variables that measure financial knowledge, behavior or attitudes. These are commonly referred to as *outcome variables* because they reflect the primary results of interest in an impact evaluation. In addition, for process evaluations or program monitoring, standard outcome variables might include the number of participants reached or participants' satisfaction with the program.

Summary statistics provide an overview of the variables and can be assigned to two different categories: statistics for *central tendency* or measures for *variability and spread* (Yoong, Mihaly, et al., 2013). Additionally, measures of association between variables – such as correlations – can also be considered summary statistics, as they capture relationships within the data. Table 1 shows the most important statistics for central tendency as well as variability and spread and their definitions.

Table 1

Statistical measures

Central Tendency		Variability and Spread	
Measure	Definition	Measure	Definition
Mean	The sum of all data values for the variable divided by the number of values for the variable.	Range	The difference between the maximum and minimum data values of the variable.
Median	The middle value of the data values of the variable when the values are arranged in order.	Quartile	Divides the data values into four equal parts, each containing 25% of the values for the variable.
Mode	The data value that occurs most frequently for the variable.	Variance	Measures how spread out the values are by measuring how tightly the data are clustered around the mean.
Minimum	The smallest data value for the variable.	Standard Deviation	Is the square root of the variance and brings the measure back to the original units of the data, making it more interpretable.
Maximum	The largest data value for the variable.	Skewness	Measures asymmetry of the variable's distribution. positive values signal a longer right tail, negative values a longer left tail.

Source: Authors' compilation.

Another important differentiation is the scale level of a variable. Important distinctions are *continuous*, *discrete* and *categorical* variables.

Continuous variables can take on an infinite number of values within a given range, like height, weight, temperature or time. *Discrete* variables can take on a finite or countable number of values. They are often counts or categories with numerical labels, like the number of students in a class or the number of correctly answered financial knowledge questions. The difference between continuous and discrete variables is that the latter cannot be broken down meaningfully into smaller parts. For example, you can count 20 children in one class but not 20.5. Often, discrete variables with sufficient granularity are treated as continuous, e.g. age measured in full years.

Categorical data can be divided into distinct groups or categories, like educational levels (primary school, secondary school, bachelor's degree, master's degree, PhD), satisfaction ratings (unsatisfied, neutral, satisfied) or gender (male, female, diverse). Categorical variables such as educational levels and satisfaction ratings are so-called *ordinal* variables as they can be put in a meaningful order. This is not possible for *nominal* variables such as gender.

² In regression analysis, as outlined in section 2, these types of variables are often called control variables.

Understanding the scale of a variable is a prerequisite for choosing the right statistical methods and graphical representations for data analysis. Nominal data are usually reported as proportions (e.g. 50% of people observed in the dataset are male). Ordinal variables can be ranked and divided into intervals of equal probability; these are called quantiles. In this context, the median, quartiles, quintiles and percentiles are commonly used. For continuous and discrete variables, all central tendency and variability measures (table 1) as well as proportions and quantiles can be used to summarize the data.

While all these statistics can be reported in text form or in tables, in many cases it may be more valuable to visualize them in charts, which makes it considerably easier to grasp patterns in data quickly, especially for a nontechnical audience (Franconeri et al., 2021; Yoong, Mihaly, et al., 2013). Core data visualization techniques and tools are shown below.

1.2 Data visualization

Data visualization is the encoding of data through shapes, colors and animations, which must be chosen so that they can be easily decoded by the reader. Through decoding, you can examine variables from a different perspective and identify patterns that you might not have seen if they were presented in a table (Yau, 2011). Visualization can be applied at various levels of data aggregation, from plotting raw data points to presenting statistical information, including both summary statistics (e.g. means) and inferential statistics (e.g. confidence intervals). Because the core techniques for visualizing these different forms of data are largely similar, this subsection will focus on the visualization of raw data and summary statistics.

When approaching the task of visualizing a variable, the initial step is to consider the nature of the data – whether they are numeric or categorical, continuous or discrete.

Next, reflecting on the goals you want your audience to achieve and the narrative you aim to convey, ask yourself questions like: Do you intend to illustrate temporal trends (*time series*)? Are you interested in conveying relative *proportions*? Do you seek to explore *relationships* between variables? Are you aiming to display the *distribution* of a variable or identify *outliers*? Depending on your answers to these questions, different types of charts and graphs are suitable for effective data representation.

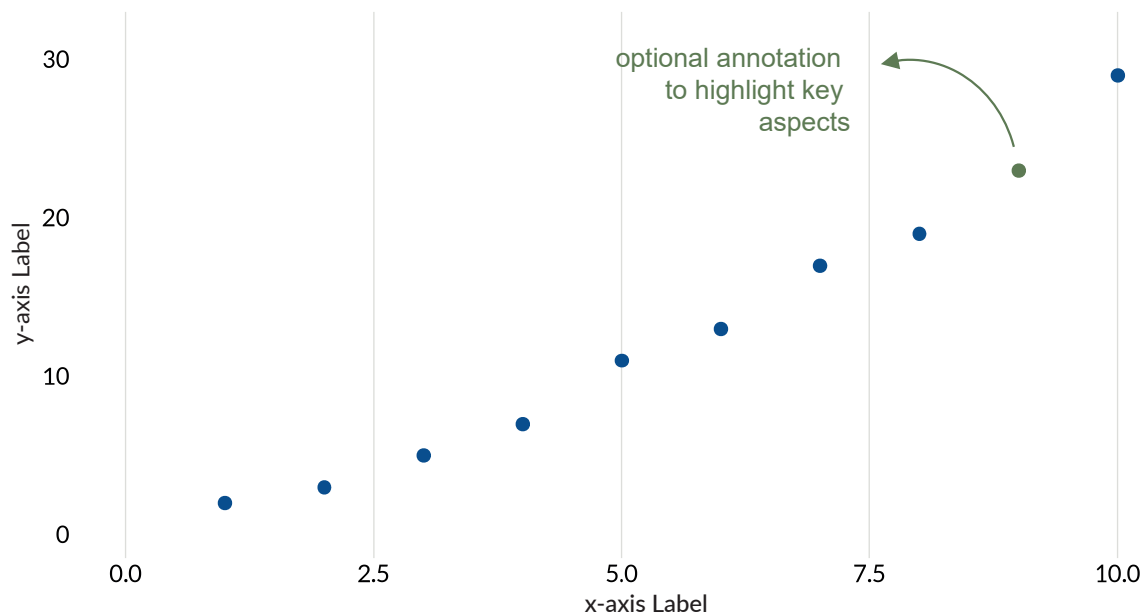
Below, we will describe the core characteristics of some popular chart types and explain which of them can best be used to answer the above questions. For more details, see *Storytelling with data* (2023) or Yau (2011). Most charts, regardless of their type or purpose, typically contain some core elements that serve to convey information effectively. These elements include:

- (1) **Title:** A clear, concise title describes what a chart shows, providing context and helping viewers understand the main message.
- (2) **Axes:** Charts usually map data to a two-dimensional coordinate system, using a horizontal axis (x-axis) and a vertical axis (y-axis) and creating a structure for mapping data points. Without axes, it is not possible to understand the exact values or relative differences in the data. Some charts, such as pie charts, present values not along axes but according to circular segments based on angles or areas.
- (3) **Data points:** Data points are the visual representations of the data values of one or more variables and are placed along the axes according to their coordinates. The representation of data points can vary with the chart type.
- (4) **Labels:** Labels are used to identify specific data points or features of a chart. They can be data point labels, axis labels or annotations.
- (5) **Source and note:** Information about the data source and any important notes or explanations related to the chart should be provided.

These core elements serve to make a chart informative and accessible to the audience (see chart 1). Depending on the chart type and the complexity of the data, there may be additional elements like a legend, subtitle, error bars, trendlines or grid lines. The choice of chart elements should be based on the data being represented and the message researchers want to convey to their audience.

Chart 1: Core elements of a chart

Title



Source: Simulated data.
Note: Sample scatter plot.

Below, we will show some typical chart types to represent 1) time series, 2) proportions, 3) relationships and 4) distributions. In the annex, we provide the R codes that reproduce the charts shown in this section.

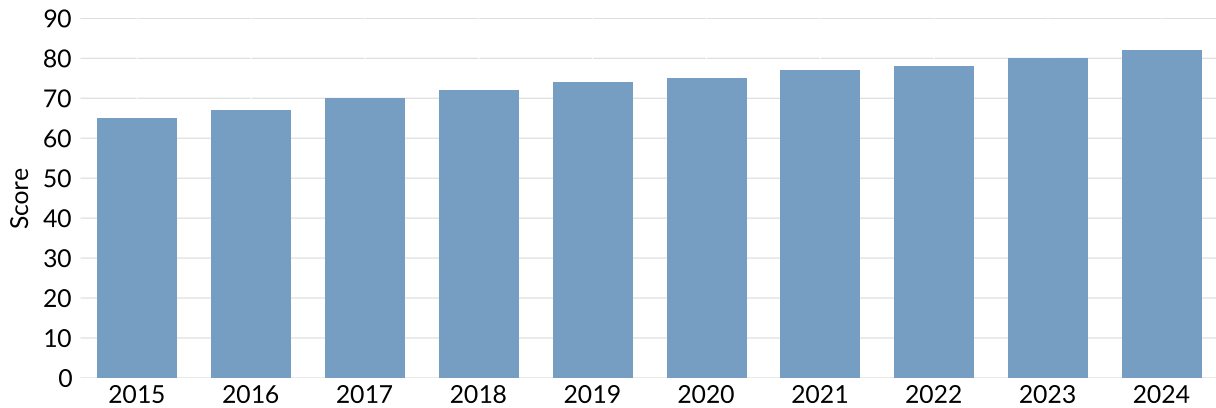
1.2.1 Time series

A time series is a collection of data points or observations recorded at regular intervals over a specified period, with each data point being associated with a particular timestamp. These data points are ordered chronologically and are used to analyze and study how a variable or phenomenon changes and evolves over time. For instance, one might examine the evolution of financial literacy scores in a group of students over several years. Typically, the time dimension is plotted on the x-axis (time axis), while the y-axis represents the variable's values. Time series are typically plotted using continuous or discrete variables. Common ways to represent time series data include bar plots and line charts. For a checklist on what to consider

when designing a bar chart, see Rapp (2022). In chart 2 and chart 3, we show how a bar chart and a line chart can be used to represent a time series.

Chart 2: Bar plot to represent time series

Financial literacy scores over the years

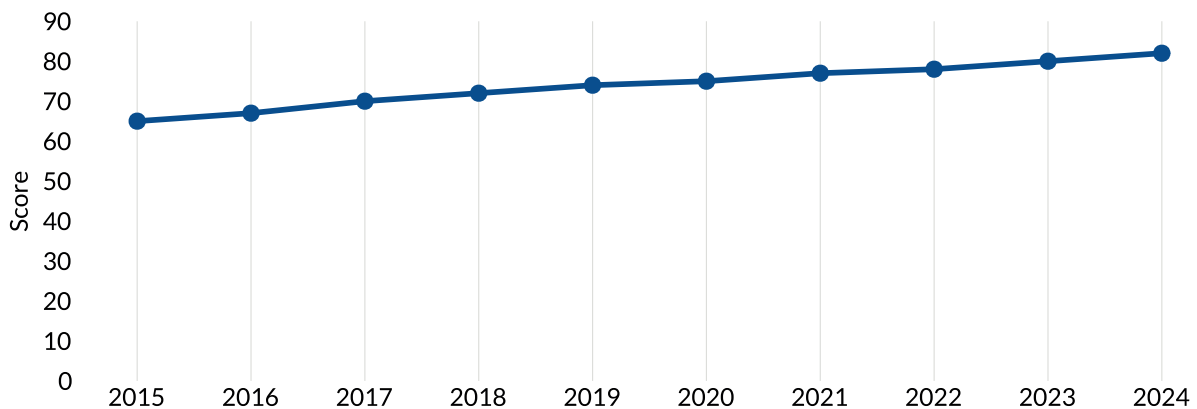


Source: Simulated data.

Note: Financial literacy scores of students over the years.

Chart 3: Line chart to represent time series

Financial literacy scores over the years



Source: Simulated data.

Note: Financial literacy scores of students over the years.

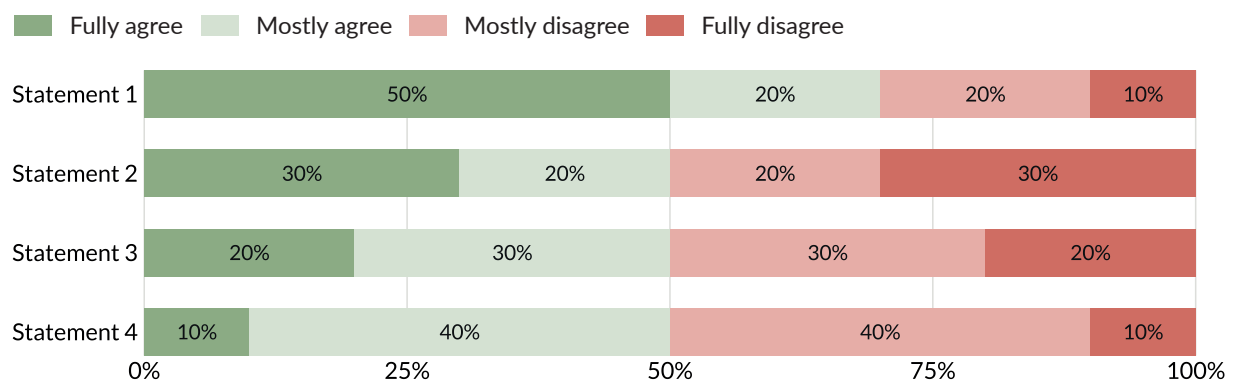
1.2.1 Proportions

Researchers frequently aim to illustrate the division of a group, entity or quantity into its individual components, with each part representing a portion of the whole to convey the relative significance of individual elements within the whole. For instance, you might analyze the distribution of a variable among men and

women within a population or the percentage of respondents who agree or disagree on a particular topic. Common ways to represent proportions include pie charts, donut plots, (stacked) bar charts and (stacked) area charts. It is worth noting that bar charts and area charts are often preferred over pie charts, as they facilitate easier comparisons, especially with numerous categories or small differences in proportions (Muth, 2018a). Square area charts are an alternative to pie charts as they offer improved comparability. Area charts are useful for plotting the change of proportions over time. The defining feature of proportion-representing charts is their ability to showcase how each component contributes to the total, whether through slices, areas or bars. Typically, categorical variables are used to represent proportions. In chart 4, we plot a stacked bar chart and in chart 5 an area chart to visualize proportions.

Chart 4: Stacked bar chart to represent proportions

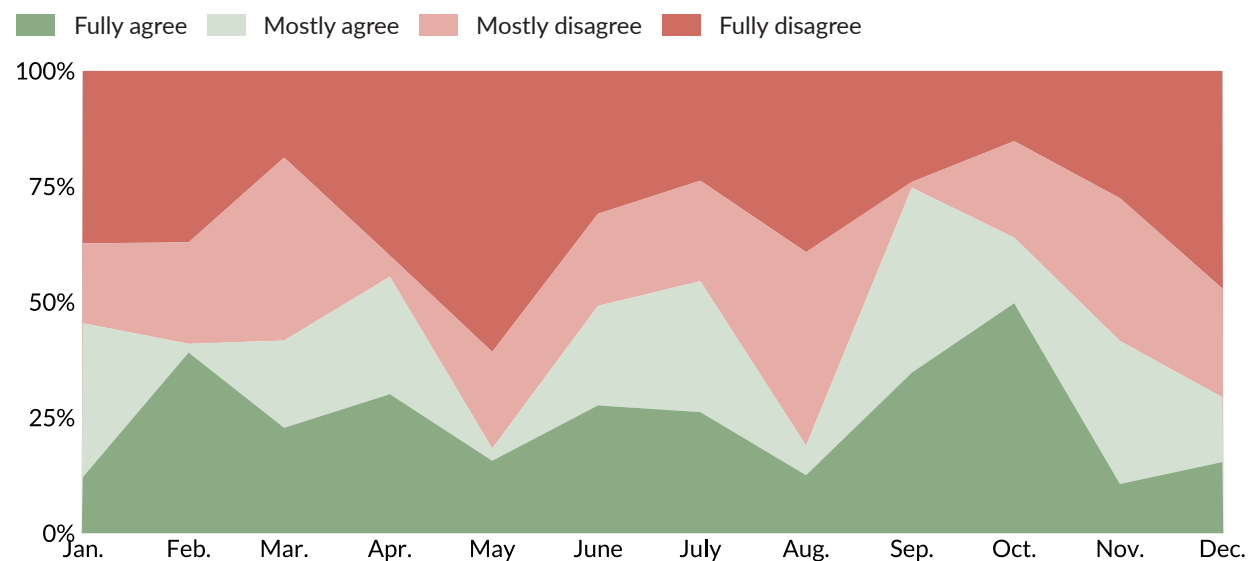
Agreement to statements



Source: Simulated data.

Chart 5: Area chart to represent proportions

Agreement to statement 4 | 2024



Source: Simulated data.

1.2.3 Relationships

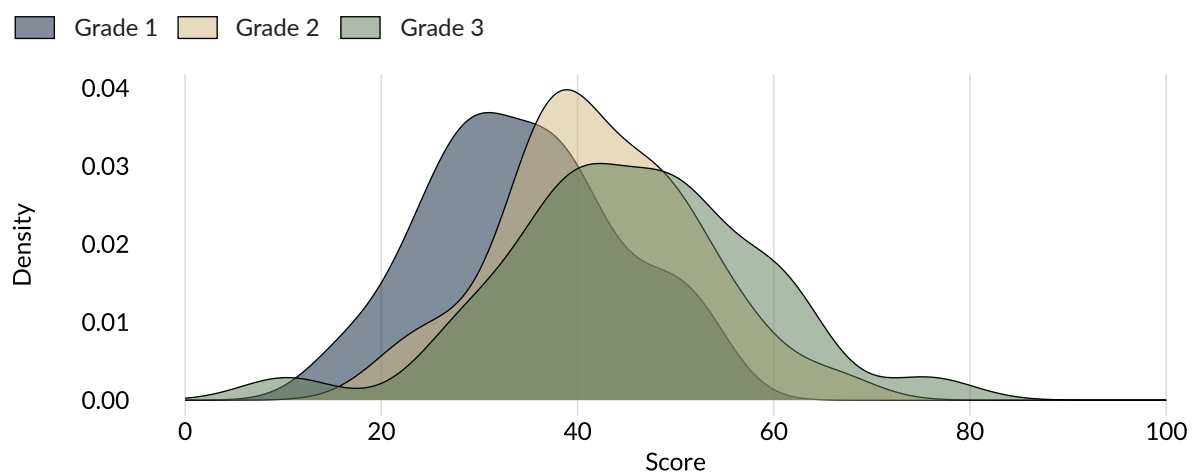
When exploring how a variable's values change with changes in another variable, we are interested in visualizing relationships or correlations. The go-to visualization for this purpose is a scatter plot (see e.g. chart 1). In a scatter plot, one variable is represented on the x-axis and the other variable is plotted on the y-axis. The choice of which variable goes on which axis depends on whether one variable influences the other. The dependent variable, which changes in response to the other (independent) variable, is typically plotted on the y-axis, while the independent variable, which influences the dependent variable, is represented on the x-axis. While scatter plots share similarities with line charts in representing two discrete or continuous values, they are different in that they emphasize individual data points as dots. Line charts, on the other hand, typically connect multiple data points into a continuous line, as seen in time series data (see chart 3). Trend lines can be added to scatter plots to highlight the underlying relationship between the variables. However, caution is advised when using scatter plots for audiences who are not familiar with statistics, as scatter plots are exploratory in nature and can be misinterpreted as showing causal relationships.

1.2.4 Distributions

When the goal is to comprehend the underlying structure of a variable, including the shape, central tendency and spread of its distribution, various visualization methods can be employed. Histograms, density plots and box plots are common choices. Histograms (not shown here) display the frequency or count of data points within different "bins" of a continuous or discrete variable. Density plots offer a continuous representation of the distribution and are suitable for continuous variables or fine-grained discrete variables. Box plots provide a summary of the distribution of continuous or discrete variables, displaying the median, quartiles and potential outliers, allowing for a quick overview of data distribution and skewness. To exemplify this, we show a density plot in chart 6 and a box plot in chart 7.

Chart 6: Density plots to represent distributions

Financial literacy scores in three grades

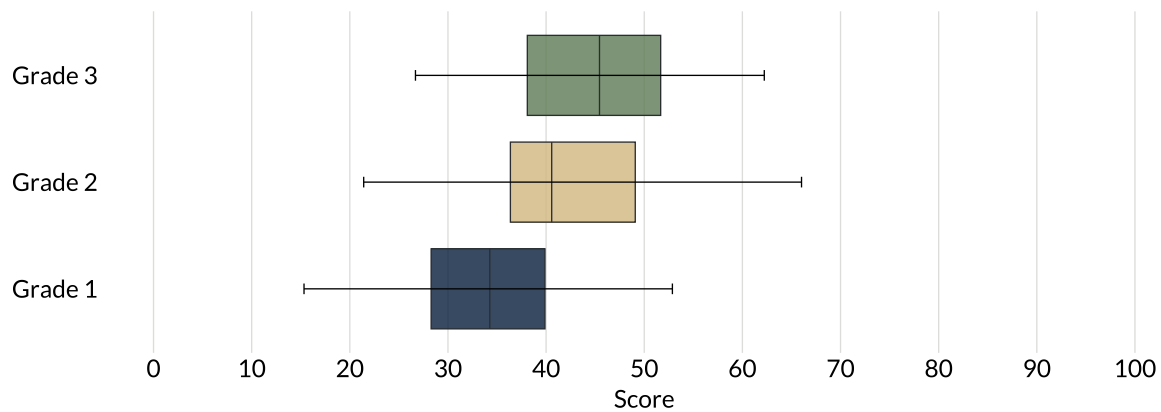


Source: Simulated data.

Note: Financial literacy scores of students.

Chart 7: Boxplots to represent distributions

Financial literacy scores in three grades



Source: Simulated data.

Note: Financial literacy scores of students.

For more details on how to find the most appropriate chart for your data and for a variety of further types of visualization, see *From Data to Viz* (2018).

1.2.5 Effective data visualization guidelines

Once the appropriate chart type has been selected, it is crucial to consider several fundamental visualization principles to ensure that the reader can easily understand the intended message. The visualization guidelines presented in this article are essential for creating effective and informative visual representations of data and support the accurate interpretation of charts. For further reading on visualization guidelines, we suggest Evergreen and Emery, (2016), Franconeri et al. (2021), Wilke (2019) and Yau (2011).

Visualizations can be misleading if chart elements, shapes and spacing are not chosen carefully. To mitigate illusion and prevent misinterpretation, axes should start at zero (Franconeri et al., 2021). Only in some cases can it be useful to start at values other than zero, especially if small differences are important for the interpretation of the data (Pearce, 2020). Also, axis intervals should maintain consistent spacing and have the same unit of measurement. For example, do not mix months and years in a time series presentation and ensure that all labels and annotations are readable, by potentially using horizontal annotation for easier reading also on the y-axis, and remove label redundancy. For instance, do not label every tick mark on your axis and do not use axis labeling and numeric labels together. Moreover, make the visualization as intuitive as possible by following a common understanding of directions (e.g., “up” and “right” indicating “more”). Differences in sizes of circles and squares are generally more difficult to capture than the area of bars. If you chose circles or squares, represent them based on their areas, not their radius, diameter or circumference as this creates distortions.

If used wisely, colors, annotations and highlighting provide a great way to improve storytelling and the readability of visualizations. You might wish to guide the audience in interpreting your visualization by providing visual hints through annotation and highlighting, especially when they may not be familiar with the data. For example, if one data point indicates a surprising result or conveys the main story of a chart,

it can be useful to add a short explanation next to the relevant data point. Regarding the use of colors, you should take care to always choose colors according to their intuitive meaning, such as red for negative or lower values and green for positive or higher values. At the same time, it may be advisable to avoid an excessive use of colors as this can exaggerate minor differences and to use accessible color schemes, preferably combining shades of red and blue (for more details, see Muth, 2020). Moreover, sufficient contrast between colors prevents an overlap of intensities. Utilizing readable fonts and font sizes in your visualizations is also helpful (for further details, see Muth, 2022).

To facilitate the interpretation of your data, it is useful to reduce the cognitive load for the reader, which may involve replacing legends with direct labels and annotations where possible. If legends are necessary, care should be taken to order them logically (Wilke, 2019). Generally speaking, it is advisable to keep visualizations simple by eliminating noninformative elements like grid lines, background colors, excessive animation or decoration. Arranging data points, such as bars, in a meaningful order, e.g. by frequency or time, also facilitates readability and understanding. If a visualization contains too much information and becomes too complex, consider dividing it into multiple charts or more digestible segments to enhance comparison. For instance, use a split bar chart instead of a stacked bar chart if there are numerous categories to visualize. For more details on stacked bar charts, see Muth (2018b.)

Visualizations should be adapted to the target audience. The focus should be on what is important, meaningful and understandable for them. For example, complex statistical elements like error bars and confidence intervals should be communicated only when the intended audience will be able to interpret them correctly. If your audience is less experienced, you might want to provide guidance and rely on chart types they are familiar with.

For ethical data visualization, it is essential to consider several key aspects. First, be mindful of data privacy, particularly when working with sensitive or personal data. Avoid misleading or manipulative data representation and always aim for truthful and accurate representation. Ethical data visualization also includes the clear communication of all sources and processing of data. It is essential to recognize and transparently convey the limitations of data visualization. This involves ensuring that viewers are well informed about potential constraints or uncertainties in the data. This may include communicating issues such as missing values or a nonrepresentative sample. Additionally, it is crucial to remain mindful of the fact that relationships depicted through descriptive statistics are inherently correlational and that they should never be assumed to imply causation (Azzam et al., 2013). If data contain issues such as missing values or if they stem from unrepresentative samples, researchers should report these limitations clearly.

1.2.6 Tools for data visualization

Tools for data visualization can be categorized as out-of-the box tools or programming tools. Out-of-the box tools are user friendly, quick to get started with and suitable for common tasks but may lack customization and flexibility. Programming tools, while more complex to learn and use, offer more extensive customization and flexibility and make it possible to create tailored solutions for a wide range of data visualizations. The choice between the two types of tools depends on the specific needs and technical expertise of the researchers who will apply the tool (Yau, 2011).

Data visualization tools

Out-of-the-box tools

- **Spreadsheet applications** such as Microsoft Excel or Google Sheets are perfect for users who want an easy-to-use solution with a standard set of graphics. They are particularly useful for small datasets, offering a straightforward way to gain rapid insights into your data.
- **Dedicated data visualization software** such as Tableau stands out for its interactivity and advanced visualization features. A potential drawback is data privacy. In the free version of Tableau, for example, data become publicly accessible once they are uploaded.

Programming software

- **Web-based tools** like HTML, JavaScript and CSS are excellent for creating interactive visualizations. They provide high levels of customization. Nevertheless, they require more prior knowledge compared to other programming tools.
- **Python** is highly effective for handling extensive volumes of data and features a straightforward syntax. While default charts may be unappealing, various libraries are available for creating attractive and customizable visualizations.
- **R** has a rich ecosystem of statistical tools and libraries, making it suitable to perform complex statistical analysis alongside data visualization (particularly in the ggplot universe). While it offers a relatively straightforward syntax, customization in R can be more time consuming than using web-based tools.
- **Interactive data visualizations** are possible in Python and R by integrating HTML, JavaScript and CSS libraries (e.g. R Shiny, highcharter, plotly) (Baruffa, 2023; Sievert, 2020) although dedicated interactive visualization programs can be superior in terms of performance and maintenance (e.g. updates).

For further reading and resources on data visualization, online references can be highly recommended, as there is an active and collaborative online open-source community dealing with the area of data visualization. This community comprises social media groups, blogs and websites that are often dedicated to specific data visualization tools and software. For R, see e.g. blogs (Rapp, 2025; Scherer, 2023), books (Baruffa, 2023; Nordmann et al. 2021) or tutorials (Scherer, 2019). Moreover, the community collaborates actively, contributing to open-source data visualization projects and sharing best practices. For instance, initiatives like the Tidy Tuesday challenges encourage participants to create visualizations and share their R codes with the broader community. Similar challenges exist for various other data visualization tools.

2 Statistical inference

2.1 Quantifying uncertainty

As outlined in the introduction of this paper, quantitative data analysis aims to present and summarize data in a way that allows for identifying patterns. Section 2 presents the fundamental principles behind such summarization, exploring variables, data types and their respective descriptive statistics. Descriptive statistics can also be regarded as the first step of statistical *inference*.

Statistical inference builds on the idea that the calculated statistics stem from a *random sample*, i.e. a randomly selected subset of the population. From the sample, researchers draw conclusions on the underlying population, e.g. on the target population of a financial education intervention.

In other words, statistical inference is the process of deriving insights about a population based on a sample from that same population. Importantly, statistical inference would be meaningless if we had reliable data on the whole (target) population; in such a case, purely descriptive statistics would be sufficient.

Researchers thus use sample data to calculate *sample statistics* that serve as estimates of the unknown true characteristic of a population, the *population parameters*. Population parameters and the respective sample statistics of interest may be any quantitative summary statistics (including the descriptive statistics presented in section 1), e.g. the average (mean) financial knowledge score, the variance of that score, the correlation between that score and financial well-being or the effect of a financial education intervention on the amount of monthly savings.

Methods of statistical inference allow us to *quantify uncertainty* from (small) samples. Suppose, for instance, that we want to test the effect of a financial education intervention on the financial knowledge of adults. We could sample only five persons from the (hypothetically very large) population of potential participants to constitute the treatment group, and five nonparticipants as the control group. However, if we compared their average financial knowledge test scores, we would have little confidence that the difference is an accurate reflection of the true difference we should expect from the intervention.

Indeed, we must expect that financial knowledge varies considerably within the population and that choosing only five people at random could result in averages far above or below the groups' actual average financial knowledge. This randomness could influence our estimated averages much more than the intervention. We could conclude that we need larger samples, and rightfully so: Increasing the sample size reduces random variability in average scores, bringing the estimated averages closer to the true average level of financial knowledge in the population.

In essence, statistical inference is used to eliminate guesswork from judging a statistic as “probably correct” based on the underlying variability in the population and the sample size. Phrased more technically, statistical inference formalizes and quantifies the uncertainty with which we draw conclusions from sample data. It is used to derive the probability that a result – or a more extreme result, e.g. a difference of 10 or greater – would occur merely due to random sampling variability even if there is no actual difference in the population. This probability is the *p-value*. The *p-value* is often interpreted as the “significance” of a result in terms of the strength of evidence. In classical hypothesis testing, however, the *p-value* is compared against a pre-set criterion of just-acceptable uncertainty (often 5%).

In the context of financial literacy, statistical inference is typically applied to examine *differences* between subsamples (e.g. between men and women) and *relationships* between variables. In effect, it is used to evaluate the *effectiveness* of interventions (e.g. the effect of financial education on financial literacy) and identify other key factors associated with an outcome. In the context of quantitative evaluation research, statistical inference is almost universally accepted as the method to assess the impact of a program in terms of statistical significance.

However, statistical significance alone is not enough when it comes to evaluating an effect as practically relevant. A difference, for example, must also be interpreted in absolute terms (e.g. whether an average 50 cent increase in savings is relevant) or relative to the overall variability as a standardized effect size (e.g. a difference equivalent to 0.20 standard deviations).

The following subsections explore in more detail the basic principles behind inferential statistics. First, we describe the importance of clearly defining the population and the relevance of the random sample. We then present a short introduction to classical hypothesis testing and explore the basic principles behind common statistics and significance tests as they are commonly used in practice, building on simple

calculations of the mean and the standard deviation. We conclude with a short overview of the many possible caveats of inferential statistics and the unavoidable omissions of this paper.

While we follow what is typically taught in introductory courses on statistical inference, we aim to provide an intuitive but nevertheless concrete understanding of the most important mathematical principles of statistical inference. More detail and many more statistical methods and their application using statistical software can be found in introductory textbooks such as Woodridge (2009), Anderson et al. (2020), Field (2013) or Janczyk and Pfister (2023).

2.2 Generalizing from a random sample to a population

A clear definition of the (*target*) *population* and a *random sample* are the basis of statistical inference. Indeed, depending on the definition of the population, the quality of the sample and the generalizability and usefulness of results may change considerably. Ultimately, however, it is up to the researchers to define the target population. For instance, 100 participants in a specific financial education program could be understood as 1) a sample of the general population, such as the adult residents of a country, 2) a sample of a population of potential program participants (e.g. due to their motivation and interests) or 3) the target population itself.

Additionally, common methods of statistical inference assume a *random* sample of a larger population. In a random sample (technically a *simple* random sample), every member of the target population has the same chance of being selected into the sample. This ensures that the selection of population members (e.g. survey participants) does not introduce any systematic patterns into the sample's properties that would lead to *bias*, i.e. systematic deviations of the estimated statistic from the true population parameter. Random sampling eliminates bias on average because differences in the distribution of characteristics between sample and population are canceled out with repeated sampling. With sufficient size, a random sample can thus be considered representative of the population.

In practice, large random (or equivalent probability) samples are difficult to achieve. Quantitative researchers are thus often concerned with *reducing bias* through (in some cases quite complex) sampling and weighting procedures.³ Conceptually, however, random samples are the foundation of statistical inference and of estimating population parameters.

Any sample can be considered one of hypothetically infinite repetitions of the same procedure. Because a random sample is free of systematic bias (and if there are no other sources of bias), the estimated statistics vary around the true population parameter. The variability of these statistics is called the random sampling variance, and it determines the expected random sampling error, i.e. the deviation between a statistic and the population parameter that is to be expected. The assumptions and methods presented here are foundational to conventional frequentist statistics, which rely on the behavior of estimators across (hypothetical) repeated samples. Other approaches exist, such as Bayesian statistics, which have a fundamentally different understanding of probabilities, inference and “truth.”

Luckily, it is possible to calculate both the statistic itself and its random sampling error (e.g. a mean and its standard error) from a single sample. This works because the variability of observations within a sample gives insight into how much the statistic would fluctuate across multiple (hypothetical) samples. From this, we can derive probabilities that indicate how likely the observed statistic (or a more extreme one) would be, assuming a certain population parameter. We can quantify, for example, how likely it is to find

³ Sampling methods are discussed in a different issue of the OeNB Financial Literacy Evaluation Series.

the observed difference between two groups if we assume that the true difference in the population is zero. This is the p-value and it is the basis for statistical hypothesis tests.

Box 2

Key terms of statistical inference

- **Statistical inference:** the method of drawing conclusions about a population from sample data
- **(Target) population:** the specific group a study aims to draw conclusions about, as defined by the researcher
- **Sample:** a smaller group selected from the population for the purpose of analysis
- **(Simple) random sample:** a sample where every member of the population has the same chance of being chosen; random samples are the basis of statistical inference
- **Population parameter:** a true value that describes a characteristic of a population, like the actual average financial knowledge of all individuals of the target population
- **Sample statistic:** a value calculated from the sample that estimates a population parameter
- **Generalizability:** the extent of how well the results from a sample apply to the larger population
- **Bias:** a systematic deviation of a sample statistic from the true population parameter, often due to flaws in sampling
- **Sampling variance:** the variability in a sample statistic that occurs because different random samples would yield different results
- **Sampling error:** the difference between a sample statistic and the true population parameter, caused by random chance in the sampling process
- **Standard error:** a measure of the expected variability of a sample statistic due to random sampling

2.3 Hypothesis testing

Hypotheses are an important part of inferential statistics. A hypothesis is a statement about a population that is tested using a sample. *Hypothesis testing* or *significance testing* is generally used as a process to assess whether differences (e.g. mean differences in financial literacy scores) or associations (e.g. between financial knowledge and financial well-being) are clear enough to be deemed statistically significant, i.e. not a product of a random sampling error. Performing hypothesis testing involves four key steps:

1. defining the hypotheses to be tested;
2. determining the criteria for deciding whether the claim being tested is true or not;
3. computing the sample statistic from the data sample and deriving its corresponding test statistic and probability value (p-value);
4. deciding to *reject* the claim as being true if there is a large discrepancy between the sample data and what we would expect to observe if the claim was true.

For step 1, we need to formulate two statements that form the two complementary and opposing hypotheses known as the *null hypothesis* (H_0) and the *alternative hypothesis* (H_1). The way these hypotheses are structured depends on whether the hypothesis test is two-tailed, examining effects in both directions (e.g. whether a mean is greater or smaller, i.e. more extreme), or one-tailed, focusing on a single direction. In most cases, two-tailed tests are chosen unless there is a clear justification for limiting the test to one direction. To simplify matters, we only provide examples of two-tailed tests in the following subsections.

The null hypothesis H_0 states that a given phenomenon does not exist in the population, e.g. that there is no association between variables or difference between groups. The alternative hypothesis H_1 , in contrast, claims the opposite. For example, a pair of hypotheses could be:

H_0 : There is no correlation between financial knowledge and financial well-being.

H_1 : There is a correlation between financial knowledge and financial well-being.

In step 2, we determine when to reject H_0 . To this end, a *level of significance* must be selected. This value, denoted by α , specifies the threshold for how small the p-value must be to justify rejecting H_0 . Common choices for α are 5% or 10%, though other values can also be applied depending on the context.

For step 3, we collect data and compute the sample statistic (e.g. the correlation coefficient) from the data sample, as well as its corresponding *test statistic*. For this test statistic, a *probability value* (p-value) is calculated. This value represents the likelihood of observing the test statistic under the assumption that H_0 is true. Specifically, a low p-value indicates that the observed result is unlikely to occur if H_0 is correct. Typically, these tests are conducted using statistical software or programming such as Stata, SPSS, R or Python. The choice of the test statistic depends on the specific context, though the general testing procedure remains the same.

In step 4, we decide whether to reject or fail to reject H_0 . Specifically, if the p-value $\leq \alpha$, we reject H_0 , meaning that the difference or association is statistically significant. Otherwise, we fail to reject H_0 .

As shown in table 2, there are four possible scenarios depending on the real effect in the population and the corresponding correct or false decision. As mentioned, the α level is selected by the researcher. Under ideal conditions, an α of 5% will lead to significant results and thus a false rejection of H_0 in exactly 5% out of every 100 (hypothetical and infinitely many) samples. Such a *false positive* result is called *type I error*. Its probability is set directly by the researchers themselves as the significance level α . Its complementary is the confidence level $1 - \alpha$, which is the probability that H_0 is *not* rejected. If the null hypothesis is false (e.g. there is a true difference in the population) but it is not rejected, this is a false negative, whose probability is denoted by β . The complementary probability $1 - \beta$ is called *power*. It indicates how likely it is to find a significant result if there is a real phenomenon in the population. Power increases with larger effect sizes (e.g. the difference) and larger sample sizes. The power is greater the larger the effect (e.g. the difference) in the population or the larger the sample size. With it, researchers can derive the minimum required sample size to find an effect of prespecified size with the desired probability $1 - \beta$.

Table 2

Possible decisions and their (error) probabilities

	H_0 not rejected	H_0 rejected (significant result)
H_0 is true	Correct with probability $1 - \alpha$	Type I error: False positive with probability α
H_0 is false	Type II error: False negative with probability β	Correct with probability $1 - \beta$ (power)

Source: Authors' compilation.

Key terms of hypothesis testing

- **Hypothesis:** a statement about a population to be tested; hypotheses are typically formulated as a null hypothesis (H_0) that assumes that no effect or difference exists in the population and an alternative hypothesis (H_1) that assumes that an effect or difference does exist
- **Hypothesis testing:** a method used to determine whether observed differences or associations in sample data are statistically significant or likely due to random chance
- **Statistical significance:** a result is statistically significant if the p-value is less than or equal to α , indicating the finding is unlikely to have occurred by chance
- **p-value:** the probability of observing the test statistic (or something more extreme) if the null hypothesis is true; a low p-value suggests the observed result is unlikely under H_0 .
- **Level of significance (α):** a threshold set by the researcher (commonly 0.05, i.e. 5%) that defines how small the p-value must be to reject the null hypothesis; it represents the risk of a type I error
- **Errors (type I and type II):** a false positive occurs when H_0 is wrongly rejected (type I error), with a probability equal to α ; a false negative happens when H_0 is wrongly not rejected (type II error), with probability β
- **Power ($1 - \beta$):** the probability of correctly rejecting the null hypothesis when it is false; higher power means greater ability to detect true effects, and this ability is influenced by effect size and sample size

2.4 Statistics and tests to estimate population parameters

This section illustrates in simple mathematical terms the principles behind estimating population parameters, using the example of the mean and standard deviation and the corresponding one-sample t-test, the difference between means and the independent two-sample t-test, the correlation coefficient and a simple regression.

The sample statistic (generally denoted with a Latin letter) is the estimate for the population parameter (generally denoted with a Greek letter). For instance, the mean $\bar{x}$ (x-bar), e.g. the mean financial knowledge of a sample, is the estimate for the mean financial knowledge of the population μ (mu).

While the formulas presented in the following subsections may appear daunting at first glance, the principles behind these statistical tests follow a simple logic: For each statistic, such as an average (mean) financial knowledge score, the *variability* of the underlying scores is calculated. This variability is generally based on the sum of all squared distances of values from the mean itself, the *sum of squares*.

Together with the sample size, the sum of squares is used to calculate variability metrics such as the variance and standard deviation of the values. Moreover, the sum of squares is used to estimate the variability of the statistic itself. With this information, the resulting statistic can be standardized and compared to a known distribution. From this comparison, we can derive the probability of a result that is equal or more extreme than the statistic, i.e. we can determine a *p-value* and whether a result is *significant*.

The presented statistics are a small part of statistical inference. There are many more different statistics to describe variables and their distributions and associations, such as the median, frequency proportions, skewness and rank correlations and countless increasingly complex methodological frameworks, all of which can be used for statistical inference. The presented statistics and test, however basic, are still prevalent in practice and are well-suited to illustrate the principles behind statistical inference.

2.4.1 The mean

The mean is likely the most important and most reported summary statistic. The mean and its properties also build the basis for the most common tests of statistical inference, such as the independent t-test.

The (arithmetic) mean of a sample (denoted as $\bar{x}$) is the sum of each observation (x_i) of a variable divided by the number of observations (n):

$$\bar{x} = \frac{1}{n} \sum x_i$$

For example, if a sample of five people score 5, 6, 6, 10 and 11 in a financial knowledge test, the mean is $38/5 = 7.6$.

The sample mean $\bar{x}$ is an *estimate* of the population mean μ , i.e. it can be used to draw conclusions about the population. The sample mean is generally an *unbiased* estimator, which means that with increasing sizes of random samples, the sample mean will get closer to the population mean.

The mean has an important property: It is the value where the sum of the *squared* distances to each sample value, the *sum of squares*, is as small as possible. Using the scores from above, the mean $\bar{x} = 7.6$ minimizes the following sum of squares:

$$(5 - \bar{x})^2 + (6 - \bar{x})^2 + (6 - \bar{x})^2 + (10 - \bar{x})^2 + (11 - \bar{x})^2$$

The sum of squares can be written as:

$$SS = \sum (x_i - \bar{x})^2$$

The mean is thus the *least squares estimator* of central tendency. As the following sections will show, calculating or minimizing the sum of squares is the basis of many statistics and statistical tests. Importantly, the sum of squares is also used to calculate the sample variance and standard deviation.

Because squared differences are used, the sum of squares gives larger differences more weight: The mean gets “pulled” more strongly toward more distant values. The mean and all other estimates and significance tests based on the sum of squares is therefore sensitive to outliers (i.e. a few very large or small values) and can severely misrepresent the center of nonsymmetrical distributions. The mean stands in contrast to the *median*, which minimizes the absolute (and not the squared) distances. The median is largely robust to outliers and can often be a better measure of central tendency when a distribution is not symmetrical.

2.4.2 The standard deviation and variance

The mean alone gives us only a measure of the central tendency but no indication about the spread or variability of the underlying values. For an approximate description of the distribution, we thus need a measure that quantifies the spread of values around the mean – the standard deviation. Its calculation is closely related to the mean itself because it is based on the sum of squares.

The standard deviation s (or *SD*) is the square root of variance s^2 . Variance s^2 is the sum of squares divided by the sample size (although 1 is subtracted from the sample size to obtain unbiased estimates). The mean is therefore also the value that minimizes the variance. The variance is calculated as:

$$s^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$$

Using the same example as above, the variance of the five scores is $s^2 = 7.3$, the standard deviation is:

$$s = \sqrt{7.3} \approx 2.70$$

2.4.3 Standardization and the p-value of a single observation

Together, the mean and standard deviation can fully describe the most important theoretical distribution of statistical inference, the *normal distribution*. If we can assume that we are indeed working with a normal distribution of our values, we can use the mean and standard deviation to transform any value of a variable (e.g. any of the five scores from above) so that they can be evaluated and compared with a theoretical normal distribution or with observations from other samples. This is called *standardization* or *z-transformation*:

$$z = \frac{x - \bar{x}}{s}$$

For instance, for one of our five knowledge scores $x = 10$, the standardized z -value is approximately 0.89, which means the value is approximately 0.89 standard deviations above the mean. If all scores are standardized, the resulting distribution of standardized values has (by definition) a mean of 0 and a standard deviation of 1.

The theoretical normal distribution with a mean of 0 and a standard deviation of 1 is called standard normal distribution. Because its properties are known, we also know the probabilities to obtain certain ranges of values. The z -value of 0.89 can also be transformed into a *probability* that indicates how likely it is to get a value equal to or more extreme than 0.89 from the standard normal distribution. For example, the probability of sampling a single score that is more extreme than $z = 0.89$ from a standard normal distribution is $p = 0.37$.

Subsection 2.4.4 illustrates why the (standard) normal distribution can be used as the benchmark distribution for statistical tests and why our small hypothetical sample of five scores may not actually be suited for the tests presented here.

2.4.4 Why do statistics follow normal distributions?

Statistical tests rely on the fact that statistics from (large) samples often follow a normal distribution. This is because, to determine probabilities, statistical tests compare a standardized sample statistic (e.g. a mean) to the theoretical standard normal distribution. This section illustrates why statistics have this important attribute and how sample size affects the distribution of sample statistics.

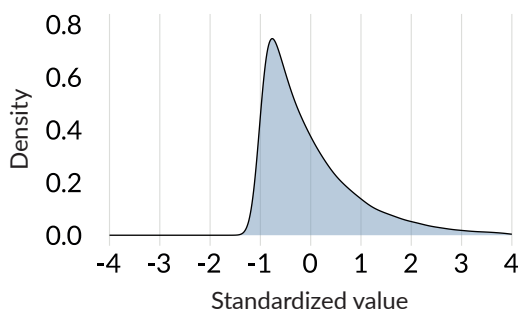
Measurements of real-world phenomena such as physical or psychological attributes (e.g. people's height or financial knowledge test scores) tend to cluster around their mean value, with fewer and fewer cases occurring with increasing differences from the mean. The measurements often tend to follow a bell-shaped distribution that is called "normal distribution."

The fact that many real-world measurements follow normal distributions is due to the central limit theorem (CLT). The CLT states that a summary score or a statistic (e.g. a sum or a mean) of many independently sampled values approach a normal distribution, regardless of the distribution of the underlying values. For example, even the sum of multiple coin toss results (where heads and tails are assigned any different values, such as 7 and 39) will approach a normal distribution if the samples are large enough. This means if we sum up the outcomes of 1,000 coin tosses and repeat this process to obtain many sums, the sums will be close to normally distributed.

Chart 8 shows the standardized distributions of means from samples of size $n=5$ or $n=50$ drawn from an asymmetric standardized distribution. As is apparent, the means from the larger samples are close to normally distributed. The means from the smaller samples, however, show considerable skewness.

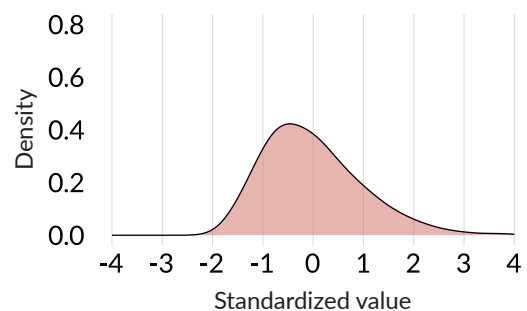
Chart 8: Illustration of the central limit theorem

Original skewed distribution



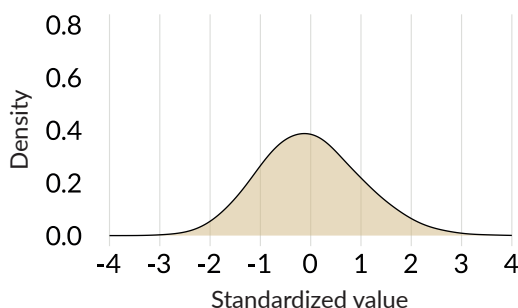
Source: Simulated data.

Distribution of means from $n=5$



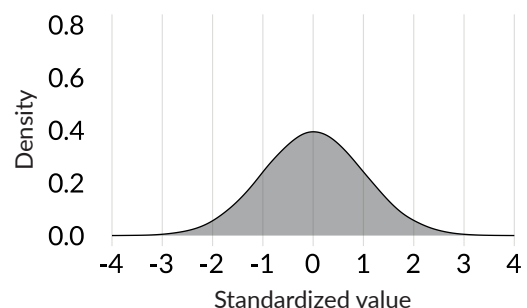
Source: Simulated data.

Distribution of means from $n=50$



Source: Simulated data.

Standard normal distribution



Source: Simulated data.

Importantly, the CLT is applicable on two levels, namely on the level of variables (such as a knowledge score) and on the level of individuals in the sample.

On the level of variables, real-world phenomena such as financial knowledge can be regarded as “summary” variables, i.e. as “statistics” on the level of an individual. Financial knowledge, for example, could be considered the result of many different mental processes. These processes may be independent enough that when we measure the financial knowledge of only one person, we effectively obtain a sample of many (unknown) *independent variables* summarized into one score. If we repeat this measurement process with many people, the scores may thus follow a normal distribution. We could therefore assume (or have empirical evidence) that the knowledge score used in our examples is normally distributed in the population.

On the level of individuals, the CLT will also come into effect when we sample multiple *independent individuals* from the population. Even if the measured variable is not normally distributed (i.e. the conditions

described above are not fulfilled), obtaining a large sample of independent individuals and thus knowledge scores will result in a summary statistic (such as the mean financial knowledge) that follows the normal distribution.

Statistical tests rely on this property. They compare a standardized sample statistics (e.g. a mean) to the theoretical standard normal distribution to determine probabilities, confidence intervals and significance levels. Statistics from large samples can thus be standardized and compared against a standardized normal distribution. The necessary sample size depends on the underlying distribution of the values: If the underlying distribution is very nonnormal, very large samples may be necessary.

The conditions for a normal distribution of summary statistics can be provided by the researcher by obtaining a large random sample. However, the CLT does *not* guarantee that values within a large sample follow a normal distribution (see Zhang et al., 2023, for an overview and typical misconceptions about the CLT).

2.4.5 The probability of a mean

In the previous section, we derived the probability to obtain a single value (or a more extreme value) from our distribution. Applying the same principle to the sample statistic, i.e. the estimate of the mean, finally allows us to draw conclusions about the population, namely about the probability of our mean estimate assuming some population mean. This is called the *one-sample t-test* and it is one of the most basic significance tests of statistical inference and the basis for confidence intervals.

To gauge the variability of the mean estimate across multiple (hypothetical) repetitions of our sample, the *standard error* of the mean is used. The standard error is the standard deviation of the mean itself and is derived from the standard deviation of the underlying values:

$$SE = \frac{s}{\sqrt{n}}$$

The standard error thus reflects the mean's variability across many (hypothetical) samples. It therefore relates to the mean *statistic* itself and not to its underlying values. For our five knowledge scores, the resulting standard error is approximately 1.21.

The mean can be standardized against its standard error, just like the single score can be standardized against the standard deviation in the example above. However, for the single score value of 10, there is a known mean to compare against. For the estimate of the mean itself, the population mean is not known. Consequently, we need to specify a population mean to compare against. The standardized value is then calculated as:

$$t = \frac{\bar{x} - \mu}{SE}$$

Because the standard error is itself only an estimate, the resulting value is not compared against a standard normal (z-)distribution, but against a *t-distribution*. t-distributions are based on standard normal distributions but compensate for the added uncertainty from estimating the standard error from the sample. There are different t-distributions for different sample sizes (technically expressed as *degrees of freedom*) for which probabilities can be looked up or calculated.

Specifying μ allows for obtaining the probability of how likely the sample mean $\bar{x} = 7.6$ is for a given mean of the population (μ). In other words, we can estimate the probability of the difference between the

estimated mean ($\bar{x}$) and a specified population mean (μ). For example, we could test the null hypothesis that assumes the population mean $\mu = 0$. In this case, the t-statistic is approximately 6.29 for our estimated mean $\bar{x} = 7.6$, which corresponds to a probability of approximately 0.003 to obtain a sample mean of 7.6 (or more extreme) with $\mu = 0$.

0.003 is the p-value of this specific test. At the conventional significance level of $\alpha = 0.05$, the mean is indeed significantly larger than zero. It is thus very unlikely that the estimated mean stems from a population where the mean is zero, and we can reject the null hypothesis.

We could also compare the sample mean to a different population mean, e.g. a minimum mean score that characterizes a population as “financially literate.” Instead of testing against zero, we would formulate the null hypothesis that the population mean is not different from this specific value.

2.4.6 The confidence interval of a mean

Instead of testing our sample mean against a specific value of μ , we may be interested in a range of values of μ that would *not* be rejected as null hypotheses and thus reflects our level of confidence for our mean estimate. By a similar logic to that used in the one-sample t-test and by rearranging the previous formula for t , we can thus calculate the upper and lower limits of a *confidence interval* for the mean as:

$$CI = \bar{x} \pm t^* \cdot SE$$

Here, t^* indicates the *critical value* from the t-distribution based on our confidence level $1 - \alpha$ (e.g. 95%). The calculation gives us the range of μ for which the null hypothesis would *not* be rejected by the one-sample t-test. 95% confidence intervals thus represent the intervals that capture the true population mean in 95% of cases if we repeated the sampling process from the population many times. If the confidence interval does not include zero, the mean is significantly different from zero at $\alpha = 0.05$. For $\bar{x} = 7.6$ the 95% confidence interval ranges from 4.25 to 10.95.

2.4.7 Comparing two group means: the independent t-test

In practice, researchers may be more interested in testing the difference of the means of two samples instead of comparing a sample mean to a hypothetical population mean. The *independent two-sample t-test* (often just called *independent t-test*) can do just that. For instance, it can be used to test the effect of a financial education measure on the financial knowledge test score. One group could be the *treatment group* that received a financial education intervention, while the other is the *control group* that did not, with means reflecting each of the groups' knowledge test scores.

The independent t-test is thus an extension of the one-sample t-test. The t-statistic weights the difference between the two estimated means against its standard error, using both subsamples' variance and sample sizes:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Like with the one-sample t-test, the p-value can be determined for the now standardized difference. We can thus test the null hypothesis that there is no difference between the two means.

2.4.8 The covariance and correlation between two variables

Researchers are often interested in the association between two different variables (x and y , e.g. the financial knowledge score and the financial well-being score. For example, if a person scored highly on the financial knowledge test, is this knowledge also associated with higher financial well-being? To measure this association, the covariance can be used:

$$\text{Cov}(x, y) = \frac{1}{n-1} \sum (x_i - \bar{x})(y_i - \bar{y})$$

Unsurprisingly, the covariance is similar to the variance. Instead of the sum of squares, however, it uses the sum of each *product* of the value pairs' distances from the mean. If a person scores above average both in knowledge (x_i) and in financial well-being (y_i), the product $(x_i - \bar{x})(y_i - \bar{y})$ will be positive. If a person scores below average in both, the product will also be positive. If a person scores above average in one and below in the other, the product will be negative. Results close to the average in one or both variables result in products close to zero. The mean of these products across all observations indicates the direction and size of the overall association and is represented by the formula above.

Although the direction of the association is clear from the covariance, the strength of the association is difficult to interpret. The result can thus be standardized as:

$$r = \frac{\text{Cov}(x, y)}{s_x s_y}$$

To get the correlation r , the covariance is weighted by the product of the two variables' standard deviations, so that it ranges from -1 (perfect negative association) to $+1$ (perfect positive association). Alternatively, we could also calculate the covariance of two standardized variables to get r .

The correlation coefficient r can be further transformed into its t-distributed test statistic to test for statistical significance, i.e. to obtain the probability of observing r or a larger (smaller) value if the association between the variables in the population is zero.

An important extension of the correlation is the *partial correlation*. For instance, researchers may suspect that the association between financial knowledge and financial well-being is not quite independent of other variables, e.g. income. They may thus want to calculate the correlation between financial knowledge and financial well-being net of their common influence from income (i.e. *controlling* for income). This can be achieved using the partial correlation. Calculating the partial correlation $r_{xy \cdot z}$ between x (e.g. financial knowledge) and y (e.g. financial well-being) net of the influence of z (e.g. income) requires the correlation coefficients between all variable pairs, where r_{xy} represents the correlation between x and y , r_{xz} represents the correlation between x and z and r_{yz} represents that between y and z :

$$r_{xy \cdot z} = \frac{r_{xy} - r_{xz}r_{yz}}{\sqrt{(1 - r_{xz}^2)(1 - r_{yz}^2)}}$$

2.4.9 Tests of correlation and difference in means are equivalent

Correlations and differences in means are conceptually and mathematically equivalent. The difference between two group means as tested with an independent t-test can also be thought of as the association between a binary (dichotomous) variable and a continuous variable.

If we assign a numeric value of 0 to the control group and a value of 1 to the treatment group — this process is called *dummy coding* — we can calculate the correlation between the numeric group variable and the knowledge test scores. This correlation has the exact same p-value as the independent t-test and it can even be interpreted as a measure of effect size. This type of correlation is called point-biserial correlation. The same principle also applies in regression analyses, as outlined in subsection 2.4.10, where both group variables (coded as 0 and 1) and continuous variables are used.

2.4.10 Regressions

Regression analyses (or simply *regressions*) can be considered the combination and generalization of all previous methods into a more comprehensive framework. This allows for comparing means, estimating associations between continuous variables and controlling for the effects of the other variables at the same time. In practice, impact evaluations often use regressions and their numerous extensions as the main statistical framework for statistical inference outcomes (Gertler et al., 2016).

In a regression, one *dependent variable* (also called *outcome variable*) and one or more *independent variables* (also called *predictor* or *control variables*) are analyzed simultaneously. As opposed to correlations and difference tests, a regression must be considered at least “directional” (i.e. one variable influences the other) in principle because the outcome is predicted or explained. In other words, a regression is a *model* of how the outcome came to be. Nevertheless, a regression model cannot per se give evidence on the causality of associations. The data and study design must be suitable for causal analysis and interpretation.

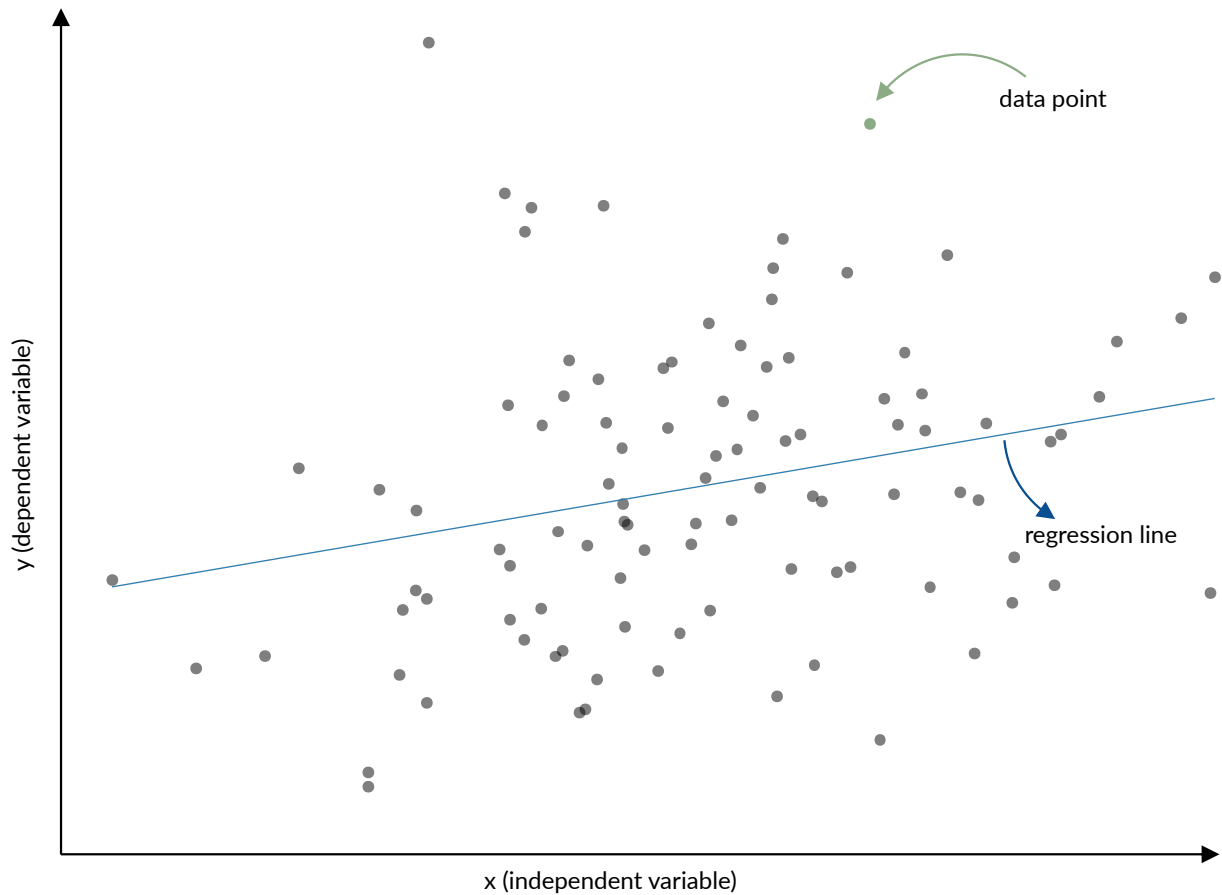
A simple linear regression with only one dependent variable and one independent variable can be represented as a geometric line in a two-dimensional coordinate system with a horizontal x-axis and a vertical y-axis (regression line). The values of the dependent variable are mapped on the vertical axis — the dependent variable is thus often denoted as y . The independent variable is mapped horizontally and denoted by x . The relationship between the two variables is modeled as a line:

$$y = b_0 + b_1x + e$$

b_1 represents the slope of the line, i.e. how much y increases with a one-unit increase of x . b_0 represents the intercept. It determines the position of the line in the coordinate system, i.e. at which level of y the line intersects the y -axis when x is zero. e is the error of the model. It is the variability of y that cannot be explained by the regression line. Chart 9 illustrates a simple regression model.

Chart 9: Illustration of a simple regression

A simple linear regression



Source: Simulated data.

Very similar to the calculation of a correlation, the regression slope can be calculated as the covariance of x and y divided by the variance of x :

$$b_1 = \frac{\text{Cov}(x, y)}{s_x^2}$$

Unlike the correlation, however, the covariance is scaled by the variance only of x so that an increase of x by 1 leads to a b_1 increase in y . However, using two standardized (z-transformed) variables in a regression gives the regression coefficient as b_1 . The intercept ensures that the line goes through the mean of y ($\bar{y}$). It is calculated as:

$$b_0 = \bar{y} - b_1 \bar{x}$$

The regression line has an important property: It minimizes the sum of squared *vertical* distances of y to the regression line. The distances are also called *residuals* in regression analysis and are denoted by e in the regression model, representing the unexplained variability of y .

The regression line can thus also be thought of as the straight line that best reflects the *mean* of y for all values of x . A regression where the only independent variable is a group variable that has been coded as 0 (e.g. for the control group) and 1 (e.g. for the treatment group) will result in a b_1 that reflects the difference in means between the two groups and is exactly equivalent to an independent t-test. Just as the mean, the independent t-test and the correlation, regressions are susceptible to outliers because they are, in effect, based on the sum of squares.

As with the statistical tests before, the b_1 and b_0 in a regression can be standardized to a t-statistic, and a p-value can be obtained for significance tests. Moreover, similar to the partial correlation, a regression can include two or more independent variables where each slope is controlled for all other variables that were entered into the model. Many more extensions to the simple regression model exist that exceed the scope of this paper. The following models are important to mention: 1) nonlinear regression models, where the slope of the line changes with x , 2) models with interaction terms, where the effect of one variable depends on the level of another variable, and 3) hierarchical or mixed models, where clustered data structures can be modeled explicitly.

2.5 Caveats and alternatives

This short introduction to statistical inference omits many important assumptions and caveats about the presented tests, leaves out countless alternative methods and approaches to hypothesis testing and statistical inference and has generally simplified a continuously growing, complex field of science.

First, it is important to recognize that *statistical significance* does not always mean *practical significance*, which refers to the actual, real-world impact of an outcome. For example, a result could be highly significant but economically or educationally irrelevant. Practical significance can, for example, be assessed using effect sizes.

Second, inferential statistical analysis is based on specific *assumptions* about the sample and population. This short guide strategically ignores many aspects. For instance, the methods presented above generally assume that variables are on an interval scale, which may not necessarily be true for many variables collected through questionnaires. Nevertheless, Likert-scale data typically collected in questionnaires are often (and only sometimes rightfully) treated as being continuous. All tests presented above are also parametric, meaning they make strict assumptions about the distributions of the variables in the population (e.g. that they are normally distributed). The presented tests are often robust to the assumption of normality with larger sample sizes due to the central limit theorem (see section 2.4.4) and other violations may not bias results severely under many circumstances. However, alternative nonparametric tests may be better suited in many situations. An example for a nonparametric test that can be used to compare the central tendency of two groups is the Mann-Whitney U test.

Third, the presented methods cover only a small subset of the vast landscape of statistical methodology. While many statistical frameworks, methods and tests are inherently related, they often use different terminology and may be used across different fields and for distinct purposes. Differences in means, for instance, can be tested using analysis of variance (ANOVA), measurement instruments can be validated using factor analysis or item response theory (IRT, see de Ayala, 2013) and complex relationships between different types of variables can be analyzed and modeled e.g. by using structural equation modeling (Kline, 2016).

Fourth, the presented approaches of classical frequentist inference are typically taught in a similar way in social sciences. However, other approaches exist that have different general assumptions about how to quantify reality. Most importantly, in Bayesian statistics, prior information about the world is integrated into the analysis and beliefs about parameters are updated as new data become available. Unlike frequentist statistics, which assumes that population parameters exist as fixed, unknown values, Bayesian statistics treats parameters as random variables with distributions that reflect this uncertainty (see Greenberg, 2012).

Fifth, statistical inference is sometimes criticized for becoming an end in itself. It can be argued that the obsession with statistically significant results has created a culture in many disciplines that puts undue emphasis on yielding a p-value smaller than 0.05. Worrisome phenomena in this context include the replication crisis (see Shrout and Rodgers, 2018), p-hacking (see Moniz et al. 2025, and Head et al., 2015), hypothesizing after the results are known (HARKing, see Kerr, 1998) and publication bias (see Maier et al., 2022), although efforts such as preregistration are attempts to counteract these issues (see Brodeur et al., 2024).

3 Summary and concluding remarks

This article underscores the crucial role of quantitative data analysis in evaluating financial literacy interventions, offering an accessible introduction to quantitative methods for evaluators, program designers and educators alike.

Section 1 focuses on *descriptive statistics*, emphasizing the importance of effective data visualization. The foundation of impactful visualization lies in selecting the right chart type. To support this, we present commonly used chart types for time series, proportions, relationships and distributions, alongside their prerequisites and associated variables. Additionally, we summarize guidelines for creating clear and meaningful visualizations that balance the need for clarity without generating an information overload. Key principles include adhering to common visualization rules, such as consistent axes and labeling, using intuitive visual cues (e.g. “up” to signify growth) and leveraging thoughtful highlights and colors (e.g. green for agreement, red for disagreement). Ethical data visualization is also addressed, with recommendations for truthful representation, consideration of privacy and transparency about data limitations to prevent misinterpretations, such as confusing correlation for causation. Section 1 concludes by introducing two types of tools for data visualization: out-of-the-box tools and programming-based tools. We summarize their characteristics to help researchers choose the best fit for their needs.

Section 2 provides an introduction to *statistical inference*. While descriptive statistics summarize data, inferential statistics allow for generalizations and the exploration of relationships. We begin by covering key concepts such as quantifying uncertainty and making inferences from a random sample to a larger population. This is followed by an introduction to hypothesis testing, a method used to determine whether differences between groups or associations among variables are statistically significant. Building on these fundamentals, we then show how statistical inference is applied in practice, by exploring how sample statistics are calculated, standardized and compared to theoretical distributions.

In evaluation research, descriptive statistics and statistical inference are essential for describing patterns in data and estimating their strength and generalizability with respect to the population. They are thus the main tools for describing and testing program impacts and identifying the key drivers of outcomes of financial education interventions.

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Annex: R code to reproduce the charts shown in this article

```

options(scipen = 999)
options(pillar.sigfig = 4)

## Library
setwd("/path")

library(readxl)
library(haven)
library(srvyr)
library(tidyverse)
library(camcorder)
library(scales)
library(ggplot2)
library(reshape2)
library(ggtext)
library(patchwork)
library(ggrepel)
library(glue)
library(unhcrthemes)
library(lubridate)
library(showtext)

#### load font ####
font_add_google("Lato", "lato")      # Regular Lato
font_add_google("Lato", "lato_bold") # Optional: bold version

# Turn on showtext
showtext_auto()

th <- theme(
  legend.title = element_blank(),
  axis.title.y = element_text(size = 8, family = "lato", color = "black"),
  axis.title.x = element_text(size = 8, family = "lato", color = "black"),
  axis.text.x = element_text(size = 8, family = "lato", colour = "black"),
  axis.text.y = element_text(size = 8, family = "lato", colour = "black"),
  plot.margin = margin(25, 25, 10, 25),
  plot.title = element_text(size = 11, family = "lato_bold", face="bold", color = "#004289",
margin=margin(0,0,10,0)),
  plot.subtitle = element_text(size = 8, family = "lato_bold", face="bold", color = "black",
margin=margin(0,0,10,0)),

```



```

plot.tag = element_text(size = 11, family = "lato"),
plot.tag.position = "topright",
plot.caption = element_text(size = 8, family = "lato", hjust = 0, margin=margin(10,0,0,0)),
plot.caption.position = "plot",
panel.background = element_rect(fill = "#FFFFFF"),
axis.ticks = element_blank(),
axis.ticks.length = unit(0, "cm"),
panel.grid.major.x = element_line(color = "#E5E4E2", size = 0.2),
panel.grid.minor.x = element_blank(),
legend.position = "top",
legend.direction = 'horizontal',
legend.text = element_text(
  size = 8,
  family = "lato",
  color = "black"),
legend.margin = margin(0, 0, 0, 0),
legend.justification.top = "left",
legend.justification.left = "top",
legend.justification.bottom = "right",
legend.justification.inside = c(1, 1),
legend.location = "plot",
plot.title.position = "plot",
legend.key.height = unit(0.3,"cm"),
legend.key.width = unit(0.5,"cm")
) +
theme(
  plot.background = element_rect(fill = "transparent", colour = NA)
)

```

set Corporate Design

```

colors <- c(
  `OeNB-Blau` = '#004289', #rgb(0,66,137,maxColorValue = 255),
  `OeNB-Sand` = '#f2e5c0', #rgb(242,229,192,maxColorValue = 255),
  `OeNB-Salbei` = '#95af83', #rgb(149,175,131,maxColorValue = 255),
  `OeNB-Türkis` = '#46aaaf', #rgb(70,170,175,maxColorValue = 255),
  `OeNB-Violett` = '#6a5988', #rgb(106,89,136,maxColorValue = 255),
  `OeNB-Koralle` = '#d36a5f', #rgb(211,106,95,maxColorValue = 255),
  `OeNB-Bronze` = '#bc8034', #rgb(188,128,52,maxColorValue = 255),
  `OeNB-Grau` = '#d1d9dd', #rgb(209,217,221,maxColorValue = 255),
  `Österreich (rot)` = '#e30613', #rgb(220,0,0,maxColorValue = 255),
  `EU (blau)` = '#164194', #rgb(22,65,148,maxColorValue = 255),

```

```
`OeNB-Blau hell`='#80A2C8',
`OeNB-Sand hell`='#faf7ef',
`OeNB-Salbei hell`='#dfe7da',
`OeNB-Türkis hell`='#9aced5',
`OeNB-Violett hell`='#9e90b6',
`OeNB-Koralle hell`='#e9b5af',
`OeNB-Bronze hell`='#dcb47f',
`OeNB-Grau hell`='#f3f5f6',

`OeNB-Blau dunkel`='#001E3D',
`OeNB-Sand dunkel`='#d9be81',
`OeNB-Salbei dunkel`='#627c50',
`OeNB-Türkis dunkel`='#0c453f',
`OeNB-Violett dunkel`='#30283e',
`OeNB-Koralle dunkel`='#903227',
`OeNB-Bronze dunkel`='#704d1f',
`OeNB-Grau dunkel`='#95a8b1')
```

scatter plot

Sample data

```
data1 <- data.frame(x = 1:10,
                    y = c(2, 3, 5, 7, 11, 13, 17, 19, 23, 29))
```

Create the plot

```
scatter <- ggplot(data1, aes(x = x, y = y)) +
  geom_point(color = "#004289") +
  labs(
    title = "Title",
    x = "x-axis Label",
    y = "y-axis Label",
    caption = "Source: Simulated data. \nNote: Sample scatter plot."
  ) +
  th +
  scale_x_continuous(limits = c(0, max(data1$x)), expand = expansion(mult = c(0.05, 0.1))) + # X-axis
  with more spacing
  scale_y_continuous(limits = c(0, max(data1$y) + 1), expand = expansion(mult = c(0.05, 0.1))) # Y-axis
  with more spacing
  theme(plot.caption = element_text(lineheight = 1.3)) +
  theme(plot.caption = element_text(size = 8, family = "lato", hjust = 0, margin=margin(10,0,-20,0))) +
  theme(plot.title = element_text(size = 11, family = "lato_bold", face="bold", color = "#004289",
    margin=margin(-20,0,10,0)))
```

```
# save plot as svg, use device = svg to save it correctly as svg
ggsave(scatter, file = "img/scatter.svg", height = 16 * 20 / 32, width = 16, units = "cm", device = svg)
```

```
#### time series - bar and line chart ####
```

```
##### bar chart #####
```

```
# Sample time series data
```

```
data2 <- data.frame(Year = 2015:2024,
  Score = c(65, 67, 70, 72, 74, 75, 77, 78, 80, 82))
```

```
# Create the bar chart
```

```
bar <- ggplot(data2, aes(x = Year, y = Score)) +
  geom_bar(stat = "identity",
    fill = "#80A2C8",
    width = 0.7) +
  labs(
    title = "Financial literacy scores over the years",
    x = "",
    y = "Score",
    caption = "Source: Simulated data. \nNote: Financial literacy scores of students over the years."
  ) +
  theme +
  scale_x_continuous(breaks = 2015:2024, limits = c(2014.5, 2024.5)) +
  scale_y_continuous(breaks = seq(0,90, by = 10), limits = c(0, 90)) +
  coord_cartesian(expand = FALSE, clip = "on") +
  theme(panel.grid.major.x = element_line(color = "white", size = 0.2), # add theme options
    panel.grid.major.y = element_line(color = "#E5E4E2", size = 0.2))
```

```
##### line chart #####
```

```
line <- ggplot(data2, aes(x = Year, y = Score)) +
  geom_line(color = "#004289", size = 1) +
  geom_point(color = "#004289", size = 2) +
  labs(
    title = "Financial Literacy Scores Over Years",
    x = "",
    y = "Score",
    caption = "Source: Simulated data. \nNote: Financial literacy scores of students over the years."
  ) +
  theme +
  scale_x_continuous(breaks = 2015:2024, limits = c(2014.5, 2024.5)) +
  scale_y_continuous(breaks = seq(0,90, by = 10), limits = c(0, 90)) +
  coord_cartesian(expand = FALSE, clip = "on") +
  coord_cartesian(expand = FALSE, clip = "on") +
  theme(panel.grid.major.x = element_line(color = "#E5E4E2", size = 0.2),
    panel.grid.major.y = element_line(color = "white", size = 0.2))
```

combine bar and line plot and add lines

```
timeseries <- patchwork::wrap_plots(bar, line,
                                     ncol = 1, nrow = 2)
```

```
ggsave(timeseries, file = "img/timeseries.svg", height = 16 * 32 / 32, width = 16, units = "cm", device =
svg)
```

proportions - stacked bar chart and area chart

stacked bar chart

sample data

```
data3 <- data.frame(
  Category = c("Statement 4", "Statement 3", "Statement 2", "Statement 1"),
  fully_Agree = c(10, 20, 30, 50),
  mostly_Agree = c(40, 30, 20, 20),
  mostly_Disagree = c(40, 30, 20, 20),
  fully_Disagree = c(10, 20, 30, 10)
)
```

Convert data to long format

```
data3_long <-
tidyr::pivot_longer(
  data3,
  cols = -Category,
  names_to = "Response",
  values_to = "Count"
)
```

Calculate percentages

```
data3_long <- data3_long %>%
  group_by(Category) %>%
  mutate(Percentage = round(Count / sum(Count), 2)) %>%
  mutate(
    Response = case_when(
      Response == "fully_Agree" ~ "Fully agree",
      Response == "mostly_Agree" ~ "Mostly agree",
      Response == "mostly_Disagree" ~ "Mostly disagree",
      Response == "fully_Disagree" ~ "Fully disagree"
    )
  ) %>%
  ungroup()
```

Create the stacked bar chart

```
stackedbar <-  
  ggplot(data3_long, aes(  
    x = factor(  
      Category,  
      levels = c("Statement 4", "Statement 3", "Statement 2", "Statement 1")  
    ),  
    y = Percentage,  
    fill = factor(  
      Response,  
      levels = c(  
        "Fully disagree",  
        "Mostly disagree",  
        "Mostly agree",  
        "Fully agree"  
      )  
    )  
  ) +  
  geom_bar(stat = "identity", width = 0.6) +  
  scale_fill_manual(  
    values = c(  
      "Fully agree" = "#95af83",  
      "Mostly agree" = "#dfe7da",  
      "Mostly disagree" = "#e9b5af",  
      "Fully disagree" = "#d36a5f"  
    )  
  ) +  
  coord_flip(expand = FALSE, clip = "off") +  
  theme(  
    title = "Agreement to statements",  
    fill = "Response",  
    caption = "Source: Simulated data.") +  
  theme(axis.title.x = element_blank()) +  
  theme(axis.title.y = element_blank()) +  
  guides(fill = guide_legend(reverse = TRUE)) +  
  scale_y_continuous(labels = scales::percent, limits = c(0, 1)) +  
  scale_x_discrete(labels = \(Category) str_wrap(Category, 40)) + # required for wrap  
  geom_text(  
    aes(label = paste0(round(Percentage * 100, 0), "%")),  
    size = 2.5,  
    position = position_stack(vjust = 0.5),  
    family = "lato_bold"  
  ) +
```

```
theme(plot.caption = element_text(size = 8, family = "lato", hjust = 0, margin=margin(10,0,20,0))) +
theme(plot.title = element_text(size = 11, family = "lato_bold", face="bold", color = "#004289",
margin=margin(-20,0,10,0)))
```

area chart

Create the 'month' column

```
months <-
  rep(seq.Date(
    from = as.Date("2022-01-01"),
    by = "month",
    length.out = 12
  ), each = 4)
```

Create the 'values' column

```
values <-
  rep(c(
    "Fully agree",
    "Mostly agree",
    "Mostly disagree",
    "Fully disagree"
  ),
  times = 12)
```

Create a random 'proportion' column such that each group of months sums up to 1

```
set.seed(123) # For reproducibility
proportions <- unlist(lapply(1:12, function(x) {
  sample_values <- runif(4)
  sample_values / sum(sample_values)
}))
```

Create the final data frame

```
data5 <-
  data.frame(month = months,
    values = values,
    proportion = proportions)
```

Plot

```
area <- ggplot(data5) +
  geom_area(aes(
    x = months,
```



```

y = proportions,
fill = factor(
  values,
  levels = c(
    "Fully disagree",
    "Mostly disagree",
    "Mostly agree",
    "Fully agree"
  )
)
)) +
labs(title = "Agreement to statement 4 | 2024",
  caption = "Source: Simulated data.") +
scale_x_date(
  breaks = seq(as.Date("2022-01-01"), by = "month", length.out = 12),
  labels = c("Jan.", "Feb.", "Mar.", "Apr.", "May", "June", "July", "Aug.", "Sep.", "Oct.", "Nov.", "Dec.")
) +
scale_fill_manual(
  values = c(
    "Fully agree" = "#95af83",
    "Mostly agree" = "#dfe7da",
    "Mostly disagree" = "#e9b5af",
    "Fully disagree" = "#d36a5f"
  )
) +
th +
theme(axis.title.y = element_blank()) +
theme(axis.title.x = element_blank()) +
scale_y_continuous(labels = scales::percent, limits = c(0, 1)) +
guides(fill = guide_legend(reverse = TRUE)) +
coord_cartesian(expand = FALSE, clip = "on") +
theme(plot.caption = element_text(size = 8, family = "lato", hjust = 0, margin=margin(10,0,20,0))) +
theme(plot.title = element_text(size = 11, family = "lato_bold", face="bold", color = "#004289",
margin=margin(-20,0,10,0)))

ggsave(stackedbar, file = "img/stackedbar.svg", height = 16 * 14 / 32, width = 16, units = "cm", device =
svg)
ggsave(area, file = "img/area.svg", height = 16 * 18 / 32, width = 16, units = "cm", device = svg)

```

distributions - density plot and boxplot

boxplot

set.seed(123) # For reproducibility

grades <- c("Grade 1", "Grade 2", "Grade 3")

```
data6 <- data.frame(Grade = factor(rep(grades, each = 30)),
  Score = c(
    rnorm(30, mean = 35, sd = 10),
    # Scores for Grade 1
    rnorm(30, mean = 40, sd = 12),
    # Scores for Grade 2
    rnorm(30, mean = 45, sd = 15) # Scores for Grade 3
  ))
```

Create the boxplot

```
box <- ggplot(data6, aes(
  x = Grade,
  y = Score,
  fill = factor(Grade, levels = c("Grade 1", "Grade 2", "Grade 3"))
)) +
  geom_boxplot(varwidth = TRUE,
    alpha = 0.8,
    outlier.shape = NA) +
  stat_boxplot(geom = 'errorbar', width = 0.1) +
  th +
  scale_color_manual(values = c(
    "Grade 1" = "#001E3D",
    "Grade 2" = "#d9be81",
    "Grade 3" = "#627c50"
  )) +
  scale_fill_manual(values = c(
    "Grade 1" = "#001E3D",
    "Grade 2" = "#d9be81",
    "Grade 3" = "#627c50"
  )) +
  theme(legend.position = "none",
    panel.grid.major.y = element_blank()) +
  labs(title = "Financial literacy scores in three grades",
    caption = 'Source: Simulated data. \nNote: Financial literacy scores of students.') +
  coord_flip(expand = TRUE, clip = "off") +
  scale_y_continuous(limits = c(0, 100),
    breaks = seq(0, 100, by = 10)) +
  theme(axis.title.y = element_blank()) +
  #geom_jitter(aes(color = Grade), size = 1) +
  theme(plot.caption = element_text(size = 8, family = "lato", hjust = 0, margin=margin(10,0,20,0))) +
```

```
theme(plot.title = element_text(size = 11, family = "lato_bold", face="bold", color = "#004289",
margin=margin(-20,0,10,0)))
```

density plot

```
density <-
ggplot(data6, aes(x = Score, fill = factor(
  Grade, levels = c("Grade 1", "Grade 2", "Grade 3")
))) +

geom_density(alpha = 0.5) +
labs(
  x = "Score",
  y = "Density",
  title = "Financial literacy scores in three grades",
  caption = 'Source: Simulated data. \nNote: Financial literacy scores of students.'
) +
th +
scale_x_continuous(limits = c(0, 100),
  breaks = seq(0, 100, by = 20)) +
scale_color_manual(values = c(
  "Grade 1" = "#001E3D",
  "Grade 2" = "#d9be81",
  "Grade 3" = "#627c50"
)) +
scale_fill_manual(values = c(
  "Grade 1" = "#001E3D",
  "Grade 2" = "#d9be81",
  "Grade 3" = "#627c50"
)) +
theme(plot.caption = element_text(size = 8, family = "lato", hjust = 0, margin=margin(10,0,20,0))) +
theme(plot.title = element_text(size = 11, family = "lato_bold", face="bold", color = "#004289",
margin=margin(-20,0,10,0)))
```

combine bar and line plot and add lines

```
distribution <- patchwork::wrap_plots(density, box,
  ncol = 1, nrow = 2)
```

```
ggsave(distribution, file = "img/distribution.svg", height = 16 * 32 / 32, width = 16, units = "cm",
device = svg)
```

print all charts

```
scatter
timeseries
stackedbar
area
distribution
```

Mixed methods – a practical guide for the gold standard of evaluation research

Theresa Lorenz¹

Mixed-method evaluations combine qualitative and quantitative research methods, offering a deep and comprehensive analysis of complex social phenomena. This approach is widely regarded as the gold standard in evaluation research. However, effectively integrating these methods can be challenging. One common issue is the tendency to treat the findings from each method separately (e.g., in distinct publications), rather than fully integrating them. This separation can hinder a holistic understanding of the evaluation results, limiting the potential of a mixed-methods approach. In this paper, I provide an overview of various mixed-methods evaluation designs and aim to mitigate some of the challenges faced by researchers employing this approach. Our goal is to equip evaluation researchers with the knowledge and tools needed to incorporate mixed methods into their methodological toolkit, enabling them to apply this approach effectively.

Mixed methods combine qualitative and quantitative research methods to investigate social phenomena. This approach is regarded as the gold standard in evaluation research for many research questions because the blend of qualitative and quantitative research methods can provide a deep and comprehensive understanding of the impacts and processes of financial education initiatives. Consequently, mixed methods may enable more robust and reliable evaluations. However, despite these advantages, mixed methods are not often used due to the additional challenges they present.

The idea of combining different methods in social and evaluation research can be traced back almost 200 years (Bazeley, 2018). Back then, researchers often employed what we would now recognize as mixed or multi-mode methods, although these terms were not explicitly used back then. Fundamental conflicts between qualitative and quantitative methods and differing views on the nature of reality and knowledge (ontology and epistemology) were also not particularly pronounced during this era (Maxwell, 2016).

It was only later, in the 1970s and 1980s, that strong conflicts between qualitative and quantitative methods emerged (Bazeley, 2018). What became known as the “paradigm wars” led to a distinct divide between qualitative and quantitative methods. Qualitative research became closely aligned with the interpretivism paradigm, characterized by inductive reasoning and interpretative techniques. In contrast, quantitative researchers typically adhered to a positivist paradigm, relying on deductive reasoning, experimental methods, and statistical analysis (Bazeley, 2018).

The late 20th and early 21st centuries finally marked the formal recognition of mixed-methods research as a distinct methodology. This shift was driven by increasing dissatisfaction with the limitations imposed by the paradigm wars and a growing acknowledgment that neither quantitative nor qualitative approaches

¹ Oesterreichische Nationalbank, financial literacy and culture division, theresa.lorenz@oenb.at. Opinions expressed by the author of studies do not necessarily reflect the official viewpoint of the OeNB, the or the Eurosystem. The author would like to thank Maximilian Zieser (OeNB), Katharina Felbermayr (OeNB) and Sandra Mauser (OeNB) for helpful comments and valuable suggestions. This paper is part of the OeNB Financial Literacy Evaluation Series. The series aims to inform researchers, policymakers and educators about the current state of research on financial literacy and education and to provide guidelines for designing and implementing comprehensive evaluation studies. The series was developed by the OeNB in collaboration with the Bank of Greece. For details and further publications of the series, see <https://www.oenb.at/en/Publications/financial-literacy/financial-literacy-evaluation-series.html>.

were sufficient on their own to comprehensively address complex social phenomena (Creswell and Clark, 2011). A new paradigm (called pragmatic paradigm) emerged as a response, advocating for the selection of methods and paradigms best suited to answer the research question at hand (Dawadi et al., 2021). While this can be seen as an effort to resolve the paradigm wars, the strict separation of methods that resulted from the paradigm wars has remained a considerable challenge in the application of mixed-methods research.

The historical divide between qualitative and quantitative research methods has led to a high degree of specialization among qualitative and quantitative researchers with limited communication among them. This presents significant challenges in mixed-methods research and is one reason why, despite being perceived as the gold standard of evaluation, mixed methods are not frequently applied, and their full potential is often not realized. Indeed, it is common practice of mixed-methods research that researchers do not work together, which gives rise to the risk of finding diverging results that are then presented in separate papers rather than a combined mixed-methods publication. The lack of formal education in mixed-methods research compounds these issues. The required skills include understanding design options, all methods used in the study, potential integration strategies, and ways to maintain focus without overextending the analysis.

Furthermore, there is a lack of guidelines that demonstrate how to practically integrate results. Despite some existing guidelines for publishing mixed-methods papers (Palinkas et al., 2019), there is also a lack of standardized, established protocols comparable to those in other research areas — like standardized protocols for randomized controlled trials or standardized protocols for structured literature reviews (Bazeley, 2018). This gap is problematic because diverse disciplinary traditions can reinforce the issue of poor integration, thus not fully leveraging the potential of mixed methods.

Lastly, mixed methods often require additional efforts and greater resources than more traditional evaluation approaches. This requires management commitment to provide the needed resources and time, and to protect these from budget and time pressures (Bamberger, 2012).

With this paper, I aim to reduce the burden on evaluation researchers who are considering mixed methods as an approach, allowing them to make this approach part of their methodological toolkit available for application when appropriate. The remainder of this paper is, thus, structured as follows: In the first section, I examine the utility of mixed methods in evaluation studies, highlighting their unique benefits and the prerequisites for their successful implementation. Subsequently, I delve into the core design features of mixed methods and common mixed-methods evaluation designs. The second section provides practical guidelines and strategies for integrating data collected in mixed-methods studies.

1 Mixed-methods designs

1.1 Defining mixed methods

To this day, mixed-methods research lacks a universally accepted definition (Schoonenboom, 2023). Bazeley (2018) addresses this issue by referring to a synthesized definition derived from 19 different descriptions of mixed methods proposed by Johnson et al. (2007). Serving as the foundation for this document, this concise definition describes mixed methods as follows:

“The type of research in which a researcher or team of researchers combines elements of qualitative and quantitative research approaches (e.g., use of qualitative and quantitative viewpoints, data collection, analysis, inference techniques) for the breadth and depth of understanding and collaboration.” (Johnson et al., 2007, p. 123).

As highlighted by the definition above, a fundamental prerequisite for a study to qualify as a mixed-methods study is some degree of integration — the process of combining different research methods to achieve a common research goal. In contrast, studies that employ both qualitative and quantitative elements but with minimal integration are typically categorized as multi-method or combined studies (Bazeley, 2018; Creswell and Clark, 2017).²

Indeed, Bazeley (2018) emphasizes that a crucial component of a mixed-methods study is “a ‘conversation’ between the different sources and/or methods used (...) within the analysis, continuing to the presentation of results and discussion of those results” (Bazeley, 2018, p. 7). The extent of “conversation” required to distinguish between mixed-methods and multi-method studies, however, remains undefined. In evaluation contexts, the use of qualitative and quantitative approaches can vary, being either applied at specific stages (e.g. only collecting quantitative and qualitative data) or integrated throughout the entire process by additionally interacting analysis and findings. The ambiguity lies in the extent of integration that needs to occur in some or all phases for a study to qualify as a mixed-methods study (Bureau of Policy & Learning, 2013; Palinkas et al., 2019).

Box 1

Two examples illustrating the ambiguity between mixed-methods and multi-method studies

For instance, qualitative methods used in planning stages of an evaluation, like focus groups for hypothesis generation or think-aloud techniques for survey item interpretation, often fall into “mixed-methods light” or “multi-method” categories if the approaches are not further integrated in analysis and interpretation. Such approaches thus primarily support the development of quantitative instruments without achieving thorough integration. However, the same study would be qualified as a mixed-methods study by most researchers if the analysis integrates both qualitative and quantitative data by comparing and synthesizing findings to draw richer, more nuanced conclusions. In such an integrated approach, the final interpretation reflects a more cohesive narrative, showing how qualitative insights explain or expand upon the quantitative results and vice versa.

Similarly, using statistical census data to contextualize qualitative findings during analysis only partially combines both methods. Full integration might involve follow-up interviews exploring insights from the census data, promoting dynamic interaction between the data sources. Moreover, the presentation of findings should intertwine results from both methods, each adding context and depth to the other.

² Studies that are purely qualitative or pure quantitative are called mono-method studies (Niglas, 2004).

1.2 Why mixed methods?

Overall, mixed methods in evaluation research are highlighted as being particularly useful for providing a more thorough, nuanced and reliable understanding of complex research topics. More specifically, the advantages can be summarized as follows (Bamberger, 2012; Bazeley, 2018; Bureau of Policy & Learning, 2013; Dawadi et al., 2021; Frechtling and Sharp, 1997):

Comprehensive understanding: Quantitative data provides robust numerical insights, while qualitative data adds rich context, depth and explanation. This integrated approach effectively compensates for the limitations of each method individually, tapping into the nuanced complexities of social environments and human behavior to paint a more holistic picture which allows a better generalization of findings (internal and external validity). In short, a mixed-methods approach “enables researchers to answer questions with sufficient depth and breadth” (Dawadi et al., 2021, p. 27).

Validation and cross-verification: Mixed methods allow for the results from one method to be validated or challenged by the other. This cross-verification enhances the reliability and validity of research findings.

Flexibility and adaptability: A mixed-methods approach is adaptable to various research settings and questions. It allows researchers to adapt their research design based on the evolving nature of the program studied.

Richer data and perspectives: Mixed-methods research captures a wider range of perspectives and data types, which can lead to more nuanced, detailed as well as unexpected and sometimes also contradictory findings. This is particularly valuable in evaluation research where understanding different stakeholders’ perspectives is crucial.

1.3 Prerequisites for applying mixed methods

Mixed methods have a number of advantages, but there are also some prerequisites to consider when applying this method, as detailed by Bamberger (2012), Bazeley (2018), Creswell and Clark (2017), and Dawadi et al. (2021).

First, mixed methods require a well-suited research question: As discussed earlier, mixed-methods approaches offer a more comprehensive perspective in various evaluation scenarios. However, there are specific research questions that benefit significantly from the integration of qualitative and quantitative methods. This is, for instance, the case when the evaluation aims to address both *processes* and *impact*. In such cases, qualitative methods excel in explaining the processes, while quantitative methods are better suited for evaluating the impact. In general, mixed methods are effective when complex interventions are investigated.

Second, mixed-methods approaches can be cost- and time-intensive. As Bamberger (2012) notes, this requires commitment to provide the needed resources and time, and to protect these from budget and time pressures. Given that researchers often specialize in their respective methods, time should also be allocated for them to understand each other’s methodologies and cultivate mutual respect and trust. The more integrated the mixed-methods approach, the more time, particularly during the planning and analysis stages of evaluation, needs to be dedicated.

Third, success in mixed-methods research requires skills in both qualitative and quantitative methods. Researchers often specialize in one or the other, making it essential for the team to collectively possess all the necessary skills. While not essential, having team members from different disciplines can be an additional benefit, enhancing the research with diverse perspectives. Each team member should, however, have at least a basic familiarity with all the methods applied in the evaluation. In addition, researchers need the skills to adequately combine different methods.

Lastly, mixed methods require openness and respect toward different perspectives that are often grounded in different quantitative and qualitative philosophical assumptions. This is crucial as diverse viewpoints can significantly enrich the quality of research, but these diverging approaches are often the reason why the full potential of mixed-methods approaches are not utilized. In a similar vein, it is critical to equally value both qualitative and quantitative methods even though methods may have different weighting in terms of how intensely they are used in the evaluation.

1.4 Main evaluation design features of mixed-methods studies

In this section, I present evaluation design aspects specific to mixed-methods studies. Aspects relevant to evaluation studies in general will be thoroughly discussed in additional parts of the evaluation series. Following the guidelines of Bamberger (2012) and Creswell and Clark (2017) three aspects need to be considered when applying a mixed-methods study, namely: (1) the integration stage, (2) timing of data collection, and (3) weighting of methods.

1.4.1 Stage of integration

Integrating qualitative and quantitative methods can happen at single, some or all stages of the evaluation cycle. Full integration across stages often yields the richest insights. Integration strategies may vary at each stage. For illustration purposes, the following are examples of integration at the planning, conducting, analysis and communication stage of an evaluation during the research/evaluation process:

At the planning stage, integration occurs at the design level - i.e. applying a mixed-methods evaluation design. For instance, this could involve planning a randomized controlled trial to quantitatively assess the outcomes of a financial literacy program together with qualitative interviews to understand program mechanisms.³

At the stage of conducting the evaluation, integration might involve collecting quantitative and qualitative data, like a quantitative survey with qualitative interview insights.

At the level of analysis, integration involves blending quantitative statistical data with qualitative data, offering a richer, more nuanced understanding through validation, contrasting themes or cases. When communicating outcomes, synthesizing findings from integrated analyses is crucial for drawing comprehensive conclusions and effectively conveying the results.

Typically, mixed method approaches involve the combination of methods during several phases, where integration at one stage is often a precondition for integration at the subsequent stage. Naturally, integrating qualitative and quantitative data during analysis is impossible if only quantitative data have been collected. However, in many studies integration does not extend beyond the data collection phase as results from different data sources are not combined but rather presented in separate qualitative and quantitative reports (Bazeley, 2018). To illustrate this, Bazeley (2018) refers to a review by Niglas (2004), noting that of 1,156 articles from educational journals, only 145 used a combined approach, and even fewer fully exploit the potential of integration. He concludes that *“while the baseline is to apply complementary methods, greater levels of integration are both possible and desirable”* (Bazeley, 2018, p.117). For more details on integration see section 2.

³ Quantitative methods, like a randomized controlled trial and qualitative methods like in-depth interviews are described in detail in additional parts of the evaluation series.

1.4.2 Timing of data collection

In terms of timing, there are two ways of combining data collection in mixed methods – either sequentially or concurrently (also called a parallel or convergent designs) (Alele and Malau-Aduli, 2023; Bazeley, 2018). In a parallel design, qualitative and quantitative data are collected at the same time, whereas in the sequential design, one type of data source (qualitative or quantitative) is collected first, followed by the other data type.

It is worth mentioning that timing as a design option is not only relevant for data collection but also for the analysis of various data sources. For instance, quantitative and qualitative data might be collected concurrently, yet their analysis could be sequential (Palinkas et al., 2019). Despite this fact, the primary focus of mixed-methods design features often revolves around the timing of data collection, which is the main emphasis of this section. The discussion on the timing of data analysis will be further elaborated in section 2.

1.4.3 Methodological weighting

The dominant method is the one that is given relative priority in answering the research question. This can either be a dominantly quantitative design, a dominantly qualitative design or a balanced design with equal weight for each methodical approach (Dawadi et al., 2021). This distinction matters because researchers are typically specialized in either qualitative or quantitative or methods. Thus, they may hold different expectations regarding the involvement of “their” methodical approach (Bamberger, 2012).

A balanced design in evaluation may be one giving both process and impact evaluation equal weight, using qualitative methods to explore processes and quantitative methods investigating impacts. Another typical approach for a balanced design is the use of qualitative and quantitative methods for the same unit of analysis (such as participants of an intervention) (Bamberger, 2012).

1.5 Design combinations

The presented design options can be combined in various ways (Palinkas et al., 2019) and adaptations and variations throughout the research process are common as new insights or requirements might emerge and mixed methods are particularly suited to such dynamic scenarios (Palinkas et al., 2019). This section summarizes common examples for combining design options based on Alele and MalauAduli (2023), Bazeley (2018), and Creswell and Clark (2017).

To systematically illustrate design options and their combinations, mixed-methods research often relies on the following notation (Morse, 1991): Concerning timing, arrows indicate a sequential design. For example, an arrow from the first to the second data collection method represents the sequence of data collection (*data collection method 1 → data collection method 2*). A parallel design, on the other hand, is

represented by a plus sign (*data collection method a + data collection method b*). With regards to weighting, it is also common practice to write the dominant method in bold, i.e. QUAN for a dominantly quantitative design and QUAL for a dominantly qualitative design. “QUAL → quan”, for example, reflects a sequential design where more weight is given to the qualitative method. For a summary on mixed-methods design options notation see Table 1.

Table 1

Mixed methods design options notation

Timing		Weighting	
Sequential data collection	→	Dominantly qualitative design	QUAL, quan
Parallel data collection	+	Dominantly quantitative design	QUAN, qual

Source: OeNB.

1.5.1 Convergent parallel design (QUAN + QUAL)

In this design, qualitative and quantitative data are collected and analyzed separately, with results then being compared, expanded, contrasted or validated. Finally, the goal is to reach coherent, valid conclusions about the research question studied. The main objective of this design is to obtain a complete understanding of the research problem. Typically, equal weight is given to qualitative and quantitative methods.

Box 2

Example 1: Convergent parallel design

In a financial literacy evaluation, this might involve conducting surveys to gather quantitative data on participants' financial competencies (impact evaluation) while simultaneously conducting interviews or focus groups for an in-depth understanding of their experiences and perceptions of the program (process evaluation). After analyzing both sets of data independently, the findings are compared and integrated. This approach allows for a comprehensive understanding of the effectiveness of the financial literacy program, combining quantitative data on financial competencies with nuanced insights into the participants' experiences and perceptions.

While this design can lead to nuanced and comprehensive results, it is time intensive and places the greatest demands on the mixed-method skills of the researchers. They might face challenges for integration when the findings from qualitative and quantitative research diverge. In such cases, researchers are advised to reanalyze the data to reconcile these discrepancies. On the other hand, these divergent findings may also reveal aspects that may otherwise be overlooked. Some guidelines on exploring dissonance and divergence can be found in section 2.4.

1.5.2 Explanatory sequential design (QUAN → qual)

This combination starts with quantitative data collection and analysis, followed by qualitative data. The qualitative findings are used to gain more in-depth information and help to explain and interpret the quantitative results. For example, if new questions arise from quantitative results they can be investigated in the subsequent qualitative phase. Also, outliers or unexpected results can be further explored. This design is usually dominantly quantitative.

Box 3

Example 2: Explanatory sequential design

A financial literacy program might first be evaluated using a quantitative approach, such as surveys to gather quantitative data on participants' financial competencies (impact evaluation). The results from this phase could highlight areas needing deeper exploration. The subsequent qualitative phase (e.g. interviews or focus groups) would aim to understand the reasons behind the quantitative findings, such as why certain financial concepts were not well understood by low-performing participants or how participants felt about the program. The initial quantitative findings guide the qualitative research, making the quantitative phase the starting point and foundation for further exploration of unexpected or highly relevant results.

1.5.3 Exploratory sequential design (QUAL → quan)

This design begins with qualitative data collection and analysis, with the findings guiding the design and development of quantitative data collection. First, this design is useful for development and validation of research instruments (e.g. a survey). Second, it can be used to explore a research topic which lacks theoretical grounds or if hypotheses are not known. Third, it can be applied to test whether qualitative results also apply to a wider population. This design is usually dominantly qualitative.

Box 4

Example 3: Exploratory sequential design

Such an approach may start with a broad qualitative exploration, perhaps through detailed interviews and/or focus groups with participants to understand their experiences and challenges with financial literacy. The insights gained would then shape the subsequent quantitative phase, such as designing a small survey to measure the prevalence of these challenges among a larger group. However, the qualitative findings are at the center of the design.

1.5.4 Embedded Design

The embedded design is characterized by integrating a smaller component of one type of data (qualitative or quantitative) within a larger framework of the other data type. The nested component plays a supportive and supplementary role. Data collection and analysis in this design can be sequential or concurrent, occurring before, during or after the primary method is implemented. This design is particularly effective when a research question demands a combination of both quantitative and qualitative data to provide context and depth. The primary focus is on one method, often quantitative, with the aim of understanding how the additional data source enhances the understanding of the research question.

Differentiation between an embedded design and other mixed-methods variations is not straightforward. Creswell and Clark (2017) point out that the key distinction lies in the standalone value of the data sources. In an embedded design, the value of the secondary, nested component is largely dependent on the context provided by the primary data source. Conversely, in other mixed-methods designs, the secondary data sources frequently possess enough standalone value to be meaningful even without the primary method.

Box 5

Example 4: Embedded Design

In a financial literacy program evaluation, the primary focus might be a quantitative survey assessing participants' financial competencies. Simultaneously, a smaller qualitative component, such as interviews with a subset of participants is conducted. These interviews are designed to provide deeper insight into specific aspects of the quantitative findings on how the program's delivery affected participants' learning. The qualitative insights are used to add depth and context to the quantitative findings, enhancing the overall understanding of the program's impact.

Each design has its specific strengths and is chosen based on the research objectives. The convergent parallel design is suitable for studies where a comprehensive understanding from both the qualitative and the quantitative perspective is needed *simultaneously* as data points are more likely to be comparable if they are collected at the same point in time. Sequential designs are preferred when the findings from one method are essential to inform or complement the other method. An embedded design is beneficial when a comprehensive understanding of the research topic is needed, but there is less time or there are fewer resources than in a convergent parallel design to devote equal emphasis to both qualitative and quantitative methods (Alele and Malau-Aduli, 2023; Creswell and Clark, 2017). For more details on common mixed-method designs see Creswell and Clark (2017).

The design options presented offer potential approaches for mixed-methods evaluation. But it is important to remember that these are no rigid formulas. The key to effective research is to flexibly combine and adapt elements of typical mixed-methods frameworks, always focusing on the overarching goal of addressing your research question effectively.

Table 2

Common mixed-methods designs

Design	Description	Timing of data collection	Dominant method	Primary objective
Convergent parallel design (QUAN + qual)	Quantitative and qualitative data are collected and analyzed separately; findings are then compared or validated.	Concurrent	Equal	Comprehensive understanding of the research problem
Explanatory sequential design (QUAN → qual)	Starts with quantitative data collection and analysis, followed by qualitative approach to explore findings further.	Quantitative first, qualitative second	Quantitative	In-depth explanation of quantitative results
Exploratory sequential design (QUAL → quan)	Begins with qualitative data collection and analysis to guide the development of quantitative instruments.	Qualitative first, quantitative second	Qualitative	Qualitative exploration guides quantitative research
Embedded Design	Integrates a smaller component of one type of data within a larger framework of the other type.	Flexible (concurrent or sequential)	Primary method (usually quantitative)	Enhance primary method's understanding

Source: OeNB.

A big challenge in mixed-methods research is choosing the right design while maintaining focus. Each stage of evaluation offers numerous possibilities for combining mixed-methods design features, leading to a complex decision-making process. There is no universal recipe for selecting and applying mixed methods which makes it hard to give universally applicable guidelines. While integration often falls short, in cases where researcher intend to integrate, they often risk losing focus (Dawadi et al., 2021).

To navigate this challenge, early and careful consideration of the optimal design can be effective. This strategy refers to the thoughtful combination of qualitative and quantitative methods which begins at the very start of the planning phase. The initial focus should be on defining the overarching goals and specific research questions, which will inform the selection of design features. Key considerations for these features, including their respective benefits and limitations, are presented above. Prior to data collection, researchers should clearly define the purpose of each data sources they intend to use for the evaluation. Additionally, they should develop an initial plan outlining how and when these data sources will be combined. This proactive planning ensures that each component of the research is aligned with the overall

objectives. Implementing a theoretical framework can also be beneficial, offering a guide for the entire research process. However, it's important to remember that while these strategies provide structure, they should not be rigid. Maintaining a degree of flexibility is crucial to foster exploration and adaptability as research unfolds (Bazeley, 2018).

2 Mixed-methods integrative data analysis

This section underscores the vital role of integration in mixed-methods research, extending beyond the initial phases of planning and execution to the crucial stages of analysis and communication. This is where the true potential of mixed methods is realized. However, challenges often lead to separate analysis and reporting of qualitative and quantitative data. Addressing this gap, I introduce three advanced techniques for combining data: the triangulation protocol, “following a thread” and the mixed-methods matrix. Building upon the work of O’Cathain et al. (2010), who proposed these three types of integration techniques, I expand on recent advancements.

The techniques will be illustrated by the following mixed-methods financial literacy evaluation example, which will be applied throughout this section:

Box 6

Financial literacy evaluation example

In this example, the evaluation aims to assess the effectiveness of a voluntary financial literacy program in improving secondary school students’ knowledge, skills and attitudes toward personal financial management.

The **quantitative** component of the evaluation is an impact evaluation where students complete surveys before and after the program to measure changes in their skills. Also, the curriculum and educational materials have been reviewed by the researchers to evaluate content coverage. Statistical analysis of survey data were used to identify significant changes in knowledge and behavior. Analysis of attendance records proved information about engagement levels. Content analysis of program materials and curriculum gave information on alignment with learning objectives.

The **qualitative** component contains in-depth interviews with a selection of students to gather insights about their experiences and perceived impact of the program. Focus groups are used to explore students’ views on the program’s content and delivery. Analysis of interview and focus group transcripts were used to derive common themes, insights and student perceptions.

2.1 Triangulation protocol

This technique was initially developed by Farmer et al. (2006) and was later adapted by O’Cathain et al. (2010) and Bazeley (2018). Triangulation is a term that has been used in different ways. It was originally meant to test the convergence of results as a quality control tool by applying more than one method in parallel. However, in mixed-methods research it has been proven that results from different methods do not necessarily give more valid results but rather provide a more complete and extended picture with often diverging results as well. Hence, finding convergence is only one element of a triangulation protocol (Bazeley, 2018).

The aim of the triangulation protocol is to compare and explain qualitative and quantitative results to gain a complete picture about the research topic to eventually develop meta-themes across the different data sources. The triangulation protocol entails several key steps:

Preparation: Separately analyze qualitative and quantitative datasets using an analytical method suitable for the data source to identify shared concepts and themes relevant to the research questions.

Convergence matrix: Organize themes/concepts in a matrix in accordance with the data sources and their respective results in a so-called convergence matrix. The matrix's rows represent the identified themes/concepts, while columns display various data sources and their findings, like qualitative quotes and quantitative frequencies. An additional column for data convergence indicates agreement, partial agreement, silence (a theme appearing in some, but not all, data sources) or dissonance (differing meanings or prominence across data sources) between the different data sources for one theme.

It is important to mention that dissonance across data sources is not a sign that something went wrong but is rather perceived as vital to get a better understanding of the research question. I will provide some strategies on how to deal with dissonance across data sources at the end of this section. Similarly, "silence" as described above might be expected as not every method is best suited to deal with every theme. However, unexpected silence should be similarly treated as dissonance across data sources and can contribute to a better understanding of the research topic. Researchers are also invited to creatively adapt and expand the convergence matrix for their particular purpose.

Collaborative analysis: If the triangulation protocol is conducted by multiple researchers, it's important to compare their individual assessments to identify any differences and to clarify interpretations of the findings. Farmer et al. (2006) advocate for an approach in which at least two researchers are involved in the triangulation process.

Synthesis: Synthesize a cohesive interpretation and description of each theme, considering the broader context of the research.

Box 7

Applying the financial literacy evaluation example

After the qualitative and quantitative data have been primarily analyzed separately, the application of the convergence matrix may look as follows:

Table 3

Convergence matrix

Themes	Quantitative data (frequencies)	Qualitative data (quotes)	Program material analysis	Data convergence
Understanding of financial concepts	75% improvement in test scores	"I feel more confident about managing money."	Curriculum covers key financial concepts.	Agreement
Application of skills	60% reported using budgeting tools	"I started tracking my expenses regularly."	Emphasis on practical budgeting tools.	Agreement
Attitude toward saving	70% reported that they had started to save more	"I'm trying to save, but it's not always easy to stick to the plan."	Materials encourage saving but respondents acknowledge challenges.	Partial agreement
Real-world relevance	Not specifically measured	"Wish there were more real-life examples."	Mostly theoretical content.	Silence
Engagement in the program	High attendance rates (above 85%)	"Many sessions were uninteresting and too technical"	Materials are dense and heavily theoretical.	Dissonance

Source: OeNB.

In cases as illustrated by the last theme, where data shows divergence, I recommend specific techniques to explore the reasons behind this variance. For instance, further investigations revealed that students' high attendance rates were not solely due to engagement with the program but were also influenced by external factors, such as monetary incentives provided by their parents.

Once the results of the convergence matrix have been discussed and a uniform interpretation has been reached by the collaborating researchers, the final step involves synthesizing the findings from the convergence matrix, which can be presented as follows:

Both the quantitative and qualitative data indicate an improvement in participant's understanding of financial concepts, which is supported by the comprehensive coverage of financial concepts in the program materials. Regarding the application of skills, there is a notable agreement between participants using budgeting tools (quantitative) and their qualitative reflections on adopting new financial habits. There is a general alignment of improved saving attitudes and behaviors, but nuances in the qualitative data and program materials suggest that the implementation of these savings practices also comes with the challenge of actually sticking to long-term goals. The lack of real-world application is noted qualitatively, but not directly measured in the quantitative data or addressed in the program materials. True engagement in the program could be improved as student feedback (qualitative) and the nature of the program materials suggest a lack of true engagement or interest in the program.

2.2 Following a thread

Initially developed by Moran-Ellis et al. (2004) and further taken up by O'Cathain et al. (2010) the "following a thread" technique offers a narrative and iterative approach to integrative data analysis. It focuses on how a single theme or issue, identified within one set of data, is explored across various datasets. This method allows for a deep and nuanced understanding of research findings and facilitates a focused exploration of specific issues across different datasets (O'Cathain et al., 2010). It provides a framework that enables both deductive (testing existing theories) and inductive (generating new theories) analysis (Dupin & Borglin, 2020).

Though less standardized than the triangulation protocol, "following a thread" may entail the following steps:

Preparation: Separately analyze qualitative and quantitative datasets using an analytical method suitable for the data source. From this initial analysis, derive concepts and themes and particular questions that require further exploration. This can be an unexpected discovery or topics that appear promising for further exploring this topic through the additionally available data sources.

Follow the thread: Then select this particular question or theme from one data source as a "thread" to explore it across other datasets. This can be a back and forth movement across datasets with the goal to iteratively interweave the findings from different datasets. This is an iterative analysis process, where initial findings guide hypothesis formation, which is then examined using other data sources. Finally, the findings from the thread lead to a pattern of findings that can also be intertwined with findings from other threads.

However, a detailed description of how to practically apply and utilize these frameworks remains insufficiently explored, as highlighted in a systematic review by Dupin and Borglin (2020).

Box 8

Applying the financial literacy evaluation example

After separately analyzing the qualitative and quantitative data, the quantitative results showed a significant improvement in survey scores related to budgeting and savings, indicating a positive change in financial skills following the completion of the program. Consequently, the first theme, “understanding of financial concepts,” was identified and traced through the qualitative data. Interviews revealed that several students began using budgeting apps or saving a portion of their allowance due to the program. Focus groups also indicated that many students found the interactive components of the program helpful. Both findings likely contributed to the positive impact on financial skills after the program.

However, some students expressed a desire for more real-life financial scenarios, leading to the identification of the second theme: “real-life scenarios are more accepted.” Attendance records showed higher attendance rates for practical financial management sessions, which included hands-on activities such as creating personal budgets and learning about financial tools. Revisiting the survey results indicated that students with higher attendance rates performed best. Therefore, the second theme is also supported by various data sources.

Intertwining both threads, we can conclude that interactive teaching methods were well-received and translated into real-life application for some students. Attendance patterns corroborate that engaging methods boost participation and lead to better learning outcomes.

2.3 Mixed-methods matrix – also called metamatrix or joint displays

This technique was notably developed and refined by Bazeley (2018) and O’Cathain et al. (2010) and has various forms. Its primary objective is to summarize and contrast trends and patterns across multiple cases or data points. While similar to the triangulation protocol, its focus shifts toward individual cases rather than overarching concepts or themes. A distinctive feature of some mixed-methods research can be the presence of qualitative and quantitative data for the same case (also called unit of analysis). Cases can be individuals, organizations, longitudinal datapoints, groups, geographical areas etc. that share some characteristics. This approach enables a very granular integration of qualitative and quantitative data sources. Cases are the unit of analysis in a mixed-methods matrix - also known as a joint display or metamatrix. Importantly, cases in the mixed-methods matrix serve as a means to illustrate derived themes but the case itself is not the main objective of the study.⁴

Box 9

Illustration of cases as the unit of analysis

For instance, a group of students who have completed a survey might also participate in detailed interviews, resulting in a specific group of cases where there is both survey data and interview transcripts available. In this example, students represent the case.

In a longitudinal study the date of data collection can be a case to investigate changes over time throughout different data sources. Different data sources can then be matched for the same case and date becomes the unit of analysis.

⁴ This is a main difference to the case study method where a few cases are the main objective of analysis (Bazeley, 2018).

The technique of the mixed-methods matrix may include the following steps:

Preparation: Even if the analysis level is individual cases, initial separate analysis of qualitative and quantitative data can also be useful for this technique. This can be coding, calculations, selecting relevant variables etc. to identify relevant cases for the research questions. Indeed, rather than examining every case available in the sample, focusing on a subset of cases that presents particularly unique or surprising results can be more insightful.

Matching: Construct a matrix where each row represents a different case, identified by an identifier. Columns correspond to various *variables or themes*, encompassing both quantitative data (like scores or demographics) and qualitative data (such as interview themes or quotes). Each matrix cell then contains the data relevant to that case and variable or theme.

Synthesis: Additional methods can be employed to further synthesize the data, integrating matched data into coherent themes or findings. A further extension of this approach is the pillar integration method by Johnson et al. (2019), which provides a more detailed systematization of application and interpretation of results. However, this advanced methodology extends beyond the scope of this discussion.

Box 10

Applying the financial literacy evaluation example

After the qualitative and quantitative data have been primarily analyzed separately and cases have been matched, the application of the mixed-methods matrix may look as follows:

Table 4

Mixed-methods matrix

Case number	Score improvement	Quoted statement	Summary of individual changes
Highest improvement in financial literacy			
Student A	60% → 85%	"Saving regularly made me more confident about my financial future."	Demonstrated the highest uptake of financial skills and concepts.
Student B	55% → 90%	"Understanding taxes and investments has opened up new opportunities for me."	Showed significant engagement and application of complex financial topics.
Lowest improvement in financial literacy			
Student C	No change, score remained at 75%	"I get the concepts in theory, but it's hard to apply them."	Minimal change indicates a struggle to apply knowledge practically.
Student D	75% → 73%	"The program was informative, but I haven't changed much about how I handle money."	Small regression suggests a disconnect between knowledge and behavior.

Source: OeNB.

Synthesizing the results, we can conclude that students with the highest improvement demonstrated a proactive approach to applying learned financial concepts, resulting in improved financial behaviors and decision-making. Students with the lowest improvement understood the financial concepts but did not implement substantial behavioral changes, indicating a need for more practical approaches.

2.4 Exploring dissonance and divergence

In the previous discussion, we have seen that mixed-methods research sometimes uncovers different, and potentially conflicting, results. This is particularly salient in studies of complex social phenomena,

where such divergent findings are not just common but can be highly informative. Uprichard and Dawney (2019) capture the importance of contrasting findings in social research with the following analogy:

“After all, we tend to assume that one method depicts one part or aspect of the object of study and if another method presents a different part or aspect, then the methods have together shown different parts or aspects of the same thing. But what if one method captures the ‘ear of the elephant’ and another method captures the ‘tail of a mouse’? Mixed methods, very successfully, capture multiple aspects of multiple parts that are entangled together instead of revealing some singular ‘thing.’” Uprichard and Dawney (2019, p. 22)

In this vein, the primary goal in mixed-methods research should not be to force convergence of findings. Instead, divergent results should be viewed as gateways to new perspectives and deeper understanding. When discrepancies arise, it is advisable to delve deeper by returning to the data or conducting additional analysis to uncover the reasons for these differences. Bazeley (2018) suggests several potential sources of divergence that could be further investigated in the data:

Sampling differences: Differences can be related to sampling, when qualitative and quantitative data do not investigate the same target population and/or a subgroup. For instance, a subsample from a larger survey may yield different results due to its unique characteristics. Additionally, qualitative and quantitative methods usually have different sampling objectives. While qualitative research often seeks diverse representation, quantitative methods aim for a more generalized, representative view.

Difference in topics: If quantitative and qualitative methods do not measure the same topic, they are likely to lead to diverging results. Also, different interviewers may have treated a topic differently (for example more or less in depth). Even if the same topic is covered in the same depth, the format how a topic has been presented in the qualitative and quantitative method can affect the results. For example, if sensitive questions are presented in a face-to-face setting participants can be more likely to give a socially expected answer than in a more anonymous online setting.

Theoretical bias: Consider whether there is a biased theoretical framework guiding the research. It can be advisable to underpin mixed methods with a common conceptual framework (more details on this see Bazeley (2018)), yet some degree of flexibility is required to revise theories if the findings don’t align with the underlying framework.

Participant variability: Different backgrounds and interpretations of survey or interview questions among participants can yield varied results. Such variability is not necessarily negative; it can be beneficial if properly understood and managed. But researchers must need to understand what mechanisms lead to different interpretations — like reactions to sensitive questions — that influence participant responses. Reanalyzing data by specific characteristics, like sociodemographic variables, are likely to contribute to a better understanding of different interpretations.

Researcher perspective: Researchers’ philosophical beliefs and/or varying perception of the research purpose can lead to different interpretations. In this case, strategic integration (Bazeley, 2018) would be advisable to avoid unnecessary dissonance already from the beginning of the project. This would involve developing a thorough understanding of how each data source will contribute to the overall research question. All data are collected and analyzed with a clear purpose, subject to the broader research objective.

3 Summary and conclusion

In response to the limited use of mixed methods, this paper aims to illuminate opportunities and potential obstacles of mixed-methods research, provide a starting point to evaluators interested in pursuing mixed-methods approaches, and consequently promote their adoption in evaluations of financial education programs.

The first section explains why the use of mixed methods is advisable and presents the key design features and common designs of mixed-methods evaluations. Even when a mixed-methods approach is chosen, the full potential of the collected data is often not realized due to a lack of integration between qualitative and quantitative data and results. This issue arises partly because effective integration requires additional effort and partly due to the lack of practical guidelines on how to achieve integration.

Therefore, the second section introduces three techniques suggested in the literature: triangulation protocol, “following a thread”, and mixed-methods matrix. These techniques are demonstrated through their application in a financial education case study. While “following a thread” offers flexibility and depth for exploring particular themes, the triangulation protocol enhances validity and reliability through cross-verification, and the mixed-methods matrix excels in case-by-case analysis.

With mixed-methods approaches, evaluations can combine the best of both worlds - quantitative and qualitative research methods - by integrating approaches of data collection and analysis, allowing researchers to capture both numerical data and rich contextual insights.

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